

SIMATIC

S7-300, M7-300, ET 200M Automation Systems I/O Modules with Intrinsically-Safe Signals

Reference Manual

This manual is part of the
documentation package
with the order number:

6ES7398-8RA00-8BA0

Preface, Contents

Mechanical Configuration of an
Automation System with SIMATIC
S7 Ex Modules

1

SIMATIC S7 Ex Digital Modules

2

SIMATIC S7 Ex Analog Modules

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SIMATIC S7 HART Analog
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Safety Guidelines

This manual contains notices which you should observe to ensure your own personal safety, as well as to protect the product and connected equipment. These notices are highlighted in the manual by a warning triangle and are marked as follows according to the level of danger:



Danger

indicates that death, severe personal injury or substantial property damage will result if proper precautions are not taken.



Warning

indicates that death, severe personal injury or substantial property damage can result if proper precautions are not taken.



Caution

indicates that minor personal injury or property damage can result if proper precautions are not taken.

Note

draws your attention to particularly important information on the product, handling the product, or to a particular part of the documentation.

Qualified Personnel

The device/system may only be set up and operated in conjunction with this manual.

Only **qualified personnel** should be allowed to install and work on this equipment. Qualified persons are defined as persons who are authorized to commission, to ground, and to tag circuits, equipment, and systems in accordance with established safety practices and standards.

Correct Usage

Note the following:



Warning

This device and its components may only be used for the applications described in the catalog or the technical description, and only in connection with devices or components from other manufacturers which have been approved or recommended by Siemens.

This product can only function correctly and safely if it is transported, stored, set up, and installed correctly, and operated and maintained as recommended.

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We have checked the contents of this manual for agreement with the hardware and software described. Since deviations cannot be precluded entirely, we cannot guarantee full agreement. However, the data in this manual are reviewed regularly and any necessary corrections included in subsequent editions. Suggestions for improvement are welcomed.

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Preface

Purpose of the manual

The information contained in this reference manual will help you

- To plan,
- To install, and
- To commission

a SIMATIC S7 explosion-proof module for an automation system in a hazardous area.

Contents of the manual

The reference manual “*S7-300, M7-300, ET 200M Automation Systems I/O Modules with Intrinsically-Safe Signals*” provides you with technical descriptions of the individual modules.

The reference manual is sub-divided into the following topics:

Mechanical structure of an automation system with SIMATIC S7 explosion-proof modules	→	Sect. 1
SIMATIC S7 Ex Digital Modules	→	Sect. 2
SIMATIC S7 Ex Analog Modules	→	Sect. 3
SIMATIC S7 HART Analog Modules	→	Sect. 4

Not in this manual

Basic information on explosion protection and the use of intrinsically-safe modules can be found in the manual “S7-300, M7-300, ET 200M Automation Systems Principles of Intrinsically-Safe Design”, which is supplied in the same documentation package.

This manual is sub-divided into the following topics:

Introduction to explosion protection

Legal principles of explosion protection

Primary explosion protection

Secondary explosion protection

Marking of explosion-protected electrical apparatus

The intrinsic safety “i” type of protection

Installation, operation and maintenance of electrical systems in hazardous areas

Validity of the manual

This reference manual is valid for all the SIMATIC S7 explosion-proof modules listed by order number in the following table.

Table 1-1 S7-300 I/O modules

SIMATIC S7 I/O module	Purchase Order Number
SM 321; DI 4 x NAMUR	6ES7 321-7RD00-0AB0
SM 322; DO 4 x 24V/10mA	6ES7 322-5SD00-0AB0
SM 322; DO 4 x 15V/20mA	6ES7 322-5RD00-0AB0
SM 331; AI 8 x 4 x TC/ 4 x RTD	6ES7 331-7SF00-0AB0
SM 331; AI 4 x 0/4...20mA	6ES7 331-7RD00-0AB0
SM 332; AO 4 x 4...20mA	6ES7 332-5RD00-0AB0
SM 331; AI 2 x 0/4...20mA HART	6ES7 331-7TB00-0AB0
SM 332; AO 2 x 0/4...20mA HART	6ES7 332-5TB00-0AB0

Note

It is essential that you note the following information on the use and configuration of the S7-300 I/O modules listed in Table 1-1.

Usage and configuration

With the exception of the SM 331; AI 2 x 0/4...20mA HART module, you can use the I/O modules listed in Table 1-1:

- In the **S7-300** (centralized configuration) with CPU 312 IFM, level 5 onwards, CPU 313, level 3 onwards, CPU 314, level 6 onwards, CPU 314 IFM, level 1 onwards, CPU 315 and CPU 315-2 DP, level 3 onwards, CPU 614, level 6 onwards.
- In the **ET 200M** (distributed configuration) with the IM 153-1 from the order number 6ES7 153-1AA02-0XB0 onwards, and with the following DP masters: IM 308 C, V3.0 onwards, and CPUs S7-41x, level 2 onwards.

You can configure the I/O modules with

- STEP 7, version 3.0 onwards or COM PROFIBUS, version 3.0 onwards

Usage and configuration of HART module

You can use the I/O module HART analog input SM 331; AI 2 x 0/4...20mA HART

- in the **ET 200M** with the IM 153-2, order number 6ES7 153-1AA02-0XB0, and with the following DP masters: IM 308 C, V3.0 onwards, and CPUs S7-41x, level 2 onwards.

You can configure the HART analog module with

- STEP 7, version 4.02 onwards or COM PROFIBUS, version 3.2 onwards.

Further manuals required:

You require the following documentation in order to understand the present manual:

- **S7-300:** Hardware and Installation /70/, Module Specifications /71/ and Instruction List /72/
- **M7-300:** Hardware and Installation /80/, Module Specifications /71/
- **ET 200M:** Distributed I/O Device /140/
- **I/O Modules S7-300, M7-300, ET 200M:** Reference Manual /150/

Accessing information in the manual

The manual contains the following orientation aids in order to help you access special information:

- At the beginning of the manual there is a complete overall table of contents as well as a list of the figures and tables contained in the complete manual.
- The individual chapters have a column in the left-hand margin which summarizes the contents of the respective section.
- After the Appendices there is a glossary in which important technical terms used in the manual are defined.
- At the end of the manual there is a detailed index which enables you to find the desired information quickly.

Electronic manuals

You can also order the documentation as an electronic manual on CD-ROM. The order number of the CD-ROM is: 6ES7 398-8RA00-8AA0

Further support

Should you have any further questions on using the products described which are not answered in the manual, please contact the Siemens representative in your area.

If you have any questions or remarks on the manual itself, please fill out the questionnaire at the end of the manual and send it to the address shown on the form. Please also enter your personal evaluation of the manual in the questionnaire.

Siemens also offers a number of training courses to introduce you to the SIMATIC S7 automation system. Please contact your regional training center or the central training center in Nuremberg, Germany for details:

D-90327 Nuremberg, Tel. (+49) (911) 895 3154.

If you require the type file or the DDB file you can download these via modem from the Interface Center in Fürth under the number +49 (911) 737972, or you can order the files on diskette.

Up-to-date information

You can obtain up-to-date information on SIMATIC products from:

- the Internet under <http://www.ad.siemens.de/>

In addition, the SIMATIC Customer Support team offers you up-to-date information and downloads which you may find useful:

- on the Internet under <http://www.ad.siemens.de/simatic-cs>
- via the SIMATIC Customer Support Mailbox under the number +49 (911) 895 - 71 00

To dial the mailbox, you require a modem with a voltage range up to V.34 (28.8 Kbps) and parameters set as follows: 8, N, 1, ANSI, or you can dial in via ISDN (x.75, 64 KBit).

You call the SIMATIC Customer Support Hotline on +49 (911) 895 – 70 00 or send a fax to +49 (911) 895 – 70 02. You can also submit inquiries by electronic mail via the Internet or by using the mailbox mentioned above.

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Mechanical Configuration of an Automation System with SIMATIC S7 Ex Modules

1

In this chapter

SIMATIC S7 Ex modules can be used in the following systems:

- S7-300,
- M7-300,
- ET 200M.

You must therefore comply with the configuration guidelines as specified in the corresponding manuals for installation purposes. In addition, further reference guidelines for SIMATIC S7 Ex modules are provided in this chapter.

Chapter overview

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1.1 Fundamental Guidelines and Specifications

Note

Ex systems may only be installed by authorized personnel!

Approval

SIMATIC S7 Ex modules have [EEx ib] IIC approval. This means they are classified as associated apparatus and must therefore be installed outside hazardous areas. Intrinsically safe electrical apparatus for Zone 1 and 2 may be connected. The approval applies to all explosive gas mixtures of Groups IIA..IIC (see Manual: *“Principles of Intrinsically-Safe Design“*, Chapter *“Secondary Explosion Protection”*, *“Marking of Explosion- Protected Electrical Apparatus”* and *The Intrinsic Safety ”i” Type of Protection“*) Refer to the Certificates of Conformity (Appendix A) for the safety-relevant limits. In Appendix A you will also find explanations of the designations used.

FM approval

SIMATIC S7 Ex modules feature the following FM approvals (see Manual: *“Principles of Intrinsically-Safe Design“*, Chapter *“Regulations for Explosion Protection Outside the CENELEC Member States”*) :

FM CL I, DIV 2, GP A, B, C, D, T4

In compliance with these approvals, the modules can be used in areas which contain volatile flammable liquids or flammable gasses which are normally within closed vessels or systems, from which they can only escape under abnormal operating or fault conditions. The approval applies to all test gasses. A surface temperature no higher than 135 °C (T4) occurs at ambient temperatures of 60 °C.

Safety Extra-Low Voltage

SIMATIC S7 Ex modules must be operated with a ”safety functional extra-low voltage”. This means that only a voltage of $U < 60 \text{ V}$ must be applied to the modules even in the event of a fault. You will find more detailed information on the safety extra-low voltage in the specifications for the power supplies to be used.

All system components which can supply electrical energy in any form whatsoever must fulfill this condition. This includes in particular:

- The power supply module PS307. It fulfills this condition.
- The MPI interface. It fulfills this condition when all users operate with safety extra-low voltage. SIMATIC automation systems and programming units fulfill this condition.
- 115/230V modules. Even if they are used in another cell or in another programmable controller they must feature safety extra-low voltage on the system side (i.e. towards the backplane bus).

Any other power circuit (24V DC) used in the system must feature safety functional extra-low voltage. Refer to the corresponding specifications or consult the manufacturer.

Also bear in mind that sensors and actuators with external power supply may be connected to I/O modules. Also ensure a safety extra-low voltage is used in this case. Even in the event of a fault, the process signal of a 24V digital module must **never reach a fault voltage $U_m > 60V$** . This also applies to non-intrinsically safe components.

Note

All voltage sources, e.g. 24V internal load voltage supplies, 24V external load voltage supplies, 5V bus voltage, must be electrically interconnected such that no voltage additions to the individual voltage sources can occur even under conditions with differences in potential thus ensuring the fault voltage U_m cannot be exceeded. You can achieve this, for example, by referring all voltage sources in the system to the functional ground. Also refer to the instructions provided in the relevant manuals (see Foreword) for this purpose. The maximum possible fault voltage U_m in the system is 60V.

Minimum thread measure

A minimum thread measure of 50 mm must be maintained between connection terminals with safety functional extra-low voltage and intrinsically safe connections. In the process connector this is achieved by the use of a line chamber (refer to Section 1.2).

It is possible that the specified thread measure cannot be maintained in individual module components. In this case, you must use the spacer module DM 370 (refer to Section 1.3) which you must set such that it does not take up an address range. If you use the ET 200M Distributed I/O, you should read Section 1.5.

Also take care with regard to the wiring to ensure this specified spacing is maintained between intrinsically safe and non-intrinsically safe terminals.

Combined use of Ex and non-Ex I/O modules

Combined use is possible, however, the minimum thread measure between conductive parts of Ex and non-Ex assemblies must be maintained in all cases. As a rule, you must install spacer modules DM 370 between Ex and non-Ex modules. You must ensure strict separation of intrinsically safe and non-intrinsically safe conductors in the wiring system. They must be routed in separate cable ducts. Mixed operation can therefore not be recommended.

Partition

The Ex partition must be fitted to achieve the minimum thread measure of 50 mm between Ex and non-Ex modules when using the bus module of the active backplane bus.

Load current circuit

The Ex sensors and Ex actuators are powered either via the Ex modules or via their own intrinsically safe power supply modules (e.g. 4-wire transducers).

The Ex I/O modules receive their power supply via the backplane bus. The 24V DC load voltage input of the front connector is required for the power supply of the Ex sensors and the Ex actuators on the majority of modules.

Connecting Ex I/O modules

The Ex I/O modules are configured in the same way as standard modules from left to right. Connect the Ex sensors and Ex actuators as well as the load voltage supply with the aid of the line chamber (see Section 1.2) to the process connector which you then plug into the module.

Note

If necessary, safety assessment of this intrinsically safe power circuit should be carried out by an expert before a sensor or actuator is connected to an Ex module.

Replacing Ex I/O modules

After being plugged in for the first time, the front connector adopts the module type coding set at the factory. This ensures that there can be no confusion with another type of module when replacing modules as the front connector can then no longer be unclipped, thus fulfilling explosion protection requirements. When replacing Ex modules, carry out the necessary steps in the order described below:

- **Removal**

1. Disconnect L+ load voltage supply
2. Unplug front connector
3. Remove module

- **Installation**

1. Install module
2. Plug in front connector
3. Connect L+ load voltage supply

1.2 Line Chamber LK393

Scope of application

With the exception of the analog input module SM 331; AI 8 x Tc/4 x RTD, all Ex I/O modules require a 24V DC load voltage supply via the process connector. Safety isolation of this signal in order to maintain the minimum thread measure between Ex and non-Ex areas is achieved by using the line chamber LK 393 (Order No. 6ES7 393-4AA00-0AA0). Process signals are carried downward while the 24V supply is routed upward in separate ducts.

Connecting the line chamber

The lines of the L+ and M terminals are cut to the required length, their insulation is stripped and wire end ferrules are fitted. The conductor ends with the ferrules are passed through the openings in the line chamber LK 393 until they are flush with the fastening pins. The conductors are then pressed into the guide ducts of the line chamber LK 393 and routed upward (secure with hot-melt adhesive if necessary). The line chamber preassembled in this way is now inserted in the terminals of the front connector. The wire end ferrules of L+ and M are screwed to the terminals 1 and 20 and the fastening pins to terminals 2 and 19. This ensures firm connection of the line chamber with the front connector thus fulfilling explosion protection safety requirements.

Figs. 1-1, 1-2, 1-3 and 1-4 illustrate the configuration.

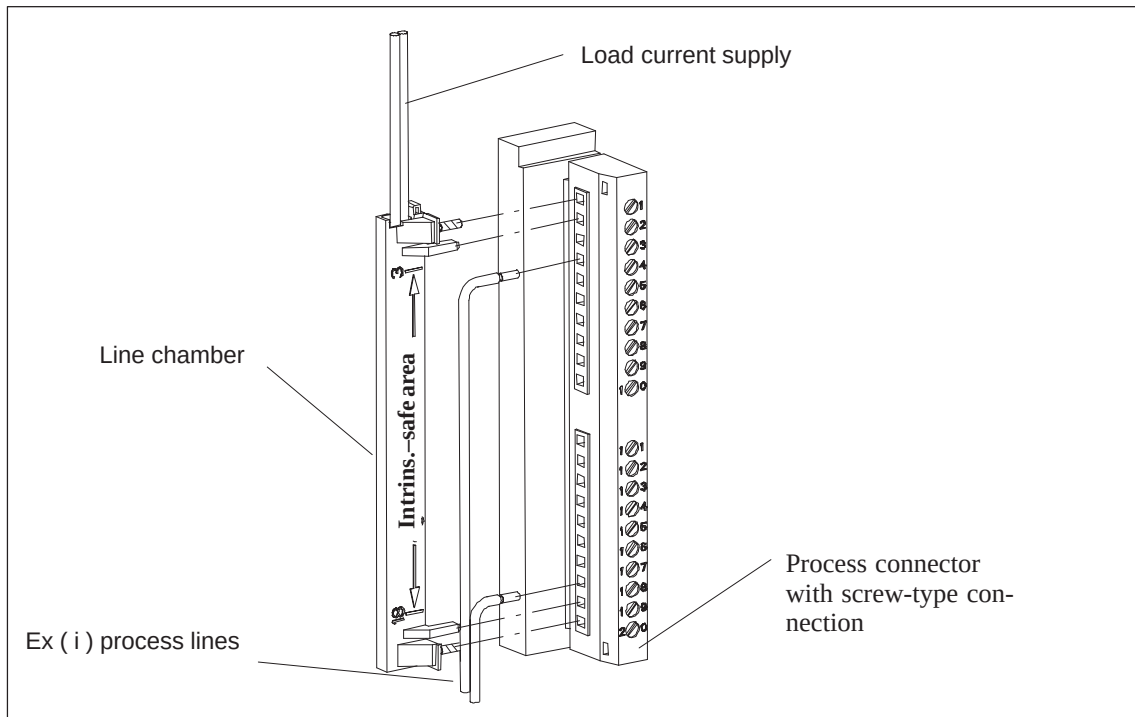


Fig. 1-1 Connecting the line chamber LK393

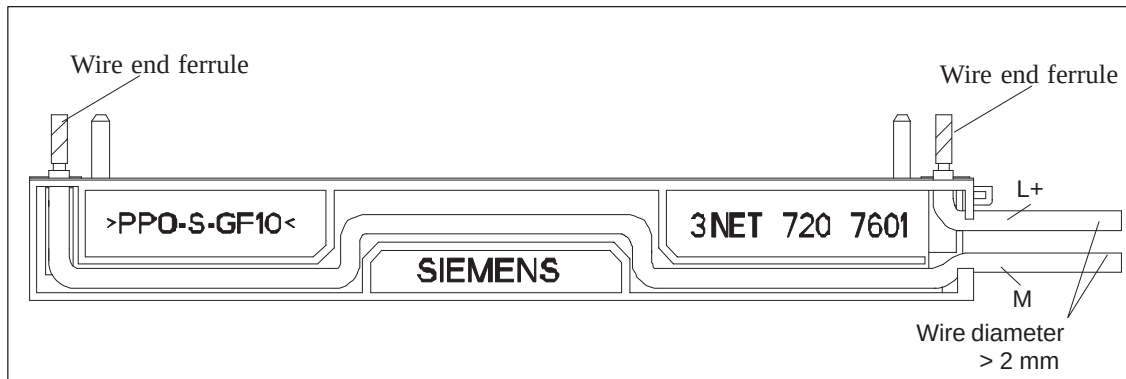


Fig. 1-2 Installing the connection lines for the load voltage in the line chamber.
Outside diameter of wires > 2 mm (view from below)

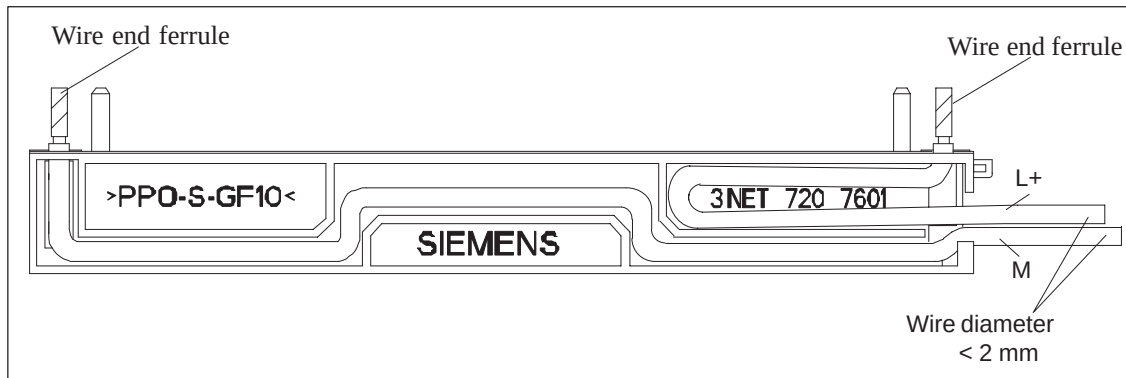


Fig. 1-3 Installing the L+ conductor in a loop in the line chamber.
Outside diameter of wires < 2 mm (view from below)

Note

Use Ex I/O modules which require a 24V load voltage only with the line chamber LK 393. It is necessary for ensuring the modules are used for their intended purpose.

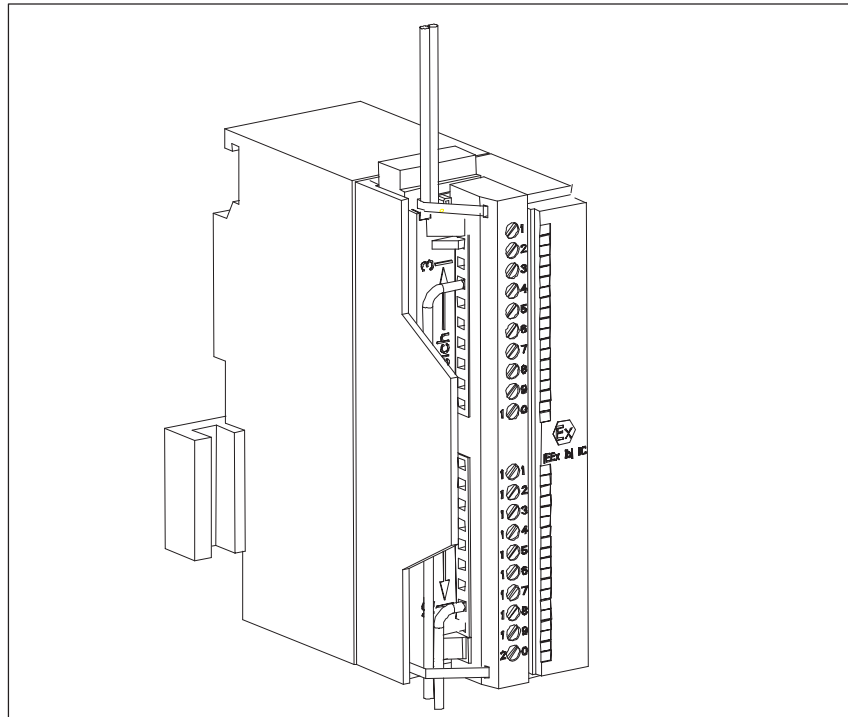


Fig. 1-4 Line chamber LK 393 when connected

You can, of course, also use Ex I/O modules for non-intrinsically safe tasks. You will not need the line chamber in this case. However, you must then clearly and permanently cancel the Ex identification symbol. Subsequent use for Ex applications is no longer possible unless you return the module to the manufacturer for testing.

1.3 Configuration of an S7-300 with Ex I/O Modules

Physical isolation of non-Ex signals from Ex signals corresponds to the requirements with regard to the configuration of explosion-protected automation technology. If the minimum distance of 50 mm between bare (uninsulated) terminals of Ex modules and bare (uninsulated) terminals of non-Ex modules can **not** be maintained, a spacer module DM 370 (order number 6ES7 370-0AA00-0AA0) must be fitted between these modules. Care must be taken to ensure that all automation systems are routed to a common ground.

This means:

- All earthing screws of the sectional rails must be referred to a common ground.
- The earthing clip of all CPUs must be locked in position.

Spacing for arrangement on several subracks

Fig. 1-5 shows the spacing dimensions between the individual subracks as well as to adjacent items of apparatus, cable ducts, cabinet panels etc. for a two-tier S7-300 configuration.

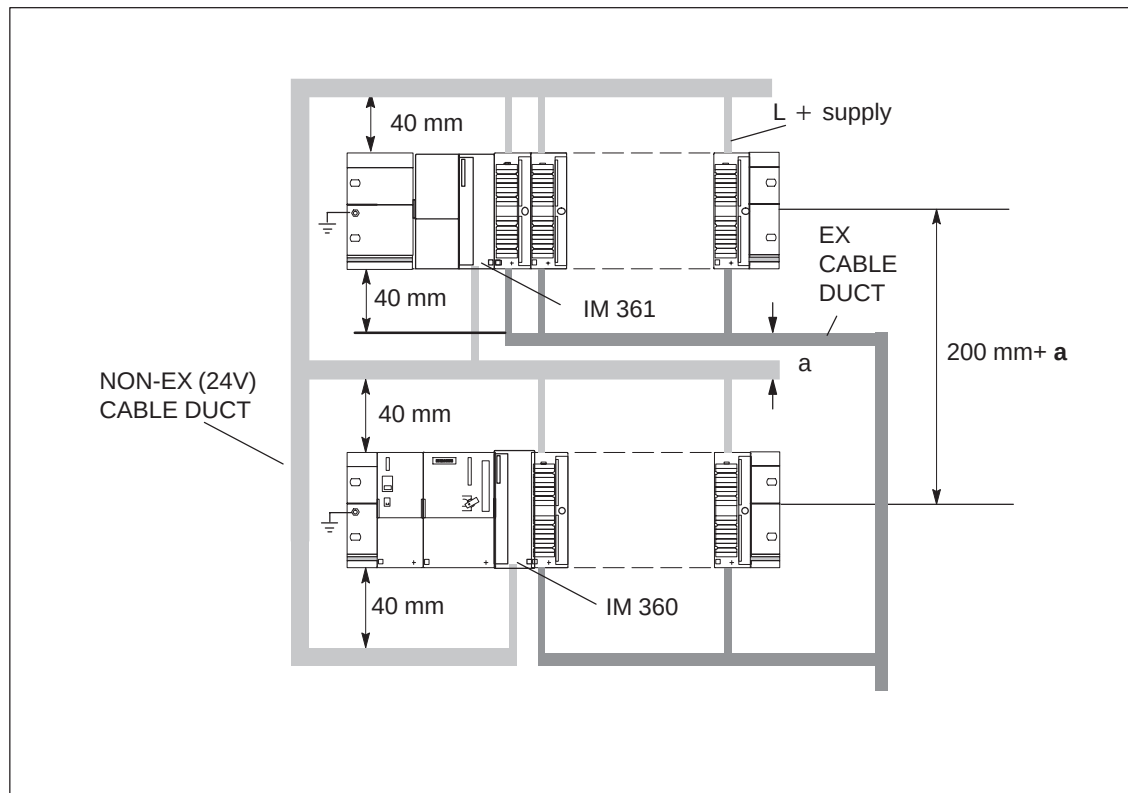


Fig. 1-5 Spacing dimensions for a two-tier S7-300 configuration

If you maintain these minimum spacing dimensions then:

- you will guarantee heat dissipation of the S7-300 modules
- you will have sufficient space to insert and remove the S7-300 modules
- you will have sufficient space for installing lines

Note

If you use a shield support element, the specified dimensions apply as from the lower edge of the shield support element.

The L+/M lines on the Ex modules can be wired directly or via connection elements.

For direct wiring, route the L+/M lines from the cable duct (if a line chamber is used, see Section 1.2) directly to the terminals of the module front connector. You can route the Ex process lines directly from the front connector to the items of apparatus.

You can use commercially available clamp-type distributors for wiring via connection elements. You then have the option of disconnecting the L+/M supply lines module by module by means of a plug connector (see Fig. 1-6).

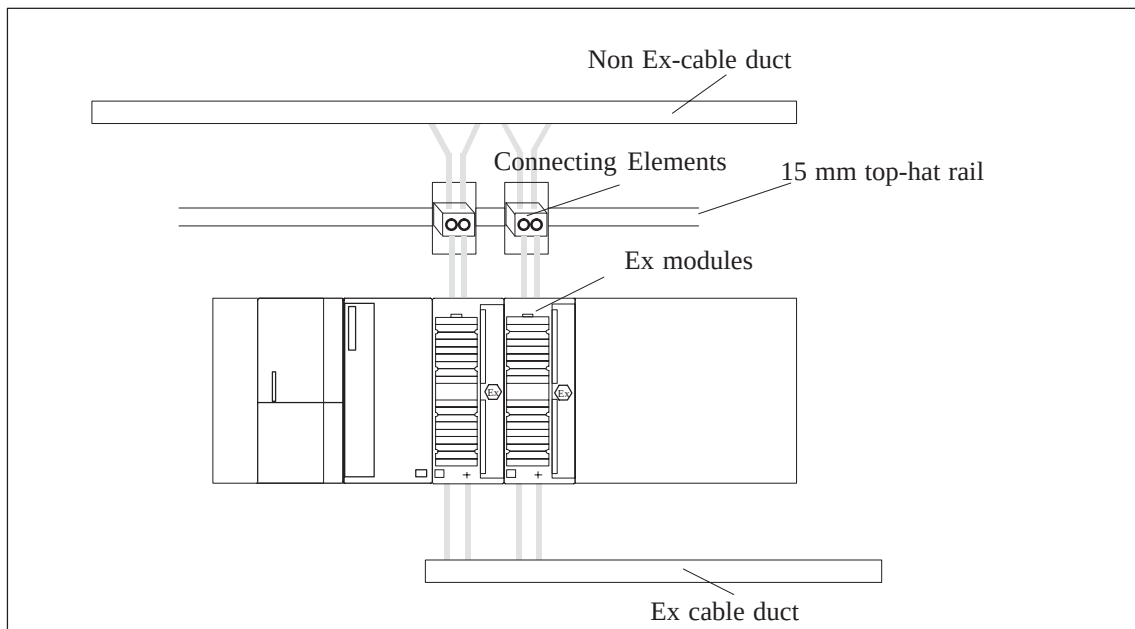


Fig. 1-6 Wiring between L+/M lines and Ex modules via connecting elements

1.4 Configuration of an M7-300 with Ex I/O Modules

Maximum configuration over four subracks

Fig. 1-7 shows an example of modules arranged in a four-tier M7 configuration.

The subrack 0 is equipped with power supply, central and interface module, a mass storage module MSM378 and up to 8 signal modules. All other subracks are each equipped with an interface module and up to 8 signal modules.

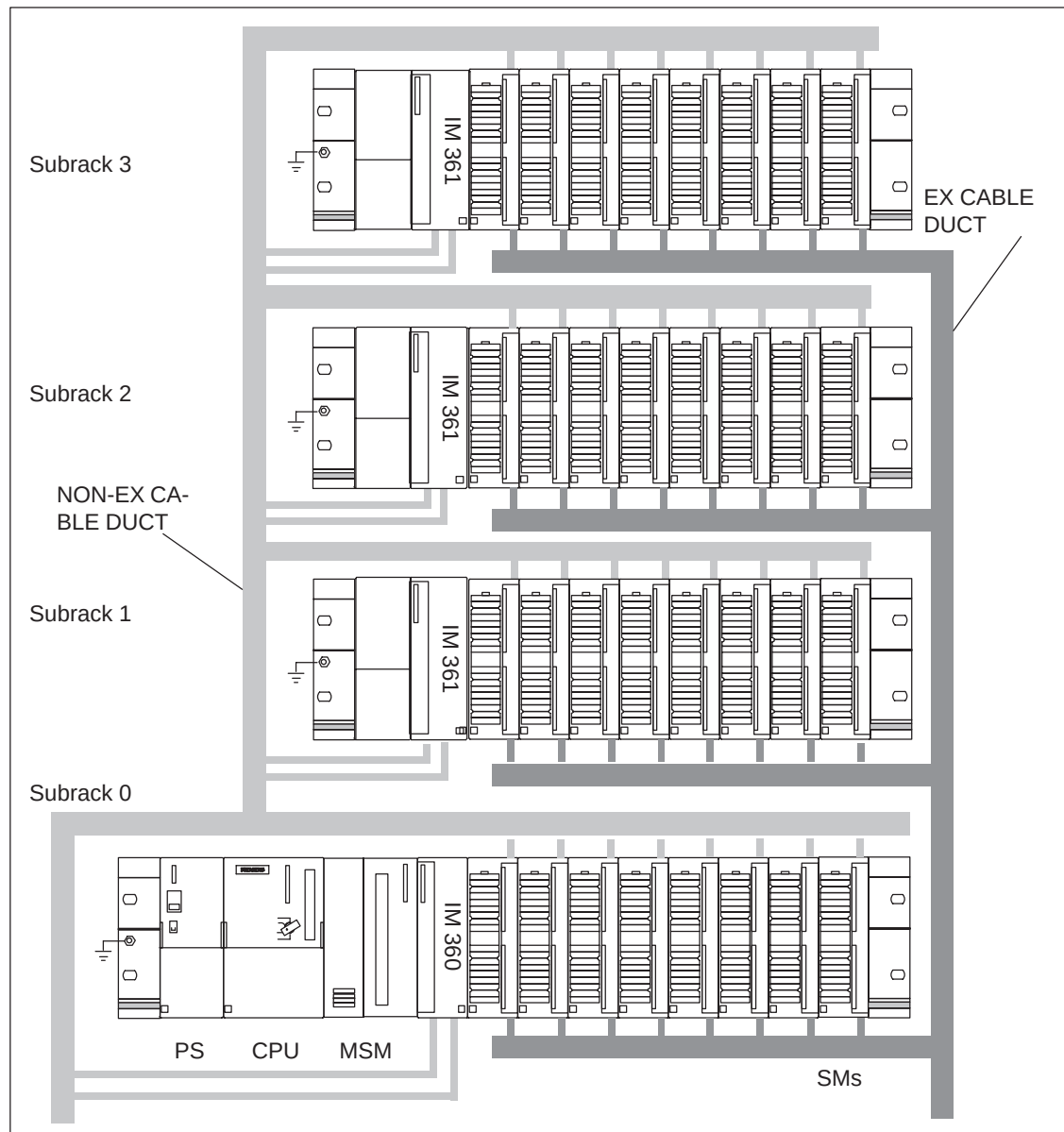


Fig. 1-7 M7-300 configuration over four subracks

1.5 Configuration of an ET 200M with Ex I/O Modules

ET 200M configuration over two subracks

Fig. 1-8 shows an example of two ET 200M configurations over two subracks. A dummy module DM 370 (6ES7 370-0AA01-0AA0) which is set such that it takes up no address space must be fitted between IM153 and the first Ex module. If the backplane bus is active, you should use the ex dividing panel/ ex barrier (Order number 6ES7 195-1KA00-0XA0) instead of the dummy module.

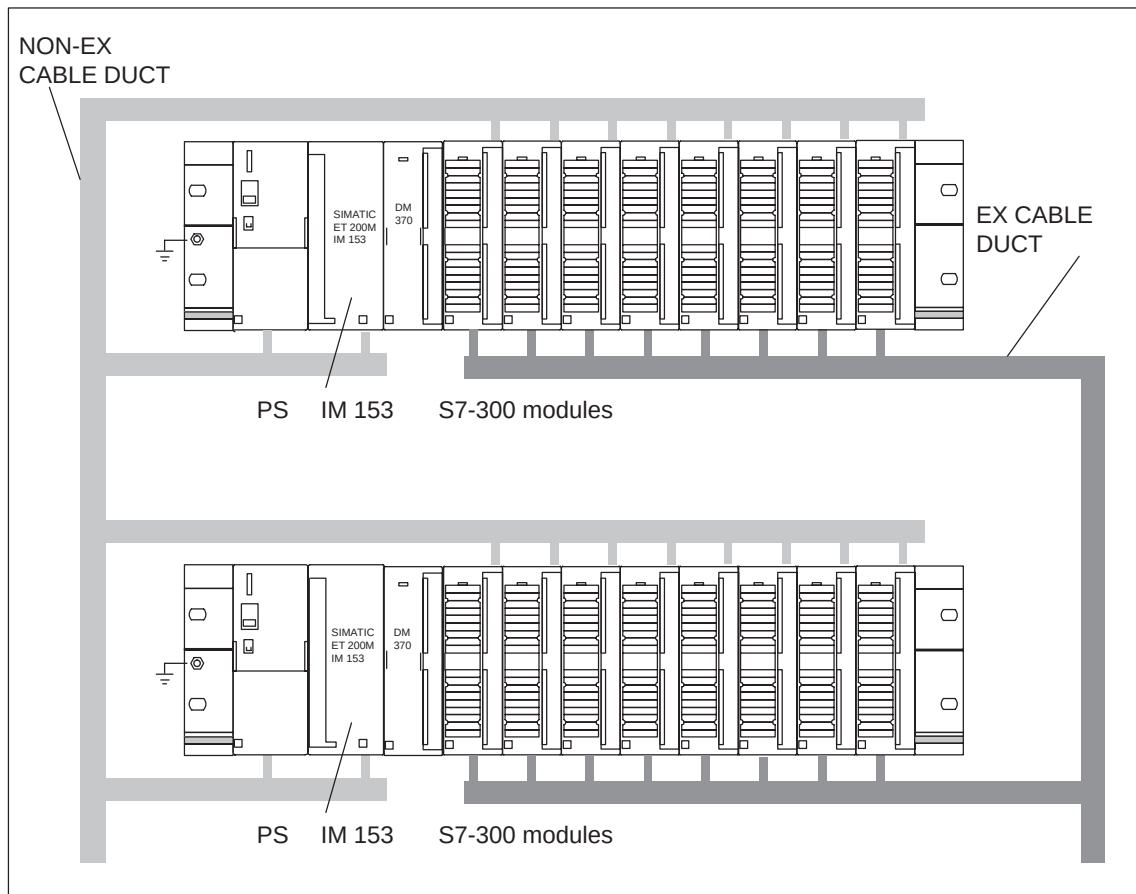


Fig. 1-8 Two subracks with ET 200M

1.6 Equipotential Bonding in Systems with Explosion Protection

Differences in potential can develop between the elements of electrical apparatus, connected with PE conductors, and conductive structural elements, piping etc. which does not pertain to the electrical apparatus. When implementing measures to bridge these differences in potential, sparks capable of causing ignition can be produced. To equalize the potentials, conductive metal parts which are accessible and can be touched must be connected to each other and to the PE conductor. Equipotential bonding with the PE conductor can be best implemented at the distribution board. The cross section of the bonding conductor must be at least that of the PE conductor. In all other cases, the equipotential bonding conductor must have a cross section of at least 10 mm² of copper.

The Ex modules feature metallic isolation between the backplane bus and the I/O circuit; there is therefore no need for connection to the equipotential bonding conductor. An exception is when a connection to the EB conductor must be made for measurement purposes. Where lightning protection devices are required in the intrinsically safe circuit (Section 1.9), they must be connected to the EB conductor at the same point as the shield of the intrinsically safe circuits.

Generally, the measures described in DIN VDE 0165 (Table 1-1) should be implemented or complied with.

Cable racks must be incorporated throughout the earthing system.

Equipotential bonding in a building

In accordance with VDE 0100, Part 410 and Part 540 and DIN VDE 0185, equipotential bonding must be provided in every building and via the overall cabling of the automation system; if this is not the case, it must be installed.

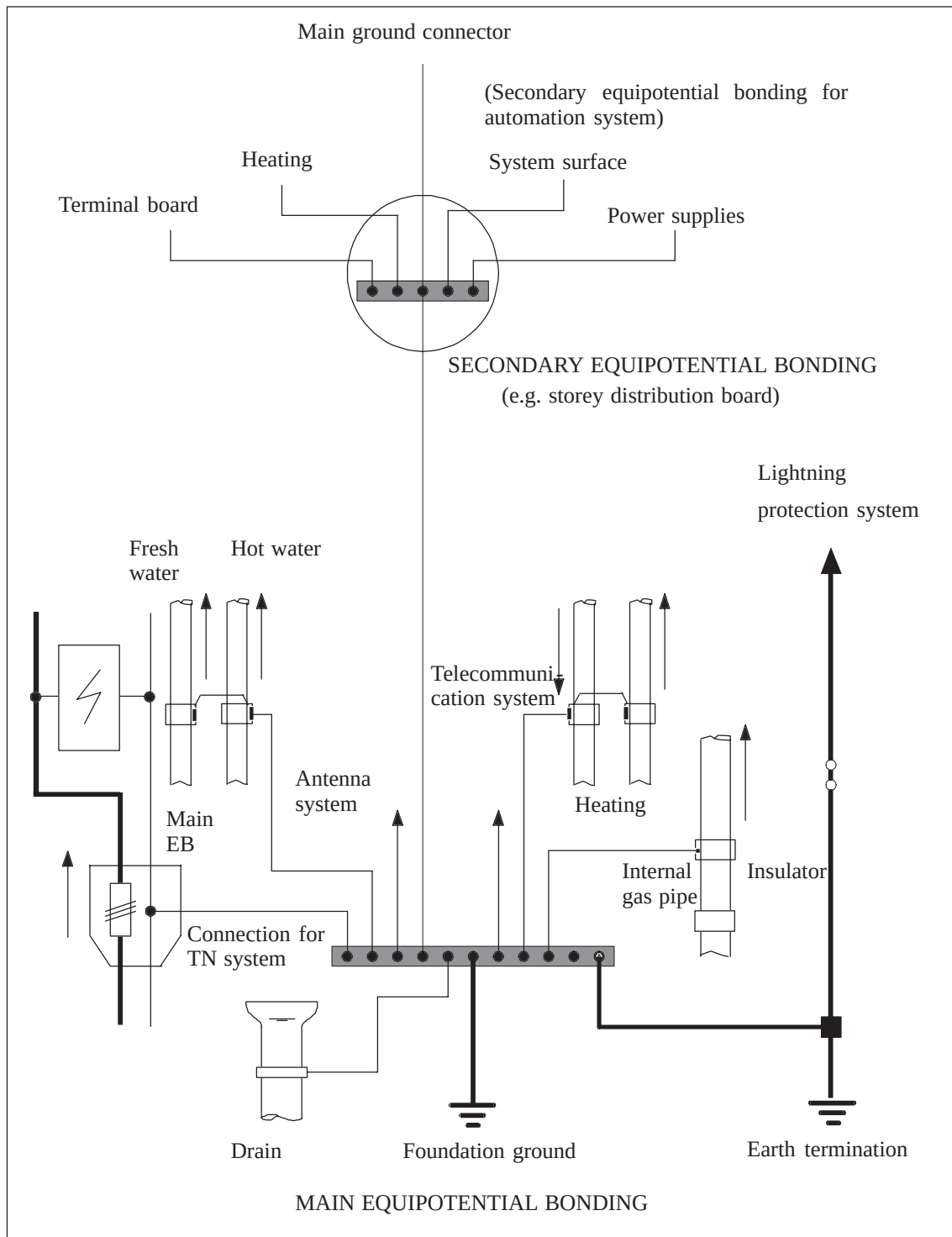


Fig. 1-9 Main and secondary equipotential bonding in accordance with VDE

Main equipotential bonding

This interconnects the following conductive elements by the EB conductor on the EB bus: $A_{PA} = 0.5 \times A_{PE-main}$

- Main PE conductor
- Main ground conductor
- Earth termination
- Main water pipes
- Main gas pipes
- Other metal piping systems
- Metal structural elements of the building (if possible)
- Power and information system cables extending beyond the building, via lightning conductor.

Additional equipotential bonding

This interconnects the following conductive elements by the EB conductor on the EB bus:

- All "extraneous conductive elements" such as structural elements, supports, containers, piping (these themselves can form EB conductors), $A_{PA} = 0.5 \times A_{PEmax}$ (A = cable cross section) from the distrib. board.
- Elements of stationary electrical apparatus which are accessible to simultaneous contact when it is connected to PEN (otherwise a PE connection is sufficient), $A_{PA} = 0.5 \times A_{PE}$ of both items of apparatus.

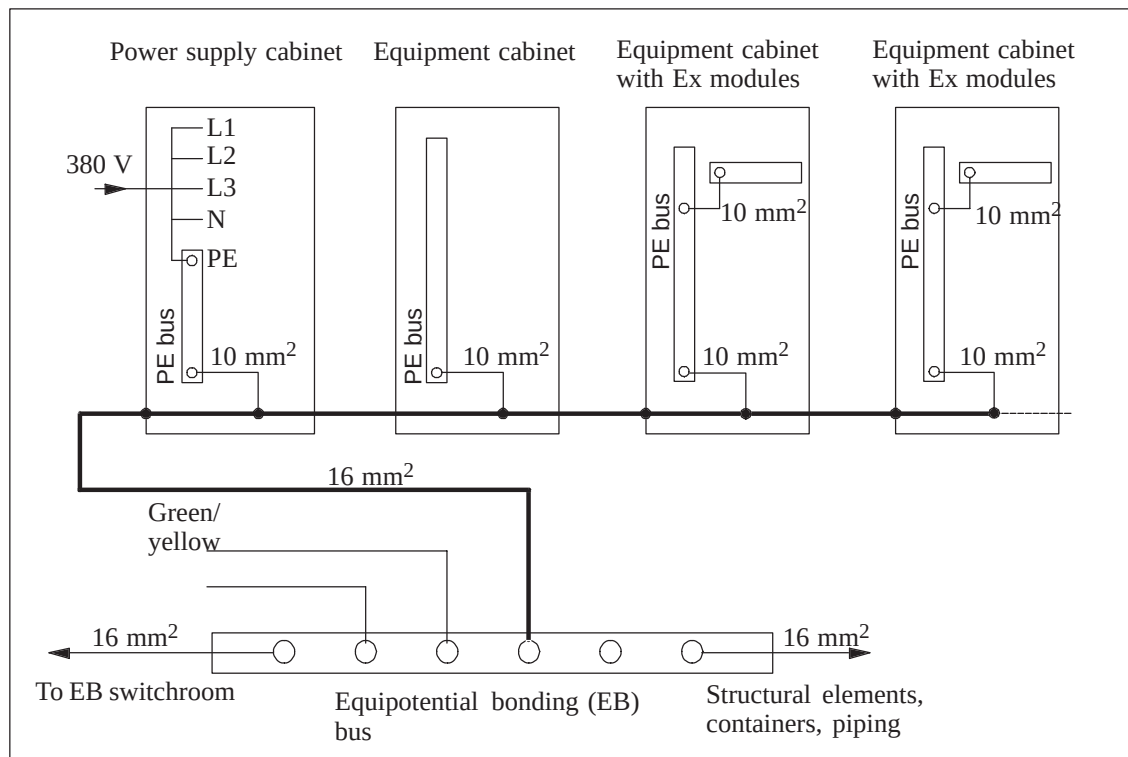


Fig. 1-10 Example of equipotential bonding for measurement and control systems

1.7 Wiring and Cabling in Ex Systems

Neither the electrical installation nor the required materials such as cables, lines and installation hardware are subject to the special test procedure of ExV with respect to their design. The responsibility of plant personnel or of an installation company for proper installation of an Ex system is particularly high, on account of the risk of explosion in the event of improper implementation.

General planning principles for cable routes are very similar to those for piping. At the drafting stage of installation plans and building layouts, areas with increased risk of fire and danger zones must be defined in accordance with ExV and VbF. Cable and piping routes should preferably be arranged only in the area of low risk. Furthermore, accessibility and ease of maintenance must be ensured, also for subsequent expansion. With all types of switchroom, steps must be taken to ensure that the cable and line routes to the hazardous area are sealed so that they do not provide escape routes for hazardous gasses or vapors to the switchroom.

Note

Laying cables in ducts in the floor should be avoided. There is a risk of

- the ingress or formation of explosive gas-air mixtures and their uncontrolled propagation;
 - the ingress of corrosive liquids.
-

In order to create intrinsically safe circuits, non-sheathed cables and single conductors in flexible cables need only have a diameter of ≥ 0.1 mm. For implementation in the Ex area, cables and lines must primarily withstand the expected mechanical, chemical and thermal effects. It is therefore always necessary to lay considerably larger cross sections and use cables and lines which are flame-retardant and oil-resistant.

Intrinsically safe and non-intrinsically safe lines (conductors, non-sheathed cables) must be laid separately or with appropriate insulation. Common routing in cables, lines and conductor bundles is not permissible.

Special care must be taken to ensure full isolation in cable ducts. This can be achieved with a continuous intermediate 1 mm layer of insulating material or by laying sheathed cables (Fig. 1-11).

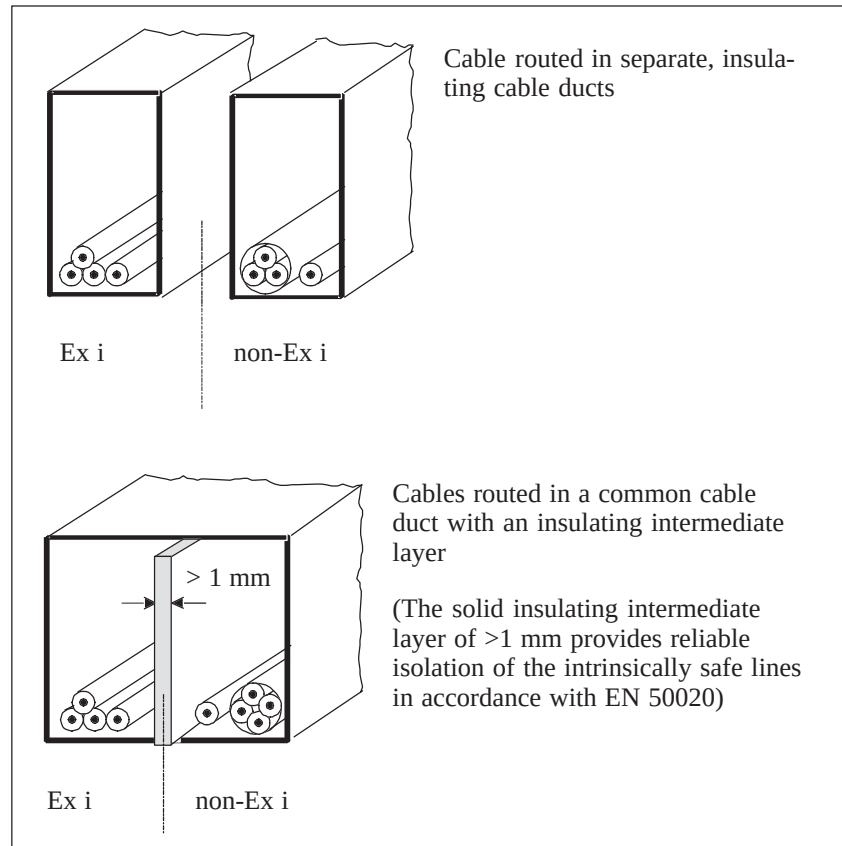


Fig. 1-11 Routing of cables for intrinsically safe circuits

Where sheathed cables of intrinsically safe and non-intrinsically safe circuits are routed together, the sheathed cable of the intrinsically safe circuit must withstand a minimum test voltage of 1500 V_{rms} AC.

The high test voltage of 1500 V AC can be dispensed with if the intrinsically safe or non-intrinsically safe circuits are enclosed in a grounded (earthed) shield. However, the test voltage of the lines for intrinsically safe circuits must be at least 500 V AC (between conductor-conductor-ground).

Intrinsically safe lines must be clearly marked. If a color is used, it must be light-blue. An exception to this rule is the routing of lines within equipment, distribution panels and switchrooms. Cables and lines thus marked must not be used for other purposes.

In general, intrinsically safe circuits must be installed in a floating arrangement. A connection to ground via a 15 kOhm resistor, e.g. to discharge electrostatic charges, does not qualify as a ground. Intrinsically safe circuits must be grounded when this is required for measurement or safety reasons. Grounding may only take place at one point by connection to the equipotential bonding conductor. Equipotential bonding must be provided throughout the entire installation area of intrinsically safe circuits.

In systems with intrinsically safe and non-intrinsically safe circuits, such as measurement and control cabinets, the connection elements must comply with the specifications of DIN EN 50020/VDE 0170/0171 Part 7/05.78, 5.4.1. The terminals of the intrinsically safe circuits must be marked as intrinsically safe. If a color is used, it must be light-blue.

1.7.1 Marking of Cables and Lines of Intrinsically Safe Circuits

Cables and lines of intrinsically safe circuits must be marked. Where jackets or sheaths are color-coded, light-blue must be chosen as the color. Cables and lines thus marked must not be used for other purposes. Equalizing conductors for thermocouples with a plastic sheath may be provided with colored longitudinal stripes as follows, according to the type of thermocouple:

- | | |
|---|-----------|
| • Copper/cupro-nickel (copper/constantan) | brown |
| • Iron/cupro-nickel (iron/constantan) | dark blue |
| • Nickel-chrome/nickel | green |
| • Platinum-rhodium/platinum | white |

In the case of equalizing conductors for thermocouples with a mineral sheath or metal braid, a light-blue strip of sufficient width must be woven in as the color code for intrinsic safety.

Within measurement and control cabinets and in the interior of switching and distribution systems, special measures must be taken where there is a risk of interchanging the lines of intrinsically safe and non-intrinsically safe circuits, e.g. where there is a blue neutral conductor in compliance with DIN 47002.

The following measures are acceptable:

- Bundling of conductors in a common light-blue sheath,
- Labelling,
- Clear arrangement and physical separation.

1.7.2 Wiring and Cabling in Cable Bedding Made of Metal or in Conduits

Cable bedding made of metal must be incorporated in the protective measures to counteract indirect contact. This can be achieved by routing an existing ground conductor made of steel strip or with a good conductive connection between individual beds.

For single laying, conduits made of metal are now only usually used where particular mechanical or thermal stress is developed. In general, PVC conduits of two different types are used depending on the expected mechanical stress. Remember, however, that PVC exhibits a linear expansion which is about 8 times of that of metal. The fixing points must therefore be such that the linear expansion is taken up.

1.7.3 Summary of Requirements of DIN VDE 0165/02.91

The following table provides, once again, an overview of the most important stipulations of DIN VDE 165/02.91 for cables and lines.

Table 1-1 Contents of DIN VDE 0165/02.91

Application	Requirements of cables and lines	
General requirements: Observe additional requirement for "i" and zone 0 (smaller cross section permissible for multicore lines with more than 5 cores, and lines for measurement and control, for example)	- Select according to mechanical, chemical and thermal influences (refer to DIN VDE 0298 and DIN VDE 0891) - Protect against fire spread (e.g. lay cables in sand; verify burning characteristics of lines in accordance with VDE0472 Part 804, test type B) - Copper or aluminum conductor material (Al only for multicore cables from 25 mm² or single-core cables from 35 mm²; use suitable connection elements) - Minimum cross sections for copper conductor: Single-core cable: 1 mm fine, 1.5 mm solid conductor Multicore cable: 0.75 mm fine, otherwise as above	
Permissible types for portable/mobile apparatus (does not apply to intrinsically safe systems)	- U ≤ 750 V - U ≤ 250 V - I ≤ 6 A - In measurement and control systems, telecommunication and telecontrol systems	TRS flexible cable H07RN or equivalent (e.g. NSHou) TRS flexible cable H07RN or equivalent (see Section 1.7.5) No severe mechanical stress Plastic-sheathed flexible cable H05VV-F min. cross section 1 mm² (not at ambient temperature below 5 °C)

Table 1-1 Contents of DIN VDE 0165/02.91, continued

Application	Requirements of cables and lines
Laying of cables and lines	• Lead-ins from Ex areas to non-Ex areas tightly sealed, e.g. with sand, mortar or similar • Unused inlets sealed with certified sealing plugs (certificate not required for zone 2) • Where there is particular thermal, mechanical or chemical stress, protect cables and lines, e.g. by laying in conduit, sheaths, metal tubing (not in enclosed conduits) • Where routed into Ex-proof enclosure, use certified cable lead-in elements.
Connection of cables and lines	• Conductor connections on the exterior of apparatus should only be crimped • Conductor connections within apparatus should use suitable clamps, multicore or fine conductor ends should be secured against separation • Crimp connections can be protected with resin fittings or shrink sleeving if they are not mechanically stressed.

1.7.4 Selecting Cables and Lines in Accordance with DIN VDE 0165

In compliance with ElexV, cables and lines laid in hazardous areas do not require a test certificate. All types which are suitable for the specific purpose may be used if they comply with the standards stipulated in DIN VDE 0165, Item 5.6. The electrical characteristic data (e.g. capacitance 200 nF/km, inductance 1 mH/km) must be specified for cables used in intrinsically safe measurement and control circuits.

The following applies within a group cable:

The insulation between lines of intrinsically safe and non-intrinsically safe circuits must withstand an alternating voltage of $2U + 1000$ V, but at least 1500 V, where U is the sum of rms voltage values of the intrinsically safe and non-intrinsically safe circuits.

Table 1-2 Minimum cross sections of copper conductors in accordance with DIN VDE 0165

Cable type	Number of cores	Flexible stranded conductor mm ²	Solid conductor mm ²	Conductor diameter mm
Power cables and lines in accordance with DIN VDE 0298, Part 1, 3	1 2 - 5 > 5	1 0.75 0.5	1.5 1.5 1	-
Wiring cables and lines in accordance with DIN VDE 0891, Parts 1, 5, 6 for voltages	> 1	0.5	0.5	0.8
< 60 V AC or < 120 V DC	2 > 2 2 (shielded)	0.5 0.28 0.28	0.5 0.28 0.28	0.8 0.6 0.6

1.7.5 Types of Cable

The cables suitable for process signals are wiring cables for industrial electronics (SIMATIC cables) with twisted pairs of color-coded bundled conductors. Cables with a solid conductor (0.5 mm² cross section, 0.8 mm diameter) have a static shield. Cables with flexible conductors (J-LIYCY) have a braided shield (C) made of copper wires.

Table 1-3 Types of cables

Cable designation		Cable for
A-Y(St) YY	nx2x0.8/1.4 BdSi	Outdoor cable (for burying in ground ¹⁾)
J-Y(St) Y	nx2x0.8/1.4 BdSi	Normal applications
J-LiYY	nx2x0.5/1.6 BdSi	Compact control stations
J-LiYCY	nx2x0.5/1.6 BdSi	Vibration and impact stress Connectors

¹⁾ Direct burying in ground is not recommended.

Type designations for lines in accordance with harmonized standards

The type designations for lines in accordance with harmonized standards are listed in the following:

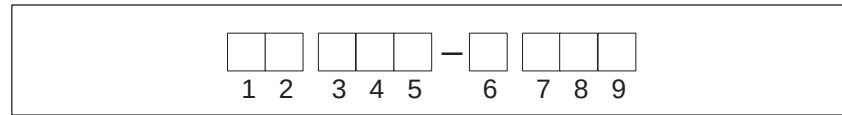


Fig. 1-12 Type designations for lines in accordance with harmonized standards

1 Basic type	H	Harmonized type
	A	National type
2 Rated voltage	03	300/300 Volt
	05	300/500 Volt
	07	450/750 Volt
3 Insulating material	V	PVC
	R	Rubber
	S	Silicon rubber
4 Sheath material	V	PVC
	R	Rubber
	N	Cloroprene rubber
	J	Glass fiber braid
	T	Fabric braid
5 Special features	H	Ribbon cable, separable
	H2	Ribbon cable, notsepar.
6 Conductor pipe	U	Solid
	R	Stranded
	K	Fine wire (permanently installed)
	F	Flexible stranded
	H	Extra fine
	Y	Tinsel
7 Number of cores	...	Number of cores
8 Protected conductor	X	Without protective conductor
	G	With protective conductor
9 Conductor cross section	...	Specified in mm ²

Type designations of telecommunication cables and lines

Type designations for telecommunication cables and lines are listed in the following:

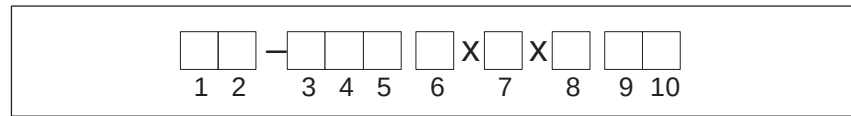


Fig. 1-13 Type designations for telecommunication cables and lines

1 Basic type	A Outdoor cable
	G Mining cable
	J Wiring cable
	L Flexible sheathed cable
	S Switchboard cable
2 Type supplement	B Lightning prot. system
	J Induction-protected
	E Electronics
3 Insulating material	Y PVC
	2Y Polyethylene
	O2Y Cellular PE
	5Y PTFE
	6Y FEP
	7Y ETFE
	P PAPER
4 Design features	F Petrolatum filler
	L Aluminium sheath
	LD Corrugated alum. sheath
	(L) Aluminium tape
	(ST) Metal foil shield
	(K) Copper tape shield
	W Corrugated steel sheath
	M Lead sheath
	Mz Special lead sheath
	B Armouring
	C Jute sheath + compound
	E Compound layer + tape (see 3. Insulation)
5 Sheath material	
6 Number of elements	n Number of stranding elements
7 Stranding element	1 Single core
	2 Pair
8 Conductor diameter	... in mm
9 Stranding element	F Star quad (railway)
	St Star quad (phantom)
	St I Star quad (long-d. cable)
	St III Star quad (local cable)
	TF Star quad for CF
	S Signal cable (railway)
	PiMF Shielded pair
10 Type of stranding	Lg Layer stranding
	Bd Unit stranding
11 Sheath color	BL blue

Table 1-4 Siemens cables for measurement and control to DIN VDE 0815

Cable designation		Order number
JE-LIYCY	2x2x0.5 BD SI BL	V45483-F25-C15
JE-LIYCY	16x2x0.5 BD SI BL	V45483-F165-C15
JE-LIYCY	32x2x0.5 BD SI BL	V45483-F325-C55
JE-Y(ST)Y	2x2x0.8 BD SI BL	V45480-F25-C25
JE-Y(ST)Y	16x2x0.8 BD SI BL	V45480-F165-C35
JE-Y(ST)Y	32x2x0.8 BD SI BL	V45480-F325-C25
JE-Y(ST)Y	100x2x0.8 BD SI BL	V45480-F1005-C15

Characteristic values of cables for intrinsically safe circuits

Example: Cable type JE-LiYCY

Coupling:	200 pF/100 m	at 800 Hz
Working capacitance	approx. 200 nF/km	at 800 Hz
Working inductance	approx. 1 mH/km	
Minimum bending radius for permanent installation:	6 x line diameter	
Temperature range, permanent installation:	- 30 °C to 70 °C	
for moveable use:	- 5 °C to 50 °C	
Test voltage:	Core/core 2000 V, Core/shield 500 V	
Loop resistance:	approx. 80 Ω/km	

1.7.6 Requirements of Terminals for Intrinsically Safe Type of Protection

These must be identifiable, for example by their type designation, and the following constructional requirements must be observed:

- Clearance in air and leakage path in accordance with EN 50014/EN 50020 between two connection elements of different intrinsically safe circuits must be at least 6 mm.
- Clearance in air and leakage path between connection elements of each intrinsically safe circuit and grounded metal parts must be not less than 3 mm.
- Marking of connection elements must be unambiguous and easily recognized. When a color is used for this purpose, it must be light blue.

The following must also be observed with regard to the use of terminals:

Connection terminals of intrinsically safe circuits must be at a distance of at least 50 mm from connection elements or bare conductors of any non-intrinsically safe circuit, or must be isolated from it by an insulating partition or grounded metal partition. When such partitions are used, they must extend at least by up to 1.5 mm from the housing panels, or must ensure a minimum clearance of 50 mm between connection elements, measured about the partition in all directions.

The insulation between an intrinsically safe circuit and the chassis of the electrical apparatus or parts which may be grounded must withstand an alternating rms voltage of twice the voltage value of the intrinsically safe circuit, but at least 500 V.

1.8 Shielding and Measures to Counteract Interference Voltage

Shielding

Shielding is a method of attenuating magnetic, electric or electromagnetic interference fields. Shielding can be subdivided into

- Equipment shielding
- Line shielding

1.8.1 Equipment Shielding

Particularly observe the following when cabinets and housings are incorporated in control system shielding:

- Cabinet covers such as side panels, rear panels, top and bottom panels, must make contact in an overlapping arrangement at adequate distances (e.g. 50 mm).
- Doors must be given additional contact with the cabinet ground. Use several grounding strips.
- Lines exiting the shielded housing should either be shielded or routed via filters.
- Where the cabinet contains sources of severe interference (transformers, lines to motors, etc.), they must be partitioned from sensitive electronic areas with metal plates. The metal plates must have several low-impedance bolted joints to the cabinet ground.

Interference voltages picked up in the programmable controller via non-Ex signal and supply lines are diverted to the central ground point (standard sectional rail).

The central ground point should have a low-impedance connection to the PE conductor via a copper conductor ($\geq 10 \text{ mm}^2$) which is as short as possible.

1.8.2 Line Shielding

Non-Ex circuits

As a rule, shielded lines must always be given a good electrical connection to cabinet potential at each end. Satisfactory suppression of all frequencies picked up can only be achieved by shielding at both ends.

Ex circuits

Three aspects must be considered with regard to the design of shielding and grounding of an S7-300 system:

- Ensuring electromagnetic compatibility (EMC)
- Explosion protection
- Person protection

Shielding of systems with optimum equipotential bonding

With regard to the electromagnetic compatibility of the systems, it is important that the system components and, in particular, the lines which connect the components are shielded and that these shields form a complete electrical enclosure wherever possible without gaps. The significance of this requirement increases with the scope of signal frequencies processed in the systems. In ideal cases, the cable shields are connected to the housings which are often metal (or corresponding shielding) of the connected field devices. Since, as a rule, they are linked to chassis ground (or to the PE conductor), the shield of the signal cable is grounded at several points. This optimum procedure with regard to electromagnetic compatibility and personal protection can be applied in these systems without any restrictions.

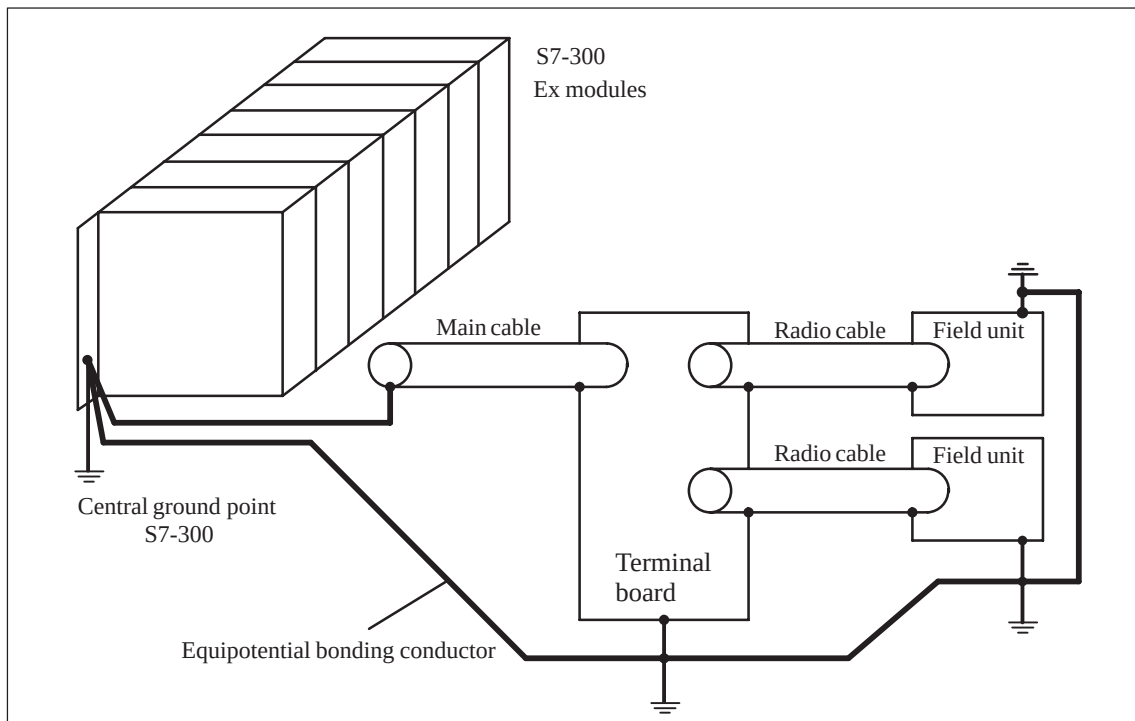


Fig. 1-14 Shielding and equipotential bonding conductors

**Shielding of
intrinsically safe
signal lines**

In Section 5.3.3, DIN VDE 0165 stipulates general equipotential bonding in hazardous areas to avoid different potentials and sparking as a result. Equipotential bonding must be rated and implemented as laid down in DIN VDE 0100.

**Grounding system
of intrinsically safe
circuits**

In accordance with DIN VDE 0165, Section 6.1.3.3, intrinsically safe circuits are generally not grounded. However, they must be grounded if required for safety reasons. They also may be grounded if required for functional reasons. Grounding may only take place at one point by connection to the equipotential bonding conductor.

Intrinsically safe signal lines and cables are shielded for measurement reasons and in order to avoid inductive coupling as, in most cases, no signal level is applied.

The following procedure must be implemented in the planning of the equipotential bonding with intrinsically safe signal lines:

- Metallic housings whose mounting arrangement provide reliable contact to structural components do not require a separate ground as they are incorporated in the equipotential bonding arrangement of the system.
- The shielding is grounded at only one point in order to avoid looping. This is implemented for systems of Zone 1, 2 and 11 outside the hazardous area, preferably in the control room.

The shield must be insulated at the device in the hazardous zone. The measured value is routed via a twisted pair signal line (single cable) to a distribution board and via a multiple cable to the control room. The shield is insulated at all intermediate points.

In zone 0, the shield is connected directly at the apparatus adapter box (mostly zone 1) to the general equipotential bonding system. The apparatus is grounded directly via the ground conductor.

Shielding of lines

Fig. 1-15 shows the shielding of Ex lines:

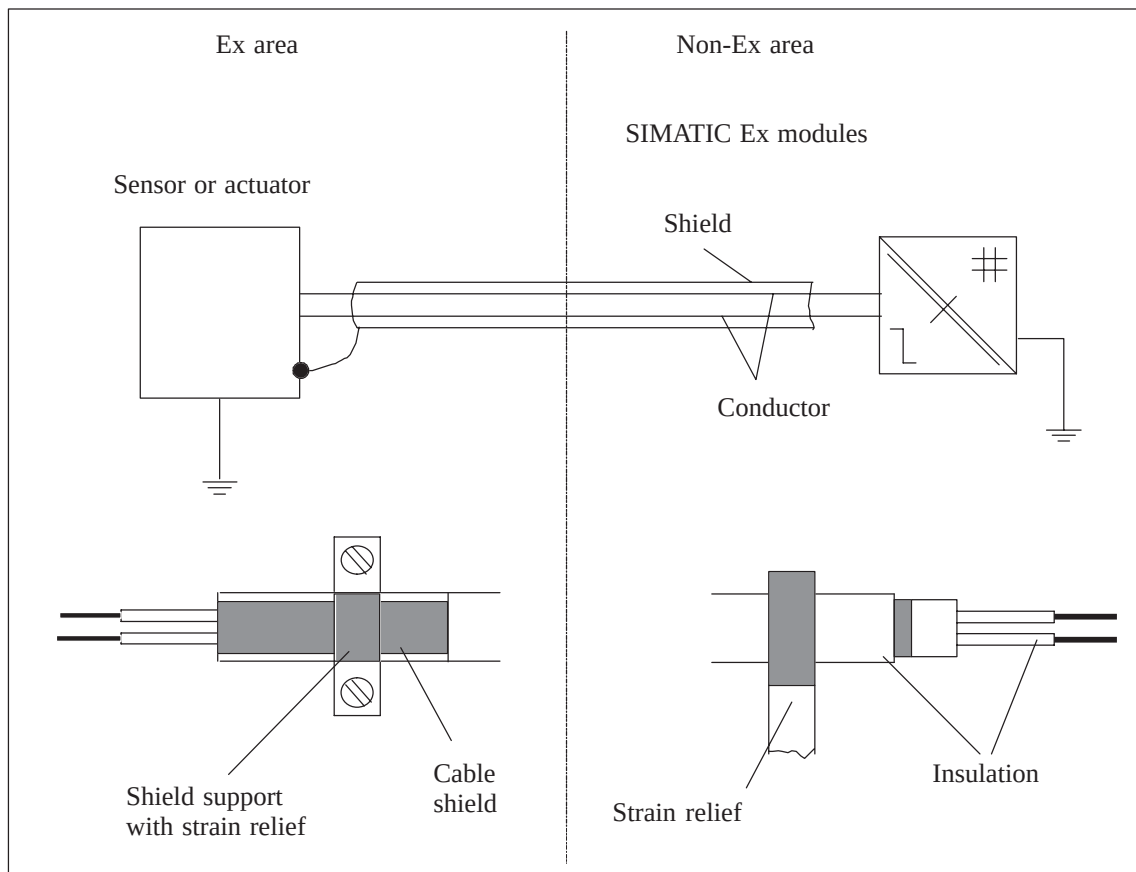


Fig. 1-15 Shielding of Ex lines

1.8.3 Measures to Counteract Interference Voltages

Measures to suppress interference voltages are often only implemented when the control system is already in operation and proper reception of a useful signal is impaired. The overhead for such measures, such as special contactors, can frequently be reduced considerably by observing the following points during configuration of your control system:

- Favorable arrangement of equipment and lines
- Grounding of all inactive metal elements
- Filtering of power cables and signal lines
- Shielding of equipment and lines
- Special interference-suppression measures

Physical arrangement of equipment and lines

Magnetic DC or AC fields of low frequency, such as 50 Hz, can only be sufficiently attenuated at great expense. In such a case, however, you can often solve the problem by providing the greatest possible distance between the interference source and sink.

Note

The analog Ex modules operate based on a method which suppresses faults caused by AC system ripple.

Grounding of inactive metal elements

Well implemented grounding is an important factor for interference-free assembly. Grounding is understood to mean a good electrical connection of all inactive metal elements (VDE 0160). The principle of surface grounding should be followed. All conductive, inactive metal elements should be grounded.

Observe the following when grounding:

- All ground connections must have a low impedance.
- All metal elements should have a large-area connection. Use particularly wide grounding strips for the connection. The surface of the ground connection and not its cross section is decisive.
- Screw-type connections should always have spring washers or lock washers.

Protection against electrostatic discharge

To protect equipment and modules from electrostatic discharge, metal housings or cabinets enclosed on all sides should be used; these should be given good electrical connection to the grounding point on site, and also connected to the main equipotential bonding conductor.

If you install your controller in a terminal box, use a cast metal or sheet metal housing if possible. Plastic housings should always have a metallized surface.

Doors or covers of housings should be connected to the grounded body of the housing with ground strips or contact springs.

If you are working on the system with the cabinet open, observe the guidelines for protective measures for electrostatically sensitive devices (ESDs).

Electrical systems must be installed such that the risk of ignition by electrostatic charges cannot be expected. Refer to "Guidelines for avoiding the risk of ignition resulting from electrostatic charges" laid down by the main association of Industrial Employers' Liability Insurance.

If electrostatic charges cannot be avoided, a charge should be kept as low as possible or safe discharge should be provided. The following measures, in particular, should be applied:

- Electrostatic grounding of all conductive elements. Solid materials can be considered as being electrostatically grounded if their leakage resistance at any point is not greater than $10^6 \Omega$. Under favorable conditions, $10^8 \Omega$ is satisfactory, particularly for small equipment of low capacitance.
- Reducing the electrical resistance of the material moved or parts moved with respect to each other.
- Incorporation of grounded metal elements in material subject to electrostatic charging.
- Increasing the relative air humidity. By increasing the relative air humidity to about 65 % with air conditioning, sprays or by hanging moist cloths, the surface resistance of most non-conductive materials can be adequately reduced. However, if the surface of plastic material is water-repellent, this measure will not succeed.
- Ionization of the air.

1.8.4 The Most Important Basic Rules for Ensuring EMC

To ensure EMC, it is often sufficient to observe some elementary rules. When assembling the control system, observe the five following basic rules.

1. When installing the programmable controllers, ensure high quality surface grounding of the inactive metal elements

- Connect all inactive metal elements over a large area and at low impedance.
- On painted and anodized metal elements, make screwed connections with special contact washers or remove the insulating protective layers.
- Provide a central connection between chassis ground and the ground/protective conductor system.

2. Follow the code of practice for line routing when wiring

- Subdivide the cabling into line groups.
(AC power cables, supply lines, Ex and non-Ex signal lines, data lines)
- Always install power cables and signal or data lines in separate ducts or bundles.
- Route the signal and data lines as closely as possible to grounded surfaces such as supporting bars, metal rails, cabinet sheet metal panels.
- Install Ex and non-Ex signal lines in separate ducts.

3. Ensure that line shields are properly secured

- Data lines should be shielded when laid. The shield should be connected at both ends.
- Analog lines should be shielded when laid. When low-amplitude signals are transmitted, it may be advantageous if the shield is connected at only one end.
- For Ex signal lines, connect the line shields only at the sensor or actuator end. Ensure that the connected shield continues without interruption as far as the module, but do not connect it there.
- Ensure that the shield has a low-impedance connection to the equipotential bonding conductor.
- Use metal or metallized plug housings for shielded data lines.

4. Implement special EMC measures for particular applications

- For all inductances, fit quenching elements provided they are not already contained in the output modules.
- Use incandescent bulbs for lighting the cabinets and avoid fluorescent lamps.

5. Provide a standard reference potential and ground all electrical apparatus if possible

- Take care to ensure specific grounding measures. Grounding of the control system is a protective and functional measure.
- System elements and cabinets should be connected in star-configuration to the ground/protective conductor system. In this way you can avoid the formation of ground loops.
- In the event of potential differences between system elements and cabinets, install adequately rated equipotential bonding conductors.

1.9 Lightning Protection

In systems with hazardous areas, the most important task, not least for reasons of explosion protection, is to avoid overvoltages; where this is not possible, they must be reduced and safely discharged.

In addition to the provision of external lightning protection, these measures cover internal lightning protection and overvoltage protection. These measures must be coordinated with the equipment-related EMC.

You will find more detailed information on the subjects of lightning protection and overvoltage protection in the manuals of the individual systems as specified in the foreword. Here, you will also find an overview of the components which can be used for this purpose.

1.9.1 External Lightning Protection/Shielding of Buildings

External lightning protection is a measure for preventing damage to buildings and fire damage. For this task, a large-mesh wire cage consisting of lightning conductors and down conductors is sufficient.

On buildings with sensitive electronic equipment such as control rooms, the external lightning protection must be supplemented by a building shield. For these purposes, where possible, metal facades and reinforcements of walls, floors and ceilings on or in the building are connected to form shield cages. Where this is not possible, the lightning conductor and down conductor should have a reduced mesh size and, where applicable, the supporting structure of the intermediate floor should be electrically interconnected.

Electrical equipment protruding above roof level must be protected against direct lightning strikes. When such equipment is metallically connected to the external lightning protection system, a partial current is picked up by the building in the event of a lightning strike; this can result in destruction of the equipment sensitive to overvoltages. The pick-up of partial lightning currents can be prevented by protecting the electrical equipment protruding above the roof from direct lightning strikes by means of rods insulated from the equipment (45 degree protective area), or by cage-type tensioned wires or cables.

The down conductors for external lightning protection and, if applicable, the reinforcements and supporting structures, should be connected to the ground system. Each individual building has its own functioning ground system. The ground systems are meshed to create a common grounding network. The voltage between the buildings is thus reduced.

1.9.2 Distributed Arrangement of Systems with S7-300, M7-300 and ET 200M

The process engineering of a plant, such as gas supply, requires a wide-ranging exchange of information between the systems with the distributed Ex I/O devices and the central, electrical or electronic measurement and control system. This necessitates a great number of cable connections, sometimes extending over several hundred meters - in the case of gas storage systems, over several thousands of meters. In the event of a lightning strike, therefore, extensive voltage pick-up occurs.

A distributed arrangement of instrumentation and control equipment with relatively short cables to the plant, and the connection of distributed I/O stations to each other and to the central controller via a bus (PROFIBUS-DP) or fiber-optic cable, are an important measure for reducing overvoltages between sections of the plant.

You will find more detailed information on this arrangement in the manuals specified in the foreword.

1.9.3 Shielding of Cables and Buildings

Overvoltages between separate plant sections or buildings cannot be avoided in practice by meshing. In the event of a lightning strike, a circulating current will flow over the path created by metal connections between the buildings or between a building and I/O device. Cable cores are ideal for this purpose. The lightning or partial lightning current must therefore be offered other conductive connections. Shielding which can be implemented in different ways is particularly suitable, for example:

- A helical current-rated metal strip or metal braid as the cable shield, e.g. NYCY or A2Y(K)Y.
 - By installing the cables in continuously connected metal conduits which are grounded at both ends.
 - By installing the cables in reinforced concrete ducts with through-connected reinforcement or on closed cable racks made of metal.
 - By laying conductors (shield conductors) in parallel with cables. This measure, however, only relieves the cables of partial lightning currents.
- or
- By laying fiber-optic cables.

Overvoltage-sensitive equipment must also be shielded to ensure the currents at the cable ends cannot destroy this equipment. This is achieved with metal housings or by installing the equipment in metal cabinets which are connected to the ground conductor.

1.9.4 Equipotential Bonding for Lightning Protection

"Internal lightning protection" covers all the additional measures which prevent the magnetic and electrical effects of the lightning current within the building to be protected. These include, in particular, the "equipotential bonding for lightning protection" which reduces the potential differences caused by the lightning current.

The principle of internal lightning protection is to incorporate in the equipotential bonding for lightning protection all the lines entering and exiting from a volume to be protected; these include, apart from all metal piping such as that for water, gas and heat, all power and information cables whose cores are connected via suitable protective devices. Since considerable, partial lightning currents can flow over such lines and must be discharged by the protective devices, they must be chosen for a suitable current carrying capacity (lightning current conductors).

1.9.5 Overvoltage Protection

The efficiency of overvoltage protection devices largely depends on the connection and cable routing. If the devices are used in hazardous areas or intrinsically safe circuits, DIN VDE 0165 must be complied with.

Since these overvoltage protection devices are passive modules in accordance with DIN VDE 0165, they require neither marking nor certificate of conformity in intrinsically safe circuits. However, the system installer must ensure compliance with the minimum ignition curves specified in DIN VDE 0170/0171 Part 7/05.78 EN 50020 and the maximum temperature rise.

Overvoltage protection in intrinsically safe circuits

Overvoltage protection devices can be used to protect intrinsically safe circuits against overvoltages. Since these overvoltage protection devices are considered as passive modules, they do not require PTB certification.

Fig. 1-16 shows how this overvoltage protection technology can be installed in an intrinsically safe circuit.

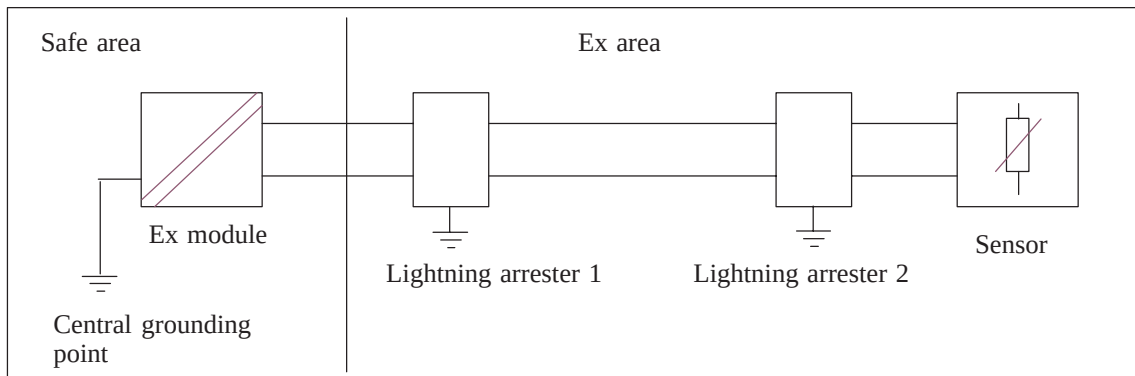


Fig. 1-16 Overvoltage protection in intrinsically safe circuits

The discussion of safety-relevant aspects is limited to the direct comparison of the data for inductance and capacity (Tables 1-5 and 1-6).

Table 1-5 Comparison of data for inductance and capacity

Ex module	Comparison	Lightning arrester 1	Cable	Lightning arrester 2	Sensor/ actuator
L_a	$\geq$	L_{BD1}	$+L_{Ltg}$	$+L_{BD2}$	$+Li$
C_a	x	C_{BD1}	$+C_{Ltg}$	$+C_{BD2}$	$+Ci$

Table 1-6 Example of the comparison of data for inductance and capacity

Ex module	Comparison	Lightning arrester 1	Cable	Lightning arrester 2	Sensor/ actuator
$L_a = 4 \text{ mH}$	$\geq$	$< 0.5 \mu\text{H}$	$< 50 \mu\text{H}$	$< 0.5 \text{ mH}$	$< 0.6 \text{ mH}$
$C_a = 270 \text{ nF}$	$\geq$	$< 1 \text{ nF}$	$< 10 \text{ nF}$	$< 6 \text{ nF}$	$< 6 \text{ nF}$

The overvoltage protection elements described in this section are only effective if used together with external lightning protection. External lightning protection measures reduce the effects of a lightning strike.

You will find suitable lightning protection elements for Ex modules in the manuals specified in the foreword.

1.9.6 Example of Lightning and Overvoltage Protection

Fig. 1-17 "Lightning/overvoltage protection for a gas compressor station" shows an example of how protective devices can be used.

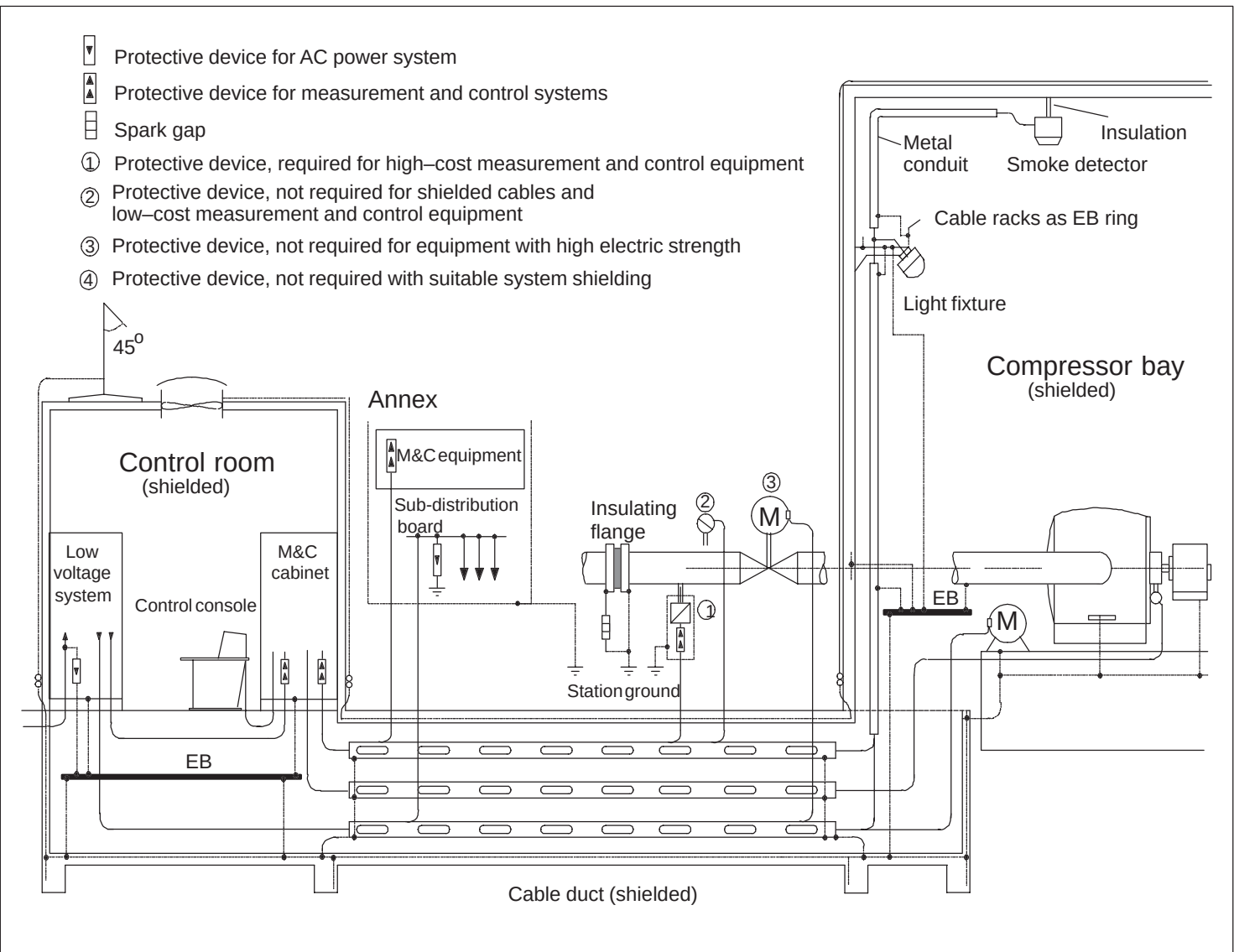


Fig. 1-17 Lightning/overvoltage protection for a gas compressor station

1.9.7 Lightning Strike

When lightning strikes an explosive atmosphere it always ignites. There is also a risk of ignition by excessive temperature raise in the lightning discharge paths. In order to prevent, at Zones 0, 1 and 10 themselves, the harmful effects of lightning strikes occurring outside the zones, surge diverters, for example, must be fitted at suitable points. Tank insulation covered with earth and made of metal materials with electrical equipment or electrically conductive system sections, which are electrically insulated with respect to the tank, require equipotential bonding; for example, in the case of measurement and control systems and filling pipes.

Note

Lightning protection equipment and grounding systems must be tested by an expert upon their completion and at regular intervals. Based on ElexV, the testing interval for electrical systems and lightning protection systems for hazardous areas is three years.

Summary:

- Enhanced external lightning protection (reduced mesh size, increased number of down conductors) on all buildings and systems.
- Meshing of grounding systems in the building to create area grounding.
- Meshing of equipotential bonding.
- Fitting of lightning conductors and surge diverters in the power system.
- Fitting of overvoltage fine-protection devices at both ends of measurement and control cables.
- Shielding of measurement and control cables.
- Measurement and control cables with twisted pairs of cores.

1.10 Installation Work in Hazardous Areas

All possible measures which eliminate the risk of explosion must be implemented not only when using programmable controllers in hazardous areas but also during the installation stage.

1.10.1 Safety Measures

Tools which tend to produce sparks must not be used for working in potentially explosive systems or system sections in operation. Copper-beryllium is a suitable material for tools such as screwdrivers, pliers, wrenches, hammers and chisels. Since this material has low wear-resistance, the tools should be used with care.

For mechanical work, the risk of sparks capable of causing ignition is

low	– when bare steel elements strike each other
possible	– when steel elements collide or drop
great	– when striking rusty steel
very great	– when striking rusty steel with an alloy coating, such as aluminum paint.

The possibility of creating sparks capable of causing ignition is substantially reduced by using non-sparking tools. An exception is when the tool is harder than the workpiece.

Measured for eliminating the risk of explosion

Safely closing off the working area, e.g. with dummy panels.

- Good ventilation of the rooms.
- Flushing with inert gas. Testing the effectiveness of the flushing (gas tester). Then working with a normal tool.

If the risk of explosion at the workplace cannot be eliminated, the following measures must be implemented:

- Avoidance of collisions and dropping of steel elements.
- Wearing antistatic shoes, e.g. leather shoes or using shoe grounding strips.
- Avoidance of rust layers and aluminum coating at impact points. If this is not possible, eliminating the risk of explosion locally, e.g. with inert gas.
- Adequate air supply and waste air disposal.
- Removing or enclosing readily flammable substances in the vicinity.
- Keeping the workplace and, if applicable, floor moist.

Table 1-7 Safety measures

Working area	Safety measures
Installations with readily flammable gas and vapor-air mixtures, e.g. hydrogen, city gas, acetylene and hydrogen sulphide	Working only allowed after implementation of special safety measures and with written permission of plant manager. Only non-sparking tools to be used (tool softer than workpiece).
Installations with gas and vapor-air mixtures such as methane, propane, butane and petrol (gasoline)	Sufficient to use non-sparking tools. Exception: For materials with rust formation and aluminum coating or similar, special protective measures required.
Installations with risk of explosion from readily flammable dust	Remove dust deposits. Keep working area wet and protect against dust formation. Normal tools may be used.

Note

Working on energized electrical installations and apparatus in hazardous industrial premises is prohibited. This also includes the disconnection of live control lines for test purposes.

As an exception, work on intrinsically safe circuits is permitted; also, in special cases, work on other electrical systems where the user has certified in writing that there is no risk of explosion for the duration of the work at the site.

If necessary, a fire permit must additionally be obtained.

Grounding and short-circuiting may only be carried out in hazardous industrial premises when there is no risk of explosion at the point of grounding and short-circuiting.

Use measuring instruments which are approved for the zones to test for no voltages.

1.10.2 Use of Ex Assemblies in Hazardous Areas

It is basically possible to install a SIMATIC assembly in a hazardous area, i.e. zone 1 or 2. However, the system installer must implement additional measures in order to protect the modules. Two types of protection are available:

- the Ex assembly is installed in a pressurized enclosure;
- the Ex assembly is installed in a flameproof enclosure.

The figure below shows a possible assembly in a flameproof enclosure with an increased-safety terminal compartment.

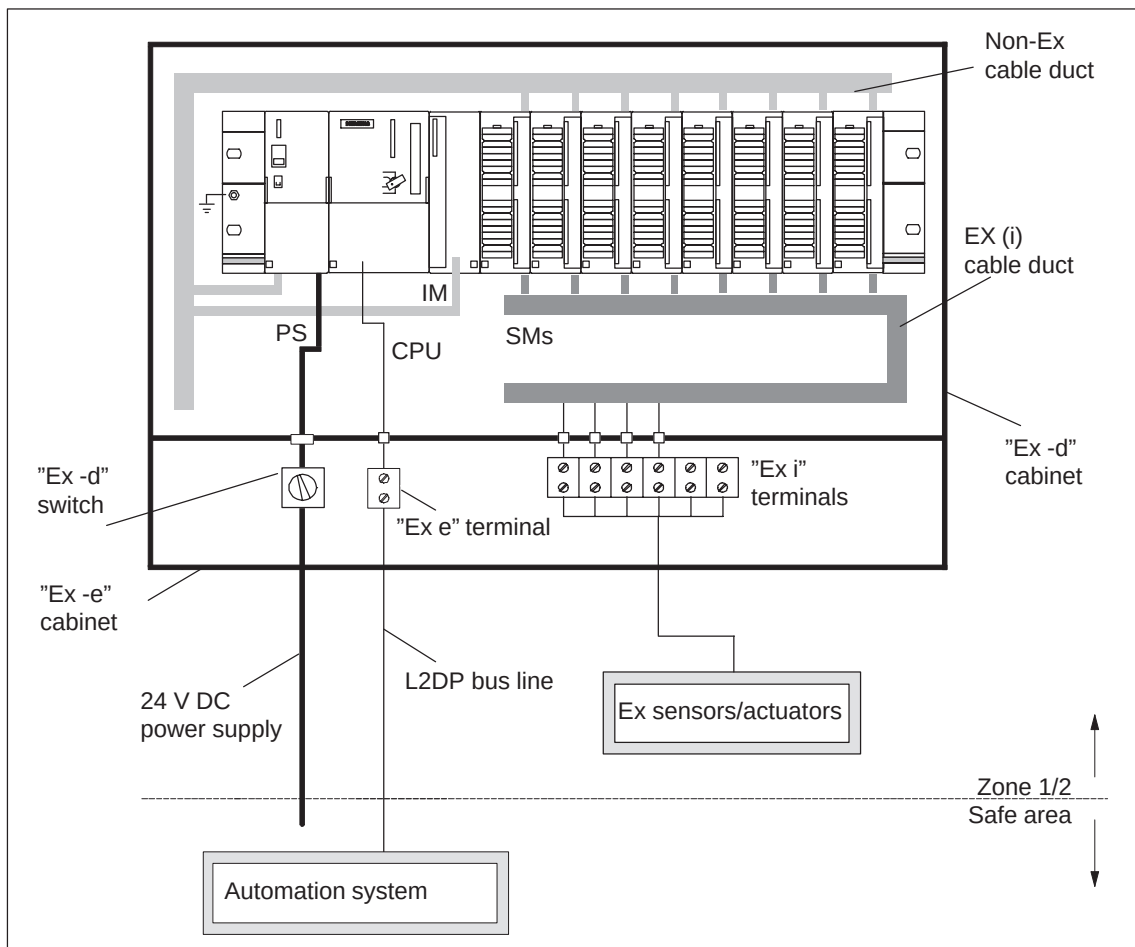


Fig. 1-18 SIMATIC Ex modules in hazardous area

Housing

The selected type of housing is characterized by the fact that it is able to withstand explosions occurring inside the housing and that an explosive gas/air mixture surrounding the housing is not ignited. In addition, the surface temperature does not exceed the limit values of the temperature classes. Cable glands that are protected against transmission of internal ignition and isolated against the housing wall must be used for routing the supply leads into the flameproof housing.

A housing with "increased safety" is used as a terminal compartment. Special screwed glands are used for the cable entries.

The housing must be certified by a testing authority to comply with the EEx d type of protection and the relevant design requirements.

Explosion protection of the housing: EEx de II T5 .. T6.

Cables

The cables must comply with the DIN EN 50014 and DIN EN 50 020 standards for intrinsically safe circuits or with DIN EN 50039 for circuits with "increased safety".

The cables for the assembly are to be installed in such a way that they are endangered neither by thermal, mechanical nor chemical load or stress.

Note

The cables should be installed in cable conduits if necessary.

Terminals

The terminal connectors for the power supply cable and the bus line should always meet the requirements of the "increased safety" type of protection. The clamping points of the intrinsically safe circuits should always be implemented according to the guidelines of "Intrinsic safety".

Protective device

The assembly is connected to a 24 V DC supply circuit fed by a power supply unit with safe electrical isolation. The supply circuit must be protected by an appropriate circuit-breaker. This circuit-breaker is installed outside the Ex zone.

Switch

The switch for enabling the system should comply with "EEx de II T6" type of protection.

Table 1-8 Working on systems to type of protection: EEx de [ib] T5 .. T6

Type of protection of apparatus used in system	Type of work to be carried out	Work within		Additional requirements and notes
EEx ib		Zone 1	Zone 2	
	Opening the housing, Ex i/e housing only	Allowed	Allowed	If no other apparatus is in the housing
	Connecting/disconnecting lines	Allowed	Allowed	
	Current, voltage and resistance measurement	Allowed with certified apparatus	Allowed	
	Soldering	Prohibited	Allowed if soldering temperature lower than ignition temperature	
EEx e		Zone 1	Zone 2	
	Opening the housing, Ex i/e housing only	Allowed	Allowed	If no other apparatus is in the housing
	Connecting/disconnecting lines	Not allowed unless in de-energized state	Only in de-energized state and if no risk of explosion	
	Current, voltage and resistance measurement	Voltage measurement with certified apparatus only	Voltage measurement with certified apparatus only	
	Soldering	Prohibited	Allowed in de-energized state if soldering temperature lower than ignition temperature	

Table 1-8 Working on systems to type of protection: EEx de [ib] T5 .. T6, continued

Type of protection of apparatus used in system	Type of work to be carried out	Work within		Additional requirements and notes
		Zone 1	Zone 2	
EEx d	Opening the housing, Ex d housing only	Prohibited	Allowed if no risk of explosion	Apparatus in flameproof enclosure are no longer protected against explosion if housing is opened
	Connecting/disconnecting lines	Not allowed unless in de-energized state	Allowed if no risk of explosion	
	Current, voltage and resistance measurement	Work not possible	Allowed if no risk of explosion	
	Soldering	Prohibited	Allowed in de-energized state if soldering temperature lower than ignition temperature	

See also “S7-300, M7-300, ET 200M Automation Systems Principles of Intrinsically-Safe Design“ Manual, Chapter”Installation, Operation and Maintenance of Electrical Systems in Hazardous Areas”, Table “Information for work to be carried out on explosion-protected apparatus” .

1.11 Maintenance of Electrical Apparatus

Replacing apparatus

Work on electrical installations and apparatus may only be carried out when a "permit" has been obtained. When replacing electrical apparatus, ensure compliance with regulations relating to temperature class, explosion group and the relevant (Ex) zone. Certificates of conformity or PTB or KEMA test certificates and design approval must have been obtained.

Repair of apparatus

Repaired electrical apparatus may only be placed in operation again after testing by a recognized expert in accordance with paragraph 15 of ExeV, and the test has been certified, unless explosion protection has not been affected by the repair. If the repair affects explosion protection, only original spare parts may be used. Improvised repairs which no longer ensure explosion protection of apparatus are not permitted.

SIMATIC S7 Ex Digital Modules

2

In this chapter

The following SIMATIC S7 Ex digital modules are described in this chapter:

- Digital input SM 321; DI 4 x NAMUR,
Order Number: 6ES7 321-7RD00-0AB0
- Digital output SM 322; DO 4 x 24V/10mA
Order Number: 6ES7 322-5SD00-0AB0
- Digital output SM 322; DO 4 x 15V/20mA
Order Number: 6ES7 322-5RD00-0AB0

Chapter overview

Section	Description	Page
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2.2	Digital Output Module SM 322; DO 4 x 24V/10mA	2-14
2.3	Digital Output Module SM 322; DO 4 x 15V/20mA	2-24

Notes

You will find information on the relevant safety standards and on other safety regulations in Appendix B.

The *General Technical Specifications* for S7-300, M7-300 modules in /71/ also apply.

2.1 Digital Input Module SM 321; DI 4 x NAMUR

Order number 6ES7 321-7RD00-0AB0

Features

The SM 321; DI 4 x NAMUR offers the following features:

- 4 inputs
 - Isolated with respect to bus
 - Isolated among each other
- Load voltage 24 V DC
- Connectable sensors
 - In compliance with DIN 19234 or NAMUR (with diagnostic evaluation)
 - Interconnected mechanical contacts (with diagnostic evaluation)
 - Open-circuited mechanical contacts (without diagnostics)
- 4 short-circuit-proof outputs for sensor power supply (8.2 V)
- Operating points: logic "1" ≥ 2.1 mA
logic "0" ≤ 1.2 mA
- Status indication (0...3) green LEDs
Fault indication red LEDs for
 - Group fault indication (SF)
 - Channel-referred fault indication for short-circuit and wire break (F0 ... F3)
- Configurable diagnostics
- Configurable diagnostic interrupt
- Configurable hardware interrupt
- Intrinsic safety of inputs in accordance with EN 50020
- 2-wire sensor connection

Wiring diagram

Fig. 2-1 shows the terminal diagram of the digital input module SM 321; DI 4 x NAMUR. The block diagram and detailed technical data can be found on the following pages.

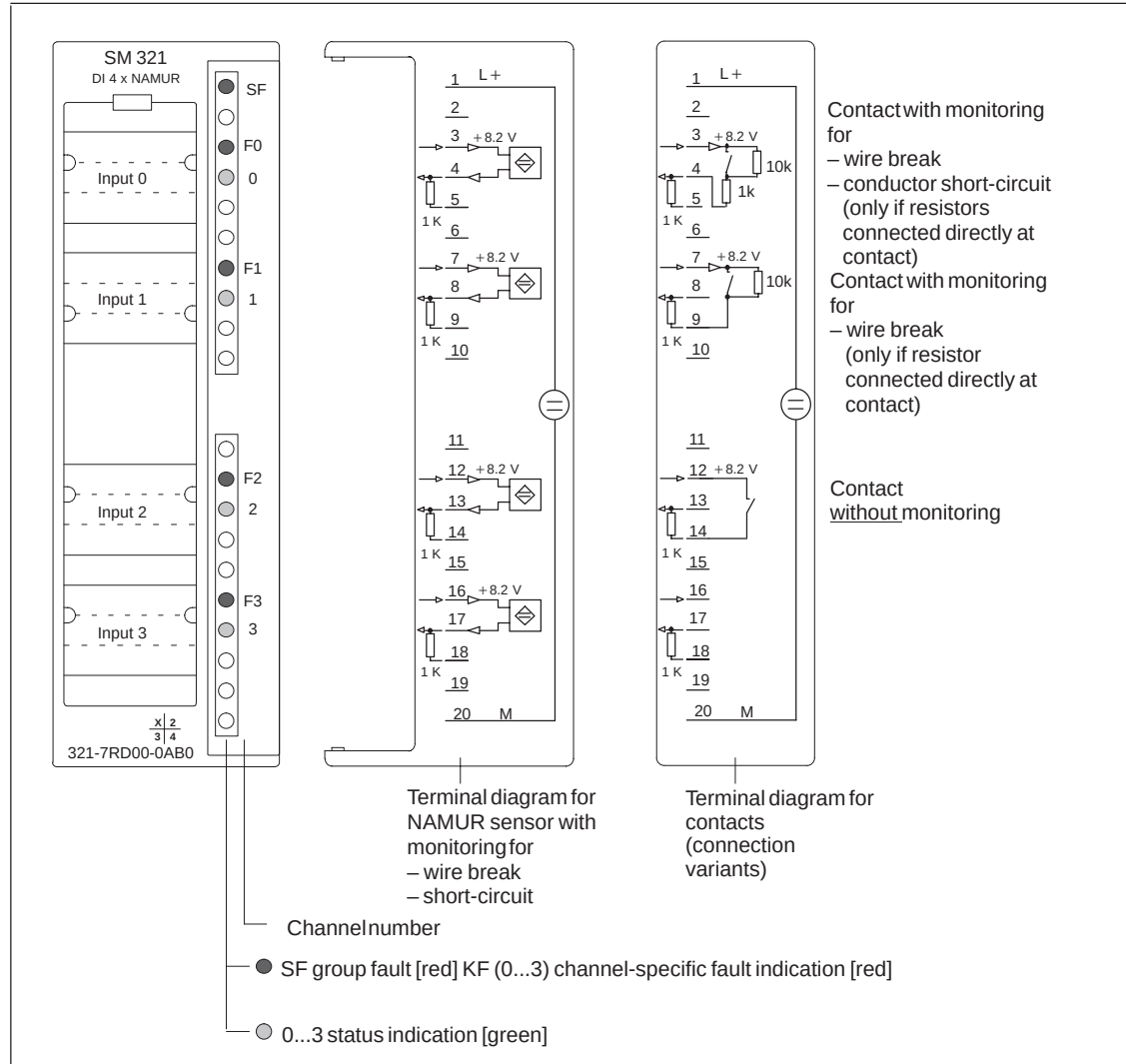


Fig. 2-1 Wiring diagram of digital input module SM 321; DI 4 x NAMUR

Notes on intrinsically-safe installation

You must connect the DM 370 dummy module between the CPU or IM 153-2 (distributed configuration) and the Ex I/O modules whose signal cables lead into the hazardous area. In a distributed configuration with an active backplane bus, you should use the explosion-proof partition instead of the dummy module. Additional information on system design can be found in Section 1.3 - 1.5.

Power supply for a intrinsically-safe structure

In order to maintain the clearances and creepage distances, L+ / M must be routed via the line chamber LK393 when operating modules with signal cables that lead to the hazardous location, see Section 1.2.

Block diagram

Fig. 2-2 shows the block diagram of the digital input module SM 321; DI 4 x NAMUR.

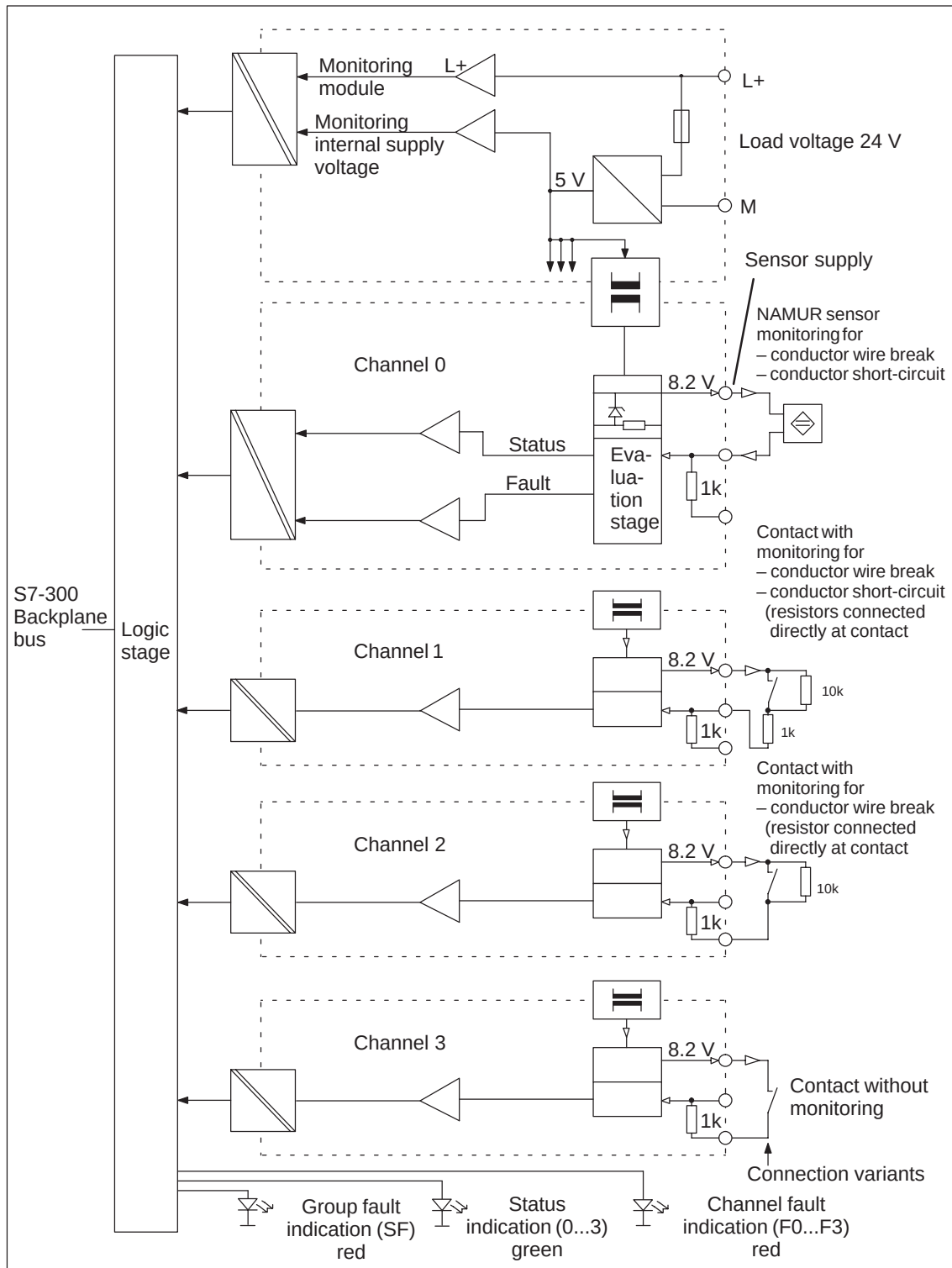


Fig. 2-2 Block diagram of digital input module SM 321; DI 4 x NAMUR

Digital input SM 321; DI 4 x NAMUR Dimensions and Weight		Voltages, currents, potentials continued	
Dimensions W x H x D (mm)	40 x 125 x 120	Permissible difference in potential (U_{ISO}) for signals from non-hazardous area	
Weight	approx. 230 g	- Between channels and backplane bus 400 V DC 250 V AC - Between channels and load voltage L+ 400 V DC 250 V AC - Between channels 400 V DC 250 V AC - Between backplane bus and load voltage L+ 75 V DC 60 V AC	
Module-specific data		Insulation tested	
Number of inputs	4	- Channels with respect to backplane bus and load voltage L+ with 1500 V AC - Channels among each other with 1500 V AC - Between load voltage L+ and backplane bus with 500 V DC	
Line length, shielded	max. 200 m	Current input	
Type of protection PTB (see Appendix A)	[EEx ib] IIC acc. to EN 50020	- From backplane bus max. 80 mA - From load voltage L+ max. 50 mA	
Test number	Ex-96.D.2094 X	Module power loss typical 1.1 W	
Type of protection FM (see Appendix B)	CL I, DIV 2, GP A, B, C, D T4	Status, interrupts, diagnostics	
Voltages, currents, potentials		Status indication	
Bus power supply	DC 5 V	Inputs green LED per channel	
Rated load voltage L+	24 V DC	Interrupts	
Reverse voltage protection	yes	- Hardware interrupt configurable - Diagnostic interrupt configurable	
Number of inputs which can be activated simultaneously	4	Diagnostic functions	
Galvanic isolation		- Group fault indication red LED (SF) - Channel fault indication red LED (F) per channel - Diagnostic functions readout possible	
Between channels and backplane bus	yes	Monitoring for	
Between channels and load voltage L+	yes	- Short-circuit $I > 8.5 \text{ mA}$ - Wire break $I \leq 0.1 \text{ mA}$	
Between channels	yes		
Between backplane bus and load voltage L+	yes		
Permissible difference in potential (U_{ISO}) of signals from hazardous area			
Between channels and backplane bus	60 V DC 30 V AC		
Between channels and load voltage L+	60 V DC 30 V AC		
Between channels	60 V DC 30 V AC		
Between backplane bus and load voltage L+	60 V DC 30 V AC		

Safety data (refer to Certificate of Conformity in Appendix A)			Data for sensor selection	
Maximum values of input circuits (per channel)			In accordance with DIN 19234 or NAMUR	
• U_0 (no-load output voltage) max. 10 V • I_0 (short-circuit current) max. 14.1 mA • P_0 (load power) max. 33.7 mW • L_0 (permissible external inductance) max. 100 mH • C_0 (permissible external capacitance) max. 3 μF • U_m (error voltage) max. 60 V DC 30 V AC • T_a (permissible ambient temperature) max. 60°C			Input current	
			• at signal "1" 2.1 to 7 mA • at signal "0" 0.35 to 1.2 mA	
			Time/frequency	
			Interrupt conditioning time for	
			• Interrupt conditioning only max. 250 μs • Interrupt and diagnostic conditioning max. 250 μs	
			Input delay (EV)	
			• configurable yes • Nominal value typical 0.1/0.5/3/15/20ms	

Parameterization

The parameters for the digital input modules SM 321; DI 4 x NAMUR are set with *STEP 7*. You must implement the settings in CPU STOP mode. The parameters set in this way are stored in the CPU during transfer from PG to S7-300. These parameters are transferred to the digital module during the status change from STOP → RUN.

Alternatively, you can also change several parameters in the user program with the SFCs 55 to 57 (refer to /235/).

The parameters for the 2 parameterization alternatives are subdivided into:

- Static parameters
- Dynamic parameters

Table 2-1 below shows the characteristics of static and dynamic parameters.

Table 2-1 Static and dynamic parameters of SM 321; DI 4 x NAMUR

Parameter	Set with	CPU status
Static	PG	STOP
Dynamic	PG	STOP
Dynamic	SFCs 55 to 57 in user program	RUN

Default settings

The SM 321; DI 4 x NAMUR features default settings for diagnostics, interrupts etc. (see Table 2-2).

These default settings are applicable when the digital input module has not been parameterized via *STEP 7*.

Configurable characteristics

The characteristics of the SM 321; DI 4 x NAMUR can be parameterized with the following parameter blocks:

- Basic settings
- Diagnostics
- Hardware interrupts

Channel group allocation

Table 2-2 shows the allocation of 4 channels to the channel groups of SM 321; DI 4 x NAMUR.

Table 2-2 Allocation of 4 digital input channels to the 4 channel groups of SM 321; DI 4 x NAMUR

Channel	Allocated channel group
Channel 0	Channel group 0
Channel 1	Channel group 1
Channel 2	Channel group 2
Channel 3	Channel group 3

Parameters of the digital input module

Table 2-3 provides an overview of the parameters of the SM 321; DI 4 x NAMUR and shows what parameters

- are static or dynamic and
- can be used for the module as a whole or for a channel group.

Table 2-3 Parameters of SM 321; DI 4 x NAMUR

Parameter	SM 321; DI 4 x NAMUR			
	Value range	Default	Type	Effective range
Basic settings				
• Input delay (ms)	0.1/0.5/3/15/20	3	static	Module
• Hardware interrupt enable	yes/no	no	dynamic	Module
• Diagnostic interrupt enable	yes/no	no	dynamic	Module
Diagnostics				
• Wire break monitoring	yes/no	no	static	Channel group
• Short to M	yes/no	no	static	Channel group
Hardware interrupts				
• Leading edge	yes/no	no	dynamic	Channel group
• Trailing edge	yes/no	no	dynamic	Channel group

Input delay

Table 2-4 shows the possible configurable input delay times for SM 321; DI 4 x NAMUR and their tolerances.

Table 2-4 Delay times of input signal for SM 321; DI 4 x NAMUR

Input delay	Tolerance
0.1 ms	75 to 150 µs
0.5 ms	0.4 to 0.8 ms
3 ms (default)	2.8 to 3.5 ms
15 ms	14.5 to 15.5 ms
20 ms	19 to 21 ms

Diagnostics

You can use the diagnostic function to determine whether signal acquisition takes place without errors.

Parameterizing diagnostics

Diagnostics is parameterized with STEP 7.

Diagnostic evaluation

When evaluating the diagnostics, a differentiation must be made between configurable and non-configurable diagnostic messages. In the case of the configurable diagnostic message "wire break" or "short to M", diagnostics is only signalled when diagnostic evaluation has been enabled by means of parameterization (parameter "wire break" or "short to M").

Non-configurable diagnostic messages are general, i.e. independent of parameterization.

A diagnostic signal results in a diagnostic interrupt being triggered providing the diagnostic interrupt has been enabled by way of parameterization.

Irrespective of the parameterization, known module errors always result in the SF LED and the corresponding channel fault LED lighting irrespective of the CPU operating status (at POWER ON).

Exception: The SF LED and the corresponding channel fault LED light in the event of a wire break only when parameterization is enabled.

Diagnostics of the digital input module

Table 2-5 provides an overview of the diagnostic messages of the SM 321; DI 4 x NAMUR. You enable diagnostics in *STEP 7* (see Table 2-3).

The diagnostics information refers to either the channel groups or the entire module.

Table 2-5 Diagnostic messages of SM 321; DI 4 x NAMUR

Diagnostic message	Effective range of diagnostics	configurable
Wire break	Channel group	yes
Short to M		
Incorrect parameters in module	Module	no
Module not parameterized		
No external auxiliary supply		
No internal auxiliary supply		
Fuse blown		
Watchdog triggered		
EPROM error		
RAM error		
CPU error		
Hardware interrupt lost		

Reading out diagnostic messages

You can read out system diagnostics with *STEP 7*. You can read detailed diagnostic messages from the module in the user program with SFC 59 (refer to /235/).

Errors and corrective measures

Table 2-6 provides a list of possible causes and corresponding corrective measures for individual diagnostic messages.

Bear in mind that, in order to detect faults which are indicated by means of configurable diagnostic messages, must also be parameterized accordingly.

Table 2-6 Diagnostic messages as well as their causes and corrective measures in SM 321; DI 4 x NAMUR

Diagnostic message	Possible fault cause	Corrective measures
Short to M ($I > 8.5 \text{ mA}$)	Short-circuit between the two sensor lines	Eliminate short-circuit
	With contacts as sensor 1 k Ω series resistor not fitted in line to contact	Connect 1 k Ω resistor in line <i>directly at contact</i>
Wire break $I \leq 0.1 \text{ mA}$)	Conductor break between module and NAMUR sensor	Make conductor connection
	Contact as sensor (wire break monitoring enabled)	10 k Ω resistor <i>not fitted or interrupted</i> directly at contact
	Contacts as sensor (without monitoring)	Disable channel by parameterization "diagnostics wire break"
	Channel not used (open)	
Incorrect parameters in module	Invalid parameters loaded in module by means of SFC	Check parameterization of module and re-load valid parameters
Module not parameterized	No parameters loaded in module	Include module in parameterization
No external auxiliary supply	No L+ supply voltage of module	Supply L+
No internal auxiliary supply	No L+ supply voltage of module	Supply L+
	Module-internal fuse defective	Replace module
Fuse blown	Module-internal fuse defective	Replace module
Watchdog triggered	In part, high electromagnetic interference	Eliminate interference sources
	Module defective	Replace module
EPROM error RAM error CPU error	In part, high electromagnetic interference	Eliminate interference sources and switch CPU supply voltage OFF/ON
	Module defective	Replace module
Hardware interrupt lost	Succession of hardware interrupt is faster than the CPU can process	Change interrupt processing in CPU and reparameterize module if necessary

Interrupts	The interrupt characteristics of the SM 321; DI 4 x NAMUR are described in the following. In principle, a differentiation is made between the following interrupts: • Diagnostic interrupt • Hardware interrupt
Parameterizing interrupts	The interrupts are parameterized with <i>STEP 7</i>.
Default setting	The interrupts are inhibited by way of default.
Diagnostic interrupt	If enabled, the module triggers a diagnostic interrupt when an fault comes or goes (e.g. wire break or short to M). Diagnostic functions inhibited by parameterization cannot trigger an interrupt. The CPU interrupts processing of the user program or low-priority classes and processes the diagnostic interrupt module (OB 82).
Hardware interrupt	Depending on the parameterization, the module can trigger a hardware interrupt for every channel optionally at leading, trailing or both edges of a signal change. You can determine which of the channels has triggered the interrupt from the local data of the OB 40 in the user program (refer to /235/). Active hardware interrupts trigger interrupt processing (OB 40) in the CPU, consequently the CPU interrupts processing of the user program or low-priority classes. If there are no higher priority classes pending processing, the stored interrupts (of all modules) are processed one after the other corresponding to the order in which they occurred.
Hardware interrupt lost	If an event occurred in one channel (edge change), this event is stored in the hardware interrupt register and a hardware interrupt is triggered. If a further event occurs on this channel before the hardware interrupt has been acknowledged by the CPU (OB 40 run) this event will be lost. A diagnostic interrupt "hardware interrupt lost" is triggered in this case. The diagnostic interrupt enable must be active for this purpose. Further events on this channel are then no longer registered until interrupt processing is completed for this channel.

Influence of supply voltage and operating status

The input values of the SM 321; DI 4 x NAMUR are dependent on the supply voltages and operating status of the CPU.

Table 2-7 provides an overview of these dependencies.

Table 2-7 Dependencies of the input values for CPU operating status and supply voltage L+ of SM 321; DI 4 x NAMUR

Operating status CPU		Supply voltage L+ at digital module	Input value of digital module
POWER ON	RUN	L+ applied	Process value
		L+ not applied	0-signal
	STOP	L+ applied	Process value
		L+ not applied	0-signal
POWER OFF	-	L+ applied	-
		L+ not applied	-

Failure of the supply voltage L+ of the SM 321; DI 4 x NAMUR is always indicated by the SF-LED on the front of the module and additionally entered in diagnostics.

In the event of the module supply voltage L+ failing, the input value is initially held for 20 to 40 ms before the "0" signal is transferred to the CPU. Dips in the supply voltage of < 20 ms do not change the process value, but they trigger a diagnostic interrupt and the group error LED is lit.

2.2 Digital Output Module SM 322; DO 4 x 24V/10mA

Order number 6ES7 322-5SD00-0AB0

Properties The SM 322; DO 4 x 24V/10mA features the following properties:

- 4 outputs
 - Isolated with respect to bus
 - Isolated among each other
- suitable for
 - intrinsically safe valves
 - acoustic interrupts
 - indicators
- Configurable diagnostics
- Configurable diagnostic interrupt
- Configurable default output
- Status indication (0...3) green LEDs
- Fault indication red LEDs for
 - Group fault signalling (SF)
 - Channel-referred fault signalling for short-circuit and wire break (wire break) (F0 ... F3)
- Intrinsic safety of outputs in accordance with EN 50020
- 2-wire connection of actuators

Wiring diagram

Fig. 2-3 shows the terminal diagram of SM 322; DO 4 x 24V/10mA.

The block diagram and detailed technical specifications for SM 322; DO 4 x 24V/10mA are provided on the following pages.

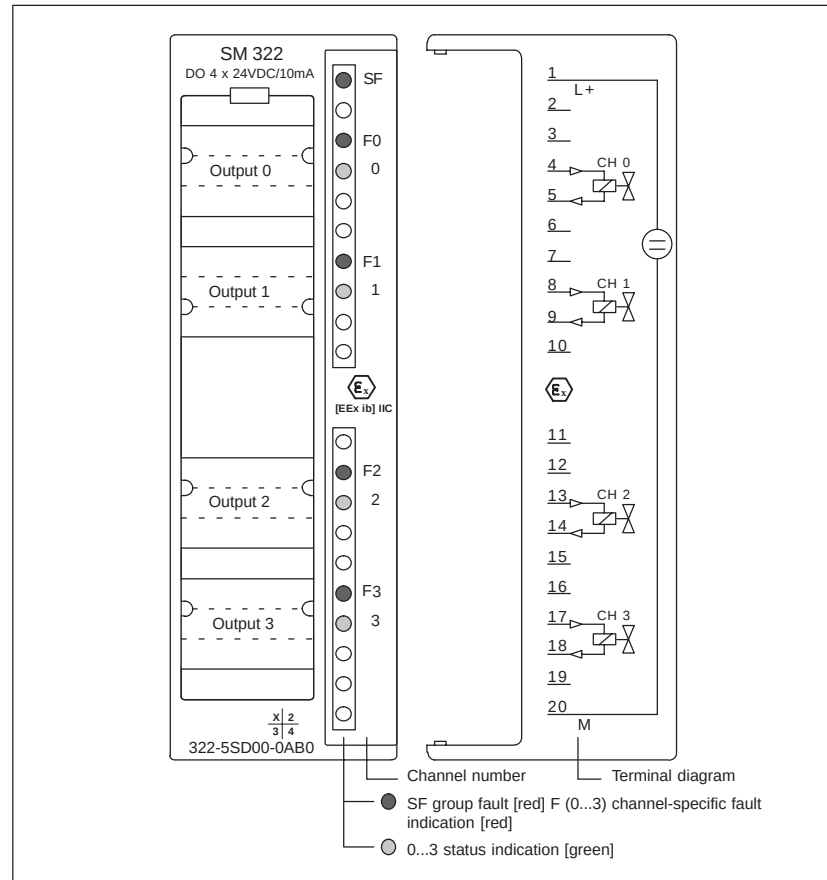


Fig. 2-3 Wiring diagram of SM 322; DO 4 x 24V/10mA

**Notes on
intrinsically-safe
installation**

You must connect the DM 370 dummy module between the CPU or IM 153-2 (distributed configuration) and the Ex I/O modules whose signal cables lead into the hazardous area. In a distributed configuration with an active backplane bus, you should use the explosion-proof partition instead of the dummy module. Additional information on system design can be found in Section 1.3 - 1.5.

**Power supply for a
intrinsically-safe
structure**

In order to maintain the clearances and creepage distances, L+ / M must be routed via the line chamber LK393 when operating modules with signal cables that lead to the hazardous location, see Section 1.2.

Block diagram

Fig. 2-4 shows the block diagram of SM 322; DO 4 x 24V/10mA.

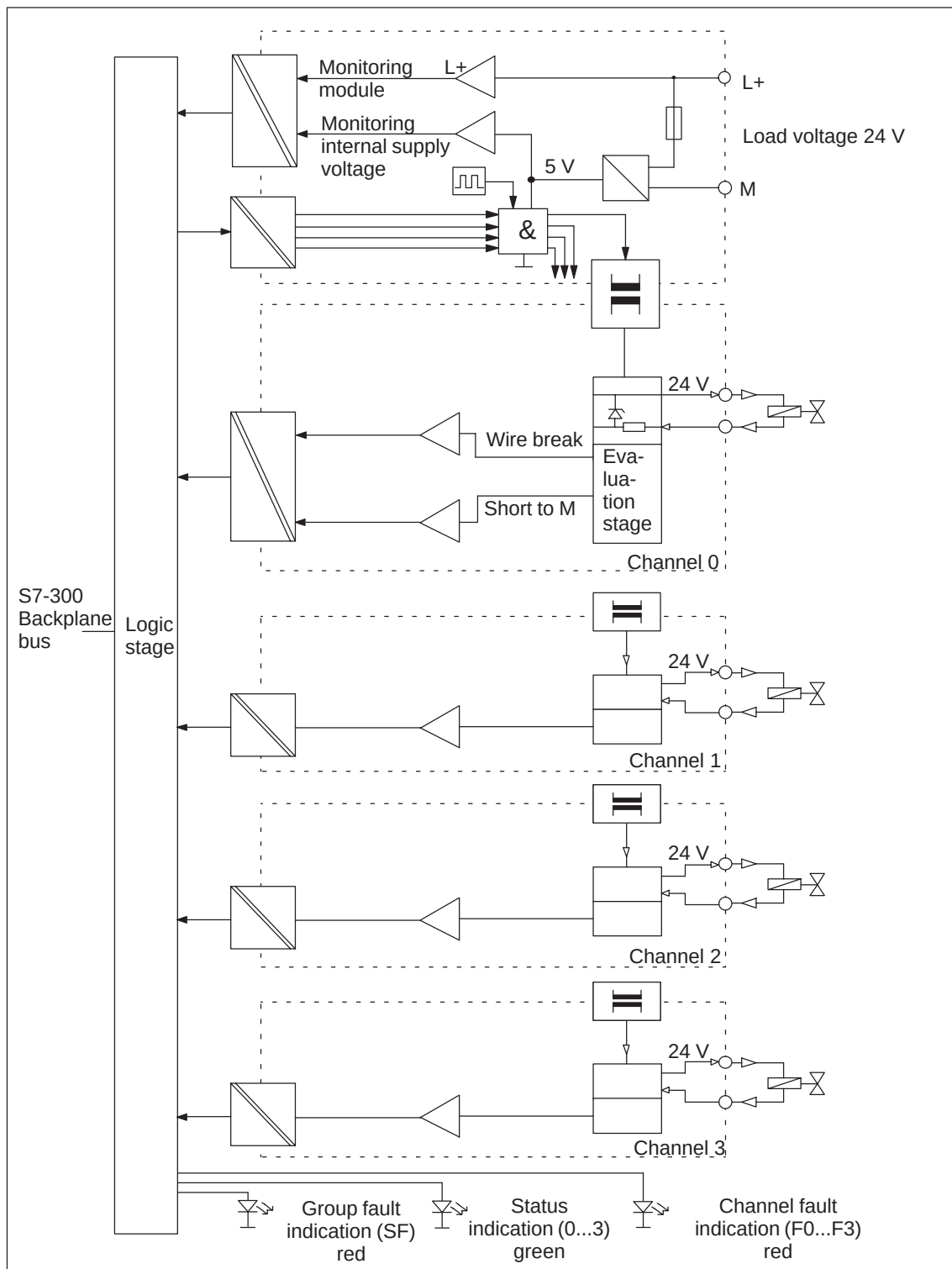
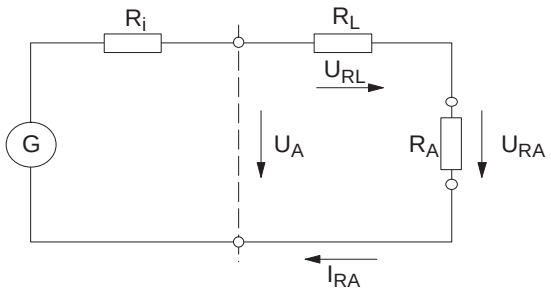
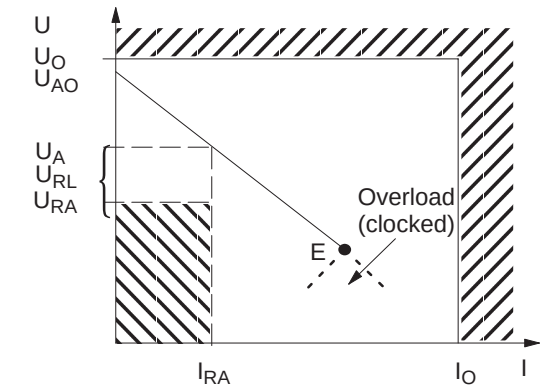


Fig. 2-4 Block diagram of digital output module SM 322; DO 4 x 24V/20mA

Digital output SM 322; DO 4 x 24V/10mA		Voltages, currents, potentials continued	
Dimensions and Weight		Permissible difference in potential (U_{ISO}) for signals from non-hazardous area	
Dimensions W x H x D (mm)	40 x 125 x 120	- Between channels and backplane bus 400 V DC 250 V AC - Between channels and load voltage L+ 400 V DC 250 V AC - Between channels 400 V DC 250 V AC - Between backplane bus and load voltage L+ 75 V DC 60 V AC	
Weight	approx. 230 g		
Module-specific data		Insulation tested	
Number of outputs	4	- Channels with respect to backplane bus and load voltage L+ with 1500 V AC - Channels among each other with 1500 V AC - Between load voltage L+ and backplane bus with 500 V DC	
Line length, shielded	max. 200 m	Current input	
Type of protection PTB (see Appendix A)	[EEx ib] IIC to EN 50020	- From backplane bus max. 70 mA - From load voltage L+ (at rated data) max. 160 mA	
Test number	Ex-96.D.2093 X	Module power loss typical 3 W	
Type of protection FM (see Appendix B)	CL I, DIV 2, GP A, B, C, D T4	Status, interrupts, diagnostics	
Voltages, currents, potentials		Status indication	
Bus power supply	5 V DC	Outputs green LED per channel	
Rated load voltage L+	24 V DC	Interrupts	
Reverse voltage protection	yes	Diagnostic interrupt configurable	
Total current of outputs		Diagnostic functions	
- Horizontal arrangement up to 60 °C - Vertical arrangement up to 40 °C	No restrictions	- Group fault indication red LED (SF) - Channel fault indication red LED (F) per channel - Diagnostic functions readout possible	
Galvanic isolation		Monitoring for	
- Between channels and backplane bus - Between channels and load voltage L+ - Between channels - Between backplane bus and load voltage L+	yes yes yes yes	- Short-circuit $I > 10 \text{ mA } (\pm 10\%)$ - Wire break $I \leq 0.15 \text{ mA}$	
Permissible difference in potential (U_{ISO}) of signals from hazardous area			
- Between channels and backplane bus - Between channels and load voltage L+ - Between channels - Between backplane bus and load voltage L+	60 V DC 30 V AC 60 V DC 30 V AC 60 V DC 30 V AC 60 V DC 30 V AC		

Safety data (refer to Certificate of Conformity in Appendix A)	Block diagram
Maximum values of output circuits (per channel) • U_0 (no-load output voltage) max. 25.2 V • I_0 (short-circuit current) max. 70 mA • P_0 (load power) max. 440 mW • L_0 (permissible external inductance) max. 6.7 mH • C_0 (permissible external capacitance) max. 90 nF • U_m (error voltage) max. 60 V DC 30 V AC • T_a (permissible ambient temperature) max. 60°C	 G: Generator R_i: Internal resistor R_L: Line resistor R_A: Load resistor U_{AO}: No-load voltage U_A: Output voltage U_{RL}: Voltage drop at line resistor U_{RA}: Voltage drop at load U_0: Max. output voltage I_0: Max. output current I_{RA}: Load current
Data for actuator selection	Output characteristic
Outputs • No-load voltage U_{AO} 24 V DC $\pm 5\%$ • Internal resistance R_i 390 Ω $\pm 5\%$ Curve vertices E • Voltage U_E 19 V DC $\pm 10\%$ • Current I_E 10 mA $\pm 10\%$ Parallel connection of 2 outputs • For redundant activation of a load Not possible • For increasing power Possible, see Manual "S7-300, M7-300, ET 200M Automation Systems Principles of Intrinsically-Safe Design" Section "Intrinsically-Safe Circuit with Two or More Items of Associated Electrical Apparatus" Switching frequency • At resistive load 100 Hz • At inductive load 100 Hz ($L < L_0$) Short-circuit protection of output • Response threshold Curve vertex E	 Area outside safety limits Output power at load E: Curve vertex (U_E, I_E) $U_E = 19 \text{ V} \pm 10\%$ $I_E = 10 \text{ mA} \pm 10\%$ Output current electronically clocked at overload. Clock ratio $\sim 1:15$

Parameterization

The parameters for the SM 322; DO 4 x 24V/10mA are set with *STEP 7*. You must implement the settings in CPU STOP mode. During transfer from the PG to the S7-300, the parameters set in this way are stored in the CPU and then transferred by the CPU to the digital module.

Alternatively, you can also change several parameters in the user program with SFCs 55 to 57 (see /235/).

The parameters for the 2 parameterization alternatives are subdivided into:

- Static parameters
- Dynamic parameters

Table 2-8 shows the characteristics of static and dynamic parameters.

Table 2-8 Static and dynamic parameters

Parameter	Set with	CPU status
Static	PG	STOP
dynamic	PG	STOP
	SFCs 55 to 57 in user program	RUN

Default settings

The digital output features default settings for diagnostics, substitute values, etc. (see Table 2-10).

These default settings are applicable when the digital module has not been parameterized with *STEP 7*.

Configurable characteristics

The characteristics of the SM 322; DO 4 x 24V/10mA can be parameterized with the following parameter blocks:

- Basic settings
- Diagnostics

Channel groups allocation

Table 2-9 shows the allocation of the 4 channels to the 4 channel groups of digital output.

Table 2-9 Allocation of the 4 channels to the 4 channel groups of SM 322; DO 4 x 24V/10mA and SM 322; DO 4 x 15V/20mA

Channel	Allocated channel group
Channel 0	Channel group 0
Channel 1	Channel group 1
Channel 2	Channel group 2
Channel 3	Channel group 3

Parameters of the digital output module

Table 2-10 provides an overview of the parameters and shows what parameters:

- are static or dynamic,
- can be used for the module as a whole or for a channel group.

Table 2-10 Parameter of SM 322; DO 4 x 24V/10mA and SM 322; DO 4 x 15V/20mA

Parameter	SM 322; DO 4 x 24 V DC/10mA or SM 322; DO 4 x 15V/20mA			
	Value range	Default	Type	Effective range
Basic settings				
• Diagnostic interrupt enable	yes/no	no	dynamic	Module
• Retain last value	yes/no	no	dynamic	Module
• Switch to substitute value	yes/no	yes	dynamic	Module
• Substitute value	0 / 1	0	dynamic	Module
Diagnostics				
• Short to chassis ground	yes/no	no	static	Channel group
• Wire break ¹⁾	yes/no	no	static	Channel group
• Supply voltage fault	yes/no	no	static	Channel group

¹⁾ If wire break diagnostic enable is not parameterized, there will be no indication by the channel fault LED in the event of wire break.

Diagnostics

You can use the diagnostic function to determine whether signal output takes place without errors.

Parameterizing diagnostics

The diagnostics is parameterized with *STEP 7*.

Diagnostic evaluation

When evaluating the diagnostics, a differentiation must be made between configurable and non-configurable diagnostic messages. In the case of the configurable diagnostic messages (e.g. short to M), diagnostics is only signalled when diagnostic evaluation has been enabled by means of appropriate parameterization (parameter "diagnostics short to M").

Non-configurable diagnostic messages are general, i.e. independent of parameterization.

A diagnostic signal results in a diagnostic interrupt being triggered providing the diagnostic interrupt has been enabled by way of parameterization.

Irrespective of the parameterization, known module errors always result in the SF LED or the corresponding channel fault LED lighting irrespective of the CPU operating status (at POWER ON).

Exception: The SF LED and the corresponding channel fault LED light in the event of a wire break only when parameterization is enabled.

Diagnostics of digital output module

Table 2-11 provides an overview of the diagnostic messages. Diagnostics is enabled in *STEP 7* (see Table 2-10).

The diagnostic information refers to either the individual channels or the entire module.

Table 2-11 Diagnostic messages of 322; DO 4 x 24V/10mA and SM 322; DO 4 x 15V/20mA

Diagnostic message	Effective range of diagnostics	configurable
M-short-circuit	Channel group	yes
Wire break		
No load voltage		
Module not parameterized	Module	no
No external auxiliary supply		
No internal auxiliary supply		
Fuse blown		
Watchdog triggered		
EPROM error		
RAM error		
CPU error		

Wire break detection

A wire break is detected at a current ≤ 0.15 mA.

Reading out diagnostic messages

You can read out system diagnostics with *STEP 7*. You can read detailed diagnostic messages from the module in the user program with SFC 59 (refer to /235/).

Faults and corrective measures

Table 2-12 provides a list of possible causes, marginal conditions for fault recognition and corresponding corrective measures for individual diagnostic messages.

Bear in mind that, in order to detect faults which are indicated by means of configurable diagnostic messages, must also be parameterized accordingly.

Table 2-12 Diagnostic messages as well as fault causes and corrective measures for SM 322; DO 4 x 24V/10mA and SM 322; DO 4 x 15V/20mA

Diagnostic message	Fault recognition at	Possible fault cause	Corrective measures
Chassis ground short-circuit	Only when output at "1"	Output overload	Eliminate overload
		Short-circuit between the two output lines	Eliminate short-circuit
Wire break	Only when output at "1"	Conductor break between module and actuator	Make conductor connection
		Channel not used (open)	Disable channel by parameterization "diagnostics wire break"
No-load voltage	Only when output at "1"	Failure of internal channel supply voltage	Replace module
Incorrect parameters in module	General	Invalid parameters loaded in module by means of SFC	Check parameterization of module and re-load valid parameters
Module not parameterized	General	Invalid parameters loaded in module by means of SFC	Check parameterization of module and re-load valid parameters
No external auxiliary supply	General	No L+ supply voltage of module	Supply L+
No internal auxiliary supply	General	No L+ supply voltage of module	Supply L+
		Module-internal fuse defective	Replace module
Fuse blown	General	Module-internal fuse defective	Replace module
Time watchdog tripped EPROM error RAM error CPU error	General	High electromagnetic interference at times	Eliminate interference sources and switch CPU supply voltage OFF/ON
		Module defective	Replace module

Interrupts The digital output can trigger a diagnostic interrupt.

Parameterizing interrupts Interrupts are parameterized with *STEP 7*.

Default setting The interrupts are inhibited as the default.

Diagnostic interrupt If enabled, the module triggers a diagnostic interrupt when a fault is recognized or is no longer applicable (e.g. short to M). diagnostic functions inhibited by parameterization cannot trigger an interrupt. The CPU interrupts processing of the user program or low-priority classes and processes the diagnostic interrupt module (OB 82).

Influence of supply voltage and operating status The output values are dependent on the supply voltages and CPU operating status.

Table 2-13 provides an overview of these dependencies.

Table 2-13 Dependencies of output values on the CPU operating status and supply voltage L+ of SM 322; DO 4 x 24V/10mA and SM 322; DO 4 x 15V/20mA

Operating status CPU		Supply voltage L+ at digital module	Output value of digital module
POWER ON	RUN	L+ applied	CPU value
		L+ not applied	0-signal
	STOP	L+ applied	Substitute value / last value Substitute value for 0-signal is default setting
		L+ not applied	0-signal
POWER OFF	–	L+ applied	0-signal
		L+ not applied	0-signal

Failure of the supply voltage in the SM 322; DO 4 x 24V/10mA is always indicated by the SF LED on the front of the module and additionally entered in diagnostics.

2.3 Digital Output Module SM 322; DO 4 x 15V/20mA

Order number 6ES7 322-5RD00-0AB0

Characteristics Refer to the description of the digital output module SM 322; DO 4 x 24V/10mA (see Section 2.2) for the module characteristics.

Wiring diagram Fig. 2-5 shows the terminal diagram of SM 322; DO 4 x 15V/20mA.

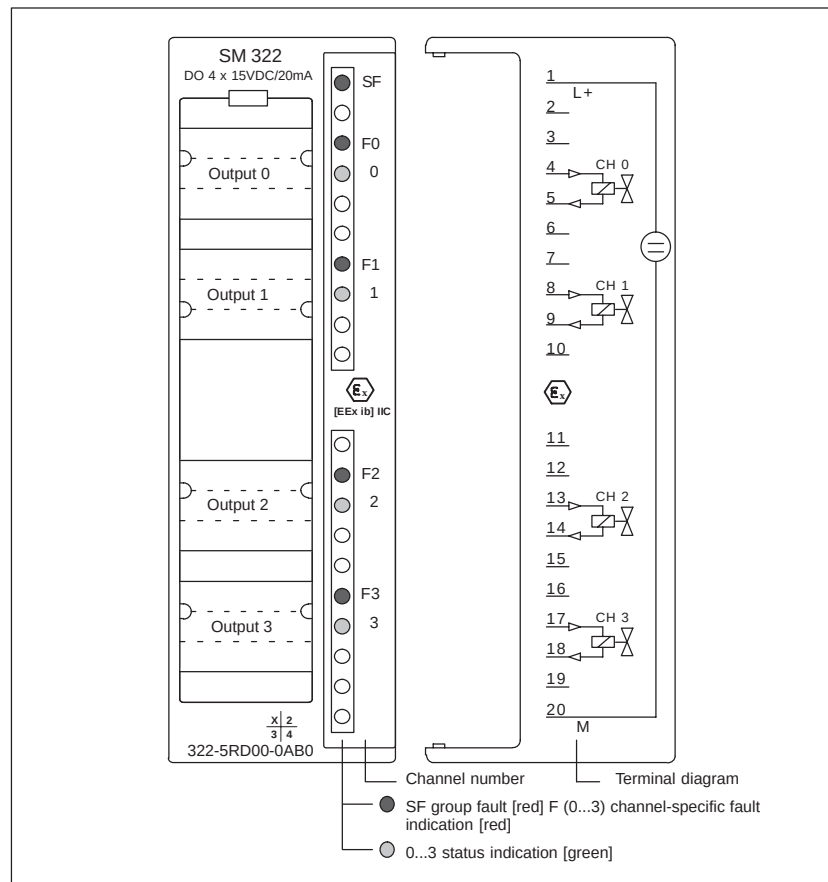


Fig. 2-5 Wiring diagram of SM 322; DO 4 x 15V/20mA

Notes on intrinsically-safe installation

You must connect the DM 370 dummy module between the CPU or IM 153-2 (distributed configuration) and the Ex I/O modules whose signal cables lead into the hazardous area. In a distributed configuration with an active backplane bus, you should use the explosion-proof partition instead of the dummy module. Additional information on system design can be found in Sections 1.3 - 1.5.

Power supply for a intrinsically-safe structure

In order to maintain the clearances and creepage distances, L+ / M must be routed via the line chamber LK393 when operating modules with signal cables that lead to the hazardous location, see Section 1.2.

Block diagram

Fig. 2-6 shows the block diagram of SM 322; DO 4 x 15V/20mA.

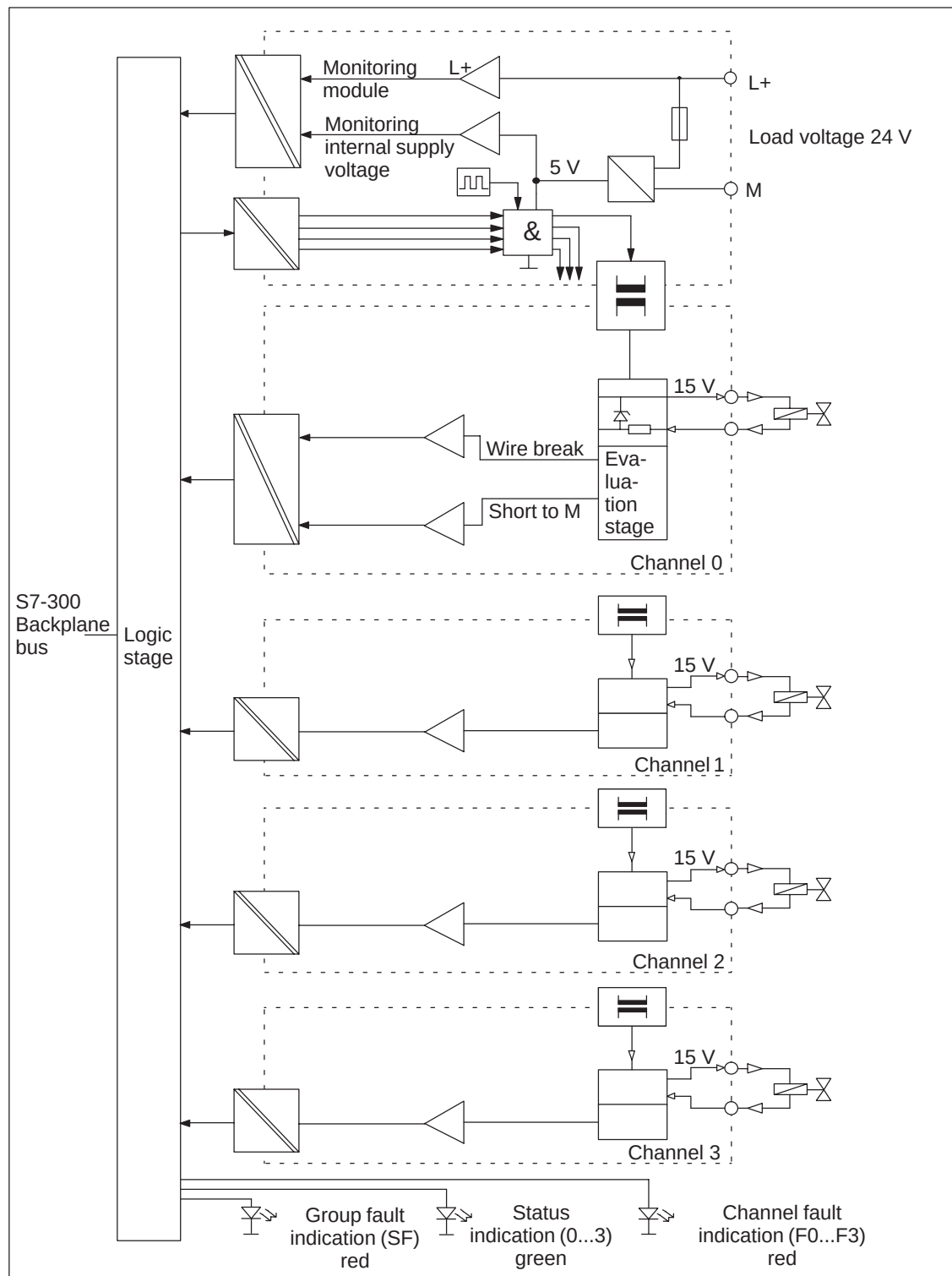


Fig. 2-6 Block diagram of digital output module SM 322; DO 4 x 15V/20mA

Digital output SM 322; DO 4 x 15V/20mA		Voltages, currents, potentials continued	
Dimensions and Weight		Permissible difference in potential (U_{ISO}) for signals from non-hazardous area	
Dimensions W x H x D (mm)	40 x 125 x 120	- Between channels and backplane bus 400 V DC 250 V AC - Between channels and load voltage L+ 400 V DC 250 V AC - Between channels 400 V DC 250 V AC - Between backplane bus and load voltage L+ 75 V DC 60 V AC	
Weight	approx. 230 g		
Module-specific data		Insulation tested	
Number of outputs	4	- Channels with respect to backplane bus and load voltage L+ with 1500 V AC - Channels among each other with 1500 V AC - Between load voltage L+ and backplane bus with 500 V DC	
Line length, shielded	max. 200 m	Current input	
Type of protection PTB (see Appendix A)	[EEx ib] IIC to EN 50020	- From backplane bus max. 70 mA - From load voltage L+ (at rated data) max. 160 mA	
Test number	Ex-96.D.2102 X	Module power loss typical 3 W	
Type of protection FM (see Appendix B)	CL I, DIV 2, GP A, B, C, D T4	Status, interrupts, diagnostics	
Voltages, currents, potentials		Status indication	
Bus power supply	5 V DC	Outputs green LED per channel	
Rated load voltage L+	24 V DC	Interrupts	
Reverse voltage protection	yes	Diagnostic interrupt configurable	
Total current of outputs		Diagnostic functions	
- Horizontal arrangement up to 60 °C - Vertical arrangement up to 40 °C	No restrictions	- Group fault indication red LED (SF) - Channel fault indication red LED (F) per channel - Diagnostic functions readout possible	
Galvanic isolation		Monitoring for	
- Between channels and backplane bus - Between channels and load voltage L+ - Between channels - Between backplane bus and load voltage L+	yes	- Short-circuit $I > 20.5 \text{ mA } (\pm 10\%)$ - Wire break $I \leq 0.15 \text{ mA}$	
Permissible difference in potential (U_{ISO}) of signals from hazardous area			
- Between channels and backplane bus - Between channels and load voltage L+ - Between channels - Between backplane bus and load voltage L+	60 V DC 30 V AC 60 V DC 30 V AC 60 V DC 30 V AC 60 V DC 30 V AC		

Safety data (refer to Certificate of Conformity in Appendix A)

Maximum values of output circuits (per channel)

- U_0 (no-load output voltage) max. 15.75 V
- I_0 (short-circuit current) max. 85 mA
- P_0 (load power) max. 335 mW
- L_0 (permissible external inductance) max. 5 mH
- C_0 (permissible external capacitance) max. 500 nF
- U_m (error voltage) max. 60 V DC
30 V AC
- T_a (permissible ambient temperature) max. 60 °C

Data for actuator selection

Outputs

- No-load voltage U_{A0} 15 V DC $\pm 5\%$
- Internal resistance R_i 200 Ω $\pm 5\%$

Curve vertex E

- Voltage U_E 10 V DC $\pm 10\%$
- Current I_E 20.5 mA $\pm 10\%$

Parallel connection of 2 outputs

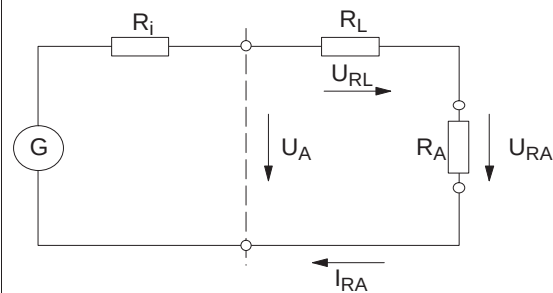
- For redundant activation of a load Not possible
- For increasing power Possible, see Manual "S7-300, M7-300, ET 200M Automation Systems Principles of Intrinsically-Safe Design" Section "Intrinsically-Safe Circuit with Two or More Items of Associated Electrical Apparatus (Requirements for Installation in Zones 0 and 1)"

Switching frequency

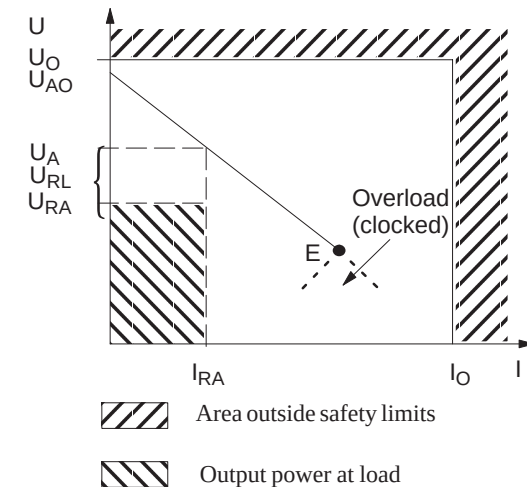
- At resistive load 100 Hz
- At inductive load ($L < L_0$) 100 Hz

Short-circuit protection of output

- Response threshold Curve vertex E

Block diagram

- G: Generator
 R_i : Internal resistor
 R_L : Line resistor
 R_A : Load resistor
 U_{A0} : No-load voltage
 U_A : Output voltage
 U_{RL} : Voltage drop at line resistor
 U_{RA} : Voltage drop at load
 U_0 : Max. output voltage
 I_0 : Max. output current
 I_{RA} : Load current

Output characteristic

- E: Curve vertex (U_E , I_E)
 $U_E = 10 \text{ V} \pm 10\%$
 $I_E = 20.5 \text{ mA} \pm 10\%$
 Output current electronically clocked at overload.
 Clock ratio $\sim 1:15$

SIMATIC S7 Ex Analog Modules

In this chapter

The following SIMATIC S7 Ex analog modules are described in this chapter:

- Analog input SM 331; AI 8 x TC/4 x RTD (6ES7 331-7SF00-0AB0)
- Analog input SM 331; AI 4 x 0/4...20 mA (6ES7 331-7RD00-0AB0)
- Analog output SM 332; AO 4 x 0/4...20 mA (6ES7 332-5RD00-0AB0)

Chapter overview

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Notes

You will find information on the relevant safety standards and on other safety regulations in Appendix B.

The *General Technical Specifications* for S7-300, M7-300 modules in /71/ also apply.

3.1 Analog Value Representation

Analog values The analog values for **all** measuring ranges and output ranges which you can use in conjunction with the S7-300 Ex analog modules are explained in this section.

3.1.1 Analog Value Representation of Analog Input and Output Values

Conversion of analog values The CPU processes the analog values only in binary form.
Analog input modules convert the analog process signal into digital form.
Analog output modules convert the digital output value into an analog signal.

Analog value representation The digitized analog value is the same for both input and output values with the same rated range.
The analog values are represented as two's complement.
Table 3-1 shows the analog value representation of analog modules:

Table 3-1 Analog value representation

Resolution	Analog value															
Bit number	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bitsignificance	Sign	2^{14}	2^{13}	2^{12}	2^{11}	2^{10}	2^9	2^8	2^7	2^6	2^5	2^4	2^3	2^2	2^1	2^0

Sign The sign of the analog value is always in bit number 15:

- "0" → +
- "1" → -

3.1.2 Analog Representation for Measuring Ranges of Analog Inputs

Introduction	The tables in this section indicate the digitized analog values for the effective measuring ranges of analog modules.
How to read the measured value tables	Tables 3-3 to 3-19 list the digitized analog values for different effective measuring ranges. Since the binary representation of analog values is always the same, these tables contain only a comparison of the measuring ranges with respect to the relevant units.
Measured value resolution	Deviating from this, a Sigma-Delta AD-converter is used with the analog input modules described in the manual. Irrespective of the configurable integration time, this converter always makes available the maximum representable 15 Bit +sign. Lower resolution ratings than indicated in the specifications are due to conversion noise based on the shorter integration times (2.5, 16²/3, 20 ms). The different integration times change nothing with regard to numerical representation of the measured values. The number of stable bits is specified in the technical specifications. The number of stable bits is the resolution, at which, despite noise, the "no missing code"-characteristics of the AD-converter are guaranteed. The bits which are no longer stable at shorter integration times are marked with "x" in the following tables.

Table 3-2 Representation of the smallest stable unit of the analog value

Stable bits (+ sign)	Smallest stable unit		Analog value	
	Decimal	Hexadecimal	High-Byte	Low-Byte
9	64	40 _H	Sign 0 0 0 0 0 0 0 0	0 1 x x x x x x
10	32	20 _H	Sign 0 0 0 0 0 0 0 0	0 0 1 x x x x x
12	8	8 _H	Sign 0 0 0 0 0 0 0 0	0 0 0 0 1 x x x
13	4	4 _H	Sign 0 0 0 0 0 0 0 0	0 0 0 0 0 1 x x
15	1	1 _H	Sign 0 0 0 0 0 0 0 0	0 0 0 0 0 0 0 1

What can you do with the noise-prone bits	At a constant input voltage, noise causes distribution of the supplied value by more than ± 1 digit. In the majority of cases, these "unsteady" values can be used as they are. In any case, this is the most effective option when subsequent processing has integral action characteristics (integrator, controller, etc.) in any form whatsoever. If this unsteady state is undesirable (e.g. for display/indication), you can • mask out the "x" bits • round up to "stable" bits • filter the successive values
--	--

With these options you must first ensure by way of interrogation that you will not change the coding for invalid measured values ($-32768 / 8000_{\text{H}}$ and $32767 / 7FFF_{\text{H}}$) or you incorporate it in the filtering process.

Voltage measuring ranges

Table 3-3 shows the representation of the digitized measured value for the voltage measuring ranges $\pm 25 \text{ mV}$, $\pm 50 \text{ mV}$, $\pm 80 \text{ mV}$, $\pm 250 \text{ mV}$, $\pm 500 \text{ mV}$ and $\pm 1 \text{ V}$.

Table 3-3 Representation of the digitized measured value of an analog input module (voltage measuring ranges)

Measuring range						Units		Range
$\pm 25 \text{ mV}$	$\pm 50 \text{ mV}$	$\pm 80 \text{ mV}$	$\pm 250 \text{ mV}$	$\pm 500 \text{ mV}$	$\pm 1 \text{ V}$	decimal	hexadecimal	
> 29.397	> 58.794	> 94.071	> 293.96	> 587.94	> 1.1750	32767	7FFF _H	Overflow
29.397	58.794	94.071	293.96	587.94	1.1750	32511	7EFF _H	Overrange
:	:	:	:	:	:	:	:	
25.001	50.002	80.003	250.02	500.02	1.0001	27649	6C01 _H	
25.000	50.000	80.000	250.00	500.00	1.0000	27648	6C00 _H	Rated range
18.750	37.500	60.000	187.50	375.00	0.7500	20736	5100 _H	
:	:	:	:	:	:	:	:	
- 18.750	- 37.500	- 60.000	- 187.50	- 375.00	- 0.7500	-20736	AF00 _H	
- 25.000	- 50.000	- 80.000	- 250.00	- 500.00	- 1.0000	-27648	9400 _H	
- 25.001	- 50.002	- 80.003	- 250.01	- 500.02	- 1.0001	-27649	93FF _H	Underrange
:	:	:	:	:	:	:	:	
- 29.398	- 58.796	- 94.074	- 293.98	- 587.96	- 1.1750	-32512	8100 _H	
$<- 29.398$	$<- 58.796$	$<- 94.074$	$<- 293.98$	$<- 587.96$	$<- 1.1750$	-32768	8000 _H	Underflow

Current measuring ranges

Table 3-4 shows the representation of the digitized measured value for the current measuring ranges 0 to 20 mA and 4 to 20 mA.

Table 3-4 Representation of the digitized measured value of analog input module SM 331; AI 4 x 0/4...20 mA and AI 2 x 0/4...20 mA HART

Measuring range from 0 to 20 mA	Measuring range from 4 to 20 mA	Units		Range
		decimal	hexadecimal	
> 23.515	> 22.810	32767	7FFF _H	Overflow
23.515	22.810	32511	7EFF _H	Overrange
:	:	:	:	
20.0007	20.0005	27649	6C01 _H	
20.000	20.000	27648	6C00 _H	Rated range
14.998	16.000	20736	5100 _H	
:	:	:	:	
0.0	4.000	0	0 _H	
< 0.0 ²⁾	3.9995	-1	FFFF _H	Wire break limit I ≤ 3.60 mA to NAMUR ¹⁾
	3.800	-345	FEA7 _H	
	3.600	-691	FD4D _H	
	:	:	:	Underrange
	1,1852	-4864	ED00 _H	Underflow
	< 1,1852	32767	7FFF _H	

¹⁾ NAMUR limits are evaluated only if wire break diagnostics is enabled. When wire break diagnostics is enabled, 7FFF_H is output if the current value drops below 3.6 mA. If the value increases again to above 3.8 mA, the wire break signal is canceled and the current value is output again.

²⁾ Negative measured values cannot be recorded. In the event of analog values < 0 mA, the relevant representation of the digital measured value of 0 mA is retained.

**Effective
measuring ranges
of resistance
sensors**

Table 3-5 shows the representation of the digitized measured value for resistance sensors with the measuring ranges 150 Ω , 300 Ω and 600 Ω .

Table 3-5 Representation of the digitized measured value of an analog input module (resistance sensor)

Measuring range 150 Ω	Measuring range 300 Ω	Measuring range 600 Ω	Units		Range
			decimal	hexadecimal	
> 176.383	> 352.767	> 705.534	32767	7FFF _H	Overflow
176.383	352.767	705.534	32511	7EFF _H	Overrange ¹⁾
:	:	:	:	:	
150.005	300.011	600.022	27649	6C01 _H	
150.000	300.000	600.000	27648	6C00 _H	Rated range
112.500	225.000	450.000	20736	5100 _H	
:	:	:	:	:	
0.000	0.000	0.000	0	0 _H	
(negative values physically not possible)					

¹⁾ The same degree of accuracy as in the rated range is guaranteed in the overrange.

Temperature range, standard, Pt 100, Pt 200

Table 3-6 shows the representation of the digitized measured value for the standard temperature range of the sensor Pt 100, Pt 200 in accordance with DIN 43760 and IEC 751.

Table 3-6 Representation of the digitized measured value of an analog input module (temperature range, standard; Pt 100, Pt 200)

Temperature range, standard 850 °C Pt 100, Pt 200 in °C	Decimal	Hexadecimal	Range
> 1300.0	32767	7FFF _H	Overflow
1300.0 : 850.1	13000 : 8501	32C8 _H : 2135 _H	Ovrange ¹⁾
850.0 : -200.0	8500 : -2000	2134 _H : F830 _H	Rated range
-200.1 : -240.0	-2001 : -2400	F82F _H : F6A0 _H	Underrange ²⁾
< -240.0	-32768	8000 _H	Underflow

- 1) The characteristic of the Pt 100, Pt 200 sensor is not defined in the overrange. The overrange has been extended to 1300°C in order to be able to incorporate future technical developments of platinum thermal resistors (thermistors). It is not possible to specify the accuracy of this range.
- 2) The characteristic of the Pt 100, Pt 200 sensor is not defined in the underrange. The rise of the characteristic curve is retained on leaving the linearized rated range. It is not possible to specify the accuracy of this range.

Temperature range, climatic, Pt 100, Pt 200

Table 3-7 shows the representation of the digitized measured value for the climatic temperature range of the sensor Pt 100, Pt 200 in accordance with DIN 43760 and DIN IEC 751.

Table 3-7 Representation of the digitized measured value of an analog input module (temperature range, climatic, Pt 100, Pt 200)

Temperature range, climatic Pt 100, Pt 200 in °C	Decimal	Hexadecimal	Range
> 325.12	32767	7FFF _H	Overflow
325.12 : 276.49	32512 : 27649	7F00 _H : 6C01 _H	Overrange ¹⁾
276.48 : -200.00	27648 : -20000	6C00 _H : B1E0	Rated range
-200.01 : -240.00	-20001 : -24000	B1E1 : A240 _H	Underrange ²⁾
< - 240.00	-32768	8000 _H	Underflow

- ¹⁾ The same degree of accuracy as in the rated range is guaranteed in the overrange Pt 100, Pt 200 climatic.
- ²⁾ The characteristic of the Pt 100, Pt 200 sensor is not defined in the underrange. The rise of the characteristic curve is retained on leaving the linearized rated range. It is not possible to specify the accuracy of these ranges.

Temperature range, standard, Ni 100

Table 3-8 shows the representation of the digitized measured value for the standard temperature range of the sensor Ni 100 in accordance with DIN 43760.

Table 3-8 Representation of the digitized measured value of an analog input module (temperature range, standard; Ni 100)

Temperature range standard Ni 100 in °C	Decimal	Hexadecimal	Range
> 295.0	32767	7FFF _H	Overflow
295.0 : 250.1	2950 : 2501	686 _H : 9C5 _H	Overrange ¹⁾
250.0 : -60.0	2500 : -600	9C4 _H : FDA8 _H	Rated range
-60.1 : -105.0	-601 : -1050	FDA7 _H : FF97 _H	Underrange ¹⁾
< - 105.0	-32768	8000 _H	Underflow

- ¹⁾ The characteristic of the Ni 100 sensor is not defined in the overrange and underrange. The rise of the characteristic curve is retained on leaving the linearized rated range. It is not possible to specify the accuracy of these ranges.

Temperature range, climatic, Ni 100

Table 3-9 shows the representation of the digitized measured value for the climatic temperature range of the sensor Ni 100 in accordance with DIN 43760.

The same value range as in the standard range of the Ni 100 sensor applies in the climatic range Ni 100 only with a higher resolution of 0.01 °C instead of 0.1 °C.

Table 3-9 Representation of the digitized measured value of an analog input module (temperature range, climatic, Ni 100)

Temperature range, climatic Ni 100 in °C	Decimal	Hexadecimal	Range
> 295.00	32767	7FFF _H	Overflow
295.00 : 250.01	29500 : 25001	733C _H : 61A9 _H	Overrange ¹⁾
250.00 : -60.00	25000 : -6000	61A8 _H : E890 _H	Rated range
-60.01 : -105.00	-6001 : -10500	E88F _H : D6FC _H	Underrange ¹⁾
< - 105.00	-32768	8000 _H	Underflow

- ¹⁾ The characteristic of the Ni 100 sensor is not defined in the overrange and underrange. The rise of the characteristic curve is retained on leaving the linearized rated range. It is not possible to specify the accuracy of these ranges.

DIN IEC 584

The basic thermal e.m.f. values specified in the following comply with DIN IEC 584.

Temperature range type T

Table 3-10 shows the representation of the digitized measured value for the temperature range, sensor type T.

Table 3-10 Representation of the digitized measured value of an analog input module (temperature range, type T)

Temperature range in °C	Decimal	Hexadecimal	Range
> 540.0	32767	7FFF _H	Overflow
540.0	5400	1518 _H	Overrange ²⁾
:	:	:	
400.1	4001	0FA1 _H	
400.0	4000	0FA0 _H	Rated range
:	:	:	
:	:	:	
-230.0 ¹⁾	-2300	F704 _H	
:	:	:	
-270.0	-2700	F574 _H	Underrange ²⁾
≤ -270.1	≤ -2701	≤ F573 _H	

In the case of incorrect wiring (e.g. polarity reversal, open inputs) or a sensor fault in the negative range (e.g. incorrect type of thermocouple), on dropping below F0C4_H the analog input module signals underflow and outputs 8000_H.

- ¹⁾ The module linearizes the range from +400 to -230°C for type T. Below -230°C, the rise of the characteristic curve decreases to such an extent that, from this point, precision evaluation is no longer possible. The rise in the characteristic curve at this point is retained until underrange is reached.
- ²⁾ The characteristic of the thermocouple is not defined in the overrange and underrange. The rise of the characteristic curve is retained on leaving the linearized range. It is not possible to specify the accuracy of these ranges.

**Temperature range
type U**

Table 3-11 shows the representation of the digitized measured value for the temperature range, sensor type U.

Table 3-11 Representation of the digitized measured value of an analog input module (temperature range, type U)

Temperature range in °C	Decimal	Hexadecimal	Range
> 850.0	32767	7FFF _H	Overflow
850.0 : 600.1	8500 : 6001	2134 _H : 0FA1 _H	Overrange ¹⁾
600.0 : : -200.0	6000 : : -2000	0FA0 _H : : F830 _H	Rated range
≤ -200.1	≤ -2001	≤ F82F _H	Underrange ¹⁾
In the case of incorrect wiring (e.g. polarity reversal, open inputs) or a sensor fault in the negative range (e.g. incorrect type of thermocouple), on dropping below F380 _H the analog input module signals underflow and outputs 8000 _H .			

- ¹⁾ The characteristic of the thermocouple is not defined in the overrange and underrange. The rise of the characteristic curve is retained on leaving the linearized range. It is not possible to specify the accuracy of these ranges.

Temperature range type E

Table 3-12 shows the representation of the digitized measured value for the temperature range, sensor type E.

Table 3-12 Representation of the digitized measured value of an analog input module (temperature range, type E)

Temperature range in °C	Decimal	Hexadecimal	Range
> 1200.0	32767	7FFF _H	Overflow
1200.0 : 1000.1	12000 : 10001	2EE0 _H : 2711 _H	Overrange ²⁾
1000.0 : : -150.0 ¹⁾ : -270.0	10000 : : -1500 : -2700	2710 _H : : FA24 _H : F574 _H	Rated range
≤ -270.1	≤ -2701	≤ F573 _H	Underrange ²⁾
In the case of incorrect wiring (e.g. polarity reversal, open inputs) or a sensor fault in the negative range (e.g. incorrect type of thermocouple), on dropping below F0C4 _H the analog input module signals underflow and outputs 8000 _H .			

- ¹⁾ The module linearizes the range from +1000 to -150°C for type E. Below -150°C, the rise of the characteristic curve decreases to such an extent that, from this point, precision evaluation is no longer possible. The rise in the characteristic curve at this point is retained until underrange is reached.
- ²⁾ The characteristic of the thermocouple is not defined in the overrange and underrange. The rise of the characteristic curve is retained on leaving the linearized range. It is not possible to specify the accuracy of these ranges.

Temperature range type J

Table 3-13 shows the representation of the digitized measured value for the temperature range, sensor type J.

Table 3-13 Representation of the digitized measured value of an analog input module (temperature range, type J)

Temperature range in °C	Decimal	Hexadecimal	Range
> 1360.0	32767	7FFF _H	Overflow
1360.0 : 1200.1	13600 : 12001	3520 _H : 2EE1 _H	Overrange ¹⁾
1200.0 : : -210.0	12000 : : -2100	2EE0 _H : : F7CC _H	Rated range
≤ -210.1	≤ -2101	≤ F7CB _H	Underrange ¹⁾
In the case of incorrect wiring (e.g. polarity reversal, open inputs) or a sensor fault in the negative range (e.g. incorrect type of thermocouple), on dropping below F31C _H the analog input module signals underflow and outputs 8000 _H .			

- ¹⁾ The characteristic of the thermocouple is not defined in the overrange and underrange. The rise of the characteristic curve is retained on leaving the linearized rated range. It is not possible to specify the accuracy of these ranges. It is not possible to specify the accuracy of these ranges.

**Temperature range
type L**

Table 3-14 shows the representation of the digitized measured value for the temperature range, sensor type L.

Table 3-14 Representation of the digitized measured value of an analog input module (temperature range, type L)

Temperature range in °C	Decimal	Hexadecimal	Range
> 1150.0	32767	7FFF _H	Overflow
1150.0 : 900.1	13500 : 9001	2CEC _H : 2329 _H	Overrange ¹⁾
900.0 : : -200.0	9000 : : -2000	2328 _H : : F830 _H	Rated range
≤ -200.1	≤ -2001	≤ F82F _H	Underrange ¹⁾
In the case of incorrect wiring (e.g. polarity reversal, open inputs) or a sensor fault in the negative range (e.g. incorrect type of thermocouple), on dropping below F380 _H the analog input module signals underflow and outputs 8000 _H .			

- ¹⁾ The characteristic of the thermocouple is not defined in the overrange and underrange. The rise of the characteristic curve is retained on leaving the linearized rated range. It is not possible to specify the accuracy of these ranges.

**Temperature range
type K**

Table 3-15 shows the representation of the digitized measured value for the temperature range, sensor type K.

Table 3-15 Representation of the digitized measured value of an analog input module (temperature range, type K)

Temperature range in °C	Decimal	Hexadecimal	Range
> 1622.0	32767	7FFF _H	Overflow
1622.0	16220	3F5C _H	Overrange ²⁾
:	:	:	
1372.1	13721	3599 _H	
1372.0	13720	3598 _H	Rated range
:	:	:	
:	:	:	
-220.0 ¹⁾	-2200	F768 _H	
:	:	:	
-270.0	-2700	F574 _H	
≤ -270.1	≤ -2701	≤ F573 _H	Underrange ²⁾
In the case of incorrect wiring (e.g. polarity reversal, open inputs) or a sensor fault in the negative range (e.g. incorrect type of thermocouple), on dropping below F0C4 _H the analog input module signals underflow and outputs 8000 _H .			

- ¹⁾ The module linearizes the range from +1372 to -220 °C for type K. Below -220 °C, the rise of the characteristic curve decreases to such an extent that, from this point, precision evaluation is no longer possible. The rise in the characteristic curve at this point is retained until underrange is reached.
- ²⁾ The characteristic of the thermocouple is not defined in the overrange and underrange. The rise of the characteristic curve is retained on leaving the rated range. It is not possible to specify the accuracy of these ranges.

**Temperature range
type N**

Table 3-16 shows the representation of the digitized measured value for the temperature range, sensor type N.

Table 3-16 Representation of the digitized measured value of an analog input module (temperature range, type N)

Temperature range in °C	Decimal	Hexadecimal	Range
> 1550.0	32767	7FFF _H	Overflow
1550.0 : 1300.1	15500 : 13001	3C8C _H : 32C9 _H	Overrange ²⁾
1300.0 : : : -220.0 ¹⁾ : -270.0	13000 : : : -2200 : -2700	32C8 _H : : : F768 _H : F574 _H	Rated range
≤ -270.1	≤ -2701	≤ F573 _H	Underrange ²⁾

In the case of incorrect wiring (e.g. polarity reversal, open inputs) or a sensor fault in the negative range (e.g. incorrect type of thermocouple), ondropping below F0C4_H the analog input module signals underflow and outputs 8000_H.

- 1) The module linearizes the range from +1300 to -220°C for type N. Below -220°C, the rise of the characteristic curve decreases to such an extent that, from this point, precision evaluation is no longer possible. The rise in the characteristic curve at this point is retained until underrange is reached.
- 2) The characteristic of the thermocouple is not defined in the overrange and underrange. The rise of the characteristic curve is retained on leaving the rated range. It is not possible to specify the accuracy of these ranges.

**Temperature range
type R**

Table 3-17 shows the representation of the digitized measured value for the temperature range, sensor type R.

Table 3-17 Representation of the digitized measured value of an analog input module (temperature range, type R)

Temperature range in °C	Decimal	Hexadecimal	Range
> 2019.0	32767	7FFF _H	Overflow
2019.0 : 1769.1	20190 : 17691	4EDE _H : 451B _H	Overrange ¹⁾
1769.0 : : -50.0	17690 : : -500	451A _H : : FE0C _H	Rated range
-50.1 : -170.0	-501 : -1700	FE0B _H : F95C _H	Underrange ¹⁾
< -170.0	-32768	8000 _H	Underflow
In the case of incorrect wiring (e.g. polarity reversal, open inputs) or a sensor fault in the negative range (e.g. incorrect type of thermocouple), on dropping below F95C _H the analog input module signals underflow and outputs 8000 _H .			

- ¹⁾ The characteristic of the thermocouple is not defined in the overrange and underrange. The rise of the characteristic curve is retained on leaving the linearized rated range. It is not possible to specify the accuracy of these ranges.

**Temperature range
type S**

Table 3-18 shows the representation of the digitized measured value for the temperature range, sensor type S.

Table 3-18 Representation of the digitized measured value of an analog input module (temperature range, type S)

Temperature range in °C	Decimal	Hexadecimal	Range
> 1850.0	32767	7FFF _H	Overflow
1850.0 : 1769.1	18500 : 17691	4844 _H : 451B _H	Overrange ¹⁾
1769.0 : : -50.0	17690 : : -500	451A _H : : FE0C _H	Rated range
-50.1 : -170.0	-501 : -1700	FE0B _H : F95C _H	Underrange ¹⁾
< -170.0	-32768	8000 _H	Underflow
In the case of incorrect wiring (e.g. polarity reversal, open inputs) or a sensor fault in the negative range (e.g. incorrect type of thermocouple), ondropping below F95C _H the analog input module signals underflow and outputs 8000 _H .			

- ¹⁾ The characteristic of the thermocouple is not defined in the overrange and underrange. The rise of the characteristic curve is retained on leaving the linearized range. It is not possible to specify the accuracy of these ranges.

Temperature range type B

Table 3-19 shows the representation of the digitized measured value for the temperature range, sensor type B.

Table 3-19 Representation of the digitized measured value of an analog input module (temperature range, type B)

Temperature range in °C type B	Decimal	Hexadecimal	Range
> 2070.0	32767	7FFF _H	Overflow
2070.0	20700	50DC _H	Overrange ²⁾
:	:	:	
1820.1	18201	4719 _H	
1820.0	18200	4718 _H	Rated range
:	:	:	
:	:	:	
200.0 ¹⁾	2000	7D0 _H	
:	:	:	
0.0	0	0 _H	
-0.1	-1	FFFF _H	Underrange ²⁾
:	:	:	
-150.0	-1500	FF24 _H	
< -150.0	-32768	8000 _H	Underflow
In the case of incorrect wiring (e.g. polarity reversal, open inputs) or a sensor fault in the negative range (e.g. incorrect type of thermocouple), on dropping below FA24 _H the analog input module signals underflow and outputs 8000 _H .			

- ¹⁾ The module linearizes the range from +1820 to -200 °C for type B. Below -200 °C, the rise of the characteristic curve decreases to such an extent that, from this point, precision evaluation is no longer possible. The rise in the characteristic curve at this point is retained until underrange is reached.

The characteristic curve of the thermocouple does not feature monotone characteristics in the temperature range between 0 and 40 °C. Measured values from this range are not distinctly allocated to a specific temperature.

- ²⁾ The characteristic of the thermocouple is not defined in the overrange and underrange. The rise of the characteristic curve is retained on leaving the linearized range. It is not possible to specify the accuracy of these ranges.

3.1.3 Analog Value Representation for the Output Ranges of Analog Outputs

Current output ranges

Table 3-20 shows the representation of the current output ranges 0 to 20 mA and 4 to 20 mA.

Table 3-20 Representation of analog output range of analog output modules (current output ranges)

Output range 0 to 20 mA	Output range 4 to 20 mA	Units		Range
		Decimal	Hexadecimal	
0.0	0.0	≥ 32512	$\geq 7F00_H$	Overflow
23.515	22.81	32511	7EFF _H	Overrange
:	:	:	:	
20.0007	20.005	27649	6C01 _H	
20.000	20.000	27648	6C00 _H	Rated range
:	:	:	:	
0.0	4.000	0	0 _H	
0.0	3.9995	-1	FFFF _H	Underrange
	:	:	:	
	0.0	- 6912	E500 _H	
	0.0	- 6913	E4FF _H	Underflow
		:	:	
		- 32768	8000 _H	

Note

In the analog output SM 332; AO 4 x 0/4...20 mA, the linearity can decrease in the overrange at load resistances $> 425 \Omega$.

3.2 Connecting Transducers to Analog Inputs

In this chapter	Depending on the measurement mode, various transducers can be connected to analog input modules: • Voltage sensor • Current sensor as – 2-wire transducer – 4-wire transducer • Resistant sensor
Line for analog signals	Shielded conductors twisted in pairs are used for the analog signals. (refer to Section 1.8; Shielding and Measures to Counteract Interference Voltage)
Isolated analog input modules	In the isolated analog input modules there is no metallic connection between M- of the measuring circuit and the M- terminal of the CPU. Isolated analog input modules are used when there is to be a difference in potential U_{ISO} between the reference point M- of the measuring circuit and the M- terminal of the CPU. Take particular care to ensure that the difference in potential U_{ISO} does not exceed the permissible value. If there is a possibility that the permissible value for U_{ISO} may be exceeded or if you cannot exactly determine the difference in potential, you must connect the reference point M- of the measuring circuit to the M- terminal of the CPU. This also refers to unused inputs.
Isolation between channels	When there is isolation between them, the channels are supplied individually by transformers and the signals are transmitted by means of optocouplers. Metallic isolation allows for high differences in potential between the channels. In addition, very good values are achieved with regard to interference voltage rejection and crosstalk between the channels. SM 331; AI 4 x 0/4...20 mA features isolation between the channels. To facilitate channel isolation, the SM 331; AI 8 x TC/4 x RTD is equipped with optical semiconductor multiplexers which ensure a high common-mode range of $U_{CM} \leq 60$ V DC between the channels. This represents a virtually equivalent solution in practical applications. Larger differences in potential are permitted when using the modules for signals from non-Ex areas (refer to technical specifications of the modules).

Abbreviations

The abbreviations used in Figs. 3-1 and 3-2 have following meanings:

M +:	Measuring conductor (positive)
M -:	Measuring conductor (negative)
U_{ISO} :	Differences in potential between inputs and ground terminal M
U_{CM} :	Differences in potential between inputs
L+:	Power supply connection 24 V DC
M:	Ground terminal for 24 V DC power supply
P5V:	Supply voltage of module logic
$M_{internal}$:	Ground of module logic

Insulated measured value sensors

Insulated measured value sensors are not connected to the local ground potential. They facilitate floating operation. Due to local conditions or interference, differences in potential U_{CM} (static or dynamic) can occur between the input channels. However, these differences in potential must not exceed the permissible values for U_{CM} . If there is a possibility that the permissible value may be exceeded, the M- terminals of the input channels must be interconnected.

If there is a possibility of exceeding the permissible value for U_{ISO} (inputs with respect to backplane bus), the M- terminals of the input channels must be connected to the M- terminal of the CPU.

Fig. 3-1 shows the connection principle of insulated transducers to an isolated analog input module.

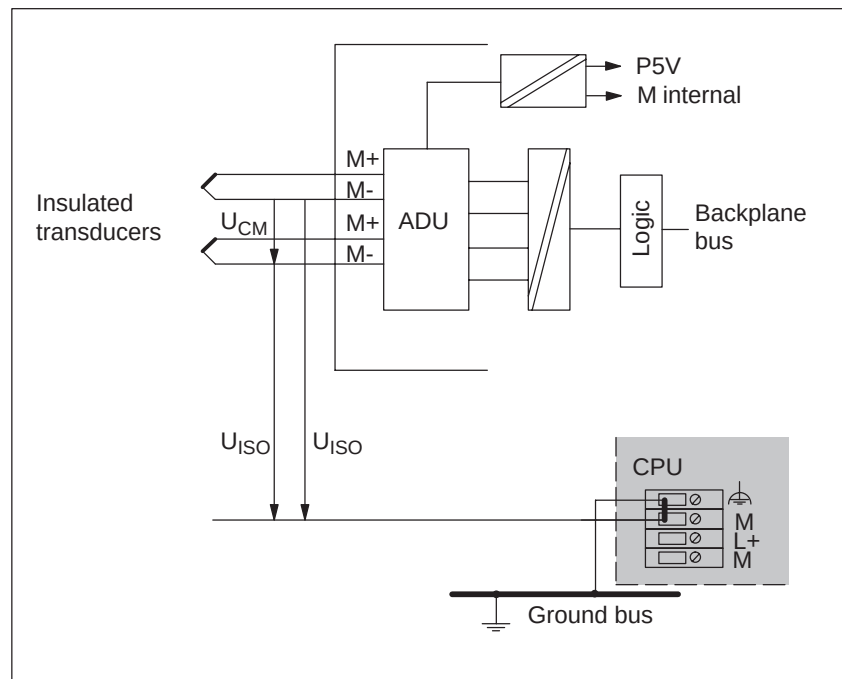


Fig. 3-1 Connection of insulated transducers to an isolated analog input module

Non-insulated transducers

Non-insulated transducers are connected to the ground potential on site. Due to local conditions or interference, differences in potential (static or dynamic) can occur between the locally distributed test points. Equipotential bonding conductors should be provided between the test points in order to avoid these differences in potential.

Fig. 3-2 shows the connection principle of non-insulated transducers to an isolated analog input module.

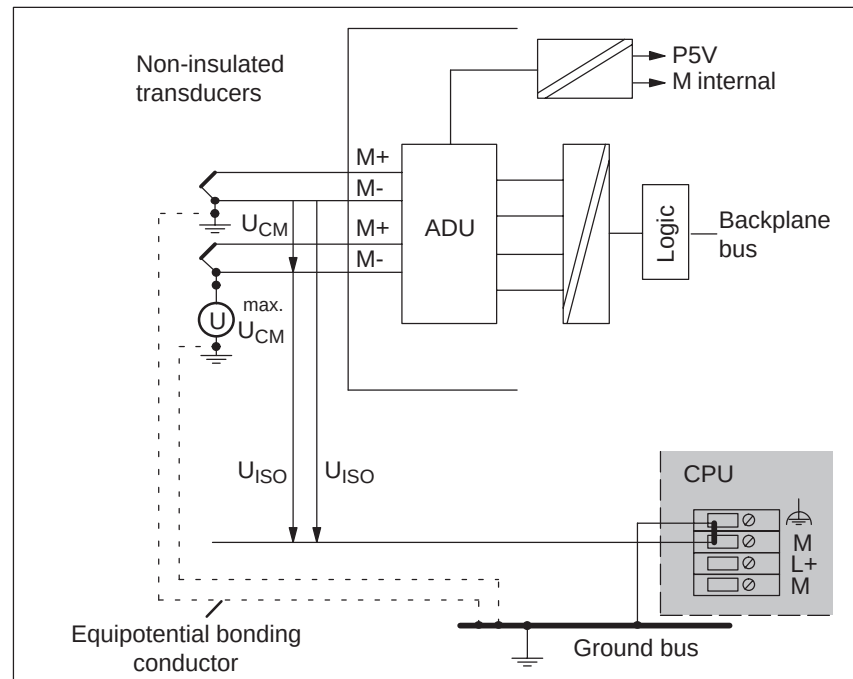


Fig. 3-2 Connection of non-insulated transducers to an isolated analog input module

3.3 Connection of Thermocouples, Voltage Sensors and Resistance Sensors to Analog Input SM 331; AI 8 x TC/4 x RTD

Overview

The following descriptions refer to the operation of transducers with the analog input module SM 331; AI 8 x TC/4 x RTD.

- A description the design and operating principle of thermocouples and the use of compensation boxes
- A description of how you connect thermocouples to analog inputs
- A description of how you connect voltage sensors to analog inputs
- A description of how you connect resistance thermometers and resistance sensors to analog inputs

3.3.1 Use and Connection of Thermocouples

Introduction

The design of thermocouples and what you must bear in mind when connecting thermocouples are described in this section.

Design of thermocouples

A thermocouple consists of

- the actual thermocouple (measuring sensor) and
- the necessary installation and connection parts.

The thermocouple is made up of two wires which are made of different metals or metal alloys and whose ends are soldered or welded together. The different material compositions produce different types of thermocouples, e.g. K, J, N. Irrespective of the type of thermocouple, the measuring principle is the same for all.

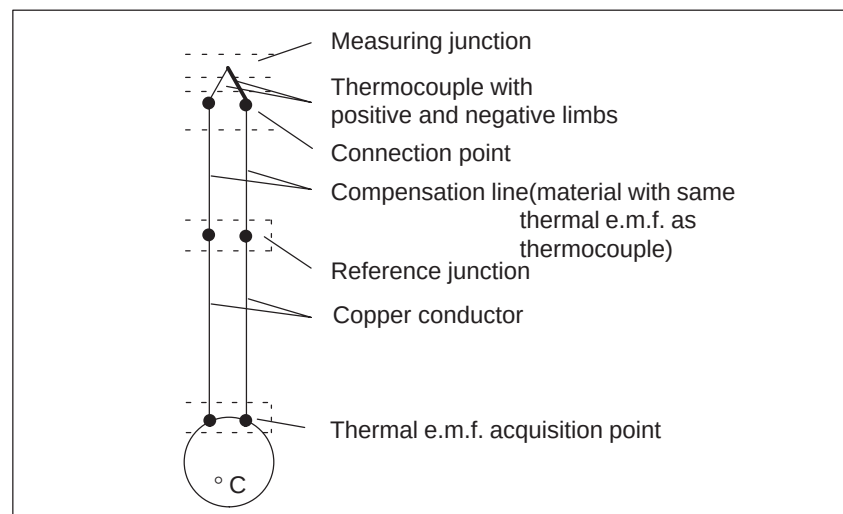


Fig. 3-3 Measuring circuit with thermocouple

Operating principle of thermocouples

If the measuring junction is subjected to a different temperature than at the free ends of the thermocouple (connection point), a voltage, i.e. the thermal e.m.f. is produced between the free ends.

The value of the thermal e.m.f. depends on the difference between the temperature at the measuring junction and the temperature at the free ends as well as on the type of material combination used for the thermocouple. Since a temperature difference is always recorded with a thermocouple, the free ends must be kept at a known temperature at a reference junction in order to determine the temperature of the measuring junction.

Extension to a reference junction

Thermocouples can be extended from their connection point by equalizing conductors up to a point with known temperature (reference junction).

The material of the equalizing conductors has the same thermal e.m.f. as the wires of the thermocouples. The conductors leading from the reference junction up to the analog module are made of copper.

Use of thermostatically controlled terminal boxes

It is possible to use temperature-compensated terminal boxes. Use boxes with reference junction temperatures of **0 °C or 50 °C** when using thermostatically controlled terminal boxes.

Compensation of thermocouples

External or internal compensation can be adopted depending on where (locally) you require the reference junction.

In the case of external compensation, the temperature of the reference junction for thermocouples is taken into consideration by means of a compensation box or thermal resistor.

In the case of internal compensation, the internal terminal temperature of the module is used for the comparison.

External compensation

The temperature of the reference junction can be compensated by means of a compensating circuit, e.g. by a compensation box.

The compensation box contains a bridge circuit which is calibrated for a certain reference junction temperature (compensating temperature). The terminal connections for the ends of the equalizing conductor of the thermocouple form the reference junction.

If the actual reference temperature deviates from the compensating temperature the temperature-dependent bridge resistance will change. A positive or negative compensation voltage is produced which is added to the thermal e.m.f.

Compensation boxes with a **reference junction temperature of 0 °C** must be used for the purpose of compensating the analog input modules.

A further external compensation option is to record the reference junction temperature with a thermal resistor in the climatic range (e.g. Pt 100).

The following conditions must be observed:

- External compensation by means of a compensation box can only be carried out for one specific type of thermocouple. This means all channels of this module which operate with external compensation must be parameterized for the same type of thermocouple.

Module diagnostic signals "incorrect parameters in module" and "reference channel error" for the corresponding channels (0..5) in the case of incorrect parameterization.

- The parameters of a channel group apply to both channels of this channel group (e.g. type of thermocouple, integration time, etc.)

Internal compensation

For the purposes of internal compensation, you can form the reference junction at the terminals of the analog input module. In this case, you must route the compensating conductors to the analog module. The internal temperature sensor senses the terminal temperature of the module. The thermocouples (also different types) connected to the module are compensated with this temperature.

Note

For the analog input module SM 331; AI 8 x TC/4 x RTD, the compensation box is connected to terminals 18 and 19.

The thermal resistor is connected to terminals 16, 17, 18 and 19 in order to register the reference junction temperature.

Thermocouple connection options

Figs. 3-4 to 3-8 show the different connection options for thermocouples with external and internal compensation.

The information provided in Section 3.2 on differences in potential U_{CM} and U_{ISO} between the individual circuits still retains its validity.

Abbreviations

The abbreviations used in the Figs. 3-4 to 3-10 have the following significance:

I_C^+ :	Positive connection of constant current output
I_C^- :	Negative connection of constant current output
M^+ :	Measuring conductor (positive)
M^- :	Measuring conductor (negative)
L^+ :	Power supply connection 24 V DC
M :	Ground terminal for 24 V DC power supply
$P5V$:	Supply voltage of module logic
$M_{internal}$:	Ground of module logic
U_V :	Isolated supply voltage for compensation box

U_{ISO} : Difference in potential between channels and M- terminal of CPU

U_{CM} : Differences in potential between channels

Thermocouples with compensation box

Necessary when all thermocouples which are connected to the inputs of a module and which have the same reference junction compensate as follows. The thermocouples which use a compensation box must be of the **same** type. Each of the thermocouples can be grounded at **any arbitrary point**.

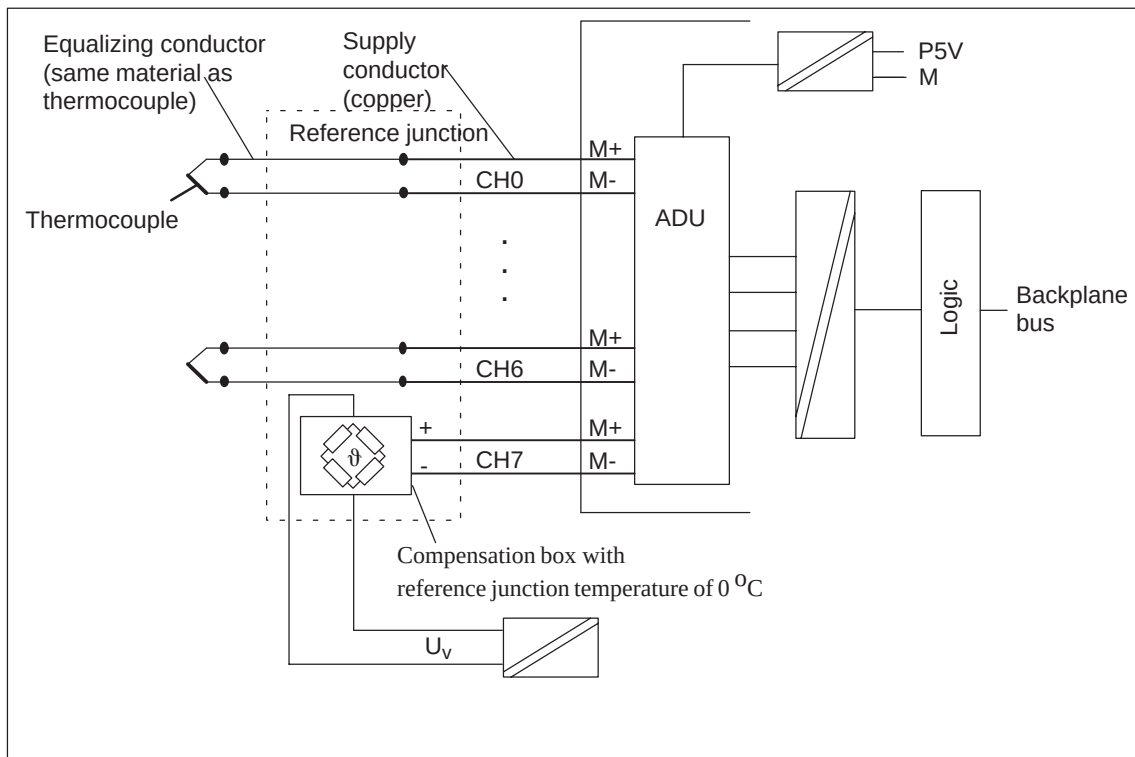


Fig. 3-4 Connection of thermocouples with external compensation box to the isolated analog input module SM 331; AI 8 x TC/4 x RTD

Thermocouples with direct looping-in of compensation box

When all thermocouples are wired **floating**, it is possible to loop the compensation box directly into the measuring circuit.

The compensation channel CH7 which is not required can now be used as an additional measurement input.

The measurement mode "**thermocouples with linearization and compensation to 0°C**" must be set for all channels. The thermocouples which use a compensation box must all be of the **same** type.

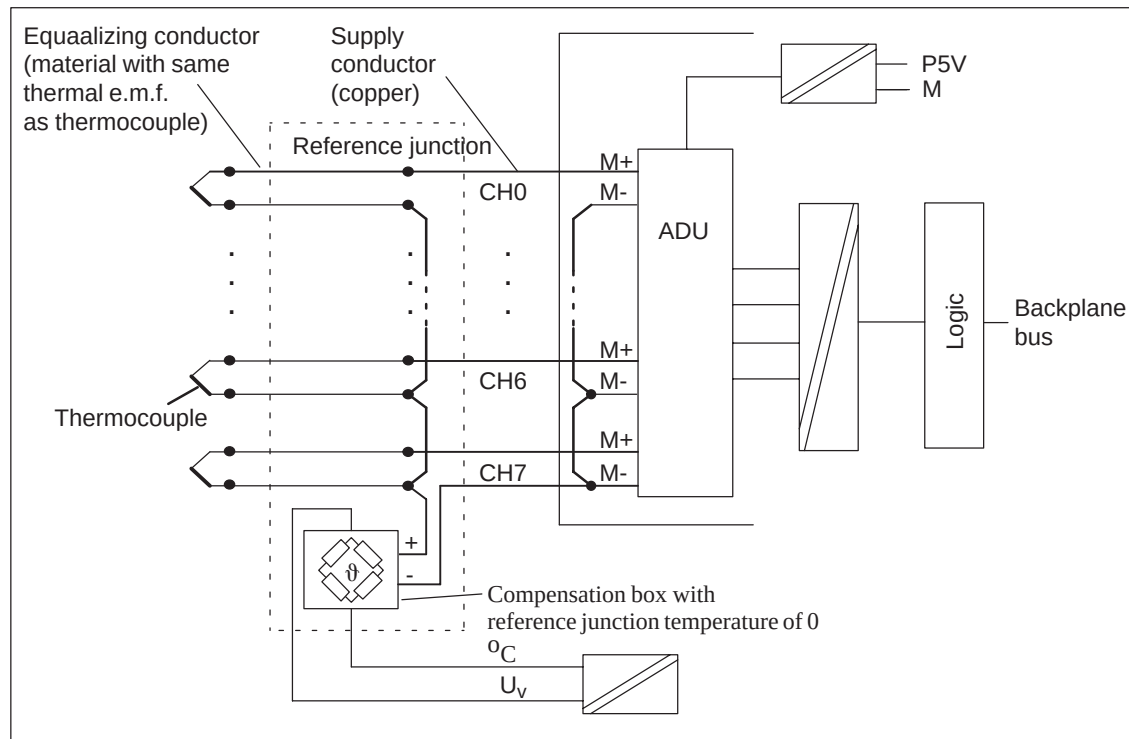


Fig. 3-5 Connection of floating thermocouples to a compensation box and measurement mode "Compensation to 0°C" with the analog input module SM 331; AI 8 x TC/4 x RTD

- Advantages:
- When using a compensation box with a reference junction temperature of 0 °C, the voltage corresponding to the reference junction temperature is subtracted **directly**.
 - **Channel 7** can be used as an **additional measuring channel** in this circuit variant.
 - The number of connection lines between the compensation box and analog input module is reduced.
 - Faults which are attributed to isolated compensation measurement do not occur.

Condition: The thermocouples which are routed to the same compensation box **must only be grounded once at one point**.

Thermocouples with temperature compensation at connection terminals

All 8 inputs are available for use as measuring channels when thermocouples are connected via a reference junction controlled to 0°C or 50°C.

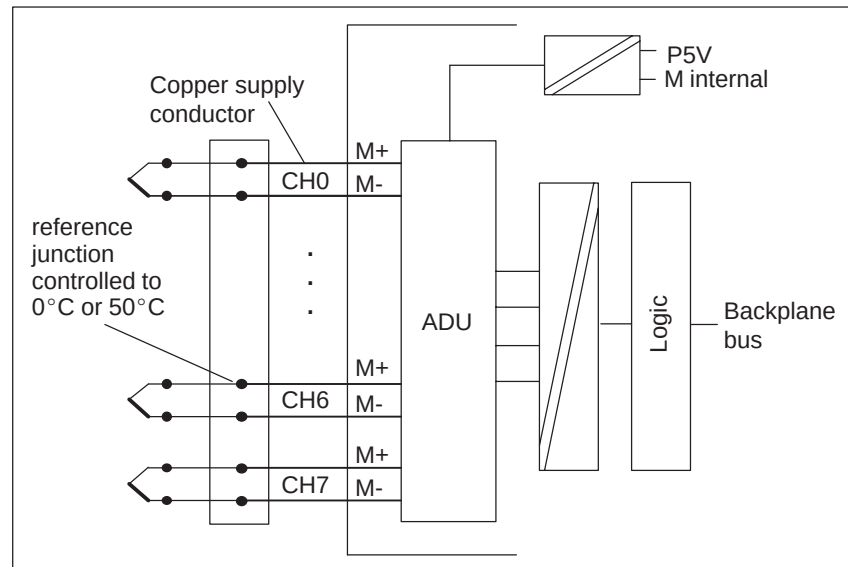


Fig. 3-6 Connection of thermocouples via a reference junction controlled to 0°C or 50°C to the analog input module SM 331; AI 8 x TC/4 x RTD

Thermocouples with thermal resistor compensation

In this type of compensation, the terminal temperature of the reference junction is determined with a thermal resistance sensor in the climatic range.

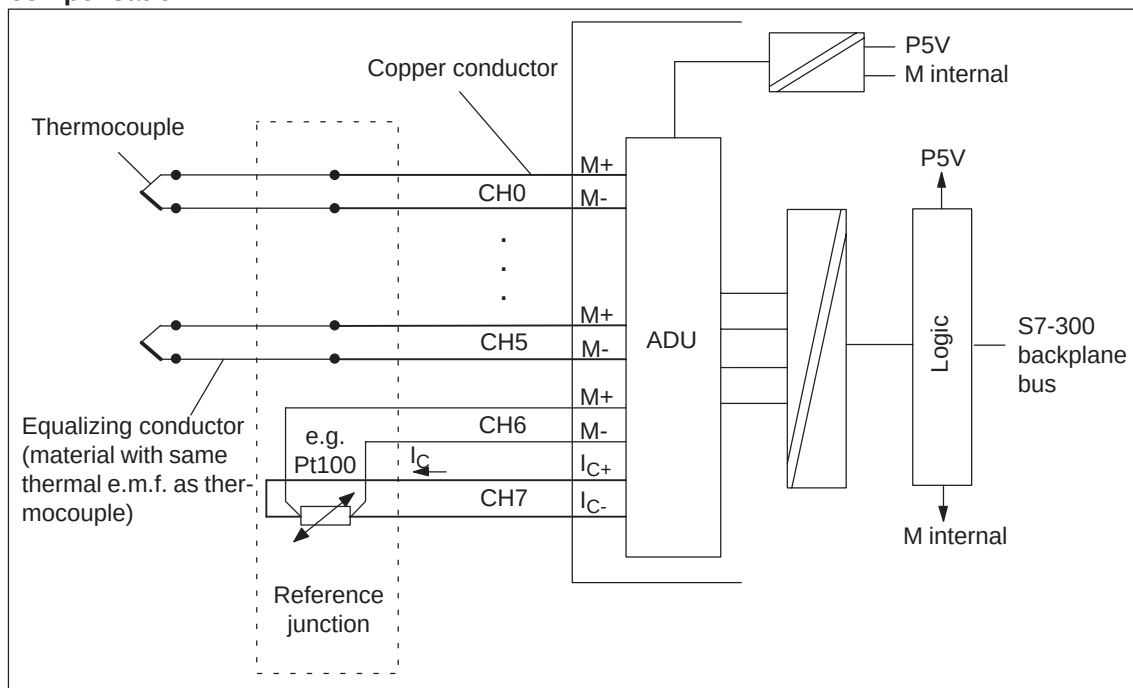


Fig. 3-7 Connection of thermocouples with external compensation with thermal resistance sensor (e.g. Pt100)

Note

The two last channels (channel 6 and 7) of the analog input module SM 331; AI 8 x TC/4 x RTD are used for temperature compensation by means of thermal resistor.

Thermocouples with internal compensation

Internal sensing of the terminal temperature must be used for compensation purposes when thermocouples are connected directly or via equalizing conductors to the module. Each channel group can use one of the supported types of thermocouple independent of the other channel groups.

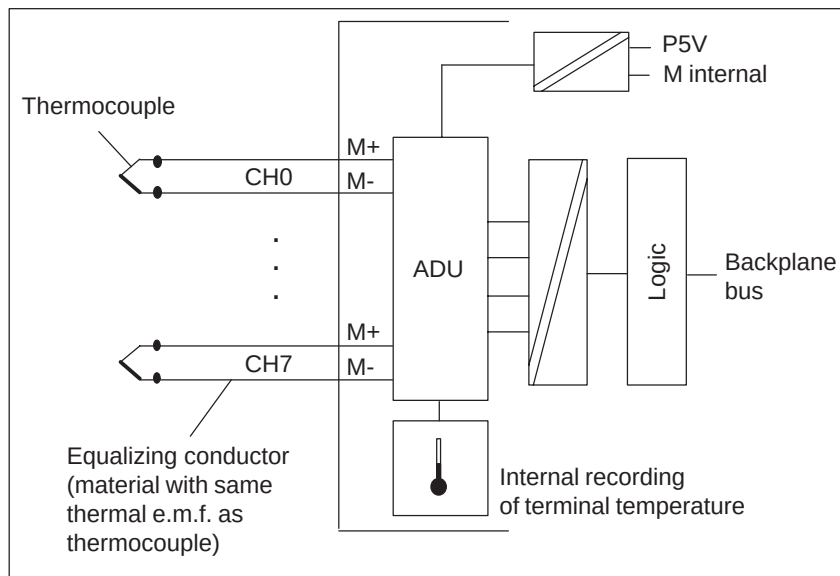


Fig. 3-8 Connection of thermocouples with internal compensation to an electrically isolated analog input module

3.3.2 Connecting Voltage Sensors

Fig. 3-9 shows the connection of voltage sensors to the isolated analog input module SM 331; AI 8 x TC/4 x RTD.

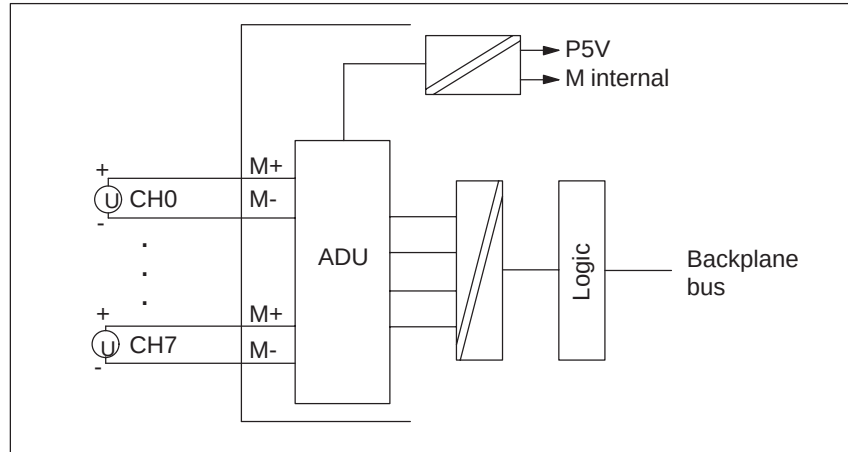


Fig. 3-9 Connection of voltage sensors to the isolated analog input module SM 331; AI 8 x TC/4 x RTD

The information provided in Section 3.2 on differences in potential U_{CM} and U_{ISO} between the individual circuits still retains its validity.

3.4 Connecting Current Sensors and Transducers to the Analog Input Module SM 331; AI 4 x 0/4...20 mA

The following description refers to the operation of transducers together with the analog input module SM 331; AI 4 x 0/4...20 mA.

Abbreviations

The abbreviations used in Figs. 3-11 to 3-12 have the following significance:

$L_{0+} \dots L_{3+}$:	Isolated transducer supply per channel
M+ :	Measuring line (positive)
M- :	Measuring line (negative)
L+ :	Power supply connection 24 V DC
M :	Ground terminal for 24 V DC power supply
U_M :	Measuring-circuit voltage
R_S :	Measuring shunt
U_{V+}, U_{V-} :	External transducer supply voltage

Connection of current sensors as 2-wire and 4-wire transducers

The 2-wire transducer is supplied short-circuit-proof via the isolated measuring transducer supply $L_{0+} \dots L_{3+}$ of the corresponding analog channel. The 2-wire transducer then converts the supplied measured variable into a current between 4...20 mA.

4-wire transducers feature a separate supply voltage connection which must be powered by an external power supply unit.

The information provided in Section 3.2 on differences in potential U_{CM} and U_{ISO} between the individual circuits still retains its validity.

Fig. 3-11 shows the connection of current sensors as 2-wire transducers to the analog input module SM 331; AI 4 x 0/4...20 mA and AI 2 x 0/4...20 mA HART.

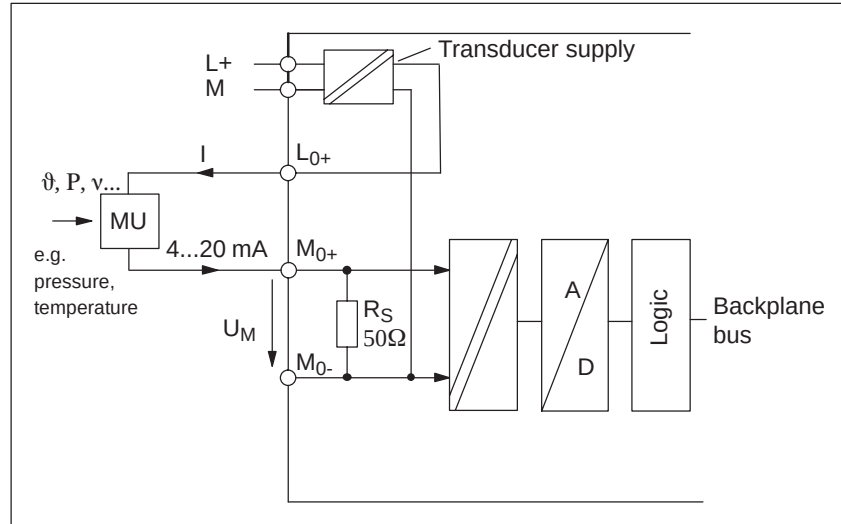


Fig. 3-11 Connection of 2-wire transducers to the analog input module SM 331; AI 4 x 0/4...20 mA and AI 2 x 0/4...20 mA HART.

Fig. 3-12 shows the connection of current sensors as 4-wire transducers with external transducer supply to the analog input module SM 331; AI 4 x 0/4...20 mA and AI 2 x 0/4...20 mA HART.

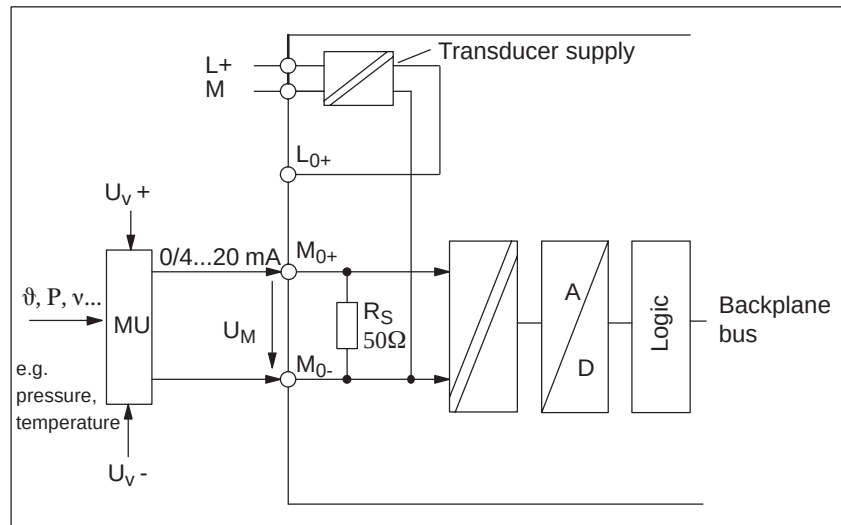


Fig. 3-12 Connection of 4-wire transducers with external supply to the analog input module SM 331; AI 4 x 0/4...20 mA and AI 2 x 0/4...20 mA HART.

3.5 Connecting Loads/Actuators to the Analog Output Module SM 332; AO 4 x 0/4...20 mA

Introduction	The analog output modules can be used to supply loads/actuators with current.												
Lines for analog signals	Shielded lines twisted in pairs are used for analog signals. So as to reduce interference influence. You should ground the shield of the analog lines at both ends. If there are differences in the potential between the line ends, an equipotential bonding current can flow across the shield and cause interference in the analog signals. In this case, the shield should only be grounded at one end of the line.												
Isolated analog output modules	There is no metallic connection between each of the reference points M_0 ... M_3 of the analog circuits and the M terminal of the CPU in the isolated analog output modules. Isolated analog output modules are used when a difference in potential U_{ISO} can occur between the reference point of the analog circuit M_0 ... M_3 and the M-terminal of the CPU. Take particular care to ensure that the difference in potential U_{ISO} does not exceed the permissible value. In cases where it is possible that the permissible value is exceeded, provide a connection between the terminals M_0 ... M_3 and the M-terminal of the CPU.												
Abbreviations	The abbreviations used in Fig. 3-13 have the following significance: <table><tr><td>QI_0 ... QI_3:</td><td>Analog outputs current</td></tr><tr><td>M_0 ... M_3:</td><td>Reference potential of analog output circuit</td></tr><tr><td>R_L:</td><td>Load/actuator</td></tr><tr><td>L+:</td><td>Power supply connection 24 V DC</td></tr><tr><td>M :</td><td>Ground terminal for 24 V DC power supply</td></tr><tr><td>U_{ISO}:</td><td>Difference in potential between reference points of channels M_0 ... M_3 or between the channels and M- terminal of the CPU.</td></tr></table>	QI_0 ... QI_3 :	Analog outputs current	M_0 ... M_3 :	Reference potential of analog output circuit	R_L :	Load/actuator	L+:	Power supply connection 24 V DC	M :	Ground terminal for 24 V DC power supply	U_{ISO} :	Difference in potential between reference points of channels M_0 ... M_3 or between the channels and M- terminal of the CPU.
QI_0 ... QI_3 :	Analog outputs current												
M_0 ... M_3 :	Reference potential of analog output circuit												
R_L :	Load/actuator												
L+:	Power supply connection 24 V DC												
M :	Ground terminal for 24 V DC power supply												
U_{ISO} :	Difference in potential between reference points of channels M_0 ... M_3 or between the channels and M- terminal of the CPU.												

Connecting loads to a current output

You must connect loads to an output current at, e.g., QI_0 and the reference point M_0- of the analog circuit.

Fig. 3-13 shows the principle connection of loads to a current output of an isolated analog output module.

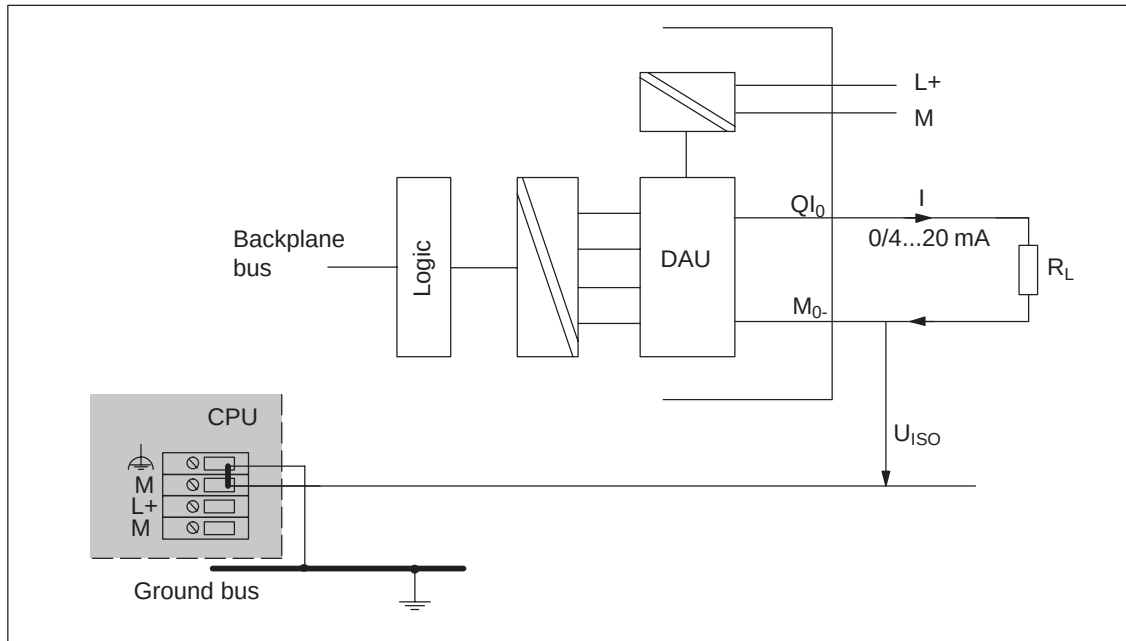


Fig. 3-13 Connection of loads to a current output of the isolated analog output module SM 332; AO 4 x 0/4...20 mA

3.6 Basic Requirements for the Use of Analog Modules

In this chapter

In this chapter you will find:

- Explanations of fundamental definitions for analog value processing.
- How to set measuring ranges of analog input channels.
- What diagnostic options the individual analog modules make available.
- The parameters you can use to set the functions of the individual analog modules.
- Characteristics of the individual analog modules

3.6.1 Conversion and Cycle Time of Analog Input Channels

Introduction

The definitions and interrelationships of conversion time and cycle time for analog input modules are described in this section.

Conversion time

The conversion time is made up of the basic conversion time and additional processing times for wire break monitoring.

The basic conversion time depends directly on the conversion method (integral action, successive approximation or sigma-delta method) of the analog input channel. In the case of integral action conversion, the integration time is included directly in the conversion time. The integration time has a direct influence on the resolution. The integration times of the individual analog modules are specified in Section 3.6.3. These times are set in STEP 7.

Cycle time

Analog/digital conversion and transfer of the digitized measured values to the memory or on the backplane bus of the S7-300 take place sequentially, i.e. the analog input channels are converted one after the other. The cycle time, i.e. the time necessary until an analog input value is converted again, is the sum of the conversion times of all activated analog input channels of the analog input module. The conversion time is based on channel groups when the analog input channels are combined in channel groups by means of parameterization. In the analog input modules SM 331; AI 8 x TC/4 x RTD, 2 analog input channels are combined to form one channel group. You must therefore subdivide the cycle time into steps of 2. Unused analog input channels should be deactivated by means of parameterization in STEP 7 in order to reduce the cycle time.

Fig. 3-14 shows an overview of how the cycle time is made up for an n-channel analog input module.

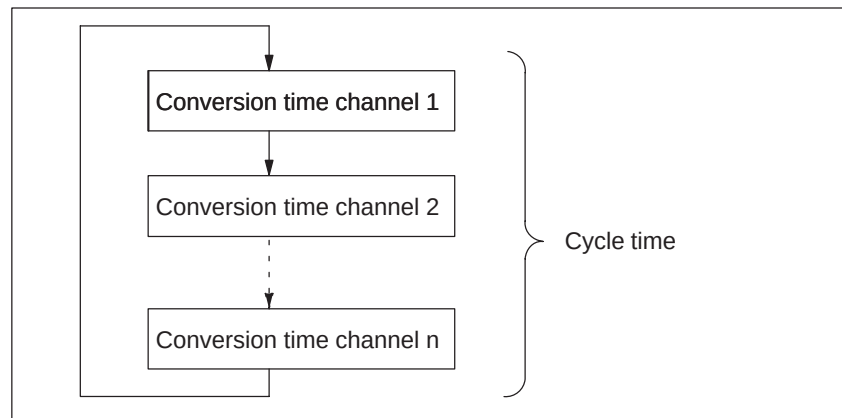


Fig. 3-14 Cycle time of an analog input module

3.6.2 Conversion, Cycle, Transient Recovery and Response Times of Analog Output Channels

Introduction

The definition and interrelationships of relevant times for analog output modules are described in this section.

Conversion time

The conversion time of analog output channels includes the transfer of digitized output values and digital/analog conversion.

Cycle time

In the SM 332; AO 4 x 0/4...20 mA, conversion of the analog output channels takes place in parallel, i.e. on receipt of the data, all four analog output channels are converted simultaneously.

The cycle time, i.e. the time required until an analog output value is re-converted, is constant and equals the conversion time.

Transient recovery time

The transient recovery time (t_2 to t_3), i.e. the time from applying the converted value up to achieving the specified value at the analog output is dependent on load. A differentiation is made between resistive, capacitive and inductive load.

Response time

In the most unfavorable case, the response time (t_1 to t_3), i.e. the time from receiving the digital output values in the module up to obtaining the specified value at the analog output is the sum of the cycle time and transient recovery time. The most unfavorable case is when channel conversion begins just before transfer of a new output value.

The digitized output values are connected simultaneously to all output channels.

Fig. 3-15 shows the response time of the analog output channels.

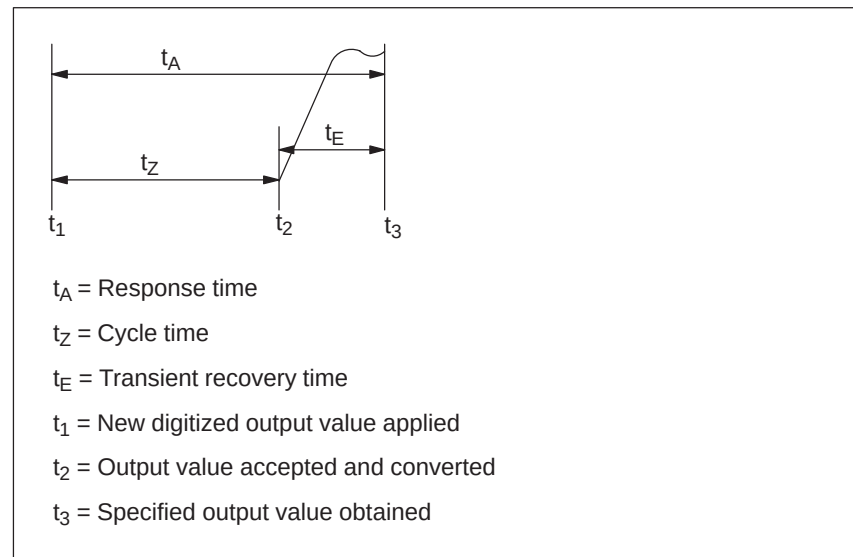


Fig. 3-15 Response time of analog output channels

3.6.3 Parameters of Analog Modules

Introduction This section contains a summary of the analog modules and their parameters.

Parameterization The parameters for the analog modules are set in *STEP7*. These settings must then be transferred in STOP mode to the CPU. During the status change from STOP → RUN, the CPU then transfers the parameters to the relevant analog modules.

Alternatively, you can also change several parameters in the user program with SFC 55. These parameters are specified in Appendix A of the *Reference Manual S7-300, M7-300 Modules* (see /71/) or in the Tables 3-21 to 3-23. With the SFCs 56 and 57, you transfer parameters set with *STEP 7* in RUN mode of the CPU to the analog module (see /235/).

The parameters are subdivided as follows for the 2 parameterization alternatives:

- Static parameters and
- Dynamic parameters

The table below shows the characteristics of static and dynamic parameters.

Parameter	Set with	CPU status
Static	PG	STOP
Dynamic	PG	STOP
	SFC 55 in user program	RUN

Configurable characteristics

The characteristics of the analog modules can be parameterized in *STEP7* with the following parameter blocks:

- For input channels
 - Basic settings (enables)
 - Limits (triggers for hardware interrupt)
 - Diagnostics
 - Measurement
- For output channels
 - Basic settings
 - Diagnostics
 - Default values
 - Output

Parameters of analog input modules

Tables 3-21 and 3-22 provide an overview of the parameters for analog input modules and show what parameters

- are static or dynamic and
- can be set for the modules as a whole or for a channel group or a channel.

Table 3-21 Parameters of analog input module **SM 331; AI 8 x TC/4 x RTD**

Parameter	Value range	Default	Type of parameters	Effective range
Basic settings				
Enable				
• Diagnostic interrupt	yes/no	no	Dynamic	Module
• Hardware interrupt on exceeding limit	yes/no	no		
• Hardw. inter. at end of cycle	yes/no	no		
Limit				
• Upper limit	from 32511 to - 32512	32767	Dynamic	Channel
• Lower limit	from - 32512 to 32511	- 32768		
Diagnostics				
• Enable	yes/no	no	Static	Channel group
• Wire break monitoring	yes/no	no		
Measurement				
• Interference frequency suppression	400 Hz; 60 Hz; 50 Hz; 10 Hz	50 Hz	Dynamic	Channel group
• Measurement mode	– Deactivated – Voltage – Resistance 4-wire configuration – Thermal resistance (RTD) with linearization 4-wire configuration – Thermocouple with linearization and compensation to 0°C – Thermocouple with linearization and compensation to 50°C – Thermocouple with linearization and internal compensation – Thermocouple with linearization and external compensation ¹⁾	Voltage	Dynamic	Channel group
• Ranges	See Tables 3-32 to 3-34	± 1V	Dynamic	Channel gr.

¹⁾ Following types of compensation are possible with this measurement method:

- Use of a compensation box
The compensation box must correspond to the connected type of thermocouple.
All thermocouples must be of the same type.
- Use of a thermal resistor for compensation (e.g. Pt 100)
The absolute terminal temperature is determined for compensation with a Pt 100 resistor in the climatic range.
In this case, the thermocouples to be compensated can be of different types.

Table 3-22 Parameters of the analog input module SM 331; AI 4 x 0/4...20 mA

Parameter	Value range	Default	Type of parameters	Effective range
Basic settings				
Enable				
• Diagnostic interrupt	yes/no	no	Dynamic	Module
• Hardware interrupt on exceeding limit	yes/no	no		
• Hardware interrupt at end of cycle	yes/no	no		
Limit				
• Upper limit	from 32511 to - 32512	32767	Dynamic	Channel
• Lower limit	from - 32512 to 32511	- 32768		
Diagnostics				
• Enable	yes/no	no	Static	Channel group
• wire break monitoring	yes/no	no		
Measurement				
• Interference frequency suppression	400 Hz; 60 Hz; 50 Hz; 10 Hz	50 Hz	Dynamic	Channel group
• Measurement mode	4-wire transducer 2-wire transducer	4-wire transducer	Dynamic	Channel group
• Measuring range	0...20 mA, 4...20 mA	4...20 mA	Dynamic	Channel group

Parameters of analog output module

Table 3-23 provides an overview of the parameters of the analog output module and shows what parameters

- are static or dynamic and
- can be set for the modules as a whole or for a channel.

Table 3-23 Parameters of the analog output module **SM 332; AO 4 x 0/4...20 mA**

Parameter	Value range	Default	Type of parameter	Effective range
Basic settings				
• Diagnostic interrupt enable	yes/no	no	Dynamic	Module
Diagnostics				
• Group diagnostics	yes/no	no	Static	Channel
• and wire break monitoring				
Default				
• Retain last value	yes/no	no	Dynamic	Channel
• Type of value	-32512 ... 32511	-6912 (0 mA)		
Output				
• Type of output	Deactivated Current	Current	Dynamic	Channel
• Output range	4...20 mA 0...20 mA	4...20 mA	Dynamic	Channel

3.6.4 Diagnostics of Analog Modules

Introduction	A comparison of the analog modules with regard to their diagnostic messages is described in this section.
What is diagnostics	With the aid of the diagnostics function, you can determine whether analog processing is faulty or free of faults and what faults have occurred. On detecting a fault, the analog modules output the signal value "7FFF _H " irrespective of the parameterization.
Parameterizing diagnostics	Diagnostics is parameterized with STEP 7.
Diagnostic evaluation	A differentiation is made with regard to diagnostic evaluation between configurable and non-configurable diagnostic messages. In the case of the configurable diagnostic messages, evaluation only takes place when diagnostics has been enabled by means of parameterization ("diagnostic enable" parameter). The non-parameterizable diagnostic messages are always evaluated irrespective of the diagnostic enable. Diagnostic messages trigger following actions: • SF LED on analog module lights, • if applicable channel fault LED, • transfer of diagnostic message to CPU, • diagnostic interrupt triggered (only if diagnostic interrupt has been enabled in the parameterization).

Diagnostics of analog input modules

Table 3-24 provides an overview of the parameterizable diagnostic messages of the analog input modules. The enable is set in the “diagnostics” parameter block (see Section 3.6.3). Diagnostic information is assigned to the individual channels or the entire module.

Table 3-24 Diagnostic messages of analog input modules SM 331;
AI 8 x TC/4 x RTD, AI 4 x 0 / 4...20 mA and AI 2 x 0/4...20 mA HART

Diagnostic message	Effective range of diagnostics	configurable
Wire break ¹⁾	Channel	yes
Underrange		yes, jointly for all 3 faults
Overrange		
Reference channel fault ²⁾		
Incorrect parameters in module		no
Incorrect parameters in module	Module	no
Module not parameterized		
No external auxiliary voltage ³⁾		
No internal auxiliary voltage ³⁾		
Fuse blown ³⁾		
Watchdog triggered		
EPROM error ⁴⁾		
RAM error ⁴⁾		
CPU error ⁴⁾		
ADU error ⁴⁾		
Hardware interrupt lost		

¹⁾ If wire break diagnostics is enabled, the modules AI 4 x 0 / 4...20 mA and AI 2 x 0/4...20 mA output the wire break message for the connected 2-wire transducer (4...20 mA) if the input current drops below a value of $I \leq 3.6$ mA (wire break limits in accordance with NAMUR). For the digital measured value, see Figure 3-4. In the case of the module AI 8 x TC/4 x RTD the line is checked by connecting a test current if wire break diagnostics is enabled. The wire break message is only deactivated (hysteresis), when the input current rises above 3.8 mA again.

²⁾ Only for thermocouples with external compensation and compensation fault.

³⁾ Only for AI 4 x 0 / 4...20 mA and AI 2 x 0/4...20 mA HART with 24 volt supply from L+.

⁴⁾ The tests are conducted during start-up and on-line.

Faults and corrective measures

Table 3-25 provides a list of possible causes and corresponding corrective measures for individual diagnostic messages.

Bear in mind that, in order to detect faults which are indicated by means of configurable diagnostic messages, the module must also be parameterized accordingly.

Table 3-25 Diagnostic messages of analog input modules SM 331; AI 8 x TC/4 x RTD, AI 4 x 0 / 4...20 mA and AI 2 x 0/4...20 mA HART their possible causes and corrective measures

Diagnostic message	Possible fault cause	Corrective measures
Wire break	Break in line between module and sensor	Connect line
	Channel not connected (open)	Deactivate channel group ("Measurement mode parameter")
Measuring range underflow - on AI 8 x TC/4 x RTD - on AI 4 x 0 / 4...20 mA	Input value below underflow range, fault possibly caused by: – Incorrect type of thermocouple – Sensor connected with reversed polarity – incorrect measuring range selected – Module does not signal measuring range underflow – Sensor connected with reversed polarity; (a digitized value is output for 0 mA)	– Check type of thermocouple – Check connection terminals – Parameterize different measuring range
Measuring range overflow	Input value exceeds overflow range	Parameterize different measuring range
Reference channel fault	Measuring channel has different type of sensor parameterized as reference channel	Parameterize different type of sensor
	Fault in reference channel (e.g. wire break) values of all measuring channels set to 7FFF _H	Eliminate fault in reference channel
Incorrect parameters in module	Module supplied with invalid parameters	Check parameterization of module and re-load valid parameters
Module not parameterized	Module not supplied with parameters	Include module in parameterization
No external auxiliary voltage	No module supply voltage L+	Provide L+ supply
No internal auxiliary voltage	No module supply voltage L+	Provide L+ supply
	Module-internal fuse defective	Replace module
Fuse blown	Module-internal fuse defective	Replace module
Time watchdog tripped	In part, high electromagnetic interference	Eliminate interference sources
	Module defective	Replace module
EPROM error RAM error CPU error ADU error	In part, high electromagnetic interference	Eliminate interference sources and switch CPU supply voltage OFF/ON
	Module defective	Replace module
Hardware interrupt lost	Successive hardware interrupts (limits exceeded, end of cycle interrupt) occur faster than the CPU can process them	Change interrupt processing in CPU and reparameterize module if necessary

Diagnostics of analog output modules

Table 3-26 provides an overview of the diagnostic messages of the analog output module which can be parameterized. The enable is set in the "diagnostics" parameter block (see Section 3.6.3).

Table 3-26 Diagnostic messages of analog output module
SM 332; AO 4 x 0/4...20 mA

Diagnostic message	Effective range of diagnostics	configurable
Wire break ²⁾	Channel group	yes
Incorrect parameters in module		no
Incorrect parameters in module	Module	no
Module not parameterized		
No internal auxiliary voltage		
No external auxiliary voltage		
Fuse blown		
Time watchdog tripped		
EPROM error ¹⁾		
RAM error ¹⁾		
CPU error ¹⁾		

¹⁾ The tests are conducted during start-up and on-line.

²⁾ Wire break recognition at output values $I > 100 \mu A$ and output voltage $> 12V$

Faults and corrective measures

Table 3-27 provides a list of possible causes and corresponding corrective measures for individual diagnostic messages.

Bear in mind that, in order to detect faults which are indicated by means of configurable diagnostic messages, the module must also be parameterized accordingly.

Table 3-27 Diagnostic messages of analog output module SM 332; AO 4 x 0/4...20 mA and their possible causes and corrective measures

Diagnostic message	Possible fault cause	Corrective measures
Wire break	Break in line between module and actuator	Connect line
	Voltage at load resistor > 12V	Lower load resistance to $\leq 500 \Omega$
	Channel not connected (open)	Deactivate channel ("Measurement mode parameter")
Incorrect parameters in module	Module supplied with invalid parameters	Check parameterization of module and re-load valid parameters
Module not parameterized	Module not supplied with parameters	Include module in parameterization
No external auxiliary voltage	No module supply voltage L+	Provide L+ supply
No internal auxiliary voltage	No module supply voltage L+	Provide L+ supply
	Module-internal fuse defective	Replace module
Fuse blown	Module-internal fuse defective	Replace module
Time watchdog tripped	In part, high electromagnetic interference	Eliminate interference sources
	Module defective	Replace module
EPROM error CPU error RAM error	In part, high electromagnetic interference	Eliminate interference sources and switch CPU supply voltage OFF/ON
	Module defective	Replace module

Reading out diagnostic messages

You can read out the detailed diagnostic messages in STEP 7 after setting diagnostics for the analog modules (refer to /231/).

3.6.5 Interrupts of Analog Modules

Introduction	The interrupt characteristics of the analog modules are described in this section. In principle, a differentiation is made between the following interrupts: • Diagnostic interrupt • Hardware interrupt
Parameterizing interrupts	The interrupts are parameterized with <i>STEP 7</i>.
Default setting	The interrupts are inhibited by way of default.
Diagnostic interrupt	If enabled, the module triggers a diagnostic interrupt when a fault comes or goes (e.g. wire break or short to M). Diagnostic functions inhibited by parameterization cannot trigger an interrupt. The CPU interrupts processing of the user program or low-priority classes and processes the diagnostic interrupt module (OB 82).
Hardware interrupt	The range is defined by parameterization of an upper and a lower limit. If the process signal (e.g. temperature of an analog input module) is outside this range, the module triggers a hardware interrupt provided limit interrupt is enabled. You can determine which of the channels has triggered the interrupt with the aid of the local data of the OB 40 in the user program (see /235/). Active hardware interrupts trigger interrupt processing (OB 40) in the CPU, consequently the CPU interrupts processing of the user program or low-priority classes. If there are no higher priority classes pending processing, the stored interrupts (of all modules) are processed one after the other corresponding to the order in which they occurred.
Hardware interrupt lost	If an event occurred in one channel (overrange/underrange of limit), this event is stored and a hardware interrupt is triggered. If a further event occurs on this channel before the hardware interrupt has been acknowledged by the CPU (OB 40 run) this event will be lost. A diagnostic interrupt "hardware interrupt lost" is triggered in this case. The diagnostic interrupt enable must be active for this purpose. Further events on this channel are then no longer registered until interrupt processing is completed for this channel.

3.6.6 Characteristics of Analog Modules

Introduction

Described in this section:

- Dependency of the analog input and output values on the supply voltage of the analog module and the operating status of the CPU.
- Characteristics of the analog modules depending on the position of the analog values in the relevant value range.
- Influence of faults on the analog modules.

Influence of supply voltage and operating status

The input and output values of the analog modules are dependent on the supply voltage of the analog module and on the operating status of the CPU.

Table 3-28 provides an overview of these dependencies.

Table 3-28 Dependencies of analog input/output values on the CPU operating status and the supply voltage L +

CPU operating status		Supply voltage L+ at analog module	Input value of analog input modules	Output value of analog output module
POWER ON	RUN	L + applied	Process value	CPU value
			7FFF _H up to first conversion after switching on or after module parameterization has been completed	Up to first conversion ... • after switch-on has been completed if signal of 0 mA is output. • after parameterization has been completed, previous value is output.
		No L +	Overflow value ¹⁾	0 mA
POWER ON	STOP	L + applied	Process value	Default/last value
			7FFF _H up to first conversion after switching on or after module parameterization has been completed	at 0...20 mA: 0 mA default at 4...20 mA: 4 mA default
		No L +	Overflow value ¹⁾	0 mA
POWER OFF	–	L + applied	–	0 mA
		No L +	–	0 mA

¹⁾ only applies to SM 331; AI 8xTC/4xRTD as no L+ supply voltage is required.

Failure of the L+ supply voltage for the analog modules is always indicated by the group fault LED on the module and additionally entered in diagnostics.

Triggering of a diagnostic interrupt is dependent on the parameterization (see Section 3.6.3).

Table 3-29 Characteristics of analog modules dependent on position of analog input value in value range

Process value in	Input value	SF LED	Diagnostics	Interrupt	Channel fault LED
Rated range	Process value	–	–	–	–
Overrange/underrange	Process value	–	–	–	–
Overflow	7FFF _H	lit	Entry made ¹⁾	Diagnostic interrupt ¹⁾	lit
Underflow	8000 _H	lit			lit
Wire break	7FFF _H	lit ¹⁾			lit ¹⁾
Outside parameterized limit	Process value	–	–	Hardware interrupt ¹⁾²⁾	–

¹⁾ depending on parameterization

²⁾ A channel diagnostic error prevents the limit hardware interrupt.

Example: An enabled wire break diagnostics renders limits below the wire break threshold ineffective.

Influence of value range for output

The characteristics of the output modules depend on what part of the value range the output values are in. Table 3-30 shows this dependency for analog output values.

Table 3-30 Characteristics of analog modules dependent on position of analog output value in value range

Output value in	Output value	SF LED	Diagnostics	Interrupt	Channel fault LED
Rated range	CPU value	–	–	–	–
Overrange/underrange	CPU value	–	–	–	–
Overflow	0 mA	–	–	–	–
Wire break	CPU value	lit ¹⁾	Entry made ¹⁾	Entry made ¹⁾	lit ¹⁾

¹⁾ depending on parameterization

Influence of faults

Faults occurring in analog modules with diagnostic capabilities and corresponding parameterization (see Section 3.6.3 "Parameters of Analog Modules") result in diagnostic entry and diagnostic interrupt. Possible faults are listed in Table 3-25 and 3-27 in Section 3.6.4.

The SF LED and, if applicable, the channel fault LED light on the analog module.

Faults which cannot be parameterized in diagnostics (e.g. fuse blown) result in an entry being made in the diagnostic range and the fault LED lighting irrespective of the CPU operating status.

3.7 Analog Input Module SM 331; AI 8 x TC/4 x RTD

Order number 6ES7 331-7SF00-0AB0

Features

The analog input module SM 331; AI 8 x TC/4 x RTD is characterized by the following features:

- 8 inputs in 4 channel groups
- Measured value resolution; adjustable per channel group (depending on set interference frequency rejection)
 - 9 Bit + sign (integration time 2.5 ms) $\triangleq$ 400 Hz
 - 12 Bit + sign (integration time $16^{2/3}$ / 20 ms) $\triangleq$ 60/50 Hz
 - 15 Bit + sign (integration time 100 ms) $\triangleq$ 10 Hz
- measurement mode selectable per channel group:
 - Voltage
 - Resistance
 - Temperature
- Arbitrary measuring range selection per channel group
- Configurable diagnostics
- Configurable diagnostic interrupt
- 2 channels with limit monitoring
- Configurable limit interrupt
- Isolated with respect to CPU
- Common mode < 60 V between channels

Resolution

The resolution of a measured value depends directly on the selected integration time, i.e. the longer the integration time for an analog input channel, the more accurate the resolution of the measured value (refer to technical specifications of the analog input module and Table 3-2).

Wiring diagram

Fig. 3-16 shows the module view and the terminal diagram of the SM 331; AI 8 x TC/4 x RTD

You will find detailed technical specifications of the analog input module SM 331; AI 8 x TC/4 x RTD on the following pages.

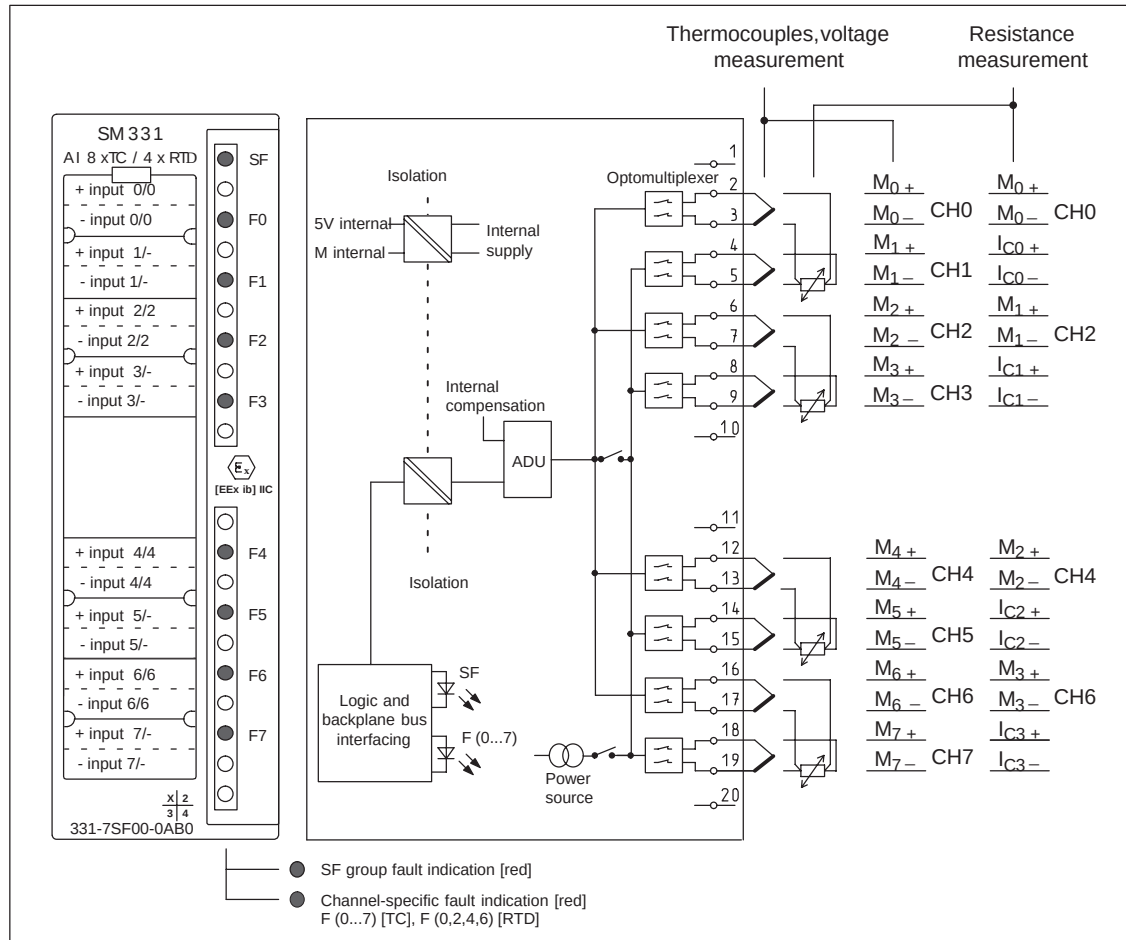


Fig. 3-16 Module view and block diagram of SM 331; AI 8 x TC/4 x RTD

Notes on intrinsically-safe installation

You must connect the DM 370 dummy module between the CPU or IM 153-2 (in a distributed configuration) and the Ex I/O modules whose signal cables lead into the hazardous location. In a distributed configuration with an active backplane bus, you should use the ex dividing panel/ ex barrier instead of the dummy module. Additional information on system design can be found in Sections 1.3 - 1.5.

Notes on the module

No external voltage supply L+ (24 V) is necessary for the analog input module SM 331; AI 8 x TC/4 x RTD.

If thermal resistors (e.g. Pt 100) are used for external compensation, connect them to channel 6 and 7.

If a compensation box is used for external compensation, connect it to channel 7.

Parameterization

The functions of the analog input module SM 331; AI 8 x TC/4 x RTD are set

- with *STEP 7* (refer to /231/) or
- in the user program with SFCs (refer to /235/).

Default settings

The analog input module features default settings for integration time, diagnostic interrupts etc. (see Table 3-21).

These default settings are valid if re-parameterization has not been carried out via *STEP 7*.

Channel groups

2 channels each of the analog input module SM 331; AI 8 x TC/4 x RTD are combined to form a channel group. Parameters can always only be assigned to one channel group, i.e. parameters which are specified for a channel group are always valid for both channels of this channel group.

Table 3-31 shows the allocation of channels to channel groups of the analog input module SM 331; AI 8 x TC/4 x RTD.

Table 3-31 Allocation of analog input channels of the SM 331; AI 8 x TC/4 x RTD to channel groups

Channel	Allocated channel group
Channel 0	Channel group 0
Channel 1	
Channel 2	Channel group 1
Channel 3	
Channel 4	Channel group 2
Channel 5	
Channel 6	Channel group 3
Channel 7	

Special feature of resistant measurement

Only one channel per channel group is required for resistance measurement. The "2nd" channel of the group is used for current injection (I_C).

The measured value is obtained on accessing the "1st" channel of the group. The "2nd" channel of the group is preset with the overflow value "7FFF_H".

During diagnostics, the 1st channel provides the actual status (in compliance with parameterization) and the 2nd channel "faultless".

Non-connected input channels

Activated and non-connected channels of the analog input module SM 331; AI 8 x TC/4 x RTD must be short-circuited to ensure optimum interference immunity for the analog input module. The non-connected channels should also be deactivated in *STEP 7* (see Section 3.6.3) in order to shorten the module cycle time.

Adjustable types of measurement

The following types of measurement can be set on the analog input module SM 331; AI 8 x TC/4 x RTD. The measurement mode is set in *STEP 7* (see Section 3.6.3).

- Voltage measurement
- Resistance measurement
- Temperature measurement

Adjustable measuring ranges

The measuring ranges, for which you can use the analog input module SM 331; AI 8 x TC/4 x RTD are specified in the Tables 3-32 to 3-34. You can set the required measuring ranges in *STEP 7* (see Section 3.6.3).

Wire break check

The analog input module SM 331; AI 8 x TC/4 x RTD carries out a wire break check, provided it is enabled by means of parameterization, for all areas. All 4 terminal wires are monitored for wire break in resistance thermometer mode (RTD).

Measuring ranges for voltage measurement

Table 3-32 contains all measuring ranges for voltage measurements.

Table 3-32 Measuring ranges for voltage measurement

Selected measurement mode	Explanation	Measuring range
Voltage	The digitized analog values are specified in Section 3.1.2 in Table 3-3 Voltage measuring ranges	$\pm 25 \text{ mV}$ $\pm 50 \text{ mV}$ $\pm 80 \text{ mV}$ $\pm 250 \text{ mV}$ $\pm 500 \text{ mV}$ $\pm 1 \text{ V}$

Measuring ranges for resistance measurement

Table 3-33 contains all measuring ranges for resistance measurements

Table 3-33 Measuring ranges for resistance measurements

Selected measurement mode	Explanation	Measuring range
Resistance 4-wire connection	The digitized analog values are specified in Section 3.1.2 in Table 3-5 Resistance measuring ranges.	150 ohms 300 ohms 600 ohms

Connectable thermocouples

Table 3-34 shows all connectable thermocouples and thermal resistors. The linearization of characteristic curves is specified for thermocouples in accordance with **DIN IEC 584**.

For thermal resistor measurements, linearization of the characteristic curves is based on **DIN 43760** and **IEC 751**.

Table 3-34 Connectable thermocouples and thermal resistors

Measurement mode	Explanation	Measuring range
– Linearization and compensation to 0°C	Digitized analog values for the specified thermocouples are listed in Section 3.1.2, Tables 3-10 to 3-12. (one unit corresponds to 0.1°C)	Type T [Cu-CuNi]
– Linearization and compensation to 50°C		Type U [Cu-CuNi]
– Linearization and compensation internal comparison ¹⁾		Type E [NiCr-CuNi]
– Linearization and compensation external comparison ²⁾		Type J [Fe-CuNi] Type L [Fe-CuNi] Type K [NiCr-Ni] Type N [NiCr-SiNiSi] Type R [Pt13Rh-Pt] Type S [Pt10Rh-Pt] Type B [Pt30Rh-Pt6Rh]
Thermal resistance + linearization 4-wire connection (temperature measurement)	The digitized analog values for the specified thermal resistors are listed in Section 3.1.2, Tables 3-6 to 3-9.	Pt 100, Pt 200, Ni 100 Standard range Pt 100, Pt 200, Ni 100 Climatic range

1) – In the case of internal compensation in the module, all 8 channels are available for temperature measurements also with different types of thermocouples.

– The terminal temperature of the module is provided at a short-circuited input.
This does not apply to thermocouple Type B which is not suitable for measurements in the ambient temperature range.

2) Following types of compensation are possible with this measurement method:

- Use of compensation box
The compensation box must correspond to the connected type of thermocouple.
Connection to channel 7.
- Use of thermal resistors in climatic range (e.g. Pt 100) for compensation.
The absolute terminal temperature is determined in the climatic range with a thermal resistor (e.g. Pt 100) for compensation purposes. In this case, the thermocouples to be compensated can be of different types.
Connection to channels 6 and 7

Analog Input SM 331; AI 8 x TC/4 x RTD Dimensions and Weight		Safety data (refer to Certificate of Conformity in Appendix A)				
Dimensions W x H x D (mm)	40 x 125 x 120	Type of protection to EN 50020		[EEx ib] IIC		
Weight	approx. 210 g	Maximum values per channel for thermocouples and thermal resistors				
Module-specific data		• U ₀ (no-load output voltage)				max. 5.9 V
Number of inputs	8	• I ₀ (short-circuit current)				max. 28.8 mA
• Resistancesensor	4	• P ₀ (load power)				max. 41.4 mW
Line length, shielded	max. 200 m	• L ₀ (permissible external inductance)				max. 40 mH
	max. 50 m for voltage ranges ≤ 80 mV and thermocouples	• C ₀ (permissible external capacitance)				max. 60 µF
Type of protection PTB (see Appendix A)	[EEx ib] IIC to EN 50020	• U _m (error voltage)				max. 60 V DC 30 V AC
Test number	Ex-96.D.2108 X	• T _a (permissible ambient temperature)				max. 60 °C
Type of protection FM (see Appendix B)	CL I, DIV 2, GP A, B, C, D T4					
Voltages, currents, potentials		Connection of an active sensor with following maximum values U _i = ± 1.2 V I _i = 20 mA deviating from above-specified values				
Bus power supply	5 V DC	• L ₀ (permissible external inductance)				max. 15 mH
Isolation		• C ₀ (permissible external capacitance)				max. 17 µF
• Between channels and backplane bus	yes					
• between channels	no					
Permissible difference in potential of signals from hazardous area						
• between channels and backplane bus (U _{ISO})	60 V DC 30 V AC					
• between channels (U _{CM})	60 V DC 30 V AC					
Insulation tested						
• Channels with respect to backplane bus	with 1500 V AC					
Current input from backplane bus	max. 120 mA					
Module power loss	typical 0.6 W					
Permissible difference in potential of signals from non-hazardous area						
• between channels and backplane bus (U _{ISO})	400 V DC 250 V AC					
• between channels (U _{CM})	75 V DC 60 V AC					

Analog value formation					
Measuring principle	SIGMA-DELTA				
Integration time/conversion time/resolution (per channel)					
• configurable	yes	yes	yes	yes	yes
• Integration time in ms	2.5	16 ² / ₃	20	100	
• Basic conversion time = 3 x integration time + transient recovery time optomultiplexer in ms	7.5 + 2.5	50 + 2.5	60 + 2.5	300 + 2.5	
• Additional conversion time for wire break recognition in ms	2.5	2.5	2.5	2.5	
• Resolution in bit (incl. overrange)	9+ sign	12+ sign	12+ sign	15+ sign	
• Interference voltage rejection for interference frequency f1 in Hz	400	60	50	10	

Interference rejection, error limits	Interference rejection, error limits continued
Interference voltage rejection for $f = n \times (f_1 \pm 1\%)$, ($f_1$ = interference frequency) • Common-mode rejection > 130 dB ($U_{ISO} < 60$ V) • Normal-mode rejection > 40 dB (interference peak value $<$ rated value of input range) Crosstalk attenuation between inputs ($U_{ISO} < 60$ V) > 70 dB Operational limit (in total temperature range, referred to input range) • ± 25 mV $\pm 0.09\%$ • ± 50 mV $\pm 0.06\%$ • ± 80 mV $\pm 0.05\%$ • ± 250 mV / ± 500 mV / ± 1 V $\pm 0.04\%$ Basic error (operational limit at 25°C, referred to input range) • ± 25 mV $\pm 0.018\%$ • ± 50 mV $\pm 0.014\%$ • ± 80 mV $\pm 0.011\%$ • ± 250 mV / ± 500 mV / ± 1 V $\pm 0.008\%$ Temperature error (referred to input range) • ± 25 mV $\pm 0.0019\%/K$ • ± 50 mV $\pm 0.0013\%/K$ • ± 80 mV $\pm 0.0011\%/K$ • ± 250 mV / ± 500 mV / ± 1 V $\pm 0.0010\%/K$ Linearity error (referred to input range) $\pm 0.003\%$ Repeatability (in steady-state condition at 25°C, referred to input range) $\pm 0.003\%$	The accuracy of temperature measurement with <i>external compensation with thermal resistors</i> is derived from: – Error for analog input of the type of thermocouple used – Accuracy ¹⁾ of the type of thermal resistor used for compensation – Error ¹⁾ of compensation input The accuracy of temperature measurement with <i>external compensation with compensation box</i> is derived from: – Error for analog input of the type of thermocouple used – Accuracy ¹⁾ of compensation box – Error ¹⁾ of compensation input The accuracy of temperature measurement with <i>compensation of the external reference junction maintained at 0°C / 50°C</i> is derived from: – Error for analog input of the type of thermocouple used – Accuracy ¹⁾ of reference junction temperature The accuracy of temperature measurement with <i>internal compensation (terminal temperature)</i> is derived from: – Error for analog input of the type of thermocouple used – Accuracy ¹⁾ of internal reference junction temperature ± 0.5 K

1) Due to the constant increase in the thermocouple characteristic at higher temperatures, the error in the compensation element is less effective than at temperatures in the vicinity of the compensation temperature. Exception: Thermocouple types J and E (relative linear progression)

Due to the little increase in the range from approx. 0°C to 40°C , the lack of compensation of the reference junction temperature has only a negligible effect in the case of **thermocouple type B**. If there is no compensation and the measurement mode "Compensation to 0°C " is set, the deviation in thermocouple type B during temperature measurement is between

700°C and 1820°C	$< 0.5^\circ\text{C}$
500°C and 700°C	$< 0.7^\circ\text{C}$

"Internal compensation" should be set if the reference junction temperature closely corresponds to the module temperature. As a result, the error for the temperature range from 700 to 1820°C is reduced to $< 0.5^\circ\text{C}$.

Error limits of analog inputs for thermocouples					Error limits of analog inputs for thermal resistors			
(at 25°C ambient temperature and 100 ms integration time)					(at 25°C ambient temperature and 100 ms integration time)			
Type	Temperature range		Basic error ¹⁾	Temperature error ²⁾ [°C/K]	Type	Temperature range		Basic error ¹⁾ Temperature error ²⁾ [°C/K]
T	-150 °C	+400 °C	± 0.2K	± 0.006	Pt 100 Climatic	-200 °C	+325 °C	± 0.05K ± 0.006
	-230 °C	-150 °C	± 1K					
U	-50 °C	+400 °C	± 0.2K	± 0.006	Pt 200 Climatic	-200 °C	+325 °C	± 0.05K ± 0.006
	-200 °C	-50 °C	± 1K					
E	-100 °C	+1000 °C	± 0.2K	± 0.0075	Ni 100 Climatic	-60 °C	+250 °C	± 0.05K ± 0.003
	-200 °C	-100 °C	± 1K					
J	-150 °C	+1200 °C	± 0.2K	± 0.02	Pt 100 Standard	-200 °C	+850 °C	± 0.2K ± 0.01
	-210 °C	-150 °C	± 0.5K					
L	-50 °C	+1200 °C	± 0.2K	± 0.02	Pt 200 Standard	-200 °C	+850 °C	± 0.2K ± 0.01
	-200 °C	-50 °C	± 1K					
K	-100 °C	+1372 °C	± 0.2K	± 0.018	Ni 100 Standard	-60 °C	+250 °C	± 0.1K ± 0.003
	-220 °C	-100 °C	± 1K					
N	-50 °C	+1300 °C	± 0.2K	± 0.025	Error limits of analog inputs for resistance sensors			
	-150 °C	-50 °C	± 1K		(at 25°C ambient temperature and 100 ms integration time)			
R	+200 °C	+1769 °C	± 0.3K	± 0.025	Type	Resistant sensor		Basic error ³⁾ Temperature error ²⁾ [%/K]
	-50 °C	+200 °C	± 1K					
S	+100 °C	+1769 °C	± 0.3K	± 0.025	150Ω	0.000 Ω... 176.383 Ω	± 0.006%	± 0.001
	-50 °C	+100 °C	± 1K		300Ω	0.000 Ω... 352.767 Ω	± 0.006%	± 0.001
B	+700 °C	+1820 °C	± 0.3K	± 0.04	600Ω	0.000 Ω... 705.534 Ω	± 0.006%	± 0.001
	+500 °C	+700 °C	± 0.5K					
	+200 °C	+500 °C	± 3K					

- 1) The basic error includes the linearization error of the voltage temperature conversion and the basic error of the analog/digital conversion at $T_u = 25^\circ\text{C}$.
- 2) The total temperature error = temperature error x max. ambient temperature change ΔT_u as temperature difference with respect to 25°C .
- 3) The basic error includes the error in % of the measuring range of the analog/digital conversion at $T_u = 25^\circ\text{C}$.

The operating error for the use of **thermocouples/thermal resistors** consists of:

- Basic error of analog input at $T_u = 25^\circ\text{C}$
- Total temperature error
- Errors which occur due to compensation of the reference junction temperature
- Error of the thermocouple/thermal resistor used

The operating error for use of **resistant sensors** consists of:

- Basic error of analog input at $T_u = 25^\circ\text{C}$
- Total temperature error
- Error of sensor used

Interrupts, Diagnostics			Data for sensor selection, continued	
Interrupts			Temperature compensation	configurable
• Limit interrupt	Configurable channels 0 and 2		• Internal temperature compensation	possible
• Diagnostic interrupt	configurable		• External temperature compensation with compensation box	possible
Diagnostic functions			• External temperature compensation with thermal resistors (e.g. Pt100)	possible
• Group fault indication	red LED (SF)		• Compensation for 0 °C reference junction temperature	possible
• Channel fault indication	red LED (F) per channel		• Compensation for 50 °C reference junction temperature	possible
• Diagnostic information readout	possible			
Data for sensor selection			1) Without line resistance correction	
Input ranges (rated values) / input resistance				
• Voltage	± 25 mV	/10 MΩ		
	± 50 mV	/10 MΩ		
	± 80 mV	/10 MΩ		
	± 250 mV	/10 MΩ		
	± 500 mV	/10 MΩ		
	± 1 V	/10 MΩ		
• Resistance	150 Ω	/10 MΩ		
	300 Ω	/10 MΩ		
	600 Ω	/10 MΩ		
• Thermocouples	Type: T, U, E, J, L, K, N, R, S, B	/10 MΩ		
• Resistance thermometer	Pt 100, Pt 200, Ni 100	/10 MΩ		
Measuring current for thermal resistors and wire break testing	approx. 0.5 mA			
Permissible input voltage for voltage input (destruction limit)	max. 35 V permanent; 75 V for max. 1 s (pulse duty factor 1:10)			
Signal generator connection				
• for voltage measurement	possible			
• for resistance measurement with 4-wire connection with 3-wire connection ¹⁾ with 2-wire connection ¹⁾	possible			
Characteristic linearization				
• for thermocouples	Type: T, U, E, J, L, K, N, R, S, B			
• for thermal resistors	Pt 100, Pt 200, Ni 100 (standard and climatic range)			

3.8 Analog Input Module SM 331; AI 4 x 0/4...20 mA

In this chapter	In this chapter you will find the characteristics and the technical specifications for the analog input module SM 331; AI 4 x 0/4...20 mA, and you will learn • how to place the analog input module into operation. • what parameters influence the characteristics of the analog input module. • what diagnostic options the analog input module offers.
Order number	6ES7 331-7RD00-0AB0
Features	The analog input module SM 331; AI 4 x 0/4...20 mA is characterized by the following features: • 4 inputs in 4 channel groups • Measured value resolution; adjustable per channel (dependent on the integration time set) – 10 Bit (integration time 2.5 ms) – 13 Bit (integration time $16^{2/3}$ / 20 ms) – 15 Bit (integration time 100 ms) • measurement mode selectable per channel: – Current – Channel deactivated • Arbitrary measuring range selection per channel – 0 ... 20 mA – 4 ... 20 mA • Configurable diagnostics and configurable diagnostic interrupt • Channel 0 and 2 with limit value monitoring and configurable limit interrupt • Channels isolated among each other and with respect to CPU and load voltage L⁺ • The analog inputs are HART compatible
Resolution	The resolution of a measured value depends directly on the selected integration time, i.e. the longer the integration time for an analog input channel, the more accurate the resolution of the measured value (refer to technical specifications for the analog input module and Table 3-2).

Wiring diagram

Fig. 3-17 shows the terminal diagram of the analog input module SM 331; AI 4 x 0/4...20 mA. You will find detailed technical specifications for the analog input module SM 331; AI 4 x 0/4...20 mA on the following pages.

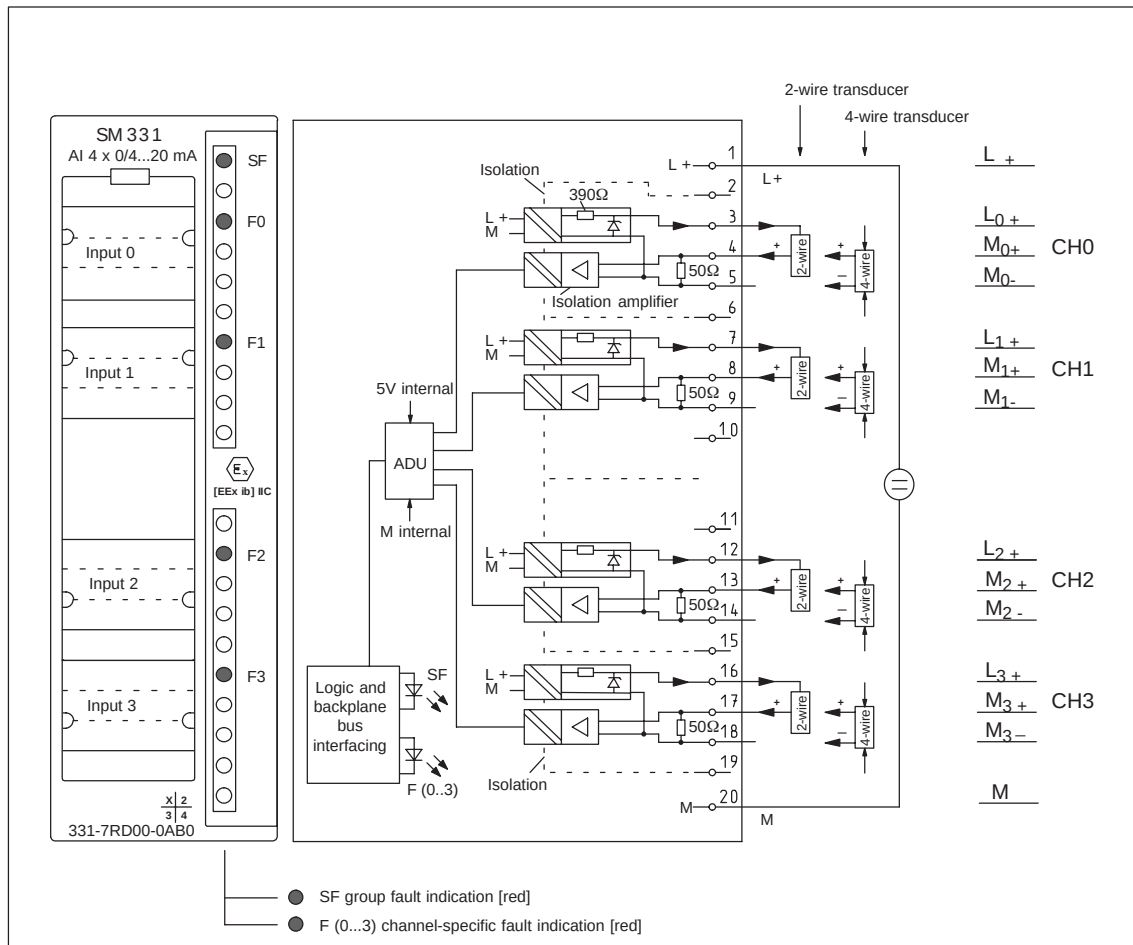


Fig. 3-17 Module view and block diagram of SM 331; AI 4 x 0/4...20 mA

Notes on intrinsically-safe installation

You must connect the DM 370 dummy module between the CPU or IM 153-2 (in a distributed configuration) and the Ex I/O modules whose signal cables lead into the hazardous location. In a distributed configuration with an active backplane bus, you should use the ex dividing panel/ ex barrier instead of the dummy module. Additional information on system design can be found in Sections 1.3 - 1.5.

Power supply for a intrinsically-safe structure

In order to maintain the clearances and creepage distances, L+ / M must be routed via the line chamber LK393 when operating I/O modules with signal cables that lead to the hazardous location, see Section 1.2.

Parameterization The functions of the analog input module SM 331; AI 4 x 0/4...20 mA are set

- with *STEP 7* (refer to /231/)
- in the user program with SFCs (refer to /235/).

Default settings The analog input module features default settings for integration time, diagnostic interrupts etc. (see Table 3-21). These default settings are valid if re-parameterization has not been carried out via *STEP 7*.

Channel groups The channel group is allocated to each input channel for parameterization of the analog input module SM 331; AI 4 x 0/4...20 mA. Advantage: You can specify separate parameters for each channel. Table 3-35 shows the allocation of channels to channel groups of the analog input module SM 331; AI 4 x 0/4...20 mA:

Table 3-35 Allocation of analog input channels of the SM 331; AI 4 x 0/4...20 mA to channel groups

Channel	Allocated channel group
Channel 0	Channel group 0
Channel 1	Channel group 1
Channel 2	Channel group 2
Channel 3	Channel group 3

Selectable measurement mode The measurement mode is set with *STEP 7* (see Section 3.6.3). The following types of measurement can be set:

- Current measurement
- Channel deactivated

Measuring ranges for 2-wire and 4-wire transducers Table 3-36 contains all measuring ranges for current measurement with 2-wire and 4-wire transducers. You can set the required measuring ranges with *STEP 7* (see Section 3.6.3).

Table 3-36 Measuring ranges for 2-wire and 4-wire transducers

Selected measurement mode	Explanation	Measuring range
2-wire transducer	The digitized analog values are specified in Section 3.1.2 in Table 3-4 Current measuring range.	from 4 to 20 mA
4-wire transducer	The digitized analog values are specified in Section 3.1.2 in Table 3-4 Current measuring range.	from 0 to 20 mA from 4 to 20 mA

Wire break check

- Wire break recognition is not possible for the current range 0 to 20 mA.
- For the current range from 4 to 20 mA, the input current dropping below $I \leq 3.6$ mA is interpreted as a wire break and, if enabled, an appropriate diagnostic interrupt is triggered.

Influencing by HART signals

If transducers with HART protocol are used, integration times of $16^{2/3}$, 20 or 100 ms should preferably be parameterized in order to maintain the influence on the measurement signal by the modulated alternating current as low as possible.

Analog Input SM 331; AI 4 x 0/4...20 mA Dimensions and Weight		Voltages, currents, potentials continued	
Dimensions W x H x D (mm)	40 x 125 x 120	Permissible difference in potential (U_{ISO}) for signals from non-hazardous area	
Weight	approx. 290 g	- between channels and backplane bus 400 V DC 250 V AC - between channels and backplane bus 400 V DC 250 V AC - Between channels 400 V DC 250 V AC - Between backplane bus and load voltage L+ 75 V DC 60 V AC	
Module-specific data			
Number of inputs	4	Insulation tested	
Line length, shielded	max. 200 m	- Channels with respect to backplane bus and load voltage L+ with 1500 V AC - Channels among each other with 1500 V AC - Backplane bus with respect to load voltage L+ with 500 V DC	
Type of protection PTB (see Appendix A)	[EEx ib] IIC to EN 50020	Current input	
Test number	Ex-96.D.2092 X	- from backplane bus max. 60 mA - from load voltage L+ max. 150 mA	
Type of protection FM (see Appendix B)	CL I, DIV 2, GP A, B, C, D T4	Module power loss typical 3 W	
Voltages, currents, potentials		Safety data (refer to Certificate of Conformity in Appendix A)	
Bus power supply	5 V DC	Type of protection to EN 50020	[EEx ib] IIC
Rated load voltage L+	24 V DC	Maximum values per channel	
Reverse voltage protection	yes	U_0 (no-load output voltage) max. 25.2 V I_0 (short-circuit current) max. 68.5 mA P_0 (load power) max. 431 mW L_0 (permissible external inductance) max. 7.5 mH C_0 (permissible external capacitance) max. 90 nF U_m (error voltage) max. 60 V DC 30 V AC T_a (permissible ambient temperature) max. 60 °C	
Power supply of transducers			
short-circuit-proof	yes		
Isolation			
- Between channels and backplane bus - Between channels and load voltage L+ - between channels - Between backplane bus and load voltage L+	yes yes yes yes		
Permissible difference in potential (U_{ISO}) of signals from hazardous area			
- Between channels and backplane bus - between channels - Between channels and load voltage L+ - Between backplane bus and load voltage L+	60 V DC 30 V AC 60 V DC 30 V AC 60 V DC 30 V AC 60 V DC 30 V AC		

Analog value formation				
Measuring principle	SIGMA-DELTA			
Integration time/conversion time/resolution (per channel)				
• configurable	yes	yes	yes	yes
• Integration time in ms	2.5	16 ² / ₃	20	100
• Basic conversion time incl. integration time in ms (several channels enabled)	7.5	50	60	300
• Basic conversion time incl. integration time in ms (one channel enabled)	2.5	16 ² / ₃	20	100
• Resolution in bit + sign (incl. overrange)	10+ sign	13+ sign	13+ sign	15+ sign
• Interference voltage rejection for interference frequency f1 in Hz	400	60	50	10
Interference rejection, error limits				
Interference voltage rejection for f = n x (f1 ± 1 %), (f1 = interference frequency)				
• Common-mode interference channels with respect to M-terminal of CPU (UISO < 60 V)	> 130 dB			
• Normal-mode interference (measured value + interference must be within the input range 0 to 22 mA)	> 60 dB			
Crosstalk attenuation between inputs (UISO < 60 V)	> 130 dB			
Operational limit (in total temperature range, referred to input range)				
• from 0/4 to 20 mA	± 0.45 %			
Basic error (operational limit at 25 °C, referred to input range)				
• from 0/4 to 20 mA	± 0.1 %			
Temperature error (referred to input range)	± 0.01%/K			
Linearity error (referred to input range)	± 0.01 %			
Repeatability (in steady-state condition at 25 °C, referred to input range)	± 0.05 %			

Interference rejection, error limits continued	
Influence of a HART signal superimposed on the input signal referred to the input range	
Error at integration time	
• 2.5 ms	± 0.25%
• 16 ² / ₃ ms	± 0.05%
• 20 ms	± 0.04%
• 100 ms	± 0.02%
Interrupts, Diagnostics	
Interrupts	
• Limit interrupt	Configurable channels 0 and 2
• Diagnostic interrupt	configurable
Diagnostic functions	
• Group fault indication	red LED (SF)
• Channel fault indication	red LED (F) per channel
• Diagnostic information readout	possible
Characteristic data for transducer supply	
• No-load voltage	< 25.2 V
• Output voltage for transducer and line at 22 mA transducer current (50 Ω measuring shunt incorporated in module)	> 13 V
Data for sensor selection	
Input ranges (rated values) / input resistance	
• Current	0 to 20 mA; /50 Ω 4 to 20 mA: /50 Ω
Permissible input current for current input (destruction limit)	40 mA
Signal generator connection	
• for current measurement	
as 2-wire transducer	possible
as 4-wire transducer	possible

3.9 Analog Output Module SM 332; AO 4 x 0/4...20 mA

In this chapter

In this chapter you will find, for the analog output module SM 332; AO 4 x 0/4...20 mA a description of its:

- characteristics
- technical specifications

and you will learn

- how to place the analog output module into operation.
- what measuring ranges the analog output module makes available
- what parameters influence the characteristics of the analog output module.

Order number

6ES7 332-5RD00-0AB0

Features

The analog output module SM 332; AO 4 x 0/4...20 mA is characterized by the following features:

- 4 current outputs in 4 groups
- Resolution 15 bit
- Configurable diagnostics
- Channels isolated among each other
- Channels isolated with respect to CPU and load voltage L+

Note

When switching the load voltage (L+) on and off, incorrect intermediate values can occur at the output for approx. 10 ms.

Wiring diagram

Fig. 3-18 shows the terminal diagram of the analog output module SM 332; AO 4 x 0/4...20 mA. You will find detailed technical specifications for the analog output module on the following pages.

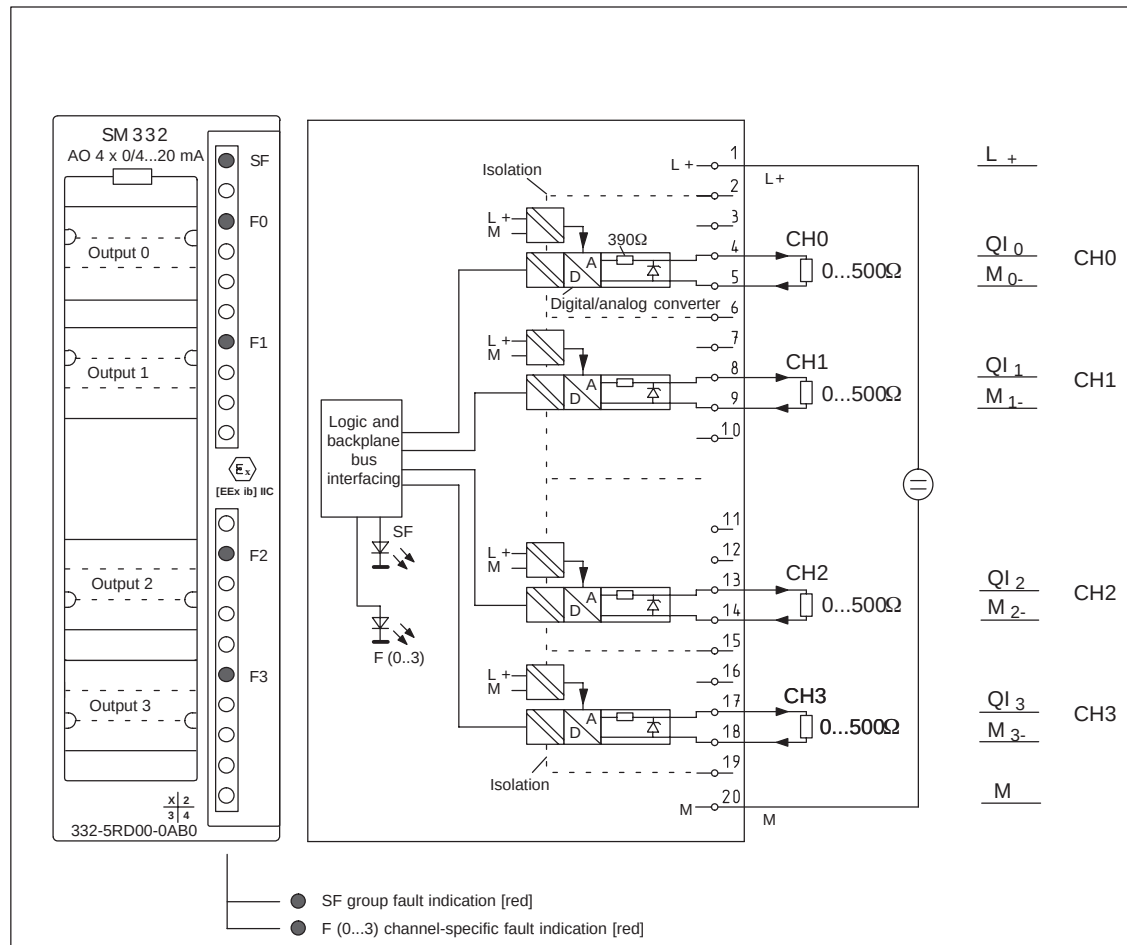


Fig. 3-18 Module view and block diagram of SM 332; AO 4 x 0/4...20 mA

Notes on intrinsically-safe installation

You must connect the DM 370 dummy module between the CPU or IM 153-2 (in a distributed configuration) and the Ex I/O modules whose signal cables lead into the hazardous location. In a distributed configuration with an active backplane bus, you should use the ex dividing panel/ ex barrier instead of the dummy module. Additional information on system design can be found in Sections 1.3 - 1.5.

Power supply for a intrinsically-safe structure

In order to maintain the clearances and creepage distances, L+ / M must be routed via the line chamber LK393 when operating I/O modules with signal cables that lead to the hazardous location, see Section 1.2.

Parameterization

The functions of the analog output module SM 332; AO 4 x 0/4...20 mA are set

- with *STEP 7* (refer to /231/) or
- in the user program with SFCs (refer to /235/).

Default setting

The analog output module features default settings for type of output, diagnostics, interrupts etc. (see Table 3-23).

These default settings are valid if re-parameterization has not been carried out via *STEP 7*.

Channel groups

Table 3-37 shows the allocation of the 4 channels to the 4 channel groups of SM 332; AO 4 x 0/4...20 mA.

Table 3-37 Allocation of 4 channels to 4 channel groups of SM 332; AO 4 x 0/4...20 mA

Channel	Allocated channel group
Channel 0	Channel group 0
Channel 1	Channel group 1
Channel 2	Channel group 2
Channel 3	Channel group 3

Non-connected output channels

Non-connected output channels of the analog output module SM 332; AO 4 x 0/4...20 mA must be deactivated to ensure no power is applied to them. You can deactivate an output channel with *STEP 7* via the "output" parameter block (see Section 3.6.3).

Analog output

You can connect the outputs as:

- Current outputs

The outputs can be set channel by channel. Output mode is parameterized with *STEP 7*.

Output ranges

You can set the various output ranges for current outputs with *STEP7*.

Table 3-38 shows the possible output ranges of the analog output module SM 332; AO 4 x 0/4...20 mA.

Table 3-38 Output ranges of analog output module SM 332; AO 4 x 0/4...20 mA

Selected output mode	Explanation	Output range
Current	The digitized analog values are specified in Section 3.1.3, Table 3-20 Current measuring range.	from 0 to 20 mA from 4 to 20 mA

Wire break check

The analog output module SM 332; AO 4 x 0/4...20 mA carries out an wire break check.

Conditions: A minimum output current of $> 100 \mu\text{A}$ must flow *and* the voltage set at the load must be $> 12 \text{ V}$ in order to signal wire break.

Influence of load voltage drop on diagnostic message

If the 24 V load voltage drops below the permissible rated range ($< 20.4 \text{ V}$) the output current can be reduced before a diagnostic message is output if a load of $> 400 \Omega$ is connected and the output currents are $> 18 \text{ mA}$.

Analog Output SM 332; AO 4 x 0/4...20 mA Dimensions and Weight		Voltages, currents, potentials continued	
Dimensions W x H x D (mm)	40 x 125 x 120	Permissible difference in potential (U_{ISO}) for signals from non-hazardous area	
Weight	approx. 280 g	- between channels and backplane bus 400 V DC 250 V AC - Between channels and load voltage L+ 400 V DC 250 V AC - Between channels 400 V DC 250 V AC - Between backplane bus and load voltage L+ 75 V DC 60 V AC	
Module-specific data		Insulation tested	
Number of outputs	4	- Channels with respect to backplane bus and load voltage L+ with 1500 V AC - Channels among each other with 1500 V AC - Backplane bus with respect to load voltage L+ with 500 V DC	
Line length, shielded	max. 200 m	Current input	
Type of protection PTB (see Appendix A)	[EEx ib] IIC to EN 50020	- from backplane bus max. 80 mA - From load voltage L+ (at rated data) max. 180 mA	
Test number	Ex-96.D.2026 X	Module power loss typical 4 W	
Type of protection FM (see Appendix B)	CL I, DIV 2, GP A, B, C, D T4	Analog value formation	
Voltages, currents, potentials		Resolution (incl. overrange)	15 Bit
Bus power supply	5 V DC	Cycle time (all channels)	9.5 ms
Rated load voltage L+	24 V DC	Transient recovery time	
• Reverse voltage protection	yes	- for resistive load 0.2 ms - for capacitive load 0.5 ms - for inductive load 0.2 ms	
Isolation		Substitute values switchable yes, configurable	
• Between channels and backplane bus	yes		
• Between channels and load voltage L+	yes		
• between channels	yes		
• Between backplane bus and load voltage L+	yes		
Permissible difference in potential (U_{ISO}) of signals from hazardous area			
• Between channels and backplane bus	60 V DC 30 V AC		
• Between channels and load voltage L+	60 V DC 30 V AC		
• between channels	60 V DC 30 V AC		
• Between backplane bus and load voltage L+	60 V DC 30 V AC		

Interference rejection, error limits		Safety data (refer to Certificate of Conformity in Appendix A)	
Crosstalk attenuation between outputs	> 130 dB	Type of protection to EN 50020	[EEx ib] IIC
Operational limit (in total temperature range, referred to output range)	± 0.55 %	Maximum values of output circuits (per channel)	
Basic error (operational limit at 25 °C, referred to output range)	± 0.2 %	• U ₀ (no-load output voltage)	max. 14 V
Temperature error (referred to output range)	± 0.01 %/K	• I ₀ (short-circuit current)	max. 70 mA
Linearity error (referred to output range)	± 0.02 %	• P ₀ (load power)	max. 440 mW
Repeatability (in steady-state condition at 25 °C, referred to output range)	± 0.005 %	• L ₀ (permissible external inductance)	max. 6.6 mH
Output ripple; range 0 to 50 kHz (referred to output range)	± 0.02 %	• C ₀ (permissible external capacitance)	max. 850 nF
Interrupts, Diagnostics		• U _m (error voltage)	max. 60 V DC 30 V AC
Interrupts		• T _a (permissible ambient temperature)	max. 60 °C
• Diagnostic interrupt	configurable	Data for actuator selection	
Diagnostic functions	configurable	Output ranges (rated values)	
• Group fault indication	red LED (SF)	• Current	from 0 to 20 mA from 4 to 20 mA
• Channel fault indication	red LED (F) per channel	Load impedance (in rated range of output)	
• Diagnostic information readout	possible	• for current outputs	
Monitoring for		– resistive load	max. 500 Ω
• Wire break	yes	– inductive load	max. 6.6 mH ¹⁾
as of output value <i>and</i>	> 0.1 mA	– capacitive load	max. 850 nF ¹⁾
output voltage	> 12 V	Current output	
		• No-load voltage	max. 14 V
		Destruction limit for externally applied voltages / currents	
		• Voltages	max. + 12 V / - 0.5 V
		• Current	max. + 60 mA / - 1 A
		Connection of actuators	
		• for current output	
		2-wire connection	possible

1) Limitation by PTB-approval

When used in non-Ex area

- resistive load max. 500 Ω
- inductive load max. 15 mH
- capacitive load max. 3 µF can be set as the load impedance.

SIMATIC S7 HART Analog Modules

In this chapter

The following SIMATIC S7 HART analog modules are described in this chapter:

- SM 331; AI 2 x 0/4...20mA HART (HART analog input module)
Order number: 6ES7 331-7TB00-0AB0
- SM 332; AO 2 x 0/4...20mA HART (HART analog output module),
Order number: 6ES7 332-5TB00-0AB0

This chapter provides you with the information you require in order to use the module as a HART interface:

- An introduction to HART, to help you familiarize yourself with the technology,
- Guidelines for installation, startup, and operation, with the aid of a sample configuration,
- HART-specific parameter assignment and diagnostics,
- Technical data for the HART analog modules.

Chapter overview

Section	Description	Page
4.1	Product overview for the use of HART analog modules	4-2
4.2	Introduction to HART	4-3
4.3	Guidelines for installation, startup, and operation	4-7
4.4	Parameters of HART analog modules	4-11
4.5	Diagnostics and interrupts of HART analog modules	4-13
4.6	HART analog input module SM 331; AI 2 x 0/4...20mA HART	4-15
4.7	HART analog output module SM 332; AO 2 x 0/4...20mA HART	4-20
4.8	Data record interface and user data	4-25

Basic characteristics

The SIMATIC S7 HART analog modules belong to the category of SIMATIC S7-Ex analog modules. Their basic properties were described in Chapter 3 and also apply here. The channel properties of the HART analog input module correspond to the properties of the module SM 331; AI 4 x 0/4...20mA. The channel properties of the HART analog output module correspond to the properties of the module SM 332; AO 4 x 0/4...20mA.

Note

The HART analog module can only be used within the ET200 M distributed I/O system with the interface module IM153-2AA01, IM153-2AB00 or IM153-2AB80 acting as a connection to the PROFIBUS DP.

4.1 Product Overview for the Use of HART Analog Modules

Product overview The following figure shows you where the HART analog modules can be used:

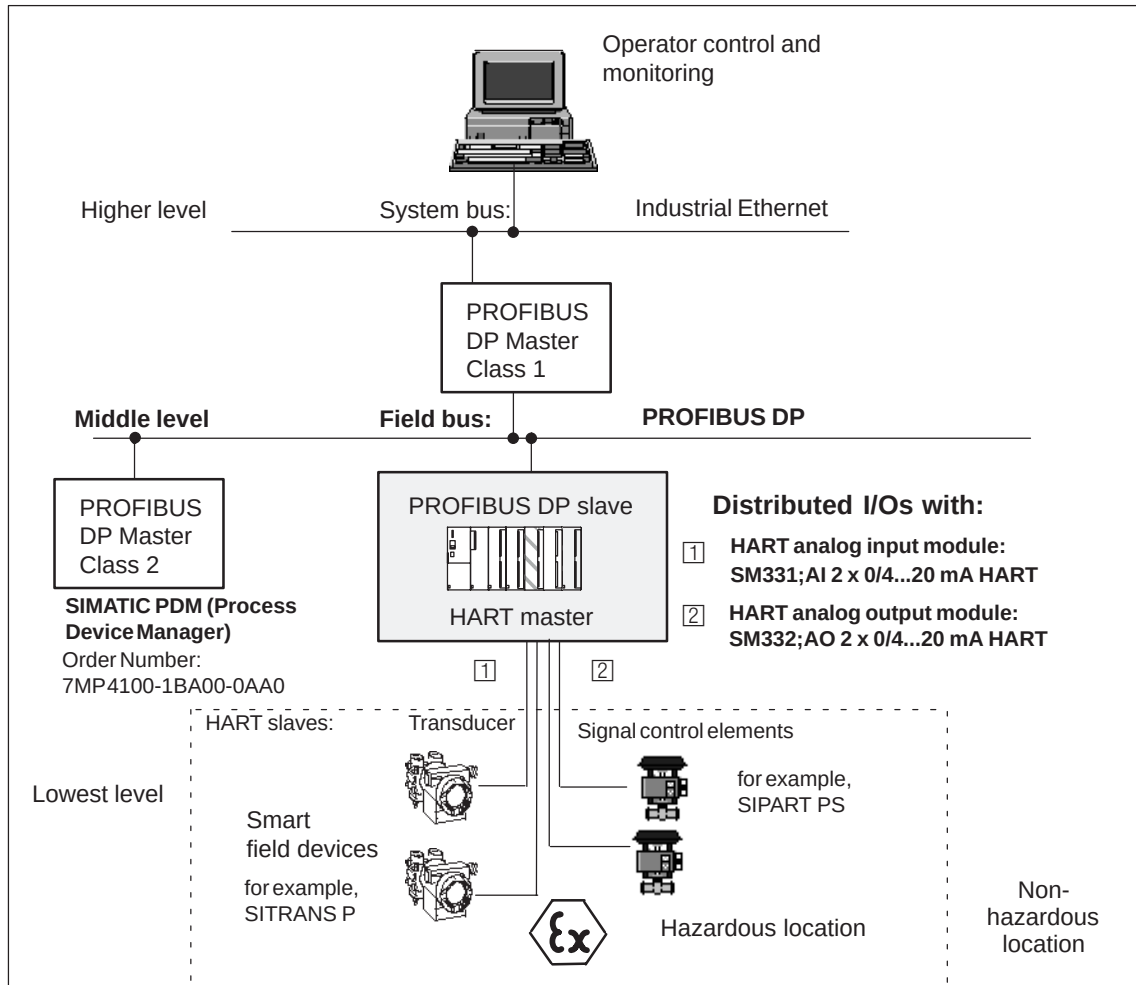


Fig. 4-1 Location of the HART analog modules in the distributed system

Using the modules in a system

The HART analog modules are used in the distributed I/Os attached to PROFIBUS DP (see Figure 4-1).

You can connect one field device to each of the two channels on a HART analog module: the module acts as HART master, the field devices as HART slaves.

Different software applications can transmit or receive data via a HART analog module. These applications can be compared to clients, for which the HART analog module acts as a server.

4.2 Introduction to HART

Introduction

This section provides you with an introduction to HART from a user's perspective:

- Definition of HART
- Advantages of HART analog modules
- Typical applications of HART

What is HART?

The HART functions enable you to operate an analog module in conjunction with digital communication. The HART protocol is generally accepted as a standard protocol for communication with smart field devices: Hart is a registered trademark of the "HART Communication Foundation" (HCF), which retains all rights for the HART protocol. You can find detailed information about HART in the HART Specification /900/ and in the booklet /901/ published by Fisher-Rosemount Ltd.

Note

The HART analog modules are designed to be used with version 5.4 of the HART protocol. Field devices which operate with an earlier version of the HART protocol are only supported to a limited extent: the command instruction format must be "long frame," with one exception: the "short frame" command format must be used for command 0 (see Table 4-2) to obtain the "long frame" address. Additional features which are introduced in Version 6 of the HART protocol have not yet been implemented.

What advantages does HART offer?

The use of HART analog modules has the following advantages:

- Compatibility with analog modules: current loop 4 - 20 mA
- Digital communication with the HART protocol
- Low power requirements, important for use in hazardous areas
- A wide range of field devices with HART functions are now available
- Integration of the HART functionality in the S7 system when using HART analog modules

What are typical applications of HART?

The following are typical applications of HART:

- Installation of field devices (central assignment of parameters)
- Modification of field device parameters online
- Display of information, maintenance data and diagnostic data for field devices
- Integration of configuration tools for field devices via the HART interface

4.2.1 How Does HART Function?

Introduction

The HART protocol describes the physical characteristics of transmission: data transfer procedures, message structure, data formats, and commands.

HART signal

Figure 4-2 shows the analog signal with the HART signal (FSK procedure). The HART signal is composed of sine waves at 1200 Hz and 2200 Hz and has a mean value of zero. It can be filtered out with an input filter, leaving the original analog signal unaffected.

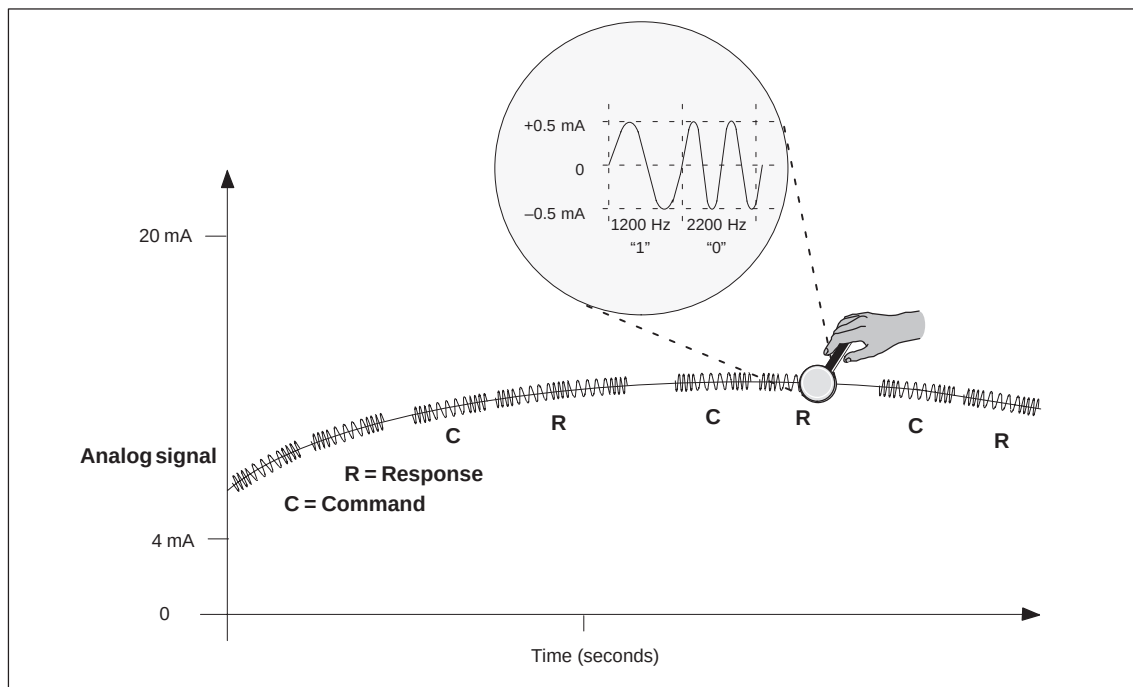


Fig. 4-2 The HART signal

HART commands and parameters

The adjustable properties of the HART field devices (HART parameters) can be set with **HART commands** and read using **HART responses**. The HART commands and their parameters are defined in three groups with the following properties:

- Universal
- Common-practice
- Device-specific

Universal commands and their parameters must be supported by all manufacturers of HART field devices; common-practice commands should also be supported. There are also device-specific commands that apply to a particular field device.

Examples of HART parameters

The following table shows the HART parameters of the different groups:

Table 4-1 Examples of HART parameters

Parameter group	HART field device parameters
Universal	Measured or manipulated value (primary variable), manufacturer name, device ID("tag"), or ID for actuator, other measured or manipulated values
Common-practice	Measuring range, filter time, interrupt parameters (message, interrupt and warning limits), output area
Device-specific	Special diagnostic information

Examples of HART commands

The following two tables provide examples of commands:

Table 4-2 Examples of universal commands

Command	Function
0, 11	Read manufacturer and device type
1	Read primary variable (PV) and units
2	Read current output and percentage of range as digital floating-point number (IEEE 754)
3	Read up to four pre-defined dynamic variables (primary variables, secondary variables, etc.)
13, 18	Read or write tag, description, date (data included)

Table 4-3 Examples of common-practice commands

Command	Function
36	Set the upper range value
37	Set the lower range value
41	Perform device self-test
43	Set primary variable to zero
109	Switch burst mode on or off

Burst mode

In burst mode, a command initiates a cyclic response from the slave device. This response is sent repeatedly until the mode is deactivated by the master device.

Data and status

HART commands are often transmitted without data, because they are used to start a processing function. HART responses always contain data. A HART response is always accompanied by status information, which you should evaluate to check that the response is correct.

4.2.2 How to Use HART

System environment

To use a smart field device with HART functionality, you require the following system environment (see Figure 4-3):

- **Current loop 4 - 20 mA**
- **HART parameter assignment tool:**
You can set the HART parameters either with an external hand-held controller (HART hand-held device) or by using a HART parameter assignment tool. The parameter assignment tool accesses the HART analog module directly, whereas the HART hand-held device is connected parallel to the field device. The PDM (Process Device Manager) can be obtained as an autonomous tool (stand alone) or it can be embedded in STEP7 HW Config. For the latter, an optional package is required.
- **How HART is linked to the system:**
The HART analog module assumes the function of a “master,” in that it receives the commands from the HART parameter assignment tool, forwards them to the field device, and then sends back the responses. The interface of the HART analog module comprises data records which are transmitted via the I/O bus. The data records must be created and interpreted by the HART parameter assignment tool.
- **Interface connection for HART parameter assignment tool:**
DP Connection which is capable of master class 1 as well as master class 2 functionality.

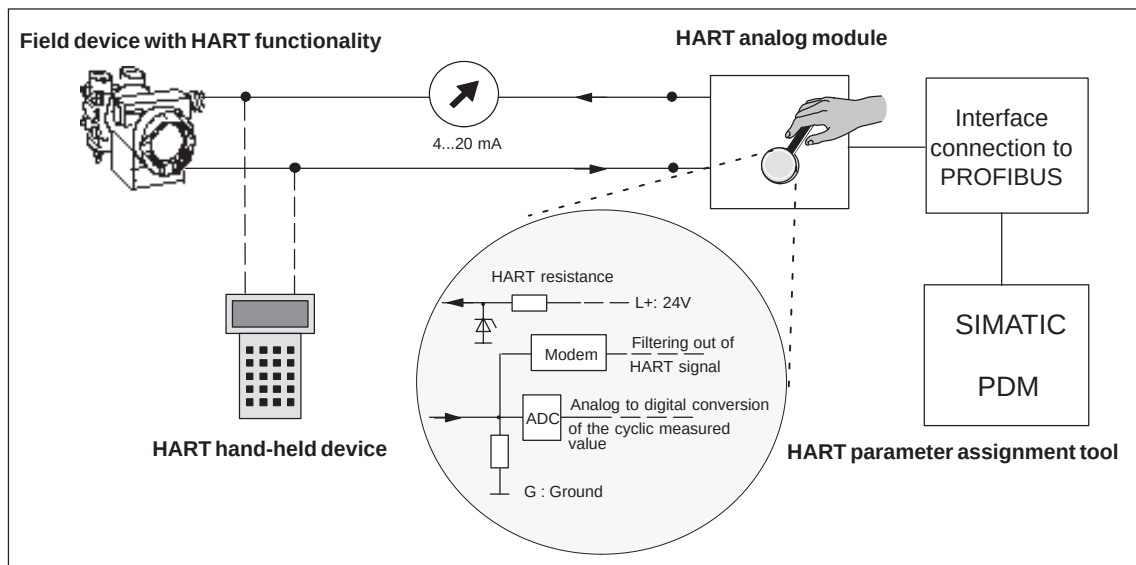


Fig. 4-3 System environment required for HART

Error handling

The two HART status bytes transmitted with each response of the field device contain error information relating to HART communication, HART commands and device status, (see HART communication data records, Section 4.8.3).

4.3 Guidelines for Installation, Startup, and Operation

Application in the system

A sample configuration is used to show you how to start up the HART analog module with the field devices connected, and the points you should take into consideration during operation. Further information can be found in the /804/ system overview of the field technology package (supplied on CD). Notes on the operation of field devices can be found in the online help on SIMATIC PDM.

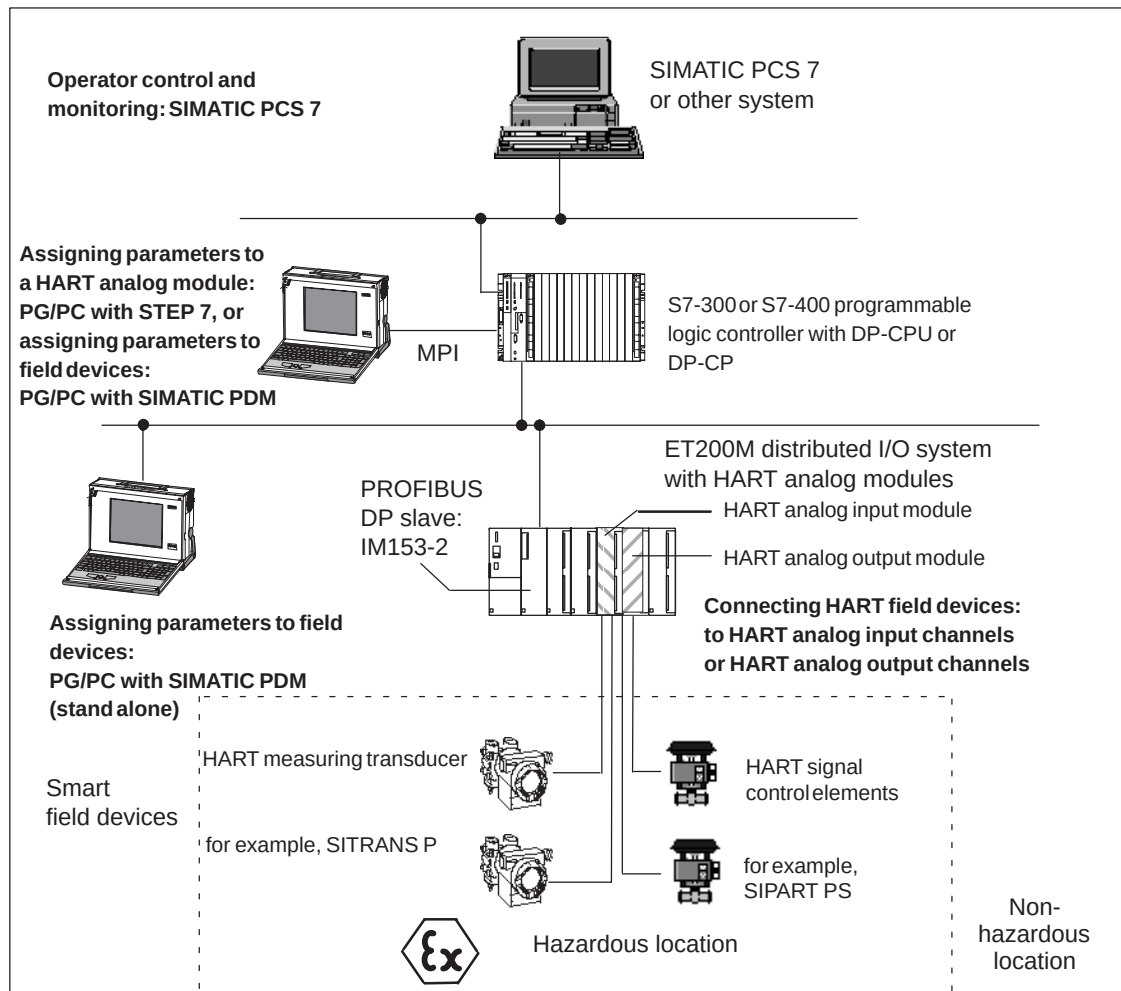


Fig. 4-4 Use of a HART analog module in a sample configuration

Notes on intrinsically-safe installation

You must connect the DM 370 dummy module between the IM 153-2 and explosion-proof I/O modules, which includes HART I/O modules, whose signal cables lead into the hazardous area. In a distributed configuration with an active backplane bus, you should use the explosion-proof partition (6ES7 195-1KA00-0XA0) instead of the dummy module. Additional information on system design can be found in Sections 1.3 - 1.5.

4.3.1 Setting Up the HART Analog Module and Field Devices

Configuring and assigning parameters

The HART analog modules are configured and assigned parameters with *STEP 7* and the connected smart field devices using the parameter assignment tool SIMATIC PDM:

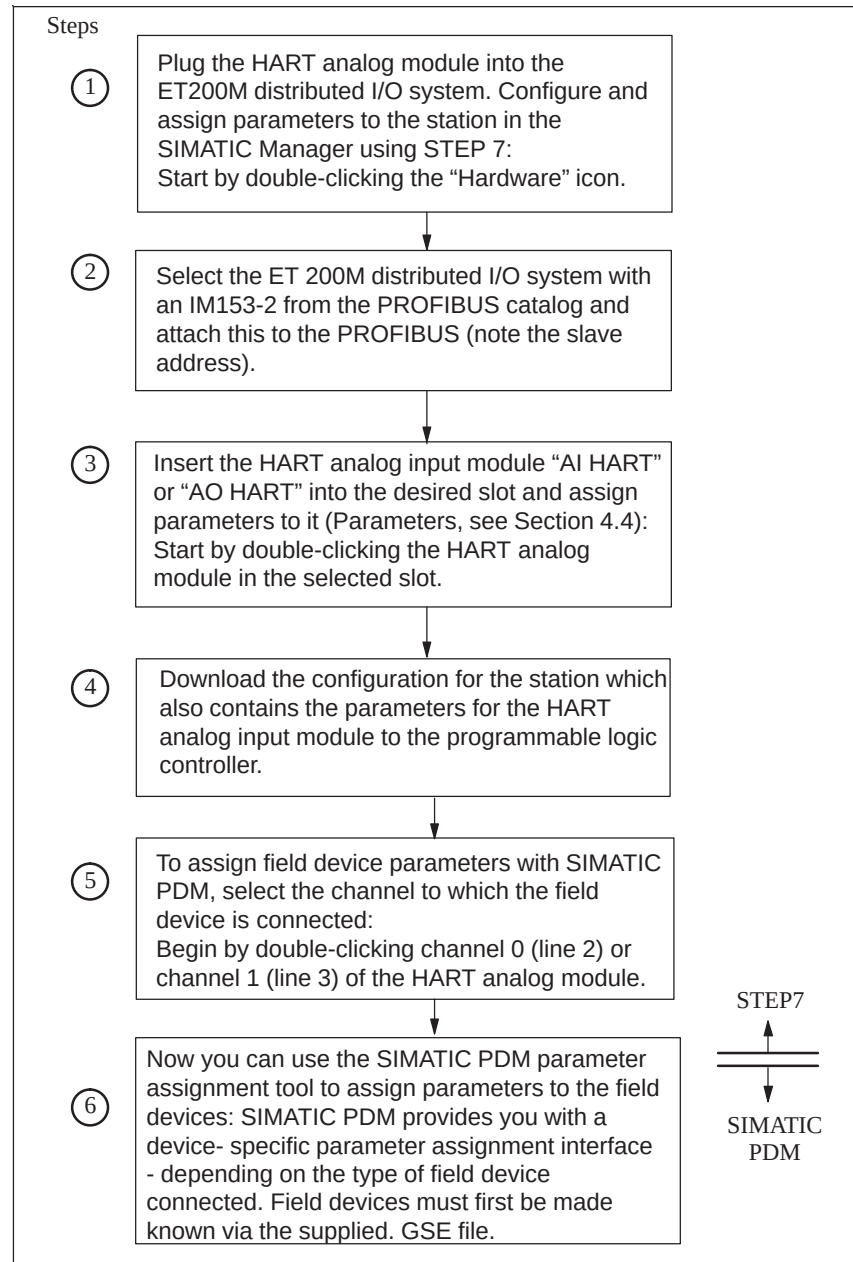


Fig. 4-5 Configuring and assigning parameters

Modifying the parameters of the field devices

Remember that the field devices signal each change in the parameters as a configuration change to the HART analog module. This leads to a diagnostic interrupt on the programmable controller, provided this option is enabled. It is advisable to disable the diagnostic interrupt during configuration and parameter assignment. You can do this when you assign parameters to the HART analog module, see Section 4.4.

4.3.2 Operating Phase of HART Analog Module and Field Devices

Operating phase

In the operating phase you must distinguish between the cyclic return of user data, acyclic HART interventions, and cyclic HART communication.

- The cyclic user data, for example measured values, are obtained from the programmable logic controller (PROFIBUS DP master class 1): The user data area exists for this purpose. In the case of the HART analog input module, this is the input area; in the case of the HART analog output module, it is the output area.
- Acyclic intervention for diagnostics and modifying the parameters of the field devices is carried out with the SIMATIC PDM parameter assignment tool (on PROFIBUS DP master class 2) or with a HART hand-held device using HART commands and HART responses.
- You can establish cyclic HART communication by writing / reading a data record in conjunction with the data ready ID.

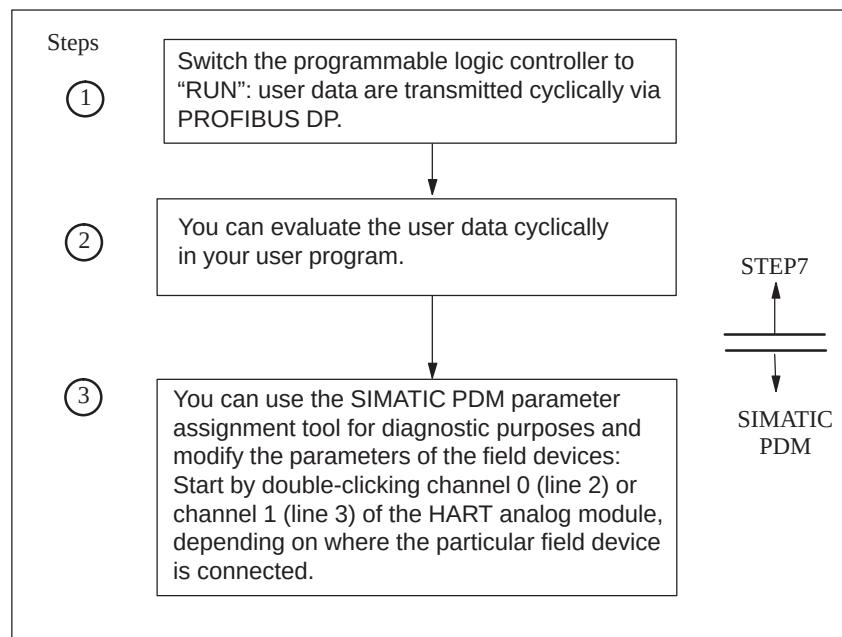


Fig. 4-6 The operating phase

Access to the field devices

The HART analog module generally accepts the modification of parameters for the field devices. Access rights can only be allocated using the parameter assignment tool.

Modifying the parameters of the field devices

To modify the parameters of the field devices connected to the HART analog modules, proceed as follows:

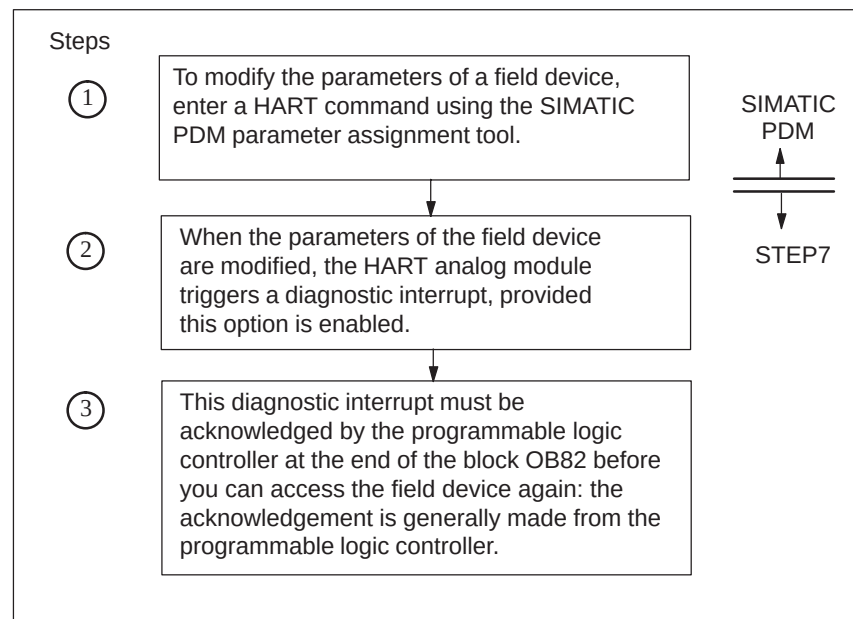


Fig. 4-7 How to modify the parameters of the field devices

Information on status

After you have modified the parameters of a HART field device, the corresponding bit is entered in the device status. This should be regarded as an indicator and not as an error and is reset by the module. For more information, see HART status bytes Section 4.5.1. You have to acknowledge the automation system diagnostic interrupt (OB 82) before you can have access to the field device again.

4.4 Parameters of HART Analog Modules

Overview of the parameters

Table 4-4 lists the parameters for the HART analog input module, Table 4-5 lists the parameters for the HART analog output module. The tables show which parameters can be set for the module as a whole and which parameters can be set separately for each channel. General information on assigning parameters can be found in the description of the SIMATIC-Ex analog modules in Chapter 3.6.3.

Table 4-4 Parameters for the analog input module SM 331; AI 2 x 0/4...20mA HART

Parameter	Range of values	Default setting	Type of parameter	Effective range
Basic settings				
Enable				
• Diagnostic interrupt	yes/no	no	dynamic	module
• Hardware interrupt on exceeding limit	yes/no	no		
• Hardware interrupt at end of cycle	yes/no	no		
Trigger for hardware interrupt				
• Upper limit	20 ...0/4 mA (from 32511 to -32512)	– (32767)*	dynamic	channel
• Lower limit	0/4 ...20 mA (from -32512 to 32511)	– (-32768)*		
Diagnostics				
• Group diagnostics	yes/no	no	static	channel
• Wire break monitoring	yes/no	no		
Measurement				
• Measurement mode	deactivated 4DMU current (4-wire transducer) 2DMU current (2-wire transducer) HART (connected to 2DMU or 4DMU)	HART	dynamic	channel
• Range of measurement	0...20mA (can only be set at 4DMU), 4...20mA	4...20mA	dynamic	channel
• Integration time	2.5ms; 16.6ms; 20ms; 100ms corresponds to interference frequency suppression of 400Hz; 60Hz; 50Hz; 10Hz	20ms	dynamic	channel
*) Values in parenthesis can be set with SFC dynamic parameterization.				

**HART
measurement type**

If you have activated the HART measurement type for a channel and HART communication is running, the green HART status display lights up. When HART starts up, the HART analog module transmits the HART command 0 to the field device, followed by HART command 13. The resulting HART response data (for example “long frame” address and “tag”), are entered in the diagnostic data record 131 or 151, see Section 4.8.4. When it is operating, the HART analog module continually sends the HART command 1 to update the value of the primary variable. This value is entered in the user data area (see Section 4.8.6).

Table 4-5 Parameters for the analog output module SM 332; AO 2 x 0/4...20mA HART

Parameter	Range of values	Default setting	Type of parameter	Effective range
Basic settings				
Enable				
• Diagnostic interrupt	yes/no	no	dynamic	module
Diagnostics				
• Group diagnostics	yes/no	no	static	channel
Behavior during CPU STOP				
• No current or voltage at outputs (NCVO)				
• Retain last value (RLV)				
• Switch substitute value (SV)	0/4...20 mA (-32512...32511)*	0/4 mA (-6912/0)* EWS	dynamic	channel
Output				
• Type of output	deactivated current HART	HART	dynamic	channel
• Range of output	4...20mA 0...20mA	4...20mA	dynamic	channel
*) Values in brackets can be set with SFC dynamic parameterization				

4.5 Diagnostics and Interrupts of HART Analog Modules

4.5.1 Diagnostic Functions of HART Analog Modules

Overview of diagnostic functions

If errors occur during configuration and parameter assignment or during the operating phase, you can use diagnostics to determine the cause of the error. The general diagnostic behavior of the HART analog module corresponds to that of the other SIMATIC S7-Ex analog modules, see Section 3.6.4.

Diagnostic messages

The diagnostic messages for the analog input modules are shown in Table 3-24 of Section 3.6.4; the diagnostic messages for the analog output modules are shown in Table 3.6.4. The additional diagnostic messages are listed in the following table:

Table 4-6 Additional diagnostic messages for the analog input module SM 331; AI 2 x 0/4...20mA HART and the analog output module SM 332; AO 2 x 0/4...20mA HART

Diagnostic message	Effective range of diagnostics	Configurable with group diagnostics
Modification of HART parameters reported by the connected field device	channel	yes
HART group error		yes

Causes of errors

The following table provides a list of possible causes and corresponding corrective measures for the individual diagnostic messages.

Table 4-7 Additional diagnostic messages, possible causes of the errors, and corrective measures

Diagnostic message	Possible cause of error / diagnostics	Corrective measures
Modification of HART parameters reported by the connected field device	The identifier for the modification of parameters to the HART field device was set in the HART device status.	If you do not want diagnostic interrupts to be triggered when parameters are modified, you should disable the diagnostic interrupt.
HART group error	Communication and command error during HART operation affecting the connected HART field devices.	For detailed information, evaluate the response data record of the relevant client (see 4.8.3) or the additional diagnostic data record (see 4.8.4)

- HART status bytes** Each HART command is followed by a HART response containing data and status bytes (see 4.8.3). The status bytes provide information on:
- Device status of the connected field device (e.g. modification of parameters)
 - Communication error during transmission between HART analog module and the connected field device
 - Command error during interpretation of the HART command by the connected field device (warning, rather than error)

The HART status bytes are entered in the response data record unchanged (see Section 4.8.3). Their significance is described in the technical specifications for HART. You can use SFC59 to read the data records in your user program.

4.5.2 Interrupts of the HART Analog Modules

Overview of the interrupts The general interrupt behavior of the HART analog module corresponds to that of the other SIMATIC S7-Ex analog modules, see Section 3.6.5. You can set parameters to enable or disable any interrupt (see Section 4.4).

Hardware interrupts with AI HART There are two types of hardware interrupt: “Hardware interrupt when limit value exceeded” and “Hardware interrupt on end of cycle.” When a hardware interrupt is triggered, you can evaluate the local data in OB40:

Table 4-8 Local data in OB40

Local data OB40	Bit 7 ...4	Bit 3	Bit 2	Bit 1	Bit 0	Limit exceeded
Byte 0	‘0’	‘0’	‘0’	Channel 1	Channel 0	Upper limit exceeded
Byte 1	‘0’	‘0’	‘0’	Channel 1	Channel 0	Lower limit exceeded
Byte 2	‘0’	‘0’	‘0’	‘0’	‘0’	Not relevant
Byte 3	‘0’	‘0’	‘0’	‘0’	‘0’	Not relevant

At the end of the cycle all the bits in bytes 0-3 of the additional information for OB40 which are not reserved for channels 0 and 1 are set to ‘1’. You can use the reserved bits to evaluate whether the upper or lower limit set has been exceeded for a particular channel: if a limit has been exceeded, a ‘1’ is displayed instead of a ‘0’.

4.6 HART Analog Input Module SM 331; AI 2 x 0/4...20mA HART

In this section	This section provides you with the properties, the technical data, and a wiring diagram.
Order number	6ES7 331-7TB00-0AB0
Features	The analog input module SM 331; AI 2 x 0/4...20mA HART has the following properties: • 2 inputs in 2 channel groups • 2 outputs to power 2-wire measuring transducers • Measured value resolution; can be set for each channel individually (see analog values and resolution on the following page). • Measurement type can be selected for each channel: – HART (2-wire transducer or 4-wire transducer for current) – 2-wire or 4-wire transducer for current (used without HART) – Channel deactivated • Measuring range selectable for each channel – 0 ... 20 mA (only for 4-wire transducers used without HART) – 4 ... 20 mA • Settings for diagnostics and diagnostic interrupt – Group diagnostics – Wire-break monitoring – Diagnostic interrupt • Settings for hardware interrupt – Channels 0 and 1 with limit monitoring; hardware interrupt can be set to trigger if limit is exceeded – Hardware interrupt can be set for cycle end • Isolation – Channels electrically isolated from each other – Channels electrically isolated from CPU and load voltage L+

Analog values and resolution

The representation of the analog values is the same as for the analog input module SM 331; AI 4 x 0/4...20mA, see Section 3.1.2. The resolution of the measured value is directly dependent on the selected integration time, i.e. the greater the integration time selected for an analog input channel, the more precise the resolution of the measured value.

- 10 bits + polarity (integration time 2.5 ms)
- 13 bits + polarity (integration time 16.6/ 20 ms)
- 15 bits + polarity (integration time 100 ms)

Table 4-9 Output range of the analog input modules SM 331; AI 2 x 0/4...20 mA HART

Selected output type	Explanation	Output range
Current	The digitalized analog values can be found in part 3.1.2 in Table 3-4 of the current measuring range.	0 to 20 mA 4 to 20 mA

Integration times when HART is used

If you use measuring transducers with the HART protocol, it is advisable to assign integration times of 16.6, 20 or 100 ms, in order to minimize the influence of the modulating alternating current on the measuring signal.

Default settings

The HART measurement mode is set as default. There are other default settings for integration time, diagnostics, interrupts (see Table 4-4). The HART analog module uses these settings, unless you modify them using *STEP 7*.

Wire break monitoring

- Wire break monitoring is not possible for the current range 0 to 20 mA.
- For the current range 4 to 20 mA, if the input current falls below $I \leq 3.6$ mA this is interpreted as a wire break and a diagnostic interrupt is triggered (provided the interrupt is enabled).

Inserting and removing modules

The HART analog modules support the function “Change modules during operation.” However, it is only possible to evaluate the insert / remove module interrupts with a S7/M7 400 CPU master and an active backplane bus in the ET 200M.

Operation with standard master

Information on operating the modules in a distributed configuration with a standard master can be found in manual /140/. The manual lists the differences to be taken into consideration if you are operating the modules with a S7/M7 DP master and a standard master (for example, IM 308C with S5).

- Parameter assignment with COM PROFIBUS (.GSE file or type file required)
- Restricted evaluation when inserting or removing modules.

Wiring diagram

Figure 4-8 shows the wiring diagram for the analog input module SM 331; AI 2 x 0/4...20 mA HART. Detailed technical data can be found on the following pages.

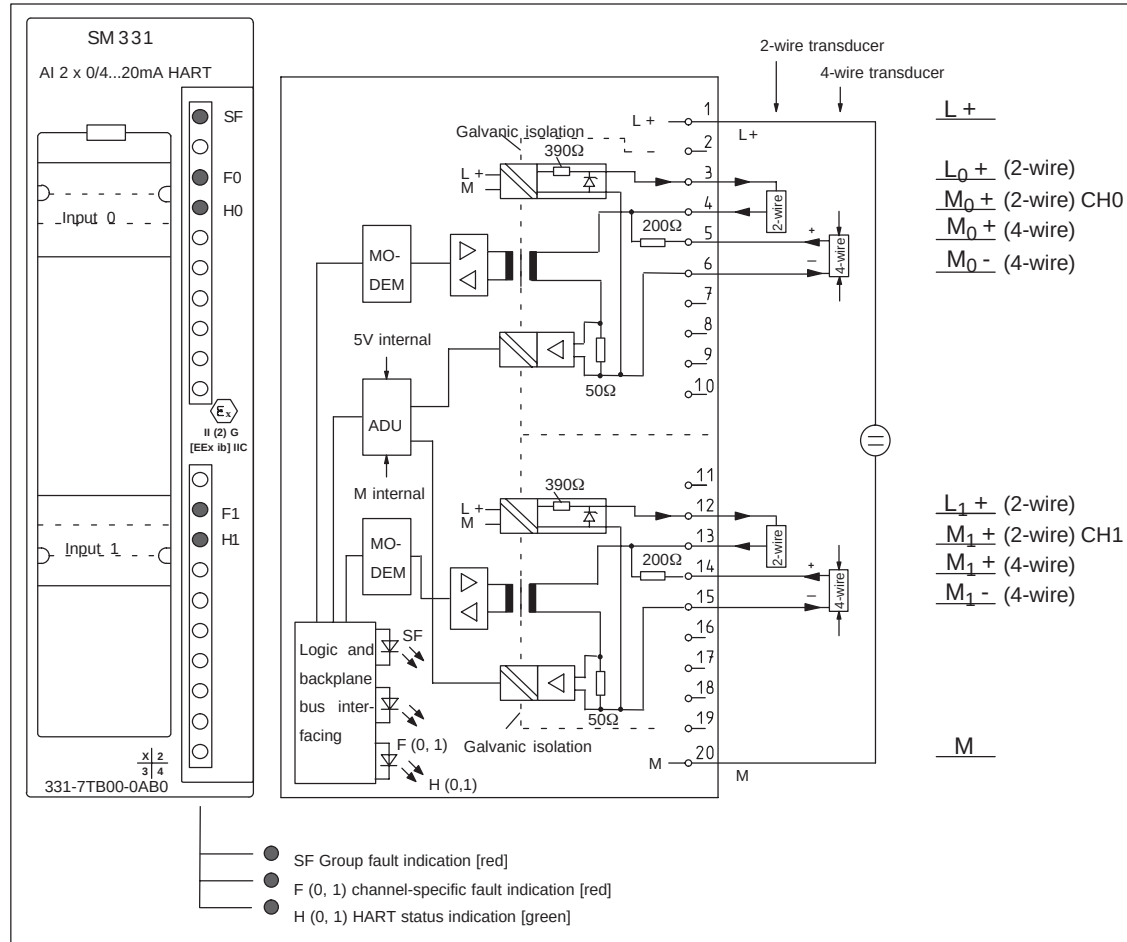


Fig. 4-8 Module view and block diagram of SM 331; AI 2 x 0/4...20mA HART

Notes on intrinsically-safe installation

Section 4.3 provides you with a summary of information on intrinsically-safe installation. Detailed information can be found in Section 1.5.

Power supply for an intrinsically-safe structure

In order to maintain the clearances and creepage distances, L+ / M must be routed via the line chamber LK393 when operating modules with signal cables that lead to the hazardous location, see Section 1.2.

SM 331; AI 2 x 0/4...20 mA HART		Voltages, currents, potentials continued	
Dimensions and weight		Permissible difference in potential (U_{ISO}) for signals from non-hazardous area	
Dimensions W x H x D (mm)	40 x 125 x 120	- Between channels and backplane bus 400 V DC 250 V AC - Between channels and load voltage L+ 400 V DC 250 V AC - Between channels 400 V DC 250 V AC - Between backplane bus and load voltage L+ 75 V DC 60 V AC	
Weight	approx. 260 g		
Module-specific data		Insulation tested	
Number of inputs	2	- Channels to backplane bus and load voltage L+ with 1500 V AC - Channels to each other with 1500 V AC - Between load voltage L+ and backplane bus with 500 V DC	
Number of power outputs	2	Current input	
Line length, shielded	max. 400 m	- From backplane bus max. 100 mA - From load voltage L+ max. 180 mA	
Type of protection KEMA (see Appendix A)	[EEx ib] IIC to EN 50020	Module power loss typically 4.5 W	
KEMA test number	97ATEX3039 X	Safety data (see Certificate of Conformity in Appendix A)	
Voltages, currents, potentials		Type of protection to EN 50020 [EEx ib] IIC	
Bus power supply	5 V DC	Maximum values per channel	
Rated load voltage L +	24 V DC	U_0 (no-load output voltage) max. 29.6 V I_0 (short-circuit current) max. 99 mA P_0 (load power) max. 553 mW L_0 (permissible external inductance) max. 3 mH C_0 (permissible external capacitance) max. 62 nF U_m (error voltage) max. 250 V DC T_a (permissible ambient temperature) 0 to 60°C	
• Reverse voltage protection	yes		
Power supply for 2-wire transducer			
• Short-circuit proof	yes (approx. 30 mA)		
Galvanic isolation			
• Between channels and backplane bus	yes		
• Between the channels	yes		
• Between channels and load voltage L+	yes		
• Between backplane bus and load voltage L+	yes		
Permissible difference in potential (U_{ISO}) for signals from a hazardous area			
• Between channels and backplane bus	60 V DC 30 V AC		
• Between channels and load voltage L+	60 V DC 30 V AC		
• Between channels	60 V DC 30 V AC		
• Between backplane bus and load voltage L+	60 V DC 30 V AC		

Analog value formation					Interference suppression, error limits, continued	
Measuring principle	SIGMA-DELTA				Influence of a HART signal modulated onto the input signal, referred to input range	
Integration time/conversion time/resolution (per channel)					Error at integration time	
• Configurable	yes	yes	yes	yes	• 2.5 ms	± 0.25%
• Integration time in ms	2.5	16 ² /3	20	100	• 16 ² /3 ms	± 0.05%
• Basic conversion time, incl. integration time in ms (one channel enabled)	2.5	16 ² /3	20	100	• 20 ms	± 0.04%
• Basic conversion time, incl. integration time in ms (two channels enabled)	7.5	50	60	300	• 100 ms	± 0.02%
• Resolution in bit + sign (incl. overrange)	10+ sign	13+ sign	13+ sign	15+ sign	Interrupts, diagnostics	
• Interference voltage suppression for interference frequency f1 in Hz	400	60	50	10	Interrupts	
Interference suppression, error limits					• Hardware interrupt configurable channels 0 and 1	
					• Diagnostic interrupt configurable	
					Diagnostic functions configurable	
					• Group fault indication red LED (SF)	
					• Channel fault indication red LED (F) per channel	
					• Diagnostic information readout possible	
					• HART communication active and OK green LED (H)	
					Data for transducer supply	
					• No-load voltage < 29.6 V	
					• Output voltage for transducer and line with 22 mA transducer current (50 Ω resistor on module taken into account) > 15 V	
					Data for sensor selection	
					Input ranges (rated values / input resistance)	
					• Current 0 to 20 mA; /50 Ω 4 to 20 mA: /50 Ω	
					Permissible input current for current input (destruction limit) 40 mA	
					Signal sensor connection	
					• for current measurement	
					as 2-wire transducer possible	
					as 4-wire transducer possible	

4.7 HART Analog Output Module SM 332; AO 2 x 0/4...20mA HART

In this section This section provides you with the properties, the technical data, and a wiring diagram.

Order number 6ES7 332-5TB00-0AB0

Features The HART analog output module SM 332; AO 2 x 0/4...20mA HART has the following properties:

- 2 outputs in 2 channel groups
- Resolution 12 bit (+ polarity)
- Measurement type can be selected for each channel:
 - Current output with HART
 - Current without HART usage
 - Channel deactivated
- Output range selectable for each channel
 - 0...20 mA (without HART usage)
 - 4...20 mA
- Settings for diagnostics and diagnostic interrupt
 - Enable group diagnostics
 - Enable/disable diagnostic interrupt
- Isolation
 - Channels electrically isolated from each other
 - Channels electrically isolated from CPU and load voltage L+
- Readback capability of the analog outputs

Analog values and resolution The representation of the analog values is the same as for the analog output module SM 332; AO 4 x 0/4...20mA, see Section 3.1.3. The resolution of the output value for the HART analog output module is, however, 12 bits.

Table 4-10 Output ranges of the analog output module SM 332; AO 4 x 0/4...20mA

Selected output type	Explanation	Output range
Current	The digitalized analog values can be found in Section 3.1.3 in Table 3-20 in the current output range.	0 to 20 mA 4 to 20 mA

Default settings	The HART output type is set as default. There are other default settings for substitute values, diagnostics, and interrupts (see Table 4-4). The HART analog output module uses these settings, unless you modify them using <i>STEP 7</i> .
Wire break monitoring	Wire break monitoring is possible for the current range 0/4 to 20 mA. Conditions: A minimum output current of >500µA is required.
Inserting and removing modules	The HART analog modules support the function “Change modules during operation.” However, it is only possible to evaluate the insert / remove module interrupts with a S7/M7 400 CPU master on an active backplane bus in the ET 200M.
Operation with standard master	Information on operating the modules in a distributed configuration with a standard master can be found in manual <i>/140/</i>. The manual lists the differences to be taken into consideration if you are operating the modules with a S7/M7 DP master and a standard master (for example IM 308C with S5). • Parameter assignment with COM PROFIBUS (.GSE file or type file required) • Restricted evaluation when inserting or removing modules
How a fall in the load voltage affects diagnostic messages	If the 24 V load voltage falls below the permitted rated range (< 20.4 V), there may be a reduction in the output current at connected loads > 650 Ω and output currents > 20 mA before a diagnostic message is transmitted.
Readback capability	The analog outputs can be readback in the user data range (see Fig. 4-20) with a resolution of 8 bits. (+polarity). Please note that the readback analog output is only available after a conversion time which varies with the precision desired.

Wiring diagram

Figure 3-18 shows the wiring diagram for the analog output module SM 332; AO 2 x 0/4...20mA HART. Detailed technical data for the analog output module can be found on the following pages.

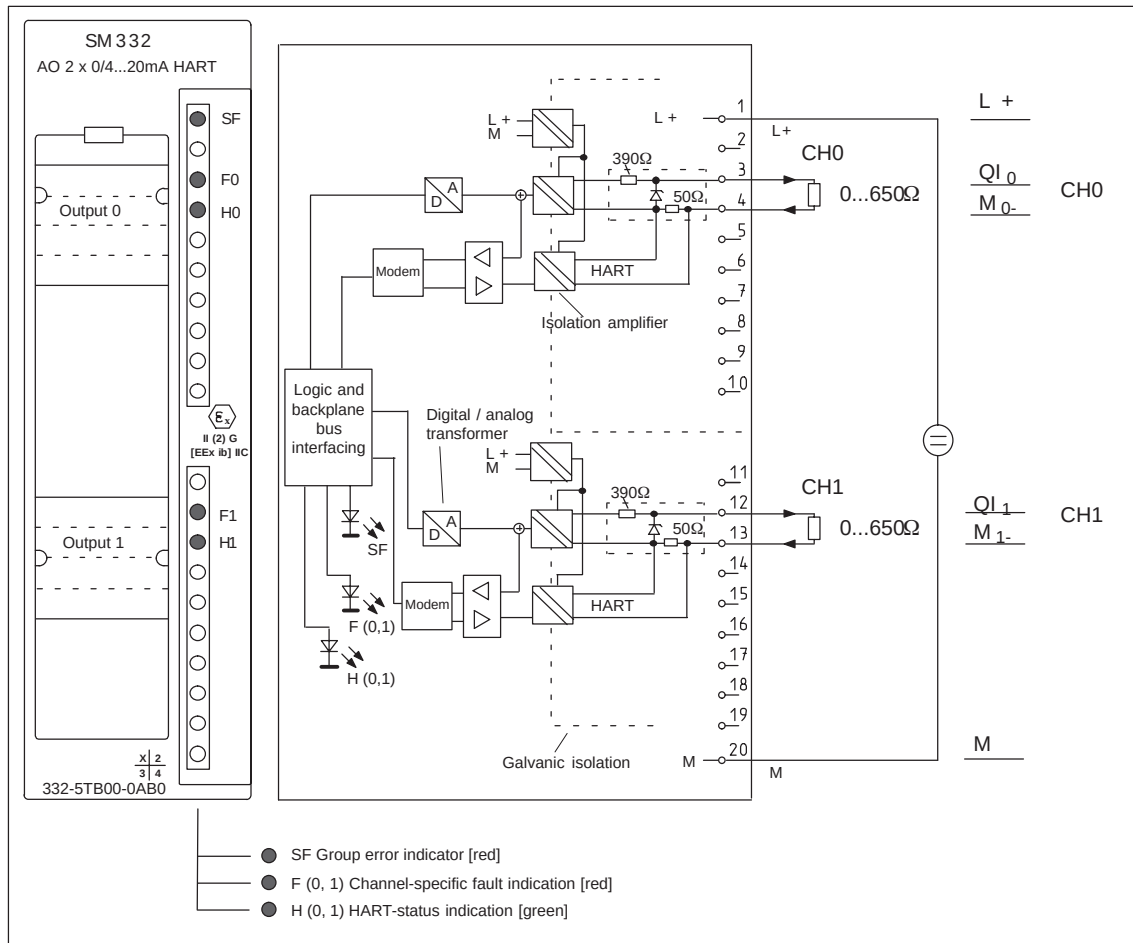


Fig. 4-9 Module view and block diagram of SM 332; AO 2 x 0/4...20mA HART

Notes on intrinsically-safe installation

Section 4.3 provides you with a summary of information on intrinsically-safe installation. Detailed information can be found in Section 1.5.

Power supply for an intrinsically-safe structure

In order to maintain the clearances and creepage distances, L+ / M must be routed via the line chamber LK393 when operating modules with signal cables that lead to the hazardous location, see Section 1.2.

Unswitched output channels

To ensure that the unswitched output channels of the analog output module SM 332; AO 2 x 0/4...20mA HART are without current or voltage, you must deactivate them. You can deactivate an output channel in *STEP 7* using the "Output" parameter block (see Section 4.4).

SM 332; AO 2 x 0/4...20mA HART		Voltages, currents, potentials continued	
Dimensions and weight		Permissible difference in potential (U_{ISO}) for signals from non-hazardous area	
Dimensions W x H x D (mm)	40 x 125 x 120	- Between channels and backplane bus 400 V DC 250 V AC - Between channels and load voltage L+ 400 V DC 250 V AC - Between channels 400 V DC 250 V AC - Between backplane bus and load voltage L+ 75 V DC 60 V AC	
Weight	approx. 280 g		
Module-specific data		Insulation tested	
Number of outputs	2	- Channels to backplane bus and load voltage L+ with 1500 V AC - Channels to each other with 1500 V AC - Between backplane bus and load voltage L+ with 500 V DC - Channels shielded with 500 V DC	
Line length, shielded	max. 400 m	Current input	
Type of protection KEMA (see Appendix A)	[EEx ib] IIC to EN 50020	- From backplane bus max. 100 mA - From load voltage L+ (at rated data) max. 150 mA	
Test number KEMA	98 ATEX2359 X	Module power loss typically 3.5 W	
Voltages, currents, potentials		Analog value formation	
Bus power supply	5 V DC	Output value	
Rated load voltage	24 V DC	Resolution (incl. overrange) 12 bit (+ polarity)	
Reverse voltage protection	yes	Readback value 8 bit	
Galvanic isolation		Cycle time (all channels) 5 ms	
- Between channels and backplane bus - Between channels - Between channels and load voltage L+ - Between backplane bus and load voltage L+	yes yes yes yes	Settling time	
Permissible difference in potential (U_{ISO}) for signals from a hazardous area		- for resistive load 2.5 ms - for inductive load 2.5 ms - for capacitive load 4 ms	
- Between channels and backplane bus - Between channels and load voltage L+ - Between channels - Between backplane bus and load voltage L+	60 V DC 30 V AC 60 V DC 30 V AC 60 V DC 30 V AC 60 V DC 30 V AC	Switch substitute values yes, configurable	
		Readback value	
		Resolution 8 bit (+ polarity)	
		Conversion time (per channel) 40 ms	

Interference suppression, error limits		Safety data (see Certificate of Conformity in Appendix A)	
Crosstalk attenuation between outputs	> 130 dB	Type of protection to EN 50020	[EEx ib] IIC
Operational limit (in total temperature range, referred to output range)	$\pm 0.55 \%$	Maximum values of the output circuits (per channel)	
Basic error limit (operational limit at 25°C, referred to output range)	$\pm 0.15 \%$	• U_0 (no-load output voltage)	max. 19 V
Temperature error (referred to output range)	$\pm 0.01 \%/K$	• I_0 (short-circuit current)	max. 66 mA
Linearity error (referred to output range)	$\pm 0.03 \%$	• P_0 (load power)	max. 506 mW
Repeatability in steady-state condition at 25°C, referred to output range)	$\pm 0.005 \%$	• L_0 (permissible external inductance)	max. 7.5 mH
Output ripple; range 0 to 50 kHz (referred to output range)	$\pm 0.02 \%$	• C_0 (permissible external capacitance)	max. 230 nF
Interrupts, diagnostics		• U_m (error voltage)	max. 60V DC
Interrupts		• T_a (permissible ambient temperature)	max. 60°C
• Diagnostic interrupt	configurable	Data for sensor selection	
Diagnostic functions	configurable	Output ranges (rated values)	
• Group fault indication	red LED (SF)	• Current	from 0 to 20 mA from 4 to 20 mA
• Channel fault indication	red LED (F) per channel	Load impedance (in rated range of output)	
• Diagnostic information readout	possible	• for current outputs	
Monitoring for		– resistive load	max. 650 Ω
• Wire break	yes	– inductive load	max. 7.5 mH ¹⁾
from output value	> 0.5 mA	– capacitive load	max. 230 nF ¹⁾
HART communication active and OK	green LED (H)	Current output	
		• No-load voltage	max. 19 V
		Destruction limit for externally applied voltages / currents	
		• Voltages	max. + 17 V / - 0.5V
		• Current	max. + 60 mA / - 1A
		Connection of actuators	
		• for current output	
		2-wire connection	yes

1) Limitation by KEMA approval

When used in a non-Ex area can be controlled as an:

- inductive load max. 15 mH
- capacitive load max. 3 μF *) can be set as the load impedance.

*) however, HART communication no longer possible

4.8 Data Record Interface and User Data

In this section...

In this section you will find the specific data which you need for parameter assignment, diagnostics and HART communication, when using standard STEP 7 applications or if you want to use your own software tool for HART communication.

The cyclic user data are described at the end of the section.

Overview of data record interface

The HART analog module uses data records as the input/output interface. The records are used for the following applications:

- Writing the parameters to the module
- Reading the diagnostic data of the module
- Transmitting the HART communication data
- Reading the additional diagnostic data for HART
- Writing the additional parameters for HART

With STEP 7

You can configure and assign parameters to the HART analog module using STEP 7. The online help will assist you with this.

Certain additional functions for writing parameters and reading diagnostic data can be integrated in your user program with SFCs. You can find detailed information about this in the reference manual /235/.

General information about data records and their structure can be found in the reference manual /71/. The manual /140/ contains information about operating the modules in a distributed configuration.

Overview of user data

The user data range of the HART analog module includes the following for both channel 0 and channel 1:

- Current as analog input value or analog output value
- Primary value in HART format (measured value or manipulated value)
- Identifiers for clients, to indicate that new data can be fetched.

Relative addresses are shown in the description of the user data. You can determine the module address to be added to the relative address using the STEP 7 application “Configuring and Assigning Parameters.”

4.8.1 Parameter Data Records

Structure of the parameter data records for the HART analog input modules

Figures 4-10 and 4-11 show data record 0 for the static parameters and data record 1 for the dynamic parameters for AI HART and AO HART. In the case of S5 and the norm master, all the parameters are transferred to data record 0.

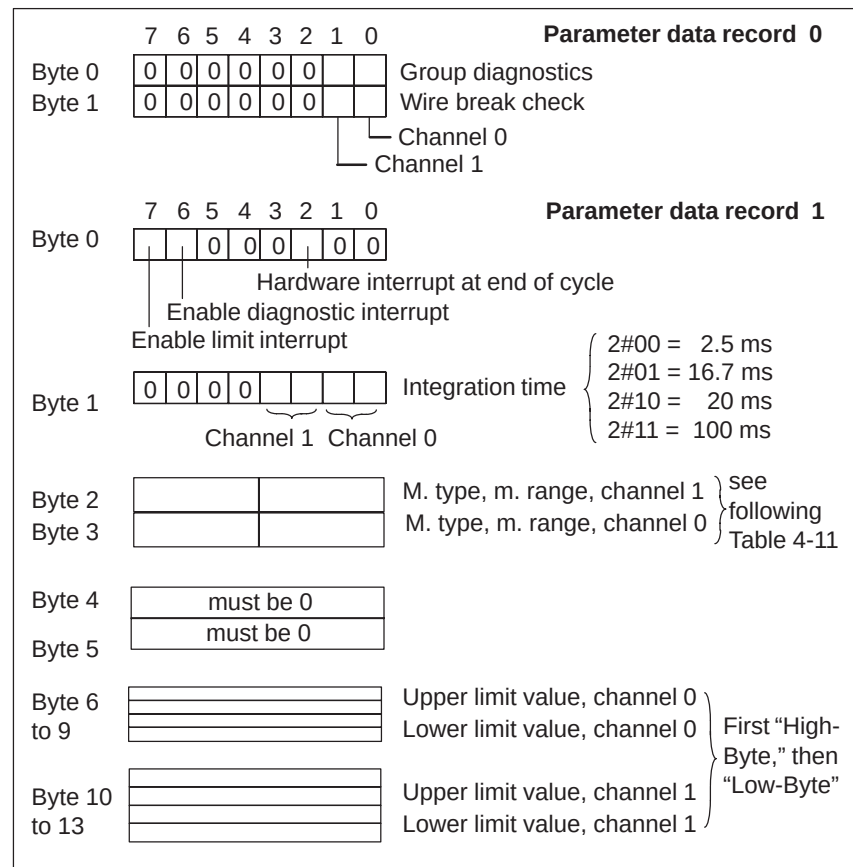


Fig. 4-10 Parameters of the HART analog input module

Table 4-11 Codes for the measurement type and measuring range for HART analog input module

Measurement type	Code	Measuring range	Code
Deactivated	2#0000	Deactivated	2#0000
4-wire transducer	2#0010	0 to 20 mA 4 to 20 mA	2#0010 2#0011
2-wire-transducer	2#0011	4 to 20 mA	2#0011
HART (2-wire or 4-wire transducer can be connected.)	2#0111	4 to 20 mA HART All commands permitted, and monodrop operation.	2#1100

Structure of the parameter data records for HART analog output modules

Figure 4-11 shows data record 0 for the static parameters and data record 1 for the dynamic parameters. In the case of S5 and the norm master, all the parameters are transferred to data record 0.

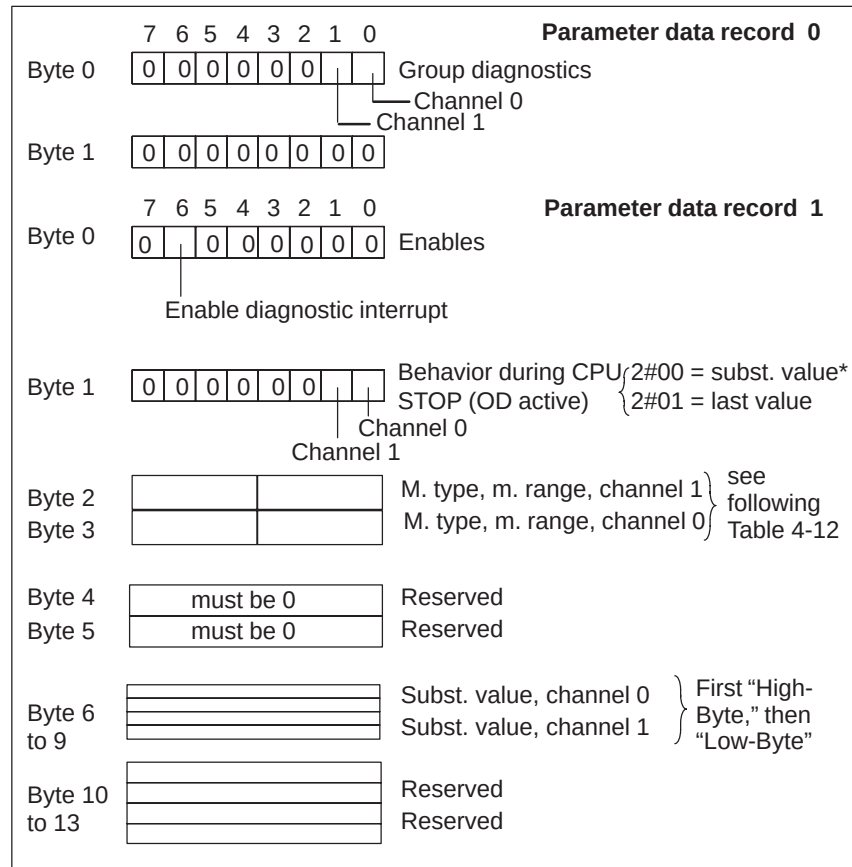


Fig. 4-11 Parameters of the HART analog output module

* For the substitute value -6912 (E500 Hex) the outputs will be disabled.

Table 4-12 Codes for the output type and output range for HART analog output modules

Output type	Code	Output range	Code
Deactivated	2#0000	Deactivated	2#0000
Current output without HART	2#0010	0 to 20 mA 4 to 20 mA	2#0010 2#0011
Current output with HART	2#0111	4 to 20 mA HART All commands permitted, and monodrop operation.	2#1100

4.8.2 Diagnostic Data Records

Structure and contents of the diagnostic data

The diagnostic data for a module can be up to 16 bytes long and consist of data records 0 and 1:

- Data record 0 contains system specific diagnostic data for the whole module: 4 bytes. It is set on a system-wide basis and applies for both HART analog input and output.
- Data record 1 contains
 - 4 bytes of diagnostic data for an S7-300 which are also in data record 0 **and**
 - Up to 12 bytes of module-class specific diagnostic data.

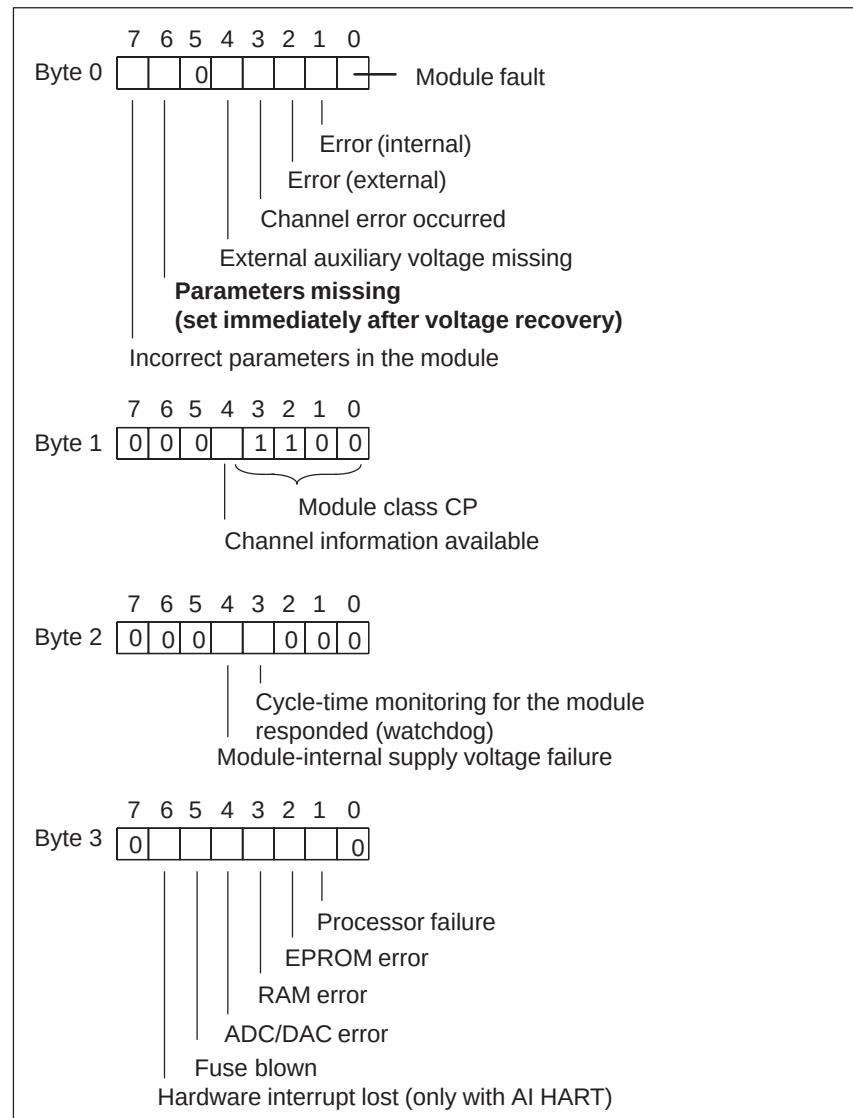


Fig. 4-12 Diagnostic data: data record 0

**Diagnostic data:
data record 1**

Figure 4-13 shows the contents of bytes 4 to 9 of the diagnostic data.

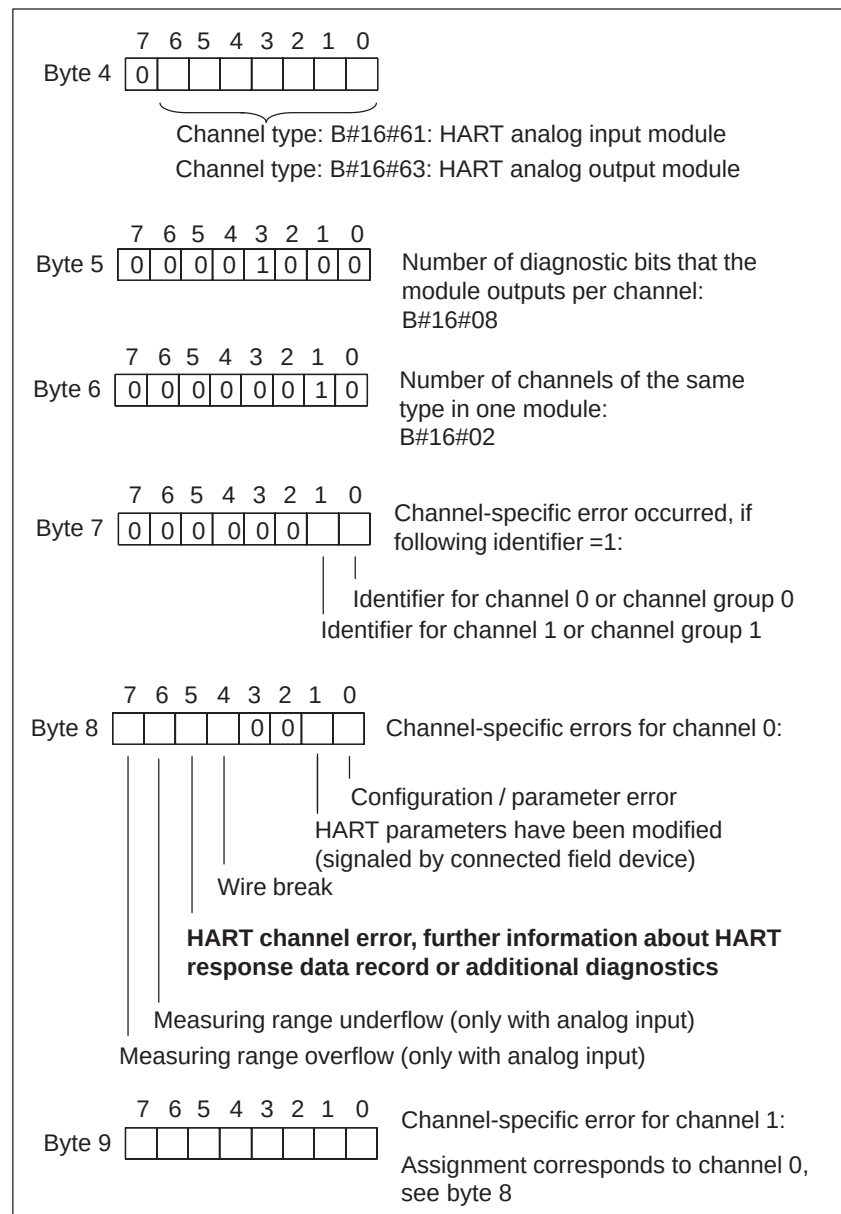


Fig. 4-13 Diagnostic data: data record 1

**Notes on the
diagnostic data**

Please note the following point:

- If a HART channel error occurs, you can obtain further information by using SFC59 to read the status in the HART response data record for the relevant client (see Section 4.8.3) or the additional diagnostic data record for the relevant channel (see Section 4.8.4).

4.8.3 HART Communication Data Records

Transfer data records

HART communication can be operated by up to 7 clients, using two separate channels each. There are 14 separate data transfer areas for this purpose, 7 for channel 0 and 7 for channel 1. Each transfer area consists of a command data record and a response data record.

Coordination rules for HART communication

- Each client / channel is allocated fixed data record numbers:

Channel	Client / Data record	1	2	3	4	5	6	7
0	Command	10	14	18	22	26	30	34
0	Response	12	16	20	24	28	32	36
1	Command	50	54	58	62	66	70	74
1	Response	52	56	60	64	68	72	76

- Each client may only use the data record numbers allocated to its transfer area.
- For example, for client 6, channel 0: the command is data record 30 and the response is data record 32.
- After a client has written a command data record, it must read the response data record before it can write another command data record.
- The transfer area of each client is allocated a data ready bit which is set when new data can be fetched (see Figure 4-20).
- In Master Class 2 the client can evaluate the “processing state” in the response data record: if the “processing state” indicates “successful” or “error,” the data record contains current response data or error bits respectively.
- The data record must always be read completely, as the the data record of the module can be changed after the first reading.
- The status section of the data record provides information on any errors that have occurred.
- The HART burst mode cannot be used by more than one client at any one time (that is, only one client can set this operating mode with a command).

Structure of command data record

The following figure shows the structure of data record +0, which you can use to write a command in the transfer area of a client. The HART analog module transmits the command to the connected HART field device.

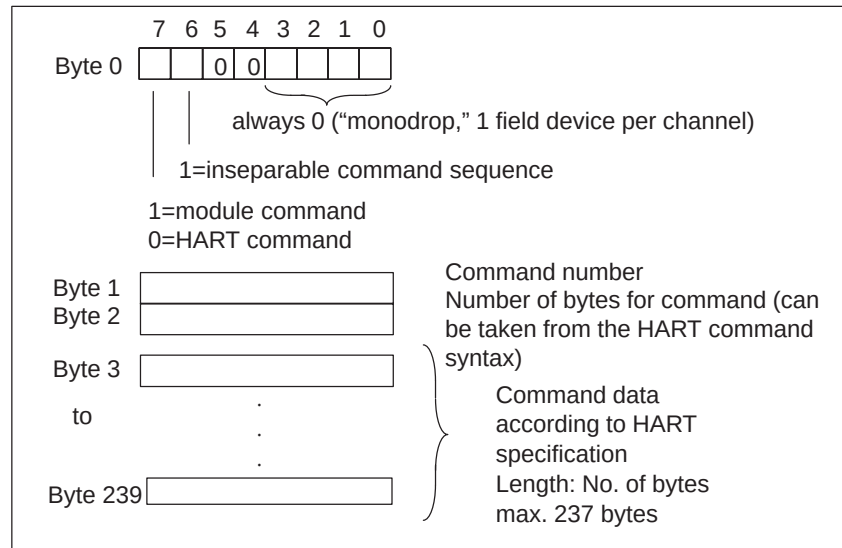


Fig. 4-14 Command data record of the HART analog module

Notes on command

The same client must not send a second command until the **response** to any previous command has been read. If you want to prevent commands from another client being processed in between, you must set the bit "inseparable command sequence" in your command:

- The inseparable command sequence is maintained as long as the bit "inseparable command sequence" is set.
- The inseparable command sequence is terminated if the bit "inseparable command sequence" is not set, or automatically after 10 seconds by the module.
- While an inseparable command sequence is set for one client, one command from each of the other clients can be stored temporarily in the buffer. The stored commands are processed once the inseparable command sequence has been terminated.

Notes on response

To read the response data record you must make sure that an up-to-date response data record has arrived:

- If the processing state in the response data record indicates "successful" or "error," the data record contains current response data or error messages respectively.
- Alternatively you can evaluate the "data ready" in the user data area: the transfer area of each client is allocated a bit in the user data area which is set when new data arrive (see Figure 4-20).

Structure of the response data record

The following figure shows the structure of the response data record, which contains the response to the HART command you sent previously and any error or status bits.

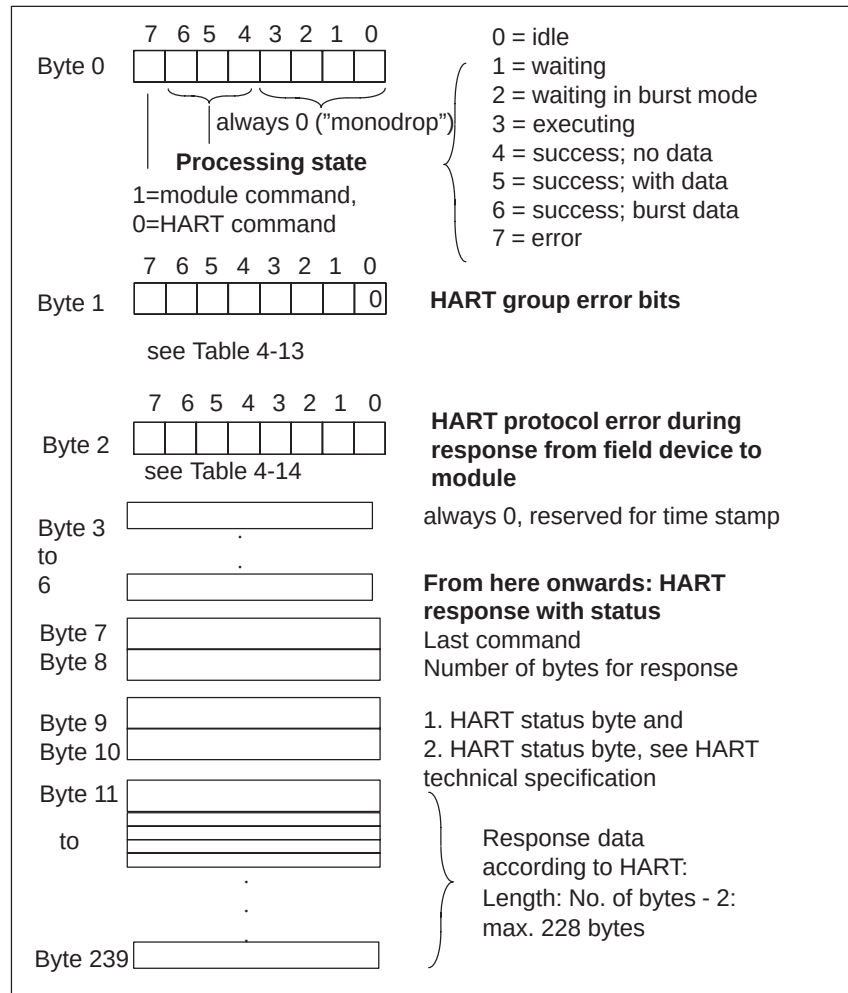


Fig. 4-15 Response data record of the HART analog module

Evaluating the response data

When you have an up-to-date response data record, you can check the following:

- You can use the "last command" byte to check that the response belongs to the command sent.
- You can evaluate the "Group error bits" (see Table 4-13) to locate individual errors.
- You can obtain more information from "HART protocol errors during response" (see Table 4-14) and both HART status bytes.
- that in the group error bytes the corresponding bits will be set to "1".

Table 4-13 HART group error displays

Bit No.	Group error display in Byte 1	Meaning
0	Always 0	Not used
1	Command rejected	Used in the following cases: For a module command which does not exist. If you try to activate the burst mode when it is already activated. If you try to deactivate the burst mode when it was activated by another client. If you try to change the polling address of the HART field device.
2	Further status information available.	Corresponds to bit 4 in the 2nd HART status byte. You can obtain further status information with HART command 48.
3	HART device status —> “Modification of parameters” entry in diagnostic data, data record 1	The field device transmits its device state. This information is found in the 2nd HART status byte which is accepted unchanged.
4	HART command status	The field device transmits displays on the receipt of the command. Information on this can be found in the 1st HART status byte .
5	Error during HART communication —> “HART group error” entry in diagnostic data, data record 1	The field device has detected a communication error while receiving the command. Information on the error can be found in the 1st HART status byte which is accepted unchanged.
6	HART protocol error during response —> “HART group error” entry in diagnostic data, data record 1	Error during HART communication between field device and module, i.e. the response was incorrectly received. Information on the cause of the error can be found in the next byte. See Table 4-14 .
7	Wire break —> Parallel entry “Wire break” in diagnostic data, data record 1	The connection to the measuring transducer or the signal control element has been broken.

Table 4-14 HART protocol error during response from field device to module

Bit No.	HART protocol error in byte 2	Meaning
0	Bad frame timing	Waiting time elapsed without response being received from field device.
1	Always 0	Not used
2	Bad character transmission timing	The pause between two bytes was not observed.
3	Checksum error in response	The checksum calculated does not match the checksum transmitted.
4	Response frame error	Error receiving HART signal (in UART)
5	Response overrun error	Error receiving HART signal (in UART)
6	Response parity error	Error receiving HART signal (in UART)
7	HART access not possible	The connection to the field device is constantly busy. This error is registered if the transmission time exceeds 10 seconds.

4.8.4 Additional Diagnostic Data Records

Additional diagnostic data

The additional diagnostic data provide information on the state of the HART communication following the last command.

- Additional diagnostic data record 128 for channel 0, 129 for channel 1
- Additional diagnostic data record 130 for channels 0 and 1: When the module is switched on, the recognized connected HART field devices and their identifiers (“tags”) are entered here.
- Additional diagnostic data records 131 for channel 0 and 151 for channel 1 with the data for the identifiers found in the additional diagnostic data record 130.

Structure of the diagnostic data records 128 and 129

The following figure shows the structure of the diagnostic data records 128 and 129.

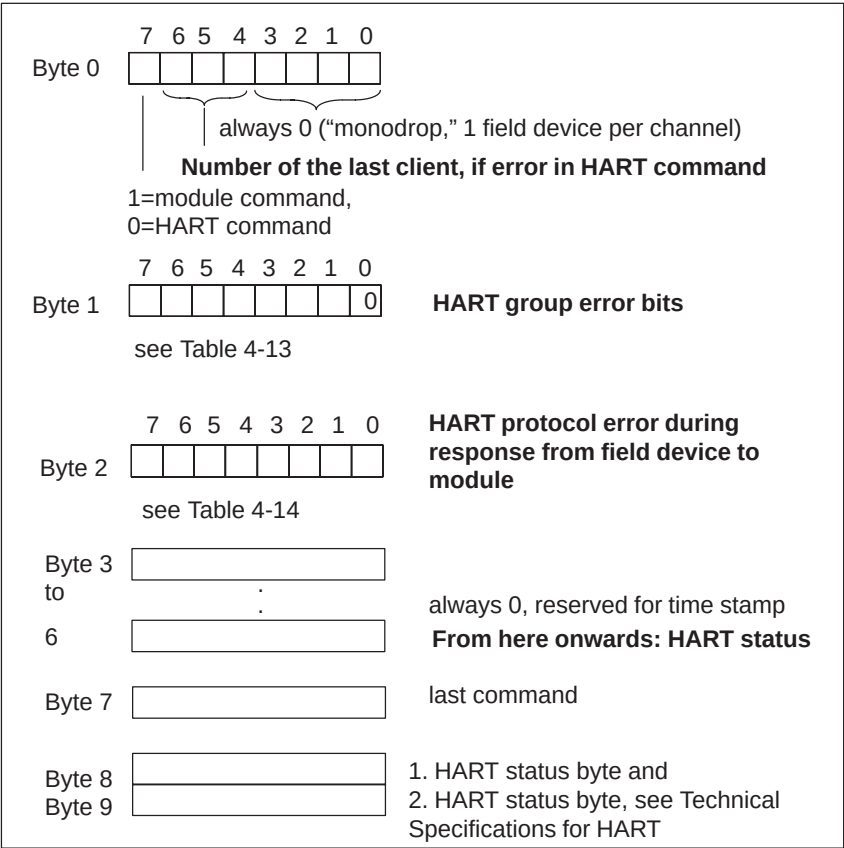


Fig. 4-16 Diagnostic data records 128 and 129 of the HART analog modules

Structure of the diagnostic data record 130

The following figure shows the structure of the diagnostic data record 130, which you can request to implement automatic recognition of the connected HART measuring transducer or the HART signal control elements.

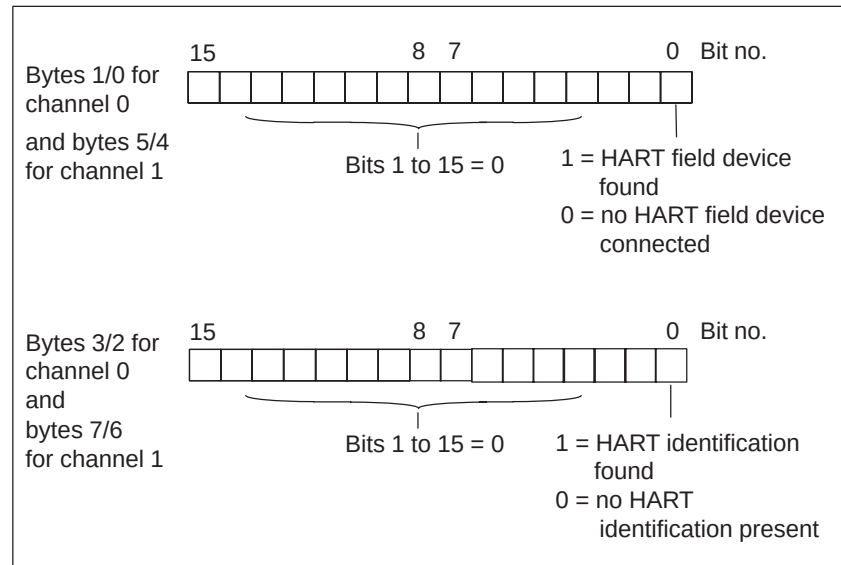


Fig. 4-17 Diagnostic data record 130 of the HART analog modules

Structure of the diagnostic data records 131 and 151

These contain the data corresponding to the identifiers marked in data record 130: the address of the HART field device which was found and the HART identification with tags or identifiers for a signal control element. The structure is illustrated in the following figure.

- Data record 131 for channel 0 (length: 38 bytes)
- Data record 151 for channel 1 (length: 38 bytes)

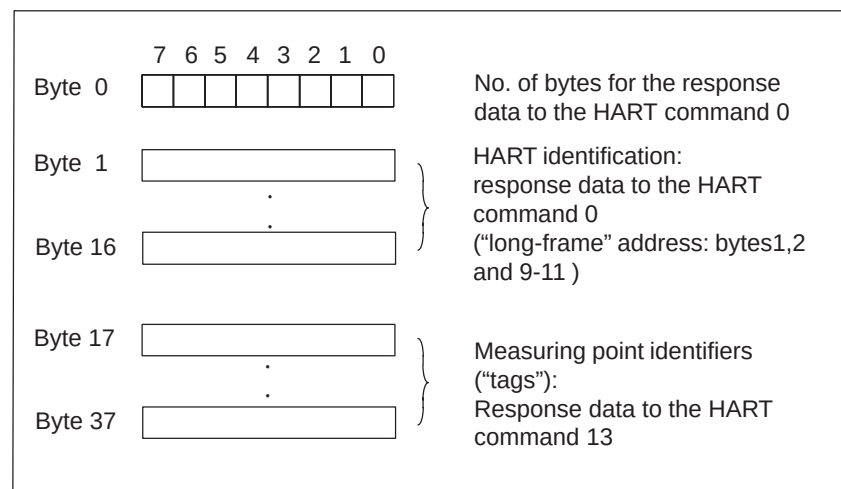


Fig. 4-18 Diagnostic data records 131 and 151 of the HART analog module

4.8.5 Additional Parameter Data Records

Structure of the parameter data records 128 and 129

The following figure shows the structure of the additional parameter data records 128 for channel 0 and 129 for channel 1. The settings affect the assigned channel.

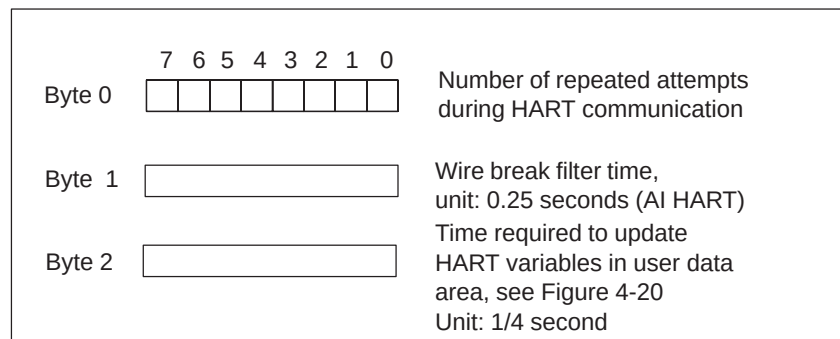


Fig. 4-19 Parameter data records 128 and 129 of the HART analog modules

Notes on the additional parameters

The additional parameters comprise parameters which you do not normally need to change, as they have already been set to a optimized value: the following table provides explanations of the parameters and the default values.

Table 4-15 Additional parameters of the HART analog module

Parameter	Explanation	Value range and default setting
Repeated attempts	If the HART analog modules transmit a command to the field device and the connection is busy, the set number of repeated attempts is started.	Value range: 0 to 255, Default setting: 3, No repeat attempts: 0
Wire break filter time ¹⁾	A wire break is only signaled if it occurs for longer than the set filter time.	Value range: 0 to 20, Default setting: 3 Δ 0.75 seconds, No filter time: 0
Update time	The HART modules send the HART command 1 automatically, to read the present value of the primary variable.	Value range: 0 to 255, Default setting: 12 Δ 3 seconds, No waiting time: 0

- 1) As some measuring transducers take longer than others to start up, you may find that several diagnostic interrupts are triggered during startup. The wire break filter time was introduced to avoid this problem.

Default parameter assignment for DP master class 2

When the HART analog modules have no parameters, for example, after a power failure, they can obtain default parameters from PROFIBUS-DP master class 2 while the programmable logic controller is deactivated. This is done with the aid of parameter data record No. 250 which consists of one byte with the value unequal 0. However, the assignment of default parameters can only be initiated when the module is in an unparameterized state. You can determine the state of the module by reading the diagnostic data record: see Figure 4-12.

4.8.6 User Data Interface Input Area (Read)

Structure of the user data

The following figure shows the structure of the user data area for the HART analog input module. The data for the user data area can be read in the desired format using “Read peripheral data” (for example, L PIW 256) and evaluated in your user program.

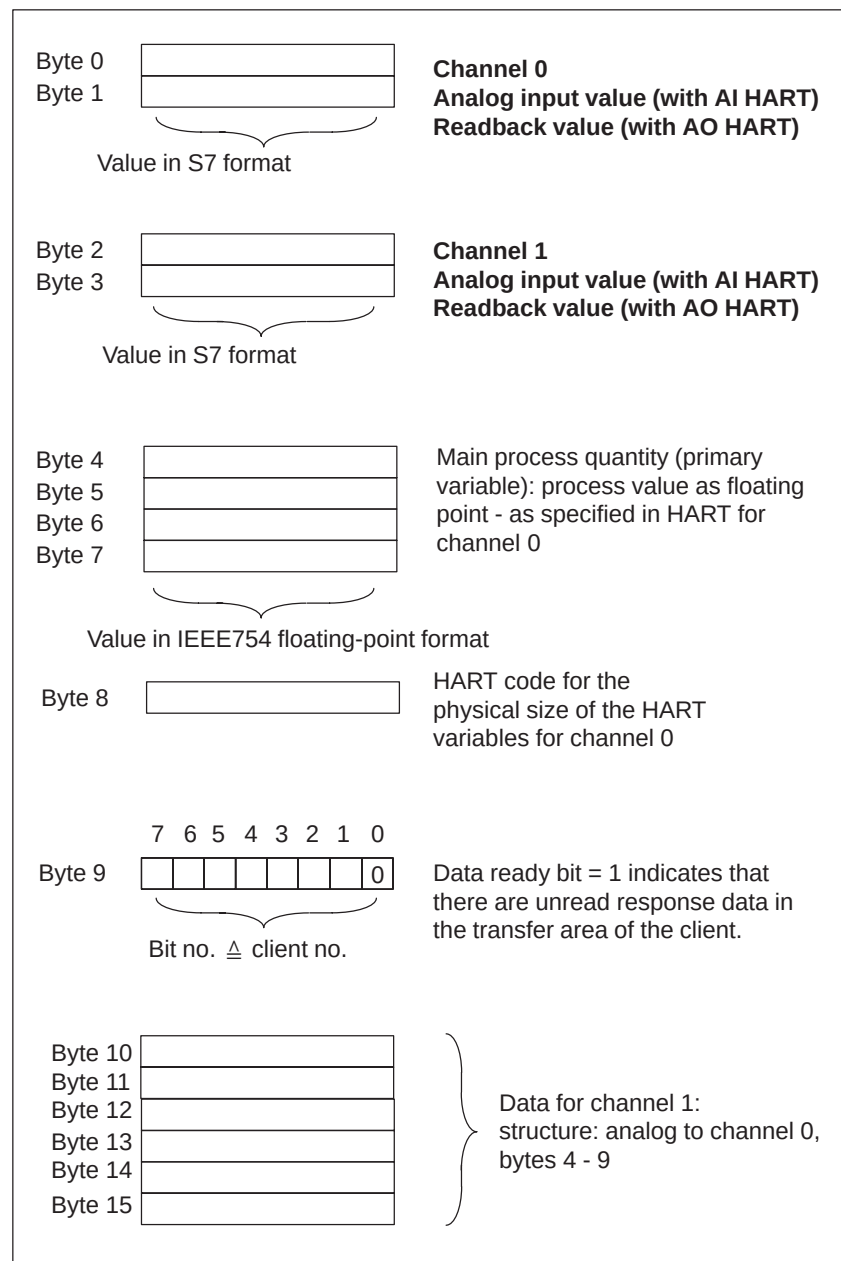


Fig. 4-20 Input user data area of the HART analog modules

4.8.7 Output Area (Write)

Structure of the user data

The following figure shows the structure of the user data area for the HART analog output module. The data for the user data area can be read in the desired format using “Write peripheral data” (for example, L PIW 256) and evaluated in your user program.

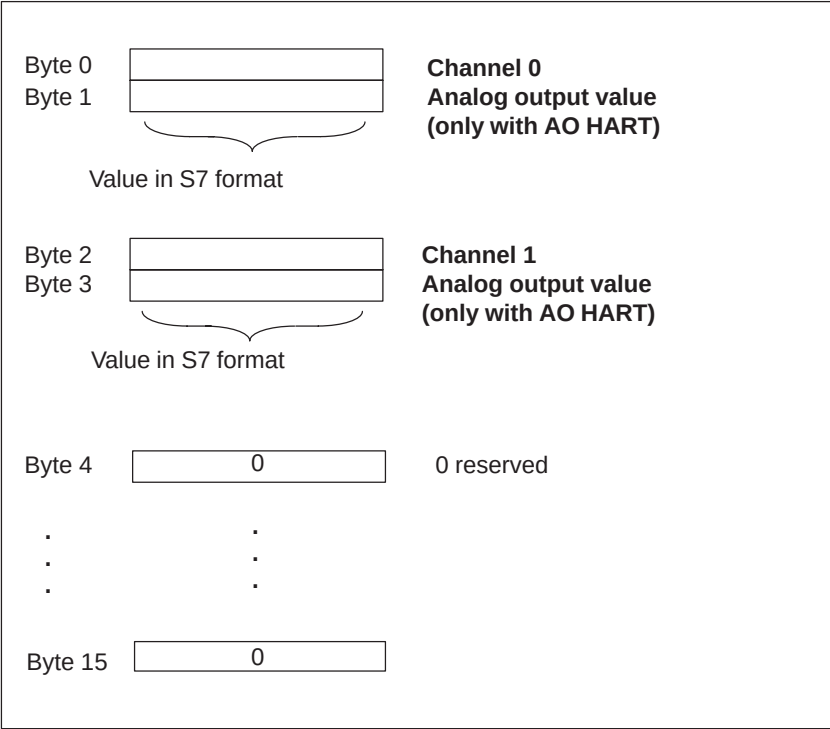


Fig. 4-21 User data area of the HART analog output module

Certificates of Conformity

A

In this appendix On the following pages you will find copies of the certificates of conformity.

Section	Module	Order Number	You Will Find	Page
A.1	SM 321; DI 4xNAMUR	6ES7 321-7RD00-0AB0	PTB Certificate of Conformity	A-3
A.1.1	SM 321; DI 4xNAMUR	6ES7 321-7RD00-0AB0	ASEV Certificate / Switzerland	A-5
A.2	SM 322; DO 4x24 V/10 mA	6ES7 322-5SD00-0AB0	PTB Certificate of Conformity	A-9
A.2.1	SM 322; DO 4x24 V/10 mA	6ES7 322-5SD00-0AB0	ASEV Certificate / Switzerland	A-11
A.3	SM 322; DO 4x15 V/20 mA	6ES7 322-5RD00-0AB0	PTB Certificate of Conformity	A-15
A.3.1	SM 322; DO 4x15 V/20 mA	6ES7 322-5RD00-0AB0	ASEV Certificate / Switzerland	A-17
A.4	SM331; AI 8xTC/4xRTD	6ES7 331-7SF00-0AB0	PTB Certificate of Conformity	A-21
A.4.1	SM331; AI 8xTC/4xRTD	6ES7 331-7SF00-0AB0	ASEV Certificate / Switzerland	A-24
A.5	SM331; AI 4x0/4...20 mA	6ES7 331-7RD00-0AB0	PTB Certificate of Conformity	A-28
A.5.1	SM331; AI 4x0/4...20 mA	6ES7 331-7RD00-0AB0	ASEV Certificate / Switzerland	A-30
A.6	SM332; AO 4x0/4...20 mA	6ES7 332-5RD00-0AB0	PTB Certificate of Conformity	A-34
A.6.1	SM332; AO 4x0/4...20 mA	6ES7 332-5RD00-0AB0	First Supplement	A-36
A.6.2	SM332; AO 4x0/4...20 mA	6ES7 332-5RD00-0AB0	ASEV Certificate / Switzerland	A-37

Section	Module	Order Number	You Will Find	Page
A.7	SM331; AI 2 x 0/4...20mA HART	6ES7 331-7TB00-0AB0	KEMA Certificate of Conformity	A-41
A.7.1	SM331; AI 2 x 0/4...20mA HART	6ES7 331-7TB00-0AB0	First Supplement	A-44
A.7.2	SM331; AI 2 x 0/4...20mA HART	6ES7 331-7TB00-0AB0	EC Declaration of Conformity	A-45
A.8	SM332; AO 2 x 0/4...20mA HART	6ES7 332-5TB00-0AB0	KEMA Certificate of Conformity	A-46
A.8.1	SM332; AO 2 x 0/4...20mA HART	6ES7 332-5TB00-0AB0	EG-Declaration of Conformity	A-49

A.1 Certificate of Conformity for Digital Input Module DI 4 x NAMUR

Physikalisch-Technische Bundesanstalt
Braunschweig und Berlin



(1) **KONFORMITÄTSBESCHEINIGUNG**

(2) **PTB Nr. Ex-96.D.2094 X**

(3) Diese Bescheinigung gilt für das elektrische Betriebsmittel
Digitaleingabebaugruppe SM 321, DI 4xNAMUR Typ 6ES7 321-7RD00-0AB0

(4) der Firma Siemens AG
D-76181 Karlsruhe

(5) Die Bauart dieses elektrischen Betriebsmittels sowie die verschiedenen zulässigen Ausführungen sind in der Anlage zu dieser Konformitätsbescheinigung festgelegt.

(6) Die Physikalisch-Technische Bundesanstalt bescheinigt als Prüfstelle nach Artikel 14 der Richtlinie des Rates der Europäischen Gemeinschaften vom 18. Dezember 1975 (76/117/EWG) die Übereinstimmung dieses elektrischen Betriebsmittels mit den harmonisierten Europäischen Normen

Elektrische Betriebsmittel für explosionsgefährdete Bereiche

EN 50 014:1977 + A1...A5 (VDE 0170/0171 Teil 1/1.87) Allgemeine Bestimmungen
EN 50 020:1977 + A1...A5 (VDE 0170/0171 Teil 7/4.92) Eigensicherheit "i"

nachdem das Betriebsmittel mit Erfolg einer Bauartprüfung unterzogen wurde. Die Ergebnisse dieser Bauartprüfung sind in einem vertraulichen Prüfprotokoll festgelegt.

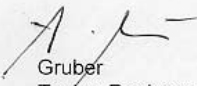
(7) Das Betriebsmittel ist mit folgender Kennzeichnung zu versehen:

[Ex ib] IIC

(8) Der Hersteller ist dafür verantwortlich, daß jedes derart gekennzeichnete Betriebsmittel in seiner Bauart mit den in der Anlage zu dieser Bescheinigung aufgeführten Prüfungsunterlagen übereinstimmt und daß die vorgeschriebenen Stückprüfungen erfolgreich durchgeführt wurden.

(9) Das elektrische Betriebsmittel darf mit dem hier abgedruckten gemeinschaftlichen Unterscheidungszeichen gemäß Anhang II der Richtlinie des Rates vom 6. Februar 1979 (79/196/EWG) gekennzeichnet werden.

Im Auftrag



Gruber
Techn. Regierungsoberamtsrat

Braunschweig, 15.07.1996



Prüfbescheinigungen ohne Unterschrift und ohne Dienstsiegel haben keine Gültigkeit.
Die Bescheinigungen dürfen nur unverändert weiterverbreitet werden.
Auszüge oder Änderungen bedürfen der Genehmigung der Physikalisch-Technischen Bundesanstalt.

Physikalisch-Technische Bundesanstalt

ANLAGE

zur Konformitätsbescheinigung PTB Nr. Ex-96.D.2094 X

Die Digitaleingabebaugruppe SM 321, DI 4xNAMUR Typ 6ES7 321-7RD00-0AB0 dient zum Anschluß von bescheinigten NAMUR Gebern im explosionsgefährdeten Bereich.

Die höchstzulässige Umgebungstemperatur beträgt 60°C.

Elektrische Daten

Versorgungs- und Busstromkreise 24 VDC (20,4 bis 28,8 V); ca. 40 mA und
(Anschluß 1 und 20 des 20-poligen Frontsteckers und die Anschlüsse des SIMATIC-Busses auf der Rückseite) 5 VDC (5,1 bis 5,3 V); ca. 60 mA
 $U_m = 60$ VDC bzw. 30 VAC

Eingangsstromkreise in Zündschutzart Eigensicherheit EEx ib IIC
(Klemmen 3 und 4 oder 5; mit folgenden Höchstwerten je Stromkreis:
7 und 8 oder 9; $U_o = 10$ V
12 und 13 oder 14; $I_o = 14,1$ mA
16 und 17 oder 18 $P_o = 33,7$ mW
des 20-poligen Frontsteckers)

Kennlinie: linear
höchstzul. äußere Induktivität 100 mH
höchstzul. äußere Kapazität 3 µF

Die Eingangsstromkreise sind untereinander und von den Versorgungs- und Busstromkreisen bis zu einem Scheitelwert der Nennspannung von 60 V sicher galvanisch getrennt.

Prüfungsunterlagen

unterschrieben am 08.05.1996

1. Beschreibung mit Blockschaltbild (10 Blatt)
2. Zeichnung Nr 3 NEP 720 7044 01 (8 Blatt)
3 NEG 720 7049 01a (6 Blatt)

Besondere Bedingungen

1. Die Digitaleingabebaugruppe SM 321, DI 4xNAMUR Typ 6ES7 321-7RD00-0AB0 muß außerhalb des explosionsgefährdeten Bereiches errichtet werden.
2. Die Baugruppe muß so errichtet werden, daß mindestens die Schutzart IP 20 gemäß IEC-Publikation 529 eingehalten wird.
3. Bei der Errichtung ist die Leitungskammer einzusetzen oder zwischen den eigensicheren und den nicht eigensicheren Stromkreisen ist durch Einfügen einer Trennwand ein Mindestabstand von 50 mm (Fadenmaß) zu realisieren bzw. die Anschlußteile sind zusätzlich zu isolieren (z.B. Schrumpfschlauch).

Im Auftrag

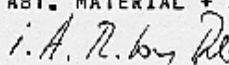
Braunschweig, 15.07.1996


Gruber
Techn. Regierungsoberamts



Blatt 1/1

**A.1.1 ASEV Certificate/Switzerland for Digital Input Module
DI 4 x NAMUR**

 Eidgenössisches Starkstrominspektorat Inspection fédérale des installations à courant fort Ispettorato federale degli impianti a corrente forte			
ZULASSUNG	ADMISSION	AMMISSIONE	
APPROVAL			
NR.	97.1 10147.07	SIEMENS SCHWEIZ AG	
GÜLTIG BIS:	26/03/2002	FREILAGERSTR. 28 8047 ZÜRICH	
KENNZEICHNUNG		787.676	
EX-KENNZEICHNUNG GEMÄSS BEILAGE			
AUFGRUND DER ANGABEN EINES PRÜFBERICHTES NR. 97.1 10147.01 VOM 23/03/1997 ERTEILT DAS EIDGENÖSSISCHE STARKSTROMINSPEKTORAT DER OBEN GENANNTEN FIRMA DAS RECHT, DAS NACHSTEHEND AUFGEFÜHRTE ERZEUGNIS IN DER SCHWEIZ IN VERKEHR ZU BRINGEN.			
ERZEUGNIS:	DIGITALEINGABEBaugruppe		
MATERIAL-CODE:	1640		
HANDELSMARKE:	SIMATIC		
TYPENBEZEICHNUNG	NENN DATEN		
6ES7 321-7BD00-0AB0	GEMÄSS BEILAGE		
BEMERKUNGEN:	BESCHREIBUNG, HINWEISE, BESONDERE BEDINGUNGEN, EX- KLASSIFIKATION UND PRÜFUNGSUNTERLAGEN GEMÄSS BEILAGE VOM 24.03.1997 ZU ZERTIFIKAT A-NR. 97.1 10147.01		
NORM:	EN 50014:1977+A1...A5, EN 50020:1977+A1...A5		
GEBÜHR:	FR.	360.00	EIDGENÖSSISCHES STARKSTROMINSPEKTORAT
FEHRALTORF,	26/03/1997	MH	ABT. MATERIAL + APPARATE
ML		 J. BRUHIN	

Schweizerischer Elektrotechnischer Verein
Association Suisse des Electriciens
Associazione Svizzera degli Elettrotecnici
Swiss Electrotechnical Association



Beilage zu Zertifikat A. Nr. 97.1 10147.01

Blatt 1/3

Siemens Schweiz AG, 8047 Zürich

Beschreibung

Die Digitaleingabebaugruppe SIMATIC Typ 6ES7 321-7RDOO-OABO dient zum Anschluss von bescheinigten NAMUR Gebern im explosionsgefährdeten Bereich.

Nenndaten

Versorgungs- und Busstromkreise (Anschluss 1 und 20 des 20-poligen Frontsteckers und die Anschlüsse des SIMATIC-Busses auf der Rückseite)	24 VDC (20,4 bis 28,8 V); ca. 40 mA und 5 VDC (5,1 bis 5,3 V); ca. 60 mA $U_m = 60$ VDC bzw. 30 VAC
Eingangsstromkreise (Klemmen 3 und 4 oder 5; 7 und 8 oder 9; 12 und 13 oder 14; 16 und 17 oder 18 des 20-poligen Frontsteckers)	in Zündschutzart Eigensicherheit EEx ib IIC mit folgenden Höchstwerten je Stromkreis: $U_o = 10$ V $I_o = 14,1$ mA $P_o = 33,7$ mW Kennlinie: linear höchstzul. äussere Induktivität 100 mH höchstzul. äussere Kapazität 3 μ F

Hinweise

1. Die Eingangsstromkreise sind untereinander und von den Versorgungs- und Busstromkreisen bis zu einem Scheitelwert der Nennspannung von 60 V sicher galvanisch getrennt.
2. Die höchstzulässige Umgebungstemperatur beträgt 60 °C.

Schweizerischer Elektrotechnischer Verein
Association Suisse des Electriciens
Associazione Svizzera degli Elettrotecnici
Swiss Electrotechnical Association



Beilage zu Zertifikat A. Nr. 97.1 10147.01

Blatt 2/3

Siemens Schweiz AG, 8047 Zürich

Besondere Bedingungen X

1. Die Digitaleingabebaugruppe SIMATIC Typ 6ES7 321-7RDOO-OABO muss ausserhalb des explosionsgefährdeten Bereiches installiert werden.
2. Die Baugruppe muss so installiert werden, dass mindestens die Schutzart IP 20 gemäss EN 60529 eingehalten wird.
3. Bei der Installation ist die Leitungskammer einzusetzen oder zwischen den eigensicheren und den nichteigensicheren Stromkreisen ist durch Einfügen einer Trennwand ein Mindestabstand von 50 mm (Fadenmass) zu realisieren bzw. die Anschlussteile sind zusätzlich zu isolieren (z.B. Schrumpfschlauch).

Ex-Klassifikation

[EEx ib] IIC

Kennzeichnung

Die Kennzeichnung muss gut sichtbar, lesbar und dauerhaft sein; sie muss folgende Angaben umfassen:

1. Name des Herstellers oder sein Warenzeichen
Typenbezeichnung
Ex-Klassifikation
Fertig.-Nr. oder Losbezeichnung
2. Zusätzliche Angaben, die in den besonderen Europäischen Normen für die betreffenden Zündschutzarten vorgeschrieben sind.
Angaben, die normalerweise durch Konstruktionsnormen für elektrische Betriebsmittel verlangt werden.

Schweizerischer Elektrotechnischer Verein
Association Suisse des Electriciens
Associazione Svizzera degli Elettrotecnici
Swiss Electrotechnical Association

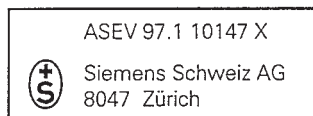


Beilage zu Zertifikat A. Nr. 97.1 10147.01

Blatt 3/3

Siemens Schweiz AG, 8047 Zürich

3. Zusätzliche Bezeichnung



Prüfungsunterlagen

Konformitätsbescheinigung PTB Nr. Ex-96.D.2094 X mit allen darin aufgeführten Prüfungsunterlagen.

Fehraltorf, 24. März 1997
EXL/ P. Bosshard
Visum:

A handwritten signature in black ink, appearing to be 'P. Bosshard'.

A.2 Certificate of Conformity for Digital Output Module DO 4 x 24 V/10 mA

Physikalisch-Technische Bundesanstalt
Braunschweig und Berlin



(1) **KONFORMITÄTSBESCHEINIGUNG**

(2) **PTB Nr. Ex-96.D.2093 X**

(3) Diese Bescheinigung gilt für das elektrische Betriebsmittel
Digitalausgabebaugruppe SM 322, DO 4x24 V/10 mA Typ 6ES7 322-5SD00-0AB0

(4) der Firma Siemens AG
D-76181 Karlsruhe

(5) Die Bauart dieses elektrischen Betriebsmittels sowie die verschiedenen zulässigen Ausführungen sind in der Anlage zu dieser Konformitätsbescheinigung festgelegt.

(6) Die Physikalisch-Technische Bundesanstalt bescheinigt als Prüfstelle nach Artikel 14 der Richtlinie des Rates der Europäischen Gemeinschaften vom 18. Dezember 1975 (76/117/EWG) die Übereinstimmung dieses elektrischen Betriebsmittels mit den harmonisierten Europäischen Normen

Elektrische Betriebsmittel für explosionsgefährdete Bereiche

EN 50 014:1977 + A1...A5 (VDE 0170/0171 Teil 1/1.87) Allgemeine Bestimmungen
EN 50 020:1977 + A1...A5 (VDE 0170/0171 Teil 7/4.92) Eigensicherheit "I"

nachdem das Betriebsmittel mit Erfolg einer Bauartprüfung unterzogen wurde. Die Ergebnisse dieser Bauartprüfung sind in einem vertraulichen Prüfprotokoll festgelegt.

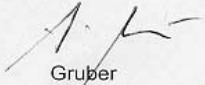
(7) Das Betriebsmittel ist mit folgender Kennzeichnung zu versehen:

[Ex ib] IIC

(8) Der Hersteller ist dafür verantwortlich, daß jedes derart gekennzeichnete Betriebsmittel in seiner Bauart mit den in der Anlage zu dieser Bescheinigung aufgeführten Prüfungsunterlagen übereinstimmt und daß die vorgeschriebenen Stückprüfungen erfolgreich durchgeführt wurden.

(9) Das elektrische Betriebsmittel darf mit dem hier abgedruckten gemeinschaftlichen Unterscheidungszeichen gemäß Anhang II der Richtlinie des Rates vom 6. Februar 1979 (79/196/EWG) gekennzeichnet werden.

Im Auftrag Braunschweig, 15.07.1996




Gruber
Techn. Regierungsoberamtsrat

Prüfbescheinigungen ohne Unterschrift und ohne Dienstsiegel haben keine Gültigkeit.
 Die Bescheinigungen dürfen nur unverändert weiterverbreitet werden.
 Auszüge oder Änderungen bedürfen der Genehmigung der Physikalisch-Technischen Bundesanstalt.

Physikalisch-Technische Bundesanstalt

ANLAGE

zur Konformitätsbescheinigung PTB Nr. Ex-96.D.2093 X

Die Digitalausgabebaugruppe SM 322, DO 4x24 V/10 mA Typ 6ES7 322-5SD00-0AB0 dient zur Ansteuerung von Magnetventilen im explosionsgefährdeten Bereich.

Die höchstzulässige Umgebungstemperatur beträgt 60 °C.

Elektrische Daten

Versorgungs- und Busstromkreise 24 VDC (20,4 bis 28,8 V); ca. 120 mA und
(Anschluß 1 und 20 des 20-poligen Frontsteckers und die Anschlüsse des SIMATIC-Busses auf der Rückseite) 5 VDC (5,1 bis 5,3 V); ca. 60 mA
 $U_m = 60$ VDC bzw. 30 VAC

Ausgangsstromkreise in Zündschutzart Eigensicherheit EEx ib IIC
(Klemmen 4, 5; 8, 9; 13, 14; 17, 18 des 20-poligen Frontsteckers) mit folgenden Höchstwerten je Stromkreis:
 $U_o = 25,2$ V
 $I_o = 70$ mA
 $P_o = 440$ mW

Kennlinie: linear
höchstzul. äußere Induktivität 6,7 mH
höchstzul. äußere Kapazität 90 nF

Die Ausgangsstromkreise sind untereinander und von den Versorgungs- und Busstromkreisen bis zu einem Scheitelwert der Nennspannung von 60 V sicher galvanisch getrennt.

Prüfungsunterlagen

unterschrieben am 26.04.1996

1. Beschreibung mit Blockschaltbild (10 Blatt)
2. Zeichnung Nr 3 NEP 720 7084 01 (8 Blatt)
3 NEG 720 7089 01 (9 Blatt)

Besondere Bedingungen

1. Die Digitalausgabebaugruppe SM 322, DO 4x24 V/10 mA Typ 6ES7 322-5SD00-0AB0 muß außerhalb des explosionsgefährdeten Bereiches errichtet werden.
2. Die Baugruppe muß so errichtet werden, daß mindestens die Schutzart IP 20 gemäß IEC-Publikation 529 eingehalten wird.
3. Bei der Errichtung ist die Leitungskammer einzusetzen oder zwischen den eigensicheren und den nichteigensicheren Stromkreisen ist durch Einfügen einer Trennwand ein Mindestabstand von 50 mm (Fadenmaß) zu realisieren bzw. die Anschlußteile sind zusätzlich zu isolieren (z.B. Schrumpfschlauch).

Im Auftrag

Braunschweig, 15.07.1996


Gruber
Techn. Regierungsoberamtsrat



Blatt 1/1

**A.2.1 ASEV Certificate/Switzerland for Digital Output Module
DO 4 x 24 V/10 mA**

 Eidgenössisches Starkstrominspektorat Inspection fédérale des installations à courant fort Ispettorato federale degli impianti a corrente forte		
ZULASSUNG	ADMISSION APPROVAL	AMMISSIONE
NR. 97.1 10147.09 GÜLTIG BIS: 26/03/2002		SIEMENS SCHWEIZ AG FREILAGERSTR. 28 8047 ZÜRICH
KENNZEICHNUNG 	787,676	
EX-KENNZEICHNUNG GEMÄSS BEILAGE		
AUFGRUND DER ANGABEN EINES PRÜFBERICHTES NR. 97.1 10147.03 VOM 25/03/1997 ERTEILT DAS EIDGENÖSSISCHE STARKSTROMINSPEKTORAT DER OBEN GENANNTE FİRMA DAS RECHT, DAS NACHSTEHEND AUFGEFÜHRTE ERZEUGNIS IN DER SCHWEIZ IN VERKEHR ZU BRINGEN.		
ERZEUGNIS:	DIGITALAUSGABEBAUGRUPPE	
MATERIAL-CODE:	1640	
HANDELSMARKE:	SIMATIC	
TYPENBEZEICHNUNG	NENNDATEN	
6ES7 322-5SD00-0AB0	GEMÄSS BEILAGE	
BEMERKUNGEN:	BESCHREIBUNG, HINWEISE, BESONDERE BEDINGUNGEN, EX- KLASSIFIKATION UND PRÜFUNGSUNTERLAGEN GEMÄSS BEILAGE VOM 24.03.1997 ZU ZERTIFIKAT A.NR. 97.1 10147.03	
NORM:	EN 50014:1977+A1...A5, EN 50020:1977+A1...A5	
GEBÜHR:	FR. 360.00	EIDGENÖSSISCHES STARKSTROMINSPEKTORAT
FEHRALTORF,	26/03/1997 MH	ABT. MATERIAL + APPARATE
Mh		<i>J. A. D. Bruhin</i> J. BRUHIN

Schweizerischer Elektrotechnischer Verein
Association Suisse des Electriciens
Associazione Svizzera degli Elettrotecnici
Swiss Electrotechnical Association



Beilage zu Zertifikat A. Nr. 97.1 10147.03

Blatt 1/3

Siemens Schweiz AG, 8047 Zürich

Beschreibung

Die Digitalausgabebaugruppe SIMATIC Typ 6ES7 322-5SD00-OABO dient zur Ansteuerung von Magnetventilen im explosionsgefährdeten Bereich.

Nenndaten

Versorgungs- und Busstromkreise (Anschluss 1 und 20 des 20-poligen Frontsteckers und die Anschlüsse des SIMATIC-Busses auf der Rückseite)	24 VDC (20,4 bis 28,8 V); ca. 120 mA und 5 VDC (5,1 bis 5,3 V); ca. 60 mA $U_m = 60$ VDC bzw. 30 VAC
Ausgangsstromkreise (Klemmen 4, 5; 8, 9; 13, 14; 17, 18 des 20-poligen Frontsteckers)	in Zündschutzart Eigensicherheit EEx ib IIC mit folgenden Höchstwerten je Stromkreis: $U_o = 25,2$ V $I_o = 70$ mA $P_o = 440$ mW Kennlinie: linear höchstzul. äussere Induktivität 6,7 mH höchstzul. äussere Kapazität 90 nF

Hinweise

1. Die Eingangsstromkreise sind untereinander und von den Versorgungs- und Busstromkreisen bis zu einem Scheitelwert der Nennspannung von 60 V sicher galvanisch getrennt.
2. Die höchstzulässige Umgebungstemperatur beträgt 60 °C.

Schweizerischer Elektrotechnischer Verein
Association Suisse des Electriciens
Associazione Svizzera degli Elettrotecnici
Swiss Electrotechnical Association



Beilage zu Zertifikat A. Nr. 97.1 10147.03

Blatt 2/3

Siemens Schweiz AG, 8047 Zürich

Besondere Bedingungen X

1. Die Digitalausgabebaugruppe SIMATIC Typ 6ES7 322-5SD00-OABO muss ausserhalb des explosionsgefährdeten Bereiches installiert werden.
2. Die Baugruppe muss so installiert werden, dass mindestens die Schutzart IP 20 gemäss EN 60529 eingehalten wird.
3. Bei der Installation ist die Leitungskammer einzusetzen oder zwischen den eigensicheren und den nichteigensicheren Stromkreisen ist durch Einfügen einer Trennwand ein Mindestabstand von 50 mm (Fadenmass) zu realisieren bzw. die Anschlusssteile sind zusätzlich zu isolieren (z.B. Schrumpfschlauch).

Ex-Klassifikation

[EEx ib] IIC

Kennzeichnung

Die Kennzeichnung muss gut sichtbar, lesbar und dauerhaft sein; sie muss folgende Angaben umfassen:

1. Name des Herstellers oder sein Warenzeichen
Typenbezeichnung
Ex-Klassifikation
Fertig.-Nr. oder Losbezeichnung
2. Zusätzliche Angaben, die in den besonderen Europäischen Normen für die betreffenden Zündschutzarten vorgeschrieben sind.
Angaben, die normalerweise durch Konstruktionsnormen für elektrische Betriebsmittel verlangt werden.

Schweizerischer Elektrotechnischer Verein
Association Suisse des Electriciens
Associazione Svizzera degli Elettrotecnici
Swiss Electrotechnical Association

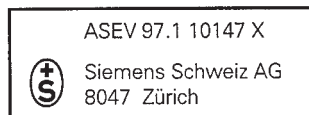


Beilage zu Zertifikat A. Nr. 97.1 10147.03

Blatt 3/3

Siemens Schweiz AG, 8047 Zürich

3. Zusätzliche Bezeichnung



Prüfungsunterlagen

Konformitätsbescheinigung PTB Nr. Ex-96.D.2093 X mit allen darin aufgeführten Prüfungsunterlagen.

Fehraltorf, 24. März 1997
EXL/ P. Bosshard
Visum:

A handwritten signature in black ink, appearing to be 'P. Bosshard'.

A.3 Certificate of Conformity for Digital Output Module DO 4 x 15 V/20 mA

Physikalisch-Technische Bundesanstalt
Braunschweig und Berlin



(1) **KONFORMITÄTSBESCHEINIGUNG**

(2) **PTB Nr. Ex-96.D.2102 X**

(3) Diese Bescheinigung gilt für das elektrische Betriebsmittel
Digitalausgabebaugruppe SM 322, DO 4x15V/20 mA Typ 6ES7 322-5RD00-0AB0

(4) der Firma Siemens AG
D-76181 Karlsruhe

(5) Die Bauart dieses elektrischen Betriebsmittels sowie die verschiedenen zulässigen Ausführungen sind in der Anlage zu dieser Konformitätsbescheinigung festgelegt.

(6) Die Physikalisch-Technische Bundesanstalt bescheinigt als Prüfstelle nach Artikel 14 der Richtlinie des Rates der Europäischen Gemeinschaften vom 18. Dezember 1975 (76/117/EWG) die Übereinstimmung dieses elektrischen Betriebsmittels mit den harmonisierten Europäischen Normen

Elektrische Betriebsmittel für explosionsgefährdete Bereiche

EN 50 014:1977 + A1...A5 (VDE 0170/0171 Teil 1/1.87) Allgemeine Bestimmungen
EN 50 020:1977 + A1...A5 (VDE 0170/0171 Teil 7/4.92) Eigensicherheit "i"

nachdem das Betriebsmittel mit Erfolg einer Bauartprüfung unterzogen wurde. Die Ergebnisse dieser Bauartprüfung sind in einem vertraulichen Prüfprotokoll festgelegt.

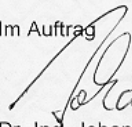
(7) Das Betriebsmittel ist mit folgender Kennzeichnung zu versehen:

[EEx ib] IIC

(8) Der Hersteller ist dafür verantwortlich, daß jedes derart gekennzeichnete Betriebsmittel in seiner Bauart mit den in der Anlage zu dieser Bescheinigung aufgeführten Prüfungsunterlagen übereinstimmt und daß die vorgeschriebenen Stückprüfungen erfolgreich durchgeführt wurden.

(9) Das elektrische Betriebsmittel darf mit dem hier abgedruckten gemeinschaftlichen Unterscheidungszeichen gemäß Anhang II der Richtlinie des Rates vom 6. Februar 1979 (79/196/EWG) gekennzeichnet werden.

Im Auftrag



Dr.-Ing. Johannes Meyer
Oberregierungsrat

Braunschweig, 26.07.1996



Prüfbescheinigungen ohne Unterschrift und ohne Dienstsiegel haben keine Gültigkeit.
 Die Bescheinigungen dürfen nur unverändert weiterverbreitet werden.
 Auszüge oder Änderungen bedürfen der Genehmigung der Physikalisch-Technischen Bundesanstalt.

Physikalisch-Technische Bundesanstalt

ANLAGE

zur Konformitätsbescheinigung PTB Nr. Ex-96.D.2102 X

Die Digitalausgabebaugruppe SM 322, DO 4x15V/20 mA Typ 6ES7 322-5RD00-0AB0 dient z.B. zur Ansteuerung von Magnetventilen im explosionsgefährdeten Bereich. Die höchstzulässige Umgebungstemperatur beträgt 60°C.

Elektrische Daten

Versorgungs- und Busstromkreise 24 VDC (20,4 bis 28,8 V); ca. 120 mA und
(Anschluß 1 und 20 des 20-poligen 5 VDC (5,1 bis 5,3 V); ca. 60 mA
Frontsteckers und die Anschlüsse $U_m = 60$ VDC bzw. 30 VAC
des SIMATIC-Busses auf der
Rückseite)

Ausgangsstromkreise in Zündschutzart Eigensicherheit EEx ib IIC
(Klemmen 4, 5; 8, 9; 13, 14; 17, 18 mit folgenden Höchstwerten je Stromkreis:
des 20-poligen Frontsteckers) $U_o = 15,75$ V
 $I_o = 85$ mA
 $P_o = 335$ mW
Kennlinie: linear
höchstzul. äußere Induktivität 5 mH
höchstzul. äußere Kapazität 500 nF

Die Ausgangsstromkreise sind untereinander und von den Versorgungs- und Busstromkreisen bis zu einem Scheitelwert der Nennspannung von 60 V sicher galvanisch getrennt.

Prüfungsunterlagen alle unterschrieben am 05.06.1996

1. Beschreibung mit Blockschaltbild (10 Blatt)
2. Zeichnung Nr.: 3 NEP 720 7064 01 A (8 Blatt)
3 NEG 720 7069 01 a (9 Blatt)

Besondere Bedingungen

1. Die Digitalausgabebaugruppe SM 322, DO 4x15V/20 mA Typ 6ES7 322-5RD00-0AB0 muß außerhalb des explosionsgefährdeten Bereiches errichtet werden.
2. Die Baugruppe muß so errichtet werden, daß mindestens die Schutzart IP 20 gemäß IEC-Publikation 529 eingehalten wird.
3. Bei der Errichtung ist die Leitungskammer einzusetzen oder zwischen den eigensicheren und den nichteigensicheren Stromkreisen ist durch Einfügen einer Trennwand ein Mindestabstand von 50 mm (Fadenmaß) zu realisieren bzw. die Anschlußteile sind zusätzlich zu isolieren (z.B. Schrumpfschlauch).

Im Auftrag

Dr.-Ing. Johannsmeyer
Oberregierungsrat



Braunschweig, 26.07.1996

Blatt 1/1

**A.3.1 ASEV Certificate/Switzerland for Digital Output Module
DO 4 x 15 V/20 mA**

 Eidgenössisches Starkstrominspektorat Inspection fédérale des installations à courant fort Ispektorato federale degli impianti a corrente forte		
ZULASSUNG	ADMISSION APPROVAL	AMMISSIONE
NR. 97.1 10147.08		SIEMENS SCHWEIZ AG
GÜLTIG BIS: 26/03/2002		FREILAGERSTR. 28 8047 ZÜRICH
KENNZEICHNUNG 	787.676	
EX-KENNZEICHNUNG GEMÄSS BEILAGE		
AUFGRUND DER ANGABEN EINES PRÜFBERICHTES NR. 97.1 10147.02 VOM 25/03/1997 ERTEILT DAS EIDGENÖSSISCHE STARKSTROMINSPEKTORAT DER OBEN GENANNTEN FIRMA DAS RECHT, DAS NACHSTEHEND AUFGEFÜHRTE ERZEUGNIS IN DER SCHWEIZ IN VERKEHR ZU BRINGEN.		
ERZEUGNIS:	DIGITALAUSGABEBAUGRUPPE	
MATERIAL-CODE:	1640	
HANDELSMARKE:	SIMATIC	
<u>TYPENBEZEICHNUNG</u>	<u>NENNDATEN</u>	
6ES7 322-5R000-0A00	GEMÄSS BEILAGE	
BEMERKUNGEN:	BESCHREIBUNG, HINWEISE, BESONDERE BEDINGUNGEN, EX- KLASSEIFIKATION UND PRÜFUNGSUNTERLAGEN GEMÄSS BEILAGE VOM 24.03.1997 ZU ZERTIFIKAT A.NR. 97.1 10147.02	
NORM:	EN 50014:1977+A1...A5, EN 50020:1977+A1...A5	
GEBÜHR:	FR. 360.00	EIDGENÖSSISCHES STARKSTROMINSPEKTORAT
FEHRALTORF, 26/03/1997 MH		ABT. MATERIAL + APPARATE
<i>MH</i>		<i>i.A. R. by Rb</i> J. BRUHIN

Schweizerischer Elektrotechnischer Verein
Association Suisse des Electriciens
Associazione Svizzera degli Elettrotecnici
Swiss Electrotechnical Association



Beilage zu Zertifikat A. Nr. 97.1 10147.02

Blatt 1/3

Siemens Schweiz AG, 8047 Zürich

Beschreibung

Die Digitalausgabebaugruppe SIMATIC Typ 6ES7 322-5RDOO-OABO dient z.B. zur Ansteuerung von Magnetventilen im explosionsgefährdeten Bereich.

Nennungen

Versorgungs- und Busstromkreise
(Anschluss 1 und 20 des 20-poligen
Frontsteckers und die Anschlüsse
des SIMATIC-Busses auf der
Rückseite)

24 VDC (20,4 bis 28,8 V); ca. 120 mA und
5 VDC (5,1 bis 5,3 V); ca. 60 mA
 $U_m = 60 \text{ VDC bzw. } 30 \text{ VAC}$

Ausgangsstromkreise
(Klemmen 4, 5; 8, 9; 13, 14; 17, 18
des 20-poligen Frontsteckers)

in Zündschutzart Eigensicherheit EEx ib IIC
mit folgenden Höchstwerten je Stromkreis:
 $U_o = 15,75 \text{ V}$
 $I_o = 85 \text{ mA}$
 $P_o = 335 \text{ mW}$

Kennlinie: linear
höchstzul. äussere Induktivität 5 mH
höchstzul. äussere Kapazität 500 nF

Hinweise

1. Die Eingangsstromkreise sind untereinander und von den Versorgungs- und Busstromkreisen bis zu einem Scheitelwert der Nennspannung von 60 V sicher galvanisch getrennt.
2. Die höchstzulässige Umgebungstemperatur beträgt 60 °C.

Schweizerischer Elektrotechnischer Verein
Association Suisse des Electriciens
Associazione Svizzera degli Elettrotecnici
Swiss Electrotechnical Association



Beilage zu Zertifikat A. Nr. 97.1 10147.02

Blatt 2/3

Siemens Schweiz AG, 8047 Zürich

Besondere Bedingungen X

1. Die Digitalausgabebaugruppe SIMATIC Typ 6ES7 322-5RDOO-OABO muss ausserhalb des explosionsgefährdeten Bereiches installiert werden.
2. Die Baugruppe muss so installiert werden, dass mindestens die Schutzart IP 20 gemäss EN 60529 eingehalten wird.
3. Bei der Installation ist die Leitungskammer einzusetzen oder zwischen den eigensicheren und den nichteigensicheren Stromkreisen ist durch Einfügen einer Trennwand ein Mindestabstand von 50 mm (Fadenmass) zu realisieren bzw. die Anschlussteile sind zusätzlich zu isolieren (z.B. Schrumpfschlauch).

Ex-Klassifikation

[Ex ib] IIC

Kennzeichnung

Die Kennzeichnung muss gut sichtbar, lesbar und dauerhaft sein; sie muss folgende Angaben umfassen:

1. Name des Herstellers oder sein Warenzeichen
Typenbezeichnung
Ex-Klassifikation
Fertig.-Nr. oder Losbezeichnung
2. Zusätzliche Angaben, die in den besonderen Europäischen Normen für die betreffenden Zündschutzarten vorgeschrieben sind.
Angaben, die normalerweise durch Konstruktionsnormen für elektrische Betriebsmittel verlangt werden.

Schweizerischer Elektrotechnischer Verein
Association Suisse des Electriciens
Associazione Svizzera degli Elettrotecnici
Swiss Electrotechnical Association



Beilage zu Zertifikat A. Nr. 97.1 10147.02

Blatt 3/3

Siemens Schweiz AG, 8047 Zürich

3. Zusätzliche Bezeichnung

ASEV 97.1 10147 X



Siemens Schweiz AG
8047 Zürich

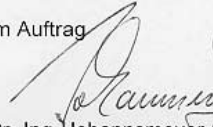
Prüfungsunterlagen

Konformitätsbescheinigung PTB Nr. Ex-96.D.2102 X mit allen darin aufgeführten Prüfungsunterlagen.

Fehraltorf, 24. März 1997
EXL/ P. Bosshard
Visum:

A handwritten signature in black ink, appearing to be 'P. Bosshard'.

A.4 Certificate of Conformity for Analog Input Module AI 8 x TC/4 x RTD

Physikalisch-Technische Bundesanstalt Braunschweig und Berlin 	
(1)	KONFORMITÄTSBESCHEINIGUNG
(2)	PTB Nr. Ex-96.D.2108 X
(3) Diese Bescheinigung gilt für das elektrische Betriebsmittel Analogeingabebaugruppe SM 331, AI 8 x TC / 4 x RTD, Typ 6ES7 331-7SF00-0AB0	
(4) der Firma Siemens AG D-76181 Karlsruhe	
(5) Die Bauart dieses elektrischen Betriebsmittels sowie die verschiedenen zulässigen Ausführungen sind in der Anlage zu dieser Konformitätsbescheinigung festgelegt.	
(6) Die Physikalisch-Technische Bundesanstalt bescheinigt als Prüfstelle nach Artikel 14 der Richtlinie des Rates der Europäischen Gemeinschaften vom 18. Dezember 1975 (76/117/EWG) die Übereinstimmung dieses elektrischen Betriebsmittels mit den harmonisierten Europäischen Normen Elektrische Betriebsmittel für explosionsgefährdete Bereiche EN 50 014:1977 + A1...A5 (VDE 0170/0171 Teil 1/1.87) Allgemeine Bestimmungen EN 50 020:1977 + A1...A5 (VDE 0170/0171 Teil 7/4.92) Eigensicherheit "I" nachdem das Betriebsmittel mit Erfolg einer Bauartprüfung unterzogen wurde. Die Ergebnisse dieser Bauartprüfung sind in einem vertraulichen Prüfprotokoll festgelegt.	
(7) Das Betriebsmittel ist mit folgender Kennzeichnung zu versehen: [EEEx ib] IIC	
(8) Der Hersteller ist dafür verantwortlich, daß jedes derart gekennzeichnete Betriebsmittel in seiner Bauart mit den in der Anlage zu dieser Bescheinigung aufgeführten Prüfungsunterlagen übereinstimmt und daß die vorgeschriebenen Stückprüfungen erfolgreich durchgeführt wurden.	
(9) Das elektrische Betriebsmittel darf mit dem hier abgedruckten gemeinschaftlichen Unterscheidungszeichen gemäß Anhang II der Richtlinie des Rates vom 6. Februar 1979 (79/196/EWG) gekennzeichnet werden.	
Im Auftrag  Dr.-Ing. Johannsmeyer Oberregierungsrat	Braunschweig, 08.08.1996 
Prüfbescheinigungen ohne Unterschrift und ohne Dienstsiegel haben keine Gültigkeit. Die Bescheinigungen dürfen nur unverändert weiterverbreitet werden. Auszüge oder Änderungen bedürfen der Genehmigung der Physikalisch-Technischen Bundesanstalt.	

Physikalisch-Technische Bundesanstalt

ANLAGE

zur Konformitätsbescheinigung PTB Nr. Ex-96.D.2108 X

Die Analogeingabebaugruppe SM 331, AI 8 x TC / 4 x RTD, Typ 6ES7 331-7SF00-0AB0 dient u.a. zum Anschluß von Thermoelementen, Widerstandsgebern und Spannungsgebern.

Die höchstzulässige Umgebungstemperatur beträgt 60 °C.

Elektrische Daten

Versorgungs- und Busstromkreise 5 VDC (5,1...5,3 V); ca. 60 mA
(Anschlüsse des $U_m = 60$ VDC bzw. 30 VAC
SIMATIC-Busses
auf der Rückseite)

Eingangsstromkreise in Zündschutzart Eigensicherheit EEx ib IIC
(Klemmen 2, 3; 4, 5; mit folgenden Höchstwerten je Stromkreis:
6, 7; 8, 9; 12, 13; $U_o = 5,9$ V
14, 15; 16, 17; 18, 19 $I_o = 28,8$ mA
des 20poligen Frontsteckers) $P_o = 41,4$ mW

Kennlinie linear

höchstzulässige äußere Induktivität 40 mH

höchstzulässige äußere Kapazität 60 µF

oder

zum Anschluß an aktive Geber mit folgenden
Höchstwerten je Stromkreis:

$U_i = \pm 1,2$ V

$I_i = 20$ mA

höchstzulässige äußere Induktivität 15 mH

höchstzulässige äußere Kapazität 17 µF

Die Eingangsstromkreise sind von den Versorgungs- und Busstromkreisen bis zu einem Scheitelwert der Nennspannung von 60 V sicher galvanisch getrennt.

Prüfungsunterlagen alle unterschrieben am 17.07.1996

1. Beschreibung mit Blockschaltbild (10 Blatt)
2. Zeichnung Nr.: 3 NEP 720 7124 02 A (7 Blatt)
3 NEG 720 7129 01 b (6 Blatt)

Blatt 1/2

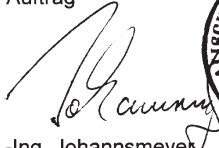
Physikalisch-Technische Bundesanstalt

Anlage zur Konformitätsbescheinigung PTB Nr. Ex-96.D.2108 X

Besondere Bedingungen

1. Die Analogeingabebaugruppe SM 331, AI 8 x TC / 4 x RTD, Typ 6ES7 331-7SF00-0AB0 muß außerhalb des explosionsgefährdeten Bereiches errichtet werden.
2. Die Baugruppe muß so errichtet werden, daß mindestens die Schutzart IP 20 gemäß IEC-Publikation 529 eingehalten wird.
3. Bei der Errichtung ist zwischen den eigensicheren und den nichteigensicheren Stromkreisen benachbarter Baugruppen eine Trennwand einzufügen (Fadenmaß ≥ 50 mm) oder die Anschlüsse sind zusätzlich zu isolieren (z.B. Schrumpfschlauch).

Im Auftrag



Dr.-Ing. Johannsmeyer
Oberregierungsrat



Braunschweig, 08.08.1996

Blatt 2/2

**A.4.1 ASEV Certificate/Switzerland for Analog Input Module
AI 8 x TC/4 x RTD**

 Eidgenössisches Starkstrominspektorat Inspection fédérale des Installations à courant fort Ispettorato federale degli impianti a corrente forte		
ZULASSUNG	ADMISSION APPROVAL	AMMISSIONE
NR. 97.1 10147.11		SIEMENS SCHWEIZ AG
GÜLTIG BIS: 26/03/2002		FREILAGERSTR. 28 8047 ZÜRICH
KENNZEICHNUNG 		787.676
EX-KENNZEICHNUNG GEMÄSS BEILAGE		
AUFGRUND DER ANGABEN EINES PRÜFBERICHTES NR. 97.1 10147.05 VOM 25/03/1997 ERTEILT DAS EIDGENÖSSISCHE STARKSTROMINSPEKTORAT DER OBEN GENANNTE FİRMA DAS RECHT, DAS NACHSTEHEND AUFGEFÜHRTE ERZEUGNIS IN DER SCHWEIZ IN VERKEHR ZU BRINGEN.		
ERZEUGNIS:	ANALOG-INGANGSBAUGRUPPE	
MATERIAL-CODE:	1640	
HANDELSMARKE:	SIMATIC	
TYPENBEZEICHNUNG	NENN DATEN	
6ES7 331-7SF00-0AB0	GEMÄSS BEILAGE	
BEMERKUNGEN:	BESCHREIBUNG, HINWEISE, BESONDERE BEDINGUNGEN, EX- KLASSIFIKATION UND PRÜFUNGSUNTERLAGEN GEMÄSS BEILAGE VOM 24.03.1997 ZU ZERTIFIKAT A.NR. 97.1 10147.05	
NORM:	EN 50014:1977+A1...A5, EN 50020:1977+A1...A5	
GEBÜHR:	FR. 360.00	EIDGENÖSSISCHES STARKSTROMINSPEKTORAT
FEHRALTORF, 26/03/1997 MH		ABT. MATERIAL + APPARATE
<i>ML</i>		<i>J. A. R. Bruhin</i> J. BRUHIN

Schweizerischer Elektrotechnischer Verein
 Association Suisse des Electriciens
 Associazione Svizzera degli Elettrotecnici
 Swiss Electrotechnical Association



Beilage zu Zertifikat A. Nr. 97.1 10147.05

Blatt 1/3

Siemens Schweiz AG, 8047 Zürich

Beschreibung

Die Analogeingabebaugruppe SIMATIC Typ 6ES7 331-7SFOO-OABO dient u.a. zum Anschluss von Thermoelementen, Widerstandsgebern und Spannungsgebern.

Nennndaten

Versorgungs- und Busstromkreise (Anschlüsse des SIMATIC-Busses auf der Rückseite)	5 VDC (5,1 bis 5,3 V); ca. 60 mA $U_m = 60$ VDC bzw. 30 VAC
Eingangsstromkreise (Klemmen 2, 3; 4, 5; 6, 7; 8, 9; 12, 13; 14, 15; 16, 17; 18, 19 des 20poligen Frontsteckers)	in Zündschutzart Eigensicherheit EEx ib IIC mit folgenden Höchstwerten je Stromkreis: $U_o = 5,9$ V $I_o = 28,8$ mA $P_o = 41,4$ mW Kennlinie linear höchstzulässige äussere Induktivität 40 mH höchstzulässige äussere Kapazität 60 μ F oder zum Anschluss an aktive Geber mit folgenden Höchstwerten je Stromkreis: $U_i = \pm 1,2$ V $I_i = 20$ mA höchstzulässige äussere Induktivität 15 mH höchstzulässige äussere Kapazität 17 μ F

Hinweise

1. Die Eingangsstromkreise sind untereinander und von den Versorgungs- und Busstromkreisen bis zu einem Scheitelwert der Nennspannung von 60 V sicher galvanisch getrennt.
2. Die höchstzulässige Umgebungstemperatur beträgt 60 °C.

Schweizerischer Elektrotechnischer Verein
Association Suisse des Electriciens
Associazione Svizzera degli Elettrotecnici
Swiss Electrotechnical Association



Beilage zu Zertifikat A. Nr. 97.1 10147.05

Blatt 2/3

Siemens Schweiz AG, 8047 Zürich

Besondere Bedingungen X

1. Die Analogeingabebaugruppe SIMATIC Typ 6ES7 331-7SFOO-OABO muss ausserhalb des explosionsgefährdeten Bereiches installiert werden.
2. Die Baugruppe muss so installiert werden, dass mindestens die Schutzart IP 20 gemäss EN 60529 eingehalten wird.
3. Bei der Installation ist die Leitungskammer einzusetzen oder zwischen den eigensicheren und den nichteigensicheren Stromkreisen ist durch Einfügen einer Trennwand ein Mindestabstand von 50 mm (Fadenmass) zu realisieren bzw. die Anschlussteile sind zusätzlich zu isolieren (z.B. Schrumpfschlauch).

Ex-Klassifikation

[EEx ib] IIC

Kennzeichnung

Die Kennzeichnung muss gut sichtbar, lesbar und dauerhaft sein; sie muss folgende Angaben umfassen:

1. Name des Herstellers oder sein Warenzeichen
Typenbezeichnung
Ex-Klassifikation
Fertig.-Nr. oder Losbezeichnung
2. Zusätzliche Angaben, die in den besonderen Europäischen Normen für die betreffenden Zündschutzarten vorgeschrieben sind.
Angaben, die normalerweise durch Konstruktionsnormen für elektrische Betriebsmittel verlangt werden.

Schweizerischer Elektrotechnischer Verein
Association Suisse des Electriciens
Associazione Svizzera degli Elettrotecnici
Swiss Electrotechnical Association



Beilage zu Zertifikat A. Nr. 97.1 10147.05

Blatt 3/3

Siemens Schweiz AG, 8047 Zürich

3. Zusätzliche Bezeichnung



Prüfungsunterlagen

Konformitätsbescheinigung PTB Nr. Ex-96.D.2108 X mit allen darin aufgeführten Prüfungsunterlagen.

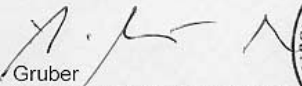
Fehraltorf, 24. März 1997

EXL/ P. Bosshard

Visum:

A handwritten signature in dark ink, appearing to be 'P. Bosshard', written over the printed name.

A.5 Certificate of Conformity for Analog Input Module AI 4 x 0/4...20 mA

Physikalisch-Technische Bundesanstalt Braunschweig und Berlin  KONFORMITÄTSBESCHEINIGUNG PTB Nr. Ex-96.D.2092 X (1) Diese Bescheinigung gilt für das elektrische Betriebsmittel Analogeingabebaugruppe SM 331, AI 4 x 0/4...20mA Typ 6ES7 331-7RD00-0AB0 (2) der Firma Siemens AG D-76181 Karlsruhe (3) Die Bauart dieses elektrischen Betriebsmittels sowie die verschiedenen zulässigen Ausführungen sind in der Anlage zu dieser Konformitätsbescheinigung festgelegt. (4) Die Physikalisch-Technische Bundesanstalt bescheinigt als Prüfstelle nach Artikel 14 der Richtlinie des Rates der Europäischen Gemeinschaften vom 18. Dezember 1975 (76/117/EWG) die Übereinstimmung dieses elektrischen Betriebsmittels mit den harmonisierten Europäischen Normen Elektrische Betriebsmittel für explosionsgefährdete Bereiche EN 50 014:1977 + A1...A5 (VDE 0170/0171 Teil 1/1.87) Allgemeine Bestimmungen EN 50 020:1977 + A1...A5 (VDE 0170/0171 Teil 7/4.92) Eigensicherheit "I" nachdem das Betriebsmittel mit Erfolg einer Bauartprüfung unterzogen wurde. Die Ergebnisse dieser Bauartprüfung sind in einem vertraulichen Prüfprotokoll festgelegt. (5) Das Betriebsmittel ist mit folgender Kennzeichnung zu versehen: [EEx ib] IIC (6) Der Hersteller ist dafür verantwortlich, daß jedes derart gekennzeichnete Betriebsmittel in seiner Bauart mit den in der Anlage zu dieser Bescheinigung aufgeführten Prüfungsunterlagen übereinstimmt und daß die vorgeschriebenen Stückprüfungen erfolgreich durchgeführt wurden. (7) Das elektrische Betriebsmittel darf mit dem hier abgedruckten gemeinschaftlichen Unterscheidungszeichen gemäß Anhang II der Richtlinie des Rates vom 6. Februar 1979 (79/196/EWG) gekennzeichnet werden. Im Auftrag Braunschweig, 15.07.1996  Gruber Techn. Regierungsoberamtsrat  Prüfbescheinigungen ohne Unterschrift und ohne Dienstsiegel haben keine Gültigkeit. Die Bescheinigungen dürfen nur unverändert weiterverbreitet werden. Auszüge oder Änderungen bedürfen der Genehmigung der Physikalisch-Technischen Bundesanstalt.	
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Physikalisch-Technische Bundesanstalt

ANLAGE

zur Konformitätsbescheinigung PTB Nr. Ex-96.D.2092 X

Die Analogeingabebaugruppe SM 331, AI 4 x 0/4...20mA Typ 6ES7 331-7RD00-0AB0 dient zum Anschluß von bescheinigten Meßumformern im explosionsgefährdeten Bereich.

Die höchstzulässige Umgebungstemperatur beträgt 60 °C.

Elektrische Daten

Versorgungs- und Busstromkreise 24 VDC (20,4 bis 28,8 V); ca.250 mA und
(Anschluß 1 und 20 des 20-poligen 5 VDC (5,1 bis 5,3 V); ca. 60 mA
Frontsteckers und die Anschlüsse $U_m = 60$ VDC bzw. 30 VAC
des SIMATIC-Busses auf der
Rückseite)

Ein- u. Ausgangsstromkreise in Zündschutzart Eigensicherheit EEx ib IIC
(Klemmen 3, 4 und 5; mit folgenden Höchstwerten je Stromkreis:
7, 8 und 9; $U_o = 25,2$ V
12, 13 und 14; $I_o = 68,5$ mA
16, 17 und 18 $P_o = 431$ mW
des 20-poligen Frontsteckers)

Kennlinie: linear
höchstzul. äußere Induktivität 7,5 mH
höchstzul. äußere Kapazität 90 nF

Die Ein- u. Ausgangsstromkreise sind untereinander und von den Versorgungs- und Busstromkreisen bis zu einem Scheitelwert der Nennspannung von 60 V sicher galvanisch getrennt.

Prüfungsunterlagen

unterschrieben am 21.05.1996

1. Beschreibung mit Blockschaltbildern (11 Blatt)
2. Zeichnung Nr. 3 NEP 720 7104 01 A (10 Blatt)
3 NEG 720 7109 01a (9 Blatt)

Besondere Bedingungen

1. Die Analogeingabebaugruppe SM 331, AI 4 x 0/4...20mA Typ 6ES7 331-7RD00-0AB0 muß außerhalb des explosionsgefährdeten Bereiches errichtet werden.
2. Die Baugruppe muß so errichtet werden, daß mindestens die Schutzart IP 20 gemäß IEC-Publikation 529 eingehalten wird.
3. Bei der Errichtung ist die Leitungskammer einzusetzen oder zwischen den eigensicheren und den nichteigensicheren Stromkreisen ist durch Einfügen einer Trennwand ein Mindestabstand von 50 mm (Fadenmaß) zu realisieren bzw. die Anschlußteile sind zusätzlich zu isolieren (z.B. Schrumpfschlauch).

Im Auftrag


Gruber
Techn. Regierungsoberratsrat



Braunschweig, 16.07.1996

Blatt 1/1

A.5.1 ASEV Certificate/Switzerland for Analog Input Module
AI 4 x 0/4...20 mA

 Eidgenössisches Starkstrominspektorat Inspection fédérale des installations à courant fort Ispettorato federale degli impianti a corrente forte		
ZULASSUNG	ADMISSION	AMMISSIONE
APPROVAL		
NR.	97,1 10147,10	SIEMENS SCHWEIZ AG
GÜLTIG BIS:	26/03/2002	FREILAGERSTR. 28 8047 ZÜRICH
KENNZEICHNUNG		787,676
EX-KENNZEICHNUNG GEMÄSS BEILAGE		
AUFGRUND DER ANGABEN EINES PRÜFBERICHTES NR. 97,1 10147,04 VOM 25/03/1997 ERTEILT DAS EIDGENÖSSISCHE STARKSTROMINSPEKTORAT DER OBEN GENANNTE FİRMA DAS RECHT, DAS NACHSTEHEND AUFGEFÜHRTE ERZEUGNIS IN DER SCHWEIZ IN VERKEHR ZU BRINGEN.		
ERZEUGNIS:	ANALOGEINGABESAUGRUPPE	
MATERIAL-CODE:	1640	
HANDELSMARKE:	SIMATIC	
TYPENBEZEICHNUNG	MERKDATEN	
-----	-----	
6ES7 331-7RD00-0AB0	GEMÄSS BEILAGE	
BEMERKUNGEN:	BESCHREIBUNG, HINWEISE, BESONDERE BEDINGUNGEN, EX- KLASSIFIKATION UND PRÜFUNGSUNTERLAGEN GEMÄSS BEILAGE VOM 24.03.1997 ZU ZERTIFIKAT A.NR. 97,1 10147,04	
NORM:	EN 50014:1977+A1...A5, EN 50020:1977+A1...A5	
GEBÜHR:	FR. 360.00	EIDGENÖSSISCHES STARKSTROMINSPEKTORAT
FERNLEITER:	26/03/1997 MH	ABT. MATERIAL + APPARATE
	<i>MH</i>	<i>i.A. D. b. Re</i> J. BRUHIN

Schweizerischer Elektrotechnischer Verein
 Association Suisse des Electriciens
 Associazione Svizzera degli Elettrotecnici
 Swiss Electrotechnical Association



Beilage zu Zertifikat A. Nr. 97.1 10147.04

Blatt 1/3

Siemens Schweiz AG, 8047 Zürich

Beschreibung

Die Analogeingabebaugruppe SIMATIC Typ 6ES7 331-7RDOO-OABO dient zum Anschluss von zertifizierten Messumformern im explosionsgefährdeten Bereich.

Nenndaten

Versorgungs- und Busstromkreise (Anschluss 1 und 20 des 20-poligen Frontsteckers und die Anschlüsse des SIMATIC-Busses auf der Rückseite)	24 VDC (20,4 bis 28,8 V); ca. 250 mA und 5 VDC (5,1 bis 5,3 V); ca. 60 mA $U_m = 60 \text{ VDC bzw. } 30 \text{ VAC}$
---	---

Ein- u. Ausgangsstromkreise (Klemmen 3, 4 und 5; 7, 8 und 9; 12, 13 und 14; 16, 17 und 18 des 20-poligen Frontsteckers)	in Zündschutzart Eigensicherheit EEx ib IIC mit folgenden Höchstwerten je Stromkreis: $U_o = 25,2 \text{ V}$ $I_o = 68,5 \text{ mA}$ $P_o = 431 \text{ mW}$
--	---

Kennlinie: linear
 höchstzul. äussere Induktivität 7,5 mH
 höchstzul. äussere Kapazität 90 nF

Hinweise

1. Die Eingangsstromkreise sind untereinander und von den Versorgungs- und Busstromkreisen bis zu einem Scheitelwert der Nennspannung von 60 V sicher galvanisch getrennt.
2. Die höchstzulässige Umgebungstemperatur beträgt 60 °C.

Schweizerischer Elektrotechnischer Verein
Association Suisse des Electriciens
Associazione Svizzera degli Elettrotecnici
Swiss Electrotechnical Association



Beilage zu Zertifikat A. Nr. 97.1 10147.04

Blatt 2/3

Siemens Schweiz AG, 8047 Zürich

Besondere Bedingungen X

1. Die Analogeingabebaugruppe SIMATIC Typ 6ES7 331-7RDOO-OABO muss ausserhalb des explosionsgefährdeten Bereiches installiert werden.
2. Die Baugruppe muss so installiert werden, dass mindestens die Schutzart IP 20 gemäss EN 60529 eingehalten wird.
3. Bei der Installation ist die Leitungskammer einzusetzen oder zwischen den eigensicheren und den nichteigensicheren Stromkreisen ist durch Einfügen einer Trennwand ein Mindestabstand von 50 mm (Fadenmass) zu realisieren bzw. die Anschlussteile sind zusätzlich zu isolieren (z.B. Schrumpfschlauch).

Ex-Klassifikation

[Ex ib] IIC

Kennzeichnung

Die Kennzeichnung muss gut sichtbar, lesbar und dauerhaft sein; sie muss folgende Angaben umfassen:

1. Name des Herstellers oder sein Warenzeichen
Typenbezeichnung
Ex-Klassifikation
Fertig.-Nr. oder Losbezeichnung
2. Zusätzliche Angaben, die in den besonderen Europäischen Normen für die betreffenden Zündschutzarten vorgeschrieben sind.
Angaben, die normalerweise durch Konstruktionsnormen für elektrische Betriebsmittel verlangt werden.

Schweizerischer Elektrotechnischer Verein
Association Suisse des Electriciens
Associazione Svizzera degli Elettrotecnici
Swiss Electrotechnical Association

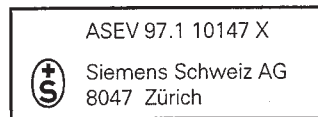


Beilage zu Zertifikat A. Nr. 97.1 10147.04

Blatt 3/3

Siemens Schweiz AG, 8047 Zürich

3. Zusätzliche Bezeichnung



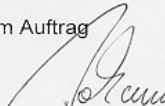
Prüfungsunterlagen

Konformitätsbescheinigung PTB Nr. Ex-96.D.2092 X mit allen darin aufgeführten Prüfungsunterlagen.

Fehraltorf, 24. März 1997
EXL/ P. Bosshard
Visum:

A handwritten signature in black ink, appearing to be 'P. Bosshard'.

**A.6 Certificate of Conformity for Analog Output Module
AO 4 x 0/4...20 mA**

Physikalisch-Technische Bundesanstalt Braunschweig und Berlin  KONFORMITÄTSBESCHEINIGUNG PTB Nr. Ex-96.D.2026 X (1) Diese Bescheinigung gilt für das elektrische Betriebsmittel Analogausgabebaugruppe SM332, AO 4x0/4...20 mA Typ 6ES7 332-5RD00-0AB0 (2) der Firma Siemens AG D-76181 Karlsruhe (3) Die Bauart dieses elektrischen Betriebsmittels sowie die verschiedenen zulässigen Ausführungen sind in der Anlage zu dieser Konformitätsbescheinigung festgelegt. (4) Die Physikalisch-Technische Bundesanstalt bescheinigt als Prüfstelle nach Artikel 14 der Richtlinie des Rates der Europäischen Gemeinschaften vom 18. Dezember 1975 (76/117/EWG) die Übereinstimmung dieses elektrischen Betriebsmittels mit den harmonisierten Europäischen Normen Elektrische Betriebsmittel für explosionsgefährdete Bereiche EN 50 014:1977 + A1...A5 (VDE 0170/0171 Teil 1/1.87) Allgemeine Bestimmungen EN 50 020:1977 + A1...A5 (VDE 0170/0171 Teil 7/4.92) Eigensicherheit "I" nachdem das Betriebsmittel mit Erfolg einer Bauartprüfung unterzogen wurde. Die Ergebnisse dieser Bauartprüfung sind in einem vertraulichen Prüfprotokoll festgelegt. (5) Das Betriebsmittel ist mit folgender Kennzeichnung zu versehen: [Ex ib] IIC (6) Der Hersteller ist dafür verantwortlich, daß jedes derart gekennzeichnete Betriebsmittel in seiner Bauart mit den in der Anlage zu dieser Bescheinigung aufgeführten Prüfungsunterlagen übereinstimmt und daß die vorgeschriebenen Stückprüfungen erfolgreich durchgeführt wurden. (7) Das elektrische Betriebsmittel darf mit dem hier abgedruckten gemeinschaftlichen Unterscheidungszeichen gemäß Anhang II der Richtlinie des Rates vom 6. Februar 1979 (79/196/EWG) gekennzeichnet werden. Im Auftrag  Dr.-Ing. Johannsmeyer Oberregierungsrat Braunschweig, 14.03.1996  Prüfbescheinigungen ohne Unterschrift und ohne Dienstsiegel haben keine Gültigkeit. Die Bescheinigungen dürfen nur unverändert weiterverbreitet werden. Auszüge oder Änderungen bedürfen der Genehmigung der Physikalisch-Technischen Bundesanstalt.	
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Physikalisch-Technische Bundesanstalt

ANLAGE

zur Konformitätsbescheinigung PTB Nr. Ex-96.D.2026 X

Die Analogausgabebaugruppe SM332, AO 4x0/4...20 mA Typ 6ES7 332-5RD00-0AB0 dient zur Signalübertragung in explosionsgefährdete Bereiche.

Die höchstzulässige Umgebungstemperatur beträgt 60 °C.

Elektrische Daten

Versorgungs- und Busstromkreise 24 VDC (20,4 V bis 28,8 V); ca. 0,25 A
(Frontstecker Klemmen 1 und 20 oder U_m = 60 VDC bzw. 30 VAC
SIMATIC-Bus auf Rückseite)

Ausgangsstromkreise in Zündschutzart Eigensicherheit EEx ib IIC
(Frontstecker Klemmen 4, 5; 8, 9; mit folgenden Werten je Stromkreis:
13, 14; 17, 18) U_o = 14 V
I_o = 70 mA
P_o = 440 mW
Kennlinien trapezförmig
höchstzulässige äußere Induktivität 6,6 mH
höchstzulässige äußere Kapazität 850 nF

Die Ausgangsstromkreise sind von den Versorgungs- und Busstromkreisen bis zu einem Scheitelwert der Nennspannung von 60 V sicher galvanisch getrennt.

Prüfungsunterlagen alle unterschrieben am 16.01.1996

1. Beschreibung (9 Blatt) und Blockschaltbilder (3 Blatt)
2. Zeichnung Nr. 3NEP 720 7144 01 A (12 Blatt)
3NEG 720 7149 01 b (9 Blatt)

Besondere Bedingungen

1. Die Analogausgabebaugruppe SM332, AO 4x0/4...20 mA Typ 6ES7 332-5RD00-0AB0 muß außerhalb des explosionsgefährdeten Bereiches errichtet werden.
2. Bei der Errichtung ist zwischen den eigensicheren und den nichteigensicheren Stromkreisen durch Nutzung der Leitungskammer oder durch Einfügen einer geeigneten Trennwand ein Mindestabstand von 50 mm (Fadenmaß) zu realisieren bzw. die Anschlußteile sind zusätzlich zu isolieren (z.B. Schrumpfschlauch).

Im Auftrag

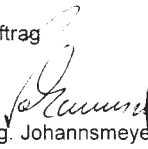
Dr.-Ing. Johannsmeyer
Oberregierungsrat



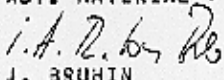
Braunschweig, 14.03.1996

Blatt 1/1

A.6.1 First Supplement for Analog Output Module AO 4 x 0/4...20 mA

Physikalisch-Technische Bundesanstalt	
1. NACHTRAG zur Konformitätsbescheinigung PTB Nr. Ex-96.D.2026 X	
der Firma Siemens AG D-76181 Karlsruhe	
Die Analogausgabebaugruppe SM 332, AO 4x0/4...20 mA Typ 6ES7 332-5RD00-0AB0 darf künftig entsprechend den unten aufgeführten Prüfungsunterlagen gefertigt werden.	
Die Änderungen betreffen den inneren Aufbau und einen Teil der "Elektrischen Daten".	
<u>Elektrische Daten</u>	
Versorgungs- und Busstromkreise (Anschluß 1 und 20 des 20-poligen Frontsteckers und die Anschlüsse des SIMATIC-Busses auf der Rückseite)	24 VDC (20,4 bis 28,8 V); ca. 0,25 A und 5 VDC (5,1 bis 5,3 V); ca. 60 mA $U_n = 60$ VDC bzw. 30 VAC
Alle übrigen Angaben und die "Besonderen Bedingungen" gelten unverändert für diesen Nachtrag.	
<u>Prüfungsunterlagen</u>	alle unterschreiben am 12.08.1996
1. Beschreibung (4 Blatt) und Blockschaltbilder (3 Blatt)	
2. Zeichnung Nr. 3 NEP 720 7144 02 A (12 Blatt) 3 NEG 720 7149 01 d (9 Blatt)	
Im Auftrag  Dr.-Ing. Johannsmeyer Oberregistrationsrat	Braunschweig, 24.10.1996 
Z.14-755 320 3.93	[EEx ib] IIC
	Blatt 1/1

**A.6.2 ASEV Certificate/Switzerland for Analog Output Module
AO 4 x 0/4...20 mA**

 Eidgenössisches Starkstrominspektorat Inspection fédérale des installations à courant fort Ispektorato federale degli impianti a corrente forte		
ZULASSUNG	ADMISSION	AMMISSIONE
APPROVAL		
NR.	97.1 10147.12	SIEMENS SCHWEIZ AG
GÜLTIG BIS:	26/03/2002	FREILAGERSTR. 28 8047 ZÜRICH
KENNZEICHNUNG		787.676
EX-KENNZEICHNUNG GEMÄSS BEILAGE		
AUFGRUND DER ANGABEN EINES PRÜFBERICHTES NR. 97.1 10147.06 VOM 25/03/1997 ERTEILT DAS EIDGENÖSSISCHE STARKSTROMINSPEKTORAT DER OBEN GENANNTEN FIRMA DAS RECHT, DAS NACHSTEHEND AUFGEFÜHRTE ERZEUGNIS IN DER SCHWEIZ IN VERKEHR ZU BRINGEN.		
ERZEUGNIS:	ANALOGAUSGABEBaugruppe	
MATERIAL-CODE:	1640	
HANDELSMARKE:	SIMATIC	
TYPENBEZEICHNUNG	NENN DATEN	
6ES7 332-5BD00-0AB0	GEMÄSS BEILAGE	
BEMERKUNGEN:	BESCHREIBUNG, HINWEISE, BESONDERE BEDINGUNGEN, EX- KLASSIFIKATION UND PRÜFUNGSUNTERLAGEN GEMÄSS BEILAGE VOM 24.03.1997 ZU ZERTIFIKAT A.NR. 97.1 10147.06	
NORM:	EN 50014:1977+A1...A5, EN 50020:1977+A1...A5	
GEBÜHR:	FR. 360.00	EIDGENÖSSISCHES STARKSTROMINSPEKTORAT
FEHRALTORF,	26/03/1997 MH	ABT. MATERIAL + APPARATE
MH		 J. BRUHIN

Schweizerischer Elektrotechnischer Verein
Association Suisse des Electriciens
Associazione Svizzera degli Elettrotecnici
Swiss Electrotechnical Association



Beilage zu Zertifikat A. Nr. 97.1 10147.06

Blatt 1/3

Siemens Schweiz AG, 8047 Zürich

Beschreibung

Die Analogausgabebaugruppe SIMATIC Typ 6ES7 332-5RDOO-OABO dient zur Signalübertragung in explosionsgefährdete Bereiche.

Nennungen

Versorgungs- und Busstromkreise
(Anschluss 1 und 20 des 20-poligen
Frontsteckers und die Anschlüsse
des SIMATIC-Busses
auf der Rückseite)

24 VDC (20,4 bis 28,8 V); ca. 0,25 A und
5 VDC (5,1 bis 5,3 V); ca. 60 mA
 $U_m = 60$ VDC bzw. 30 VAC

Ausgangsstromkreise
(Frontstecker Klemmen 4, 5; 8, 9;
13, 14; 17, 18)

in Zündschutzart Eigensicherheit EEx ib IIC
mit folgenden Werten je Stromkreis:

$U_o = 14$ V
 $I_o = 70$ mA
 $P_o = 440$ mW

Kennlinien trapezförmig
höchstzulässige äussere Induktivität 6,6 mH
höchstzulässige äussere Kapazität 850 nF

Hinweise

1. Die Eingangsstromkreise sind untereinander und von den Versorgungs- und Busstromkreisen bis zu einem Scheitelwert der Nennspannung von 60 V sicher galvanisch getrennt.
2. Die höchstzulässige Umgebungstemperatur beträgt 60 °C.

Schweizerischer Elektrotechnischer Verein
Association Suisse des Electriciens
Associazione Svizzera degli Elettrotecnici
Swiss Electrotechnical Association



Beilage zu Zertifikat A. Nr. 97.1 10147.06

Blatt 2/3

Siemens Schweiz AG, 8047 Zürich

Besondere Bedingungen X

1. Die Analogausgabebaugruppe SIMATIC Typ 6ES7 332-5RDOO-OABO muss ausserhalb des explosionsgefährdeten Bereiches installiert werden.
2. Die Baugruppe muss so installiert werden, dass mindestens die Schutzart IP 20 gemäss EN 60529 eingehalten wird.
3. Bei der Installation ist die Leitungskammer einzusetzen oder zwischen den eigensicheren und den nichteigensicheren Stromkreisen ist durch Einfügen einer Trennwand ein Mindestabstand von 50 mm (Fadenmass) zu realisieren bzw. die Anschlussteile sind zusätzlich zu isolieren (z.B. Schrumpfschlauch).

Ex-Klassifikation

[Ex ib] IIC

Kennzeichnung

Die Kennzeichnung muss gut sichtbar, lesbar und dauerhaft sein; sie muss folgende Angaben umfassen:

1. Name des Herstellers oder sein Warenzeichen
Typenbezeichnung
Ex-Klassifikation
Fertig.-Nr. oder Losbezeichnung
2. Zusätzliche Angaben, die in den besonderen Europäischen Normen für die betreffenden Zündschutzarten vorgeschrieben sind.
Angaben, die normalerweise durch Konstruktionsnormen für elektrische Betriebsmittel verlangt werden.

Schweizerischer Elektrotechnischer Verein
Association Suisse des Electriciens
Associazione Svizzera degli Elettrotecnici
Swiss Electrotechnical Association

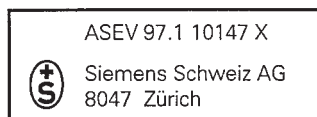


Beilage zu Zertifikat A. Nr. 97.1 10147.06

Blatt 3/3

Siemens Schweiz AG, 8047 Zürich

3. Zusätzliche Bezeichnung



Prüfungsunterlagen

Konformitätsbescheinigung PTB Nr. Ex-96.D.2026 X mit 1. Nachtrag einschliesslich allen darin aufgeführten Prüfungsunterlagen.

Fehraltorf, 24. März 1997
EXL/ P. Bosshard
Visum

A.7 KEMA Certificate of Conformity for Analog Input Module AI 2 x 0/4...20 mA HART



KEMA
REGISTERED QUALITY



(1) **EG-BAUMUSTERPRÜFBESCHEINIGUNG**

(2) Geräte und Schutzsysteme zur bestimmungsgemäßen Verwendung in explosionsgefährdeten Bereichen - Richtlinie 94/9/EG

(3) EG-Baumusterprüfbescheinigung Nummer: **KEMA 97ATEX3039 X**

(4) Gerät oder Schutzsystem: **Analogeingabebaugruppe SM331, AI HART 2x0/4...20 mA, Typ 6ES7 331-7TB00-0AB0**

(5) Hersteller: **Siemens AG**

(6) Anschrift: **Werner von Siemens Strasse 50, 92209 Amberg, Deutschland**

(7) Die Bauart dieses Gerätes sowie die verschiedenen zulässigen Ausführungen sind in der Anlage zu dieser Baumusterprüfbescheinigung festgelegt.

(8) KEMA bescheinigt als benannte Stelle Nr. 0344 nach Artikel 9 der Richtlinie 94/9/EG des Rates der Europäischen Gemeinschaften vom 23. März 1994 die Erfüllung der grundlegenden Sicherheits- und Gesundheitsanforderungen für die Konzeption und den Bau von Geräten und Schutzsystemen zur bestimmungsgemäßen Verwendung in explosionsgefährdeten Bereichen gemäß Anhang II der Richtlinie.

Die Ergebnisse der Prüfung sind in dem vertraulichen Prüfbericht Nr. 73039 festgelegt.

(9) Die grundlegenden Sicherheits- und Gesundheitsanforderungen werden erfüllt durch Übereinstimmung mit:

EN 50014: 1992 + prA1
EN 50020: 1994

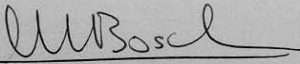
(10) Falls das Zeichen "X" hinter der Bescheinigungsnummer steht, wird auf besondere Bedingungen für die sichere Anwendung des Gerätes in der Anlage zu dieser Bescheinigung hingewiesen.

(11) Diese EG-Baumusterprüfbescheinigung bezieht sich nur auf die Konzeption und den Bau des festgelegten Gerätes oder Schutzsystemes. Falls erforderlich, sind weitere Anforderungen dieser Richtlinie für die Herstellung und das Inverkehrbringen dieser Geräte oder Schutzsysteme zu erfüllen.

(12) Die Kennzeichnung des Gerätes oder Schutzsystemes muß die folgenden Angaben enthalten:

 **II (2) G** **[EEEx ib] IIC**

Arnhem, 19. Februar 1998
im Auftrag der Direktion der N.V. KEMA



C.M. Boschloo
Certification Manager

* Diese Bescheinigung darf nur ungekürzt und unverändert weiterverbreitet werden

N.V. KEMA
Utrechtseweg 310, 6812 AR Arnhem,
Postfach 9035, 6800 ET Arnhem, Die Niederlande
Telefon +31 26 3 56 27 46, Telefax +31 26 3 51 01 78

egb.ges
970820

Blatt 1/3



(13)

ANLAGE

(14)

zur EG-Baumusterprüfbescheinigung KEMA 97ATEX3039 X

(15) **Beschreibung**

Die Analogeingabebaugruppe SM331, AI HART 2x0/4...20 mA, Typ 6ES7 331-7TB00-0AB0 ist eine steckbare Pheripheriebaugruppe aus dem Automatisierungssystem SIMATIC S7-300 und enthält zwei Eingangs- und Meßumformerversorgungsstromkreise zum Anschluß von bescheinigten eigensicheren Meßumformern.

Umgebungstemperaturbereich 0 °C ... +60 °C.

Elektrische Daten

Versorgungs- und Busstromkreise..... nicht eigensicherer Stromkreis, geeignet zum Anschluß an elektrischen Betriebsmittel mit einer Betriebsspannung bis zu 250 V.
(Klemmen 1 und 20)

Ein- und Ausgangsstromkreis in Zündschutzart Eigensicherheit EEx ib IIC mit folgenden Höchstwerten:
(Klemmen 3 bis 6 und 12 bis 15)

$$\begin{aligned} U_o &= 29,6 \text{ V} \\ I_o &= 102 \text{ mA} \\ P_o &= 553 \text{ mW} \end{aligned}$$

Höchstzulässige äußere Kapazität $C_o = 62 \text{ nF}$

Höchstzulässige äußere Induktivität $L_o = 3 \text{ mH}$

Die eigensicheren Ein- und Ausgangsstromkreise sind von den nichteigensicheren Versorgungs- und Busstromkreisen bis zu einer Summe der Scheitelwerte der Nennspannungen von 375 V sicher galvanisch getrennt.

Stückprüfung

Der Transformator soll, vor dem Einbau in dem Betriebsmittel, einer Prüfspannung von 2500 V während einer Minute zwischen der primären und sekundären Wicklung widerstehen.

(16) **Prüfbericht**

KEMA Nr. 73039

(17) **Besondere Bedingungen**

Die Analogeingabebaugruppe ist außerhalb des explosionsgefährdeten Bereiches in ein geeignetes Gehäuse einzubauen das mindestens die Schutzart IP20 nach EN 60529 gewährleistet.

Die Analogeingabebaugruppe ist innerhalb des explosionsgefährdeten Bereiches in ein geeignetes Gehäuse einzubauen. Diese Zusammenstellung muß gesondert geprüft werden.

Blatt 2/3



(13) **A N L A G E**

(14) **zur EG-Baumusterprüfbescheinigung KEMA 97ATEX3039 X**

(17) **Besondere Bedingungen (Fortsetzung)**

Nach Einbau der Analogeingabebaugruppe sollen alle zutreffende Trennungen und Verbindungsleitungen und Anschlüsse die Bedingungen von Abschnitt 6.4 nach EN 50 020 - 1994 entsprechen.

Bei der Installation ist die Leitungskammer einzusetzen oder zwischen den eigensicheren und den nichteigensicheren Stromkreisen ist durch Einfügen einer Trennwand ein Mindestabstand von 50 mm (Fadenmaß) zu realisieren bzw. die Anschlussteile sind zusätzlich zu isolieren.

(18) **Grundlegende Sicherheits- und Gesundheitsanforderungen**

Grundlegende Sicherheits- und Gesundheitsanforderungen welche nicht abgedeckt sind von den unter (9) erwähnten Normen	
Abschnitt	Thema
1.0.4	Umgebungsbedingungen
1.0.5	Kennzeichnung
1.0.6	Betriebsanleitung
1.2.8	Überlastung von Geräten
1.4	Gefahren durch äußere Störungseinflüsse
1.6.4	Risiken durch Anschlüsse

Diese Grundlegende Sicherheits- und Gesundheitsanforderungen sind geprüft worden und die Prüfergebnisse sind festgelegt worden in dem unter (16) erwähnten Prüfbericht.

(19) **Prüfungsunterlagen**

unterschrieben

- | | | |
|--|---|------------|
| 1. Product compliance report ANNEX II, Rev. A |) | |
| |) | |
| 2. Beschreibung (20 Seiten) |) | |
| |) | |
| 3. Zeichnung Nr. 3NEG-720-7209-01 c (10 Blatt) |) | 10.02.1998 |
| 3NEP 720 7204 01 D |) | |
| 4S 720 7202 01 F (6 Blatt) |) | |
| HEXBLO2.DRW-3 |) | |
| HEXBLO3.DRW-3 |) | |
| AIHARTCE.DOC-2 |) | |
| 4. Muster | | |

A.7.1 First Supplement for Analog Input Module AI 2 x 0/4...20 mA HART



1. NACHTRAG

zur EG-BAUMUSTERPRÜFBESCHEINIGUNG Nummer: KEMA 97ATEX3039 X

Hersteller: **Siemens AG**

Anschrift: **Werner von Siemens Strasse 50, 92209 Amberg, Deutschland**

Beschreibung

Die Analogeingabebaugruppe SM331, AI HART 2x0/4...20 mA, Typ 6ES7 331-7TB00-0AB0 darf künftig auch entsprechend den unten aufgeführten Unterlagen gefertigt werden.

Die Änderungen betreffen:

- das Typenschild,
- die elektrischen Daten.

Elektrische Daten

Ein- und Ausgangsstromkreis
(Klemmen 3 bis 6 und 12 bis 15)

in Zündschutzart Eigensicherheit EEx ib IIC mit
folgenden Höchstwerten:

$$\begin{aligned} U_o &= 29,6 \text{ V} \\ I_o &= 99 \text{ mA} \\ P_o &= 553 \text{ mW} \end{aligned}$$

Alle übrigen Daten bleiben unverändert.

Prüfungsunterlagen

unterscriben am

- | | | | |
|------------------|-----------------------------------|---|------------|
| 1. Beschreibung | 26.02.98 (Seite 11,
12 und 20) |) | 26.02.1998 |
| | |) | |
| 2. Zeichnung Nr. | 3NEG-720-7209-01 e (10 Blatt) |) | |

Arnhem, den 2. März 1998
im Auftrag der Direktion der N.V. KEMA

C.M. Boschloo
Certification Manager

Kennzeichen: II (2) G

[EEx ib] IIC

[98.0870]

© Dieser Nachtrag darf nur ungekürzt und unverändert weiterverbreitet werden

Blatt 1/1

A.7.2 EC Declaration of Conformity

SIEMENS

Automatisierungstechnik

Rev. A _____

EC Declaration of Conformity

We

Siemens AG, A&D AS

(name, etc.)

92224 Amberg, Werner-von-Siemens-Straße 50

(address)

declare under our sole responsibility that the product:

Analog input module SM331, AI 2x0/4...20mA HART, Typ 6ES7331-7TB00-0AB0

(name, type or model, lot, batch or serial number, possibly sources and numbers of items)

to which this declaration relates is in conformity with the following standards or normative documents:

EN 50014: 1992 + prA1, EN 50020: 1994, EN 61010: 1994

(title and / or number and date of issue of the standards or other normative documents)

As set out in Article 9 of the European Community guideline 94/9/EC of 23 March 1994, the fundamental safety and health requirements for the conception and construction of devices and protection systems for agreed use in hazardous areas are confirmed in accordance with Appendix II of the guideline.

Name of notified body - EC type evaluation certificate No.:

KEMA 97ATEX3039 X 0344

Amberg, 1998-02-19

(place and date of issue)



Klein/AS E 4

(name and signature or equivalent marking of authorized person)



Scheuerer/AS EWA QSD

Geschäftsgebiet
Industrie-Automatisierungssysteme

Leitung:
Hubert Ovenhausen

Bereichsvorstand
Automatisierungstechnik:
Manfred v. Raven, Vorsitzender
Günther Fritsch
Klaus Wucherer

Briefadresse:
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92209 Amberg

Hausadresse :
Werner-von-Siemens-Straße 50
92224 Amberg
Telefon (09621) 80- 1
Telefax (09621) 80-2136
Telex 631231

Siemens Aktiengesellschaft Vorsitzender des Aufsichtsrats: Hermann Franz Vorstand: Heinrich v. Pierer, Vorsitzender Mitglieder: Karl-Hermann Baumann, Adolf Hüttl, Volker Jung, Edward G. Krubasik, Walter Kunerth, Horst Langer, Werner Maly, Wolfram O. Martinsen, Peter Pribilla, Jürgen Radomski, Claus Weyrich, Günter Wilhelm
Sitz der Gesellschaft: Berlin und München Registergericht: Berlin-Charlottenburg, HRB 12300; München, HRB 6684

A.8 KEMA Certificate of Conformity for Analog Output Module AO 2 x 0/4...20mA HART



KEMA
REGISTERED QUALITY



(1) **EG-BAUMUSTERPRÜFBESCHEINIGUNG**

(2) Geräte und Schutzsysteme zur bestimmungsgemäßen Verwendung in explosionsgefährdeten Bereichen - Richtlinie 94/9/EG

(3) EG-Baumusterprüfbescheinigung Nummer: **KEMA 98ATEX2359 X**

(4) Gerät oder Schutzsystem: **Analogausgabebaugruppe SM332, AO HART 2x0/4...20 mA, Typ 6ES7 332-5TB00-0AB0**

(5) Hersteller: **Siemens AG**

(6) Anschrift: **Werner von Siemens Strasse 50, 92209 Amberg, Deutschland**

(7) Die Bauart dieses Gerätes sowie die verschiedenen zulässigen Ausführungen sind in der Anlage zu dieser Baumusterprüfbescheinigung festgelegt.

(8) KEMA bescheinigt als benannte Stelle Nr. 0344 nach Artikel 9 der Richtlinie 94/9/EG des Rates der Europäischen Gemeinschaften vom 23. März 1994 die Erfüllung der grundlegenden Sicherheits- und Gesundheitsanforderungen für die Konzeption und den Bau von Geräten und Schutzsystemen zur bestimmungsgemäßen Verwendung in explosionsgefährdeten Bereichen gemäß Anhang II der Richtlinie.

Die Ergebnisse der Prüfung sind in dem vertraulichen Prüfbericht Nr. 82359 festgelegt.

(9) Die grundlegenden Sicherheits- und Gesundheitsanforderungen werden erfüllt durch Übereinstimmung mit:

EN 50014 : 1992 + prA1**EN 50020 : 1994**

(10) Falls das Zeichen "X" hinter der Bescheinigungsnummer steht, wird auf besondere Bedingungen für die sichere Anwendung des Gerätes in der Anlage zu dieser Bescheinigung hingewiesen.

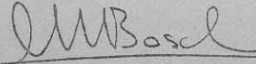
(11) Diese EG-Baumusterprüfbescheinigung bezieht sich nur auf die Konzeption und den Bau des festgelegten Gerätes oder Schutzsystemes. Falls erforderlich, sind weitere Anforderungen dieser Richtlinie für die Herstellung und das Inverkehrbringen dieser Geräte oder Schutzsysteme zu erfüllen.

(12) Die Kennzeichnung des Gerätes oder Schutzsystemes muß die folgenden Angaben enthalten:

**II (2) G**

[EEx ib] IIC

Arnhem, 29. Januar 1999
im Auftrag der Direktion der N.V. KEMA


C.M. Boschloo
Certification Manager

* Diese Bescheinigung darf nur ungekürzt und unverändert weiterverbreitet werden

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**AKKREDITIERT DURCH
DEN NIEDERLÄNDISCHEN
AKKREDITIERUNGSRAT**



egb.ges
97-08-20
Blatt 1/3



(13)

A N L A G E

(14)

zur EG-Baumusterprüfbescheinigung KEMA 98ATEX2359 X**(15) Beschreibung**

Die Analogausgabebaugruppe SM332, AO HART 2x0/4...20 mA, Typ 6ES7 332-5TB00-0AB0 ist eine steckbare Pheripheriebaugruppe aus dem Automatisierungssystem SIMATIC S7-300 und enthält zwei eigensichere Ausgangsstromkreise zum Anschluß von zwei bescheinigten eigensicheren Stellungsreglern.

Umgebungstemperaturbereich 0 °C ... +60 °C.

Elektrische Daten

Versorgungs- und Busstromkreise..... 24 Vdc (20,4 V bis 28,8 V); ca. 0,15 A
(Klemmen 1 und 20) $U_m = 60 \text{ Vdc bzw. } 30 \text{ Vac}$

Ausgangsstromkreise in Zündschutzart Eigensicherheit EEx ib IIC mit
(Klemmen 3, 4 12, 13) folgenden Höchstwerten:

$$\begin{aligned} U_o &= 19 \text{ V} \\ I_o &= 66 \text{ mA} \\ P_o &= 506 \text{ mW} \end{aligned}$$

Höchstzulässige äußere Kapazität $C_o = 230 \text{ nF}$
Höchstzulässige äußere Induktivität $L_o = 7,5 \text{ mH}$

Die eigensicheren Ausgangsstromkreise sind von den nichteigensicheren Versorgungs- und Busstromkreisen bis zu einer Summe der Scheitelwerte der Nennspannungen von 60 Vdc bzw. 30 Vac sicher galvanisch getrennt.

Stückprüfung

Der Transformator soll, vor dem Einbau in dem Betriebsmittel, die Spannungsprüfung von Abschnitt 8.1.5 und 11.2 nach EN 50020 - 1994 widerstehen.

(16) Prüfbericht

KEMA Nr. 82359

(17) Besondere Bedingungen

Die Analogausgabebaugruppe ist außerhalb des explosionsgefährdeten Bereiches in ein geeignetes Gehäuse einzubauen, das mindestens die Schutzart IP20 nach EN 60529 gewährleistet.

Die Analogausgabebaugruppe ist innerhalb des explosionsgefährdeten Bereiches in ein geeignetes Gehäuse einzubauen. Diese Zusammenstellung muß gesondert geprüft werden.

Blatt 2/3



(13)

ANLAGE

(14)

zur EG-Baumusterprüfbescheinigung KEMA 98ATEX2359 X

(17) **Besondere Bedingungen** (Fortsetzung)

Nach Einbau der Analogausgabebaugruppe sollen alle zutreffenden Trennungen und Verbindungsleitungen und Anschlüsse den Bedingungen von Abschnitt 6.4 nach EN 50020 - 1994 entsprechen.

Bei der Installation ist die Leitungskammer einzusetzen oder zwischen den eigensicheren und den nichteigensicheren Stromkreisen ist durch Einfügen einer Trennwand ein Mindestabstand von 50 mm (Fadenmaß) zu realisieren bzw. die Anschlussteile sind zusätzlich zu isolieren.

Es soll sicher gestellt werden daß die metallische geerdete Rückwand-Profileschiene eine feste elektrische Verbindung mit dem Potentialausgleichssystem bildet.

(18) **Grundlegende Sicherheits- und Gesundheitsanforderungen**

Grundlegende Sicherheits- und Gesundheitsanforderungen welche nicht abgedeckt sind von den unter (9) erwähnten Normen	
Abschnitt	Thema
1.0.5	Kennzeichnung
1.0.6 (b)	Betriebsanleitung

Diese Grundlegende Sicherheits- und Gesundheitsanforderungen sind geprüft worden und die Prüfergebnisse sind festgelegt worden in dem unter (16) erwähnten Prüfbericht.

(19) **Prüfungsunterlagen**

unterscrieben

1. Product compliance report ANNEX II, Rev. A)
2. Beschreibung (24 Seiten))
3. Zeichnung Nr. 3NEG-7207229-01 (10 Blatt))
- 3NEP720722401 (13 Blatt))
- 3NEP720722201 (7 Blatt))
4. Muster

21.12.1998

Blatt 3/3

A.8.1 EC Declaration of Conformity

SIEMENS

Automatisierungs- und Antriebstechnik

Rev. A _____

EC Declaration of Conformity

We

Siemens AG, A&D AS

(name, etc.)

92224 Amberg, Werner-von-Siemens-Straße 50

(address)

declare under our sole responsibility that the product:

Analog output module SM332, A0 2x0/4...20mA HART, Typ 6ES7332-5TB00-0AB0

(name, type or model, lot, batch or serial number, possibly sources and numbers of items)

to which this declaration relates in conformity with the following standards or normative documents:

EN 50014: 1992 + prA1, EN 50020: 1994, EN 61010: 1994

(title and / or number and date of issue of the standards or other normative documents)

As set out in Article 9 of the European Community guideline 94/9/EC of 23 March 1994, the fundamental safety and health requirements for the conception and construction of devices and protection systems for agreed use in hazardous areas are confirmed in accordance with Appendix II of the guideline.

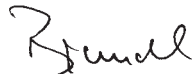
Name of notified body – EC type evaluation certificate No.:

KEMA 98ATEX2359 X 0344

Amberg, 1999-02-10

(place and date of issue)


Klein/AS E 4


Brandl/AS EWA QSD

(name and signature or equivalent marking of authorized person)

Industrie-Automatisierungssysteme

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Sitz der Gesellschaft: Berlin und München - Registergericht: Berlin-Charlottenburg, HRB 12300; München, HRB 6684

Safety Standards, FM Approval

B

In this appendix

On the following pages you will find:

- The Ex-relevant safety standards and other safety regulations
- FM approval

Safety standards used

The following safety standards apply to all EX modules:

- EN50014 (1977 + A1 .. A5):
Electrical equipment for hazardous locations:
General specifications.
- EN50020 (1977 + A1.. A5):
Electrical equipment for hazardous locations:
Intrinsic safety "i".
- DIN EN 61010 (Teil 1 v. 3/94):
Section 6.3.1 and Appendix D.2 Table D.6
Safety requirements for electrical measuring, control and laboratory equipment.
- DIN EN 61131 (Teil 2 v. 5/95):
Programmable logic controllers, operational equipment requirements and testing.
- DIN EN 60204 (Teil 1 v. 6/93):
Electrical equipment of machines:
General requirements.

The designations of safety characteristic values have been adapted in the course of harmonization of the standards EN50012 .. EN50020. The most important characteristic data for the relevant operational equipment are assigned as follows:

U_o, U_{max}, U_a	→ U_o	Maximum output voltage
I_o, I_a, I_k	→ I_o	Maximum output current
U_m	→ U_m	Maximum r.m.s. power-frequency voltage or maximum direct voltage
C_o, C_a	→ C_o	Maximum external capacitance
L_o, L_a	→ L_o	Maximum external inductance
P, P_{max}	→ P_o	Maximum output power
C, C_i	→ C_i	Maximum internal capacitance
L, L_i	→ L_i	Maximum internal inductance

FM approval

The assemblies are identified as follows for the purpose of arranging the explosion protection classes in groups for the American market:



CL I, DIV 2, GP A, B, C, D, T 4, $T_a \leq 60^\circ\text{C}$

APPROVED

Explosive atmospheres can occur temporarily in CL I, DIV 2. If modules are operated in this zone, they must **not be unplugged or connected during operation.**

Bibliography

C

In this appendix

On the following pages you will find the bibliography of general literature which is relevant to the use of Ex I/O Modules in the system environment.

S7-300

- /70/ Manual: *S7-300 Programmable Controller, Hardware and Installation*
- /71/ Reference Manual: *S7-300 and M7-300 Programmable Controllers, Module Specifications*
- /72/ Instruction List: *S7-300 Programmable Controller, CPU 312 IFM, 314 IFM, 313, 314, 315-2DP*

S7-400

- /100/ Manual: *S7-400, M7-400 Programmable Controllers, Hardware and Installation*
- /101/ Reference Manual: *S7-400, M7-400 Programmable Controllers, Module Specifications*
- /102/ Reference Guide: *S7-400 Instruction List, CPU 412, 413, 414, 416, 417*

ET-200

- /140/ Manual: *ET 200M Distributed I/O Device*
- /150/ Manual: *Automation Systems S7-300, M7-300, ET 200M, Principles of Intrinsically-Safe Design, vol.1*
- /150/ Reference Manual: *Automation Systems S7-300, M7-300, ET 200M, I/O Modules with Intrinsically-Safe Signal, vol.2*

STEP 7 Standard Package/Software

- /231/ Manual: *Configuring Hardware and Communication Connections, STEP 7 V5.0*
- /232/ Manual: *Statement List (STL) for S7-300 and S7-400, Programming*
- /233/ Manual: *Ladder Logic (LAD) for S7-300 and S7-400, Programming*
- /234/ Manual: *Programming with STEP 7 V5.0*
- /235/ Reference Manual: *System Software for S7-300 and S7-400, System and Standard Functions*
- /236/ Manual: *Function Block Diagram (FBD) for S7-300 and S7-400, Programming*

M7-300/400

- /80/ Manual: *M7-300 Programmable Controller, Hardware and Installation*
- /280/ Programming Manual: *System Software for M7-300 and M7-400, Program Design*
- /281/ Reference Manual: *System Software for M7-300 and M7-400, System and Standard Functions*
- /282/ User Manual: *System Software for M7-300 and M7-400, Installation and Operation*

Programming Devices

- /650/ Manual: *PG 720 Programming Device*
- /651/ Manual: *PG 740 Programming Device*
- /652/ Manual: *PG 760 Programming Device*

Manuals on CD only

- /803/ Reference Manual: *Standard Software for S7-300 and S7-400, Standard Functions Part 2 (CD only)*
- /804/ Package: *Field Technology, System Description*

Glossary

A

AS → Programmable logic controller (PLC)

ATEX 100a AT for atmosphere, EX for explosive. The suffix 100a refers to the legal basis, article 100a of the EEC agreement.

B

Backplane bus The backplane bus is a serial data bus which enables the modules to communicate with one another and which supplies them with the required power. The modules are connected together by means of bus connectors.
The → I/O bus is part of the backplane bus.

Backplane bus, active Backplane bus of the distributed I/O → ET 200M which is constructed from active bus modules. This is the precondition for a structure in which the use “Insert and Remove” modules is allowed.
Baud rates between 9.6 kbaud and 12 Mbaud are possible for the ET 200.

Bus Common transmission path to which all devices are connected; it has two defined ends.
The bus used for the ET 200 is either a two-wire cable or an optical fiber cable.

Bus node A device that can send, receive or amplify data via the → bus, for example, → DP master, → DP slave, RS 485 repeater, or an active star coupler.

C

CELENEC	“Comité Européen de Normalisation Electrotechnique.” The countries of the European Union as well as Norway and Switzerland are members.
Chassis ground	The chassis ground comprises all interconnected inactive parts of a device which cannot carry any dangerous touch voltage even in the event of a fault.
Client	A client can request a → server to perform a service. A client can be a program, a central processing unit (CPU), or a station (for example, a PC). The exchange of data between client and server can take place, for example, via → PROFIBUS_DP, in accordance with the → master-slave principle. If there are several clients, the data exchange between client and server can be coordinated by allocating a separate transfer area to each client.
Configuration	Assignment of modules to subracks/slots and addresses. A distinction is made between the actual configuration (modules which are actually connected) and the nominal configuration. The nominal configuration is defined by the user in STEP 7 or COM PROFIBUS (or COM ET 200 Windows). The operating system is thus able to detect incorrectly connected modules when they are started up.
Configuration, central	A configuration is considered to be central if the process I/O units and the central processing unit are accommodated either in the same subrack or in extension units in the same or an adjacent cubicle.
Configuration, distributed	A configuration is considered to be distributed if the process I/O units are not accommodated directly next to the central processing unit either in the same subrack or in the same or an adjacent cabinet, but are rather physically separate from it and connected together by means of a communication bus (e.g. a field bus).
CPU	Central processing unit of the S7 automation system, comprising a processor, an arithmetic and logic unit, a memory, an operating system and an interface for the programming unit.

D

Diagnostic interrupt

Modules with a diagnostics capability report any system faults or errors they have identified to the → CPU by means of diagnostic interrupts.

In SIMATIC S7/M7: When a fault (e.g. a wire break) is detected or when it disappears again, the module outputs a diagnostic interrupt, providing diagnostics have been enabled for it. The CPU stops processing the user program and any events with lower priority classes, and processes the diagnostic interrupt block instead (OB 82).

In SIMATIC S5: The diagnostic interrupt is simulated as part of the device-specific diagnostics. You can detect faults (e.g. a wire break) by cyclically interrogating the diagnostic bits of this diagnostics.

Diagnostic buffer

The diagnostic buffer is a backed-up memory area in the CPU where diagnostic events are stored in the order they occur.

Diagnostics

Detection, localization, classification, indication and other forms of evaluation of errors, faults, malfunctions and interrupts.

'Diagnostics' includes monitoring functions which are activated automatically whenever the system is operational. The system availability is increased as a result, and commissioning and down times are reduced.

The ET 200 incorporates various diagnostic functions, from information about the DP slave which has reported the diagnostics to monitoring of individual channels.

Distributed I/O device

An input/output unit which is installed not in the central processing unit, but at a decentralized location remote from it, e.g.:

- ET 200M, ET 200B, ET 200C, ET 200U
- DP/AS-I link
- S5-95U with PROFIBUS-DP slave interface
- Other DP slaves from Siemens or equivalent vendors

The distributed I/O devices are connected to the DP master by means of the PROFIBUS-DP.

DP address

Each bus device must be given a DP address, to enable it to be uniquely identified on the PROFIBUS-DP.

The DP address of the PC/PU or the handheld ET 200 is "0".

The DP master and the DP slaves have DP addresses between 1 and 125.

DP master

A → master which complies with EN 50170, Volume 2, PROFIBUS, is referred to as a DP master.

DP slave A → slave which is operated on the PROFIBUS with the PROFIBUS-DP protocol and which complies with EN 50170, Volume 2, PROFIBUS, is referred to as a DP slave.

DP standard The bus protocol of the ET 200 distributed I/O system; it complies with EN 50170, Volume 2, PROFIBUS.

E

Error handling via OB When the operating system detects an error (for example, STEP 7 access error), it calls the specific → organization block (error OB) for this error, where the further response of the → CPU can be specified.

Error indication One of the possible responses of the operating system to a → delay error. The other possible responses are: → error response in the user program, STOP status of the IM 153.

ET 200 The ET 200 distributed I/O system with the PROFIBUS-DP protocol is a bus designed for connecting distributed I/O units to a CPU or a suitable DP master. ET 200 is distinguished by its fast response times, since only small volumes of data (bytes) are transferred.

ET 200 is based on EN 50170, Volume 2, PROFIBUS.

ET 200 operates according to the master-slave principle. The DP master may be an IM 308-C master interface, for example, or a CPU 315-2 DP.

The DP slaves may be distributed I/O units (ET 200B, ET 200C, ET 200M, ET 200U) or DP slaves from Siemens or other vendors.

F

Field Either, an area of a plant outside the control room where measured values can be acquired through communication or manipulated values can be sent to actuators.

Or part of a message, for example an address field or command field, which has been allocated a particular function. The size or other rules for the interpretation of each field are part of the protocol specification.

Field device A → transducer which is located in the → field and exchanges data with the CPU via communication.

FM	Factory Mutual. Quality assurance system in the USA.
Frequency shift keying	The FSK procedure is a data modulation technique which is suitable for data transport via normal cables. Two audio frequencies are used to encode the binary values "0" and "1" in the frequency range 300 - 3000 Hz. In the → HART protocol the FSK signal is transmitted via a current loop.
FSK	Frequency shift keying
G	
Ground	The conductive earth whose potential can be assumed to be zero at any point. In the vicinity of → grounding electrodes, the ground may have a potential other than zero. The term "reference ground" is frequently used in this connection.
Grounding electrode	One or more conductive part(s) which make good contact with the ground.
H	
HART	Highway Addressable Remote → Transducer. HART is a registered trademark of the → HART Communication Foundation.
HART analog modules	Analog modules (→ analog input or → analog output) which can carry out HART communication in addition to their analog value. HART analog modules can be used as a → HART interface for the HART field devices.
HART commands	The HART field device works as a slave and is controlled by the master by means of HART commands. The master sets the → HART parameters or requests data in the form of → HART responses.
HART communication	Transfer of data between a master (for example, HART analog module) and a slave (→ HART field device) via the → HART protocol.
HART Communication Foundation	The HART Communication Foundation (HCF) was founded in 1993 to disseminate information on the HART protocol and to develop the protocol further. The HCF is a non-profit-making organization which is financed by its members.

HART field device	Smart field device which has special functions in accordance with the HART norm. This enables the field device to understand → HART communication.
HART hand-held device	The HART hand-held device is the original parameter assignment tool produced by Fisher-Rosemount Ltd. for → HART field devices. It is connected directly to the ports of the field devices. The HART hand-held device is used to set the → HART parameters.
HART interface	Part of system via which a → HART field device can be connected. The HART interface represents the master for the field device. As far as the system is concerned, however, the HART interface is a slave which can be fed by various masters on the system. The → HART parameter assignment tool is one example of a master. The PLC itself is another master.
HART parameter assignment tool	The HART parameter assignment tool enables you to set the → HART parameters. It can be a HART hand-held device or a parameter assignment tool which is integrated into the system, for example, SIMATIC SIPROM.
HART parameters	The HART parameters describe the configurable properties of → HART field devices which can be modified via the → HART protocol. The settings can be made with a → HART parameter assignment tool.
HART protocol	The → HART protocol is the industrial standard for extended communication with smart → field devices. It contains the → HART commands and → HART responses.
HART responses	The HART field device transfers data at the request of the master. These data are measurement results or manipulated values, or the values of → HART parameters. A HART response always contains status information in the form of → HART status bytes.
HART signal	Analog signal on a current loop of 4 - 20 mA, where the sine waves for the HART protocol are superimposed with the aid of the → FSK procedure - 1200 Hz for the binary "1" and 2200 Hz for the binary "0."
HART status byte	The status information which consists of the 1st and 2nd status byte of the HART response and which the HART field device uses to provide information on the HART communication, the receipt of the HART command, and the device status.

HART transfer area Area of data records which is provided for writing HART commands and reading HART responses. The HART transfer area consists of data records. Each → client is allocated its own area of data records, via which the → server and it can exchange data.

HCF → HART Communication Foundation

I

Interrupt The operating system of the → CPU has 10 different priority classes which control execution of the user program. These priority classes include interrupts, for example, hardware interrupts. When an interrupt occurs, the operating system automatically calls a corresponding organization block where the user can program the reaction desired.

I/O bus Part of the → S7-300 backplane bus in the automation system; it is optimally designed for fast signal exchanges between the IM 153 and the signal modules. Both → user data (e.g. the digital input signals of a signal module) and system data (e.g. the default parameter records of a signal module) are transferred on the I/O bus.

Isolated In isolated input/output modules, the → reference potentials of the control circuit and the load circuit are galvanically isolated from one another, for example by means of optocouplers, relay contacts, or transformers. The input/output circuits can be connected to a common potential.

K

KEMA Product Certification Center.

L

Load power pack Power supply for the signal and function modules and the process I/O connected to them.

M

Master	A device which is able to send data to other devices and request data from them (= active device) when in possession of the token. Examples of → DP masters include the CPU 315-2 DP and the IM 308-C.
Master class 1	Master responsible for the exchange of user data. Master class 1 is also used for parameter assignment and diagnostics of the distributed I/O.
Master class 2	Master responsible for control, setup and configuration tasks, for example, parameter assignment and diagnostics of the field devices which are connected to the distributed I/O.
Master-slave principle	Bus access method whereby only one device at a time is the → DP master and all the other devices are → DP slaves.
Measuring point identifier	Unique identifier for the measuring point, consisting of 8 characters. It is stored in the → HART field device and can be changed and displayed using HART commands.
Modem	A modem (MODulator / DEModulator) is a device which converts binary digital signals into → FSK signals and vice versa. A modem does not encode data, rather it changes the physical form of the signals.
Monodrop	In a monodrop communication system a maximum of two devices are connected on the same transmission link, for example, a channel from the HART analog module and → smart field device. The → HART protocol and the analog signal can be used simultaneously for this procedure.
Module parameters	Module parameters are values that can be set by the user in order to control the behavior of a module. They can be either static or dynamic.

N

Non-isolated	In non-isolated input/output modules, there is an electrical connection between the reference potentials of the control circuit and the load circuit.
---------------------	--

O

OB → Organization block

Organization blocks Organization blocks (OBs) represent the interface between the operating system of the CPU and the user program. The sequence of user program processing is defined in the organization blocks.

P

Parameter assignment Setting values to control the behavior of a module or a → field device.

Parameter assignment tool A software tool which can be used to set the → parameters, for example, of a → smart field device.

Parameter, dynamic Unlike static parameters, the dynamic parameters of a module can be altered online in the user program.

Parameters, static Unlike dynamic parameters, the static parameters of a module can only be altered in STEP 7 or COM PROFIBUS and not in the user program.

Primary variable Variable for the chief measured value of a → HART analog input, for example, pressure. Other measurements can also be implemented for the → HART field devices, for example, temperature. The results are stored in the secondary variable, tertiary variable, quaternary variable, etc. In the case of a → HART analog output, the primary variable contains the manipulated value.

Hardware interrupt	A hardware interrupt is tripped by interrupt-capable S7-300 modules as a result of a specific event occurring during the process. This interrupt is reported to the CPU. The assigned organization block is then processed according to the interrupt priority. In SIMATIC S7/M7: An operating range is defined by parameterizing an upper limit value and a lower limit value, for example. If the process signal (e.g. temperature) of an analog input module leaves this range, the module outputs a hardware interrupt, providing hardware interrupts have been enabled for it. The CPU stops processing the user program and any events with lower priority classes and processes the hardware interrupt block instead (OB 40). In SIMATIC S5: The hardware interrupt is simulated as part of the device-specific diagnostics. You can detect hardware interrupts (e.g. an overrange condition) by cyclically interrogating the diagnostic bits of this diagnostics.
Process image	A special memory area in the automation system. The signal states of the input modules are copied to the process image of the inputs at the start of the cyclic program. At the end of the cyclic program, the process image of the outputs is copied to the output modules as the signal state.
Programmable logic controller	A programmable logic controller (PLC) is an electronic control circuit whose automation function is stored as a software program. Accordingly, the configuration and wiring of the PLC are not dependent on the automation assignment. The PLC is constructed as a computer; it consists of a CPU module with memory, → I/O modules and an internal bus system. The I/O modules and the programming language are tailored to the needs of automation programs.
PROFIBUS	PROcess FIeld BUS, the German standard for this type of bus, which is defined in EN 50170. It lays down the functional, electrical and mechanical characteristics of a bit-serial field bus system. PROFIBUS is a bus system which enables PROFIBUS-compatible automation systems and I/O units at the cell and field levels to be networked together. It operates with the following protocols: DP (= distributed I/O), FMS (= field bus message specification) and TF (= process function).
PROFIBUS-DP	PROFIBUS bus system with the DP protocol. DP is the German abbreviation for distributed I/O. The ET 200 distributed I/O system is based on EN 50 170, Volume 2, PROFIBUS.
PTB	Physikalisch-Technische Bundesanstalt. Product certification center in Germany.

R

Reference potential	Potential on which examinations and/or measurements of the voltages in specific circuits are based.
Response time	The average time which elapses between a change at an input and the change at the corresponding output.
Run-time errors	Errors that occur in the → programmable logic controller (that is, not in the process) during execution of the user program.

S

Server	A server performs a service on request. A server can be, for example, a program, a module, or a station (for example, a PC). The exchange of data between client and server can take place, for example, via → PROFIBUS_DP in accordance with the → master-slave principle.
SFC	→ System function
Signal module	Signal modules (SM) form the interfaces between the process and the automation system. There are digital input and output modules (input/output module, digital) and analog input and output modules (input/output module, analog).
Slave	A slave is only allowed to exchange data with a → master if it has been requested to do so. All DP slaves, such as ET 200B, ET 200C, ET 200M, etc., are considered to be slaves.
Smart field device	A complex field device containing a micro processor. Its settings can be set by the control room using a corresponding → parameter assignment tool.
System diagnostics	System diagnostics comprises the recognition, evaluation and signaling of errors which occur within the programmable logic controller. Examples of such errors include: program errors or module failures. System errors can be indicated via LEDs or displayed in <i>STEP 7</i> .
System function	A system function (SFC) is a function integrated in the operating system of the → CPU, which can be called in the STEP 7 user program if required.

Substitute value A value which is output to the process if a signal output module is faulty, or which is used in the user program instead of a process value if a signal input module is faulty. Substitute values can be defined by the user (e.g. 'hold last value').

S7-300 backplane bus A serial data bus which is used by the modules to communicate with one another and which supplies them with the necessary voltage. The connections between the modules are made with bus connectors.

T

Terminating resistance A resistance for power matching on the bus cable; terminating resistances are always required at the end of a cable or segment.

The terminating resistances of the ET 200 are connected and disconnected in the → bus connector.

Time-out If an expected event does not occur within a specified period of time, this time is known as a "time-out." In the → HART protocol there are time-outs for the response of a slave to a message from the master, and for the pause after each transaction.

Transducer Sensor (measuring transducer) or actuator (signal control element). A transducer can be implemented by a → smart field device.

Transmission rate The transmission rate is the speed at which data are transmitted and indicates the number of bits transmitted per second (transmission rate = bit rate).

Transmission rates of 9.6 Kbps to 12 Mbps are possible for the → ET 200.

U

User data User data can be exchanged between a CPU and a signal module, a function module, or a communications processor via process image or via direct access. User data can be digital and analog input/output signals from signal modules or control and status information from function modules.

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